LE910R1

AT Commands Reference Guide

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Applicability Table

Table 1: Applicability Table - Products

Products
LE910R1-EU
LE910R1-EA
LE910R1-EAG

Table 2: Applicability Table - Software

Software
44.00.xx6

Introduction

Scope

This document is aimed to provide a detailed specification and a comprehensive listing for reference for the entire AT command set.

Audience

Readers of this document should be familiar with Telit modules and their ease of controlling by means of AT Commands.

Contact Information, Support

For technical support and general questions please e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use:

https://www.telit.com/contact-us/

Product information and technical documents are accessible 24/7 on our web site:

https://www.telit.com

Conventions

Note:

Provide advice and suggestions that may be useful when integrating the module.

Danger:

This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

ESD Risk:

Notifies the user to take proper grounding precautions before handling the product.

Warning:

Alerts the user on important steps about the module integration.

	Danger: This information MUST be followed or catastrophic equipment failure or personal injury may occur.
	Warning: Alerts the user on important steps about the module integration. If these points are not followed, the module and end user equipment may fail or malfunction.
	Note/Tip: This section provides all information related to the AT command involved. Each note can provide a different level of information: danger, caution/warning and tip/information
	Note/Tip: Provides advice and suggestions that may be useful when integrating the module.
	Set: This section provides all information related to SET functionality of the AT command involved. If there is strictly and relevant SET information, it can be found at the end of the section.

Read: This section provides all information related to READ functionality of the AT command involved. If there is strictly and relevant READ information, it can be found at the end of the section.

Test: This section provides all information related to TEST functionality of the AT command involved. If there is strictly and relevant TEST information, these can be found at the end of the section.

Additional info: This section provides any kind of additional and useful information related to the AT command section as well as command exceptions or special behavior cases.

Reference: This section provides useful references (standards or normative) related to the AT command involved.

Example: This section provides useful examples related to the AT command involved.

All dates are in ISO 8601 format, that is YYYY-MM-DD

Terms and Conditions

Refer to

https://www.telit.com/terms-conditions/

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Commands

The Telit family of wireless modules can be controlled via the serial interface using the standard AT commands.

It complies with:

- Hayes standard AT command set, to maintain the compatibility with existing software programs.
- 3GPP TS 27.007 specific AT command and GPRS specific commands.
- 3GPP TS 27.005 specific AT commands for SMS (Short Message Service) and CBS (Cell Broadcast Service)

Moreover, the Telit family of wireless modules also supports Telit proprietary AT commands for special purposes.

The following is a description of how to use the AT commands with the Telit family of wireless modules.

Note:

The AT is an ATTENTION command and is used as a prefix to other parameters in a string. The AT command, combined with other parameters, can be set up in the communications package or typed manually as a command line instruction.

Definitions

The following syntactic definitions apply:

<CR> Carriage return character. It is the command line and terminator character of the result code, whose value, in decimal ASCII, ranges between 0 and 255, as specified within parameter S3. The default value is 13.

<LF> Linefeed character. It is the character recognized as line feed character. Its value, in decimal ASCII ranges between 0 and 255, as specified within parameter S4. The default value is 10. The line feed character is output after the carriage return character if detailed result codes are used (V1 option). Otherwise, if numeric format result codes are used (V0 option), it will not appear in the result codes.

<...> A name enclosed in angle brackets is a syntactic element. These elements are not displayed on the command line.

[...] The optional sub parameter of a command or an optional part of the TA information response is enclosed in square brackets. The brackets themselves are not displayed on the command line. When the sub parameter is not given in AT commands that have a Read command, ts new value is equal to its previous value. In AT commands that do not store the values of any of their subparameters and therefore do not have a Read command (called action type commands), an action should be performed according to the recommended default setting of the subparameter.

AT Commands Syntax

The syntax rules followed by Telit implementation of either Hayes AT commands, GSM commands are very similar to those of standard basic and extended AT commands.

There are two types of extended command:

- Parameter type commands. This type of commands may be "set" (to store a value or values for later use), "read" (to determine the current value or stored values), or "tested" (to determine ranges of values supported). Each of them has a test command (trailing =?) to provide information about the type of its subparameters; they also have a Read command (trailing ?) to check the current values of subparameters.
- Action type commands. This type of command may be "executed" or "tested".
  - "executed" to invoke a function of the equipment, which generally involves more than the simple storage of a value for later use
  - "tested" to determine:
    - if subparameters are associated with the action, the ranges of subparameter values that are supported. If the command has no subparameters, issuing the corresponding Test command (trailing =?) raises the result code "ERROR".

      Note: issuing the Read command (trailing ?) causes the command to be executed.
    - whether the equipment implements the Action Command or not. In this case issuing the correspondent Test command (trailing =?) returns the OK result code). If subparameters are associated with the action, the ranges of subparameter values that are supported.

Action commands do not store the values of any of their possible subparameters.

Moreover:

The response to the Test Command (trailing =?) may be modified in the future by Telit to allow for the description of new values/functionalities.

If all the subparameters of a parameter type command +CMD are optional, issuing AT+CMD=<CR> causes the result code to be returned to OK and the previous values of the omitted subparameters to be retained.

String Type Parameters

A string, either enclosed between quotes or not, is a valid string type parameter input. According to V25.ter space characters are ignored on the command line and may be used

freely for formatting purposes, unless they are embedded in numeric or quoted string constants.

Therefore, a string containing a space character must be enclosed between quotes to be considered a valid string type parameter (e.g. typing AT+COPS=1,0,"A1" is the same as typing AT+COPS=1,0,A1; typing AT+COPS=1,0,"A BB" is different from typing AT+COPS=1,0,A BB).

A string is always case-sensitive.

A small set of commands always requires to write the input string parameters in quotes: this is explicitly reported in the specific descriptions.

Command Lines

A command line is made up of three elements: the prefix, the body and the termination character.

The command line prefix consists of the characters "AT" or "at", or, to repeat the execution of the previous command line, the characters "A/" or "a/" or AT#/ or at#/.

The termination character may be selected by a user option (parameter S3), the default being <CR>.

The basic structures of the command line are:

- ATCMD1<CR> where AT is the command line prefix, CMD1 is the body of a basic command (note: the name of the command never begins with the character "+") and <CR> is the command line terminator character
- ATCMD2=10<CR> where 10 is a subparameter
- AT+CMD1;+CMD2=, ,10<CR> These are two examples of extended commands (note: the name of the command always begins with the character "+"2F2F). They are delimited by semicolon. In the second command the subparameter is omitted.
- +CMD1?<CR> This is a Read command for checking current subparameter values
- +CMD1=?<CR> This is a Test command for checking possible subparameter values

Note:

The set of proprietary AT commands differs from the standard one because the name of each of them begins with either "@", "#", "$" or "*". Proprietary AT commands follow the same syntax rules as extended commands.

These commands might be performed in a single command line as shown below:

ATCMD1 CMD2=10+CMD1;+CMD2=, ,10;+CMD1?;+CMD1=?<CR>

However, it is always preferable to separate basic and extended commands in different command lines. Furthermore, it is recommended to avoid placing several action commands in the same command line because if one of them fails, then an error message is received but it is not possible to determine which one of them has failed the execution.

If command V1 is enabled (detailed responses codes) and all commands in a command line have been performed successfully, the result code <CR><LF>OK<CR><LF> is sent from the TA to the TE, if subparameter values of a command are not accepted by the TA, or the command itself is invalid, or the command cannot be executed for some reason, result code <CR><LF>ERROR<CR><LF> is sent and no subsequent commands in the command line are processed.

If command V0 is enabled (numeric responses codes), and all commands in a command line have been executed successfully, the result code 0<CR> is sent from the TA to the TE. If subparameter values of a command are not accepted by the TA, or the command itself is invalid, or command cannot be executed for some reason, result code 4<CR> and no subsequent commands in the command line are processed.

In the case of errors depending on ME operation, ERROR (or 4) response may be replaced by +CME ERROR: <err> or +CMS ERROR: <err>.

Note:

The command line buffer accepts a maximum of 400 characters. If this number is exceeded, none of the commands will be executed, and TA returns ERROR.

ME Error Result Code - +CME ERROR: <err>

This is NOT a command; it is the error response to +Cxxx 3GPP TS 27.007 commands.

Syntax: +CME ERROR: <err>

Parameter: <err> - error code can be either numeric or detailed (see +CMEE). The possible values of <err> are reported in the table:

Note:

"Numeric Format" Not all modules support the error codes shown in the table.

Note:

"Verbose Format" There could be small variations in the message depending on the module in use.

Table 3: ME Error Result Code

Numeric Format	Verbose Format
0	phone failure
1	no connection to phone
2	phone adaptor link reserved
3	operation not allowed
4	operation not supported
5	PH-SIM PIN required
6	PH-FSIM PIN required
7	PH-FSIM PUK required
10	SIM not inserted
11	SIM PIN required
12	SIM PUK required
13	SIM failure
14	SIM busy
15	SIM wrong
16	incorrect password
17	SIM PIN2 required
18	SIM PUK2 required
20	memory full
21	invalid index
22	not found
23	memory failure

24	text string too long
25	invalid characters in text string
26	dial string too long
27	invalid characters in dial string
30	no network service
31	network timeout
32	network not allowed - emergency calls only
34	numeric parameter instead of text parameter
35	text parameter instead of numeric parameter
36	numeric parameter out of bounds
37	text string too short
38	The GPIO Pin is already used
40	network personalization PIN required
41	network personalization PUK required
42	network subset personalization PIN required
43	network subset personalization PUK required
44	service provider personalization PIN required
45	service provider personalization PUK required
46	corporate personalization PIN required
47	corporate personalization PUK required
49	EAP method not supported

50	Invalid EAP parameter
51	Parameter length error for all Auth commands
52	Temporary error for all Auth command
53	not verified hidden key
100	unknown
103	Illegal MESSAGE
106	Illegal ME
107	GPRS services not allowed
111	PLMN not allowed
112	Location area not allowed
113	Roaming not allowed in this location area
132	service option not supported
133	requested service option not subscribed
134	service option temporarily out of order
148	unspecified GPRS error
149	PDP authentication failure
150	invalid mobile class
257	network rejected request
258	retry operation
259	invalid deflected to number

260	deflected to own number
261	unknown subscriber
262	service not available
263	unknown class
264	unknown network message
273	Minimum TFT per PDP address error
274	Duplicate TFT eval prec index
275	Invalid TFT param combination
277	Invalid number of parameters
278	Invalid Parameter
320	Call index error
321	Call state error
322	Sys state error
323	Parameters error
550	generic undocumented error
551	wrong state
552	wrong mode
553	context already activated

554	stack already active
555	activation failed
556	context not opened
557	can not setup socket
558	can not resolve DN
559	time-out in opening socket
560	can not open socket
561	remote disconnected or time-out
562	connection failed
563	tx error
564	already listening
565	socket disconnection
566	can not resume socket
567	ip version type incompatible
568	ipv6 not enabled
569	
600	Generic undocumented error
601	wrong state
602	Can not activate
603	Can not resolve name
604	Can not allocate control socket

605	Can not connect control socket
606	Bad or no response from server
607	Not connected
608	Already connected
609	Context down
612	Resource used by other instance
613	Data socket yet opened in cmdmode
614	FTP CmdMode data socket closed
615	FTP not connected
616	FTP disconnected
617	FTP read command closed
618	FTP read command error
619	FTP write command closed
620	FTP write command error
621	FTP read data closed
622	FTP read data error
623	FTP write data closed
624	FTP write data error
625	FTP host not found
626	FTP accept failure
627	FTP listen failure
628	FTP bind failure

629	FTP file create failure
630	FTP file get failure
631	FTP file put failure
632	FTP file not found
633	FTP timed out
634	FTP login incorrect
635	FTP close error
636	FTP server not ready
637	FTP server shutdown
638	FTP unexpected reply
639	FTP user ID and password don't match
640	FTP user ID and password don't match
641	FTP user already logged in
642	FTP open channel timeout
643	FTP communication timeout
644	FTP unknown error
657	Network survey error (No Carrier)
658	Network survey error (Busy)
659	Network survey error (Wrong request)
660	Network survey error (Aborted)
680	LU processing

681	Network search aborted
682	PTM mode
683	Network search terminated
684	CSG Search processing
690	Active call state
691	RR connection established
770	SIM invalid
900	No Response for AT Command
1000	SSL not activated
1001	SSL certs and keys wrong or not stored
1002	SSL generic error
1003	SSL already activated
1004	SSL error during handshake
1005	SSL socket error
1006	SSL invalid state
1007	SSL cannot activate
1008	SSL not connected
1009	SSL already connected
1010	SSL error enc/dec data
1011	SSL disconnected
1012	SSL version mismatch

1100	Model not recognized
1101	Model information missing
1102	Unable to open the file
1103	Unable to close the file
1104	Unable to read the nv file
1105	Unable to write the nv file
1106	Input pattern is wrong
1113	Call establishment failed
1114	File name already exist
1150	GTP error timeout
1151	PDP is not active
1152	GTP is Disabled
1153	operation not supported
1154	Reboot failed. Reboot manually

ME Error Result Code for MQTT - +CME ERROR: <err>

If MQTT commands are used detailed error codes for possible values of <err> are reported in the following table:

Table 3.1: ME Error Result Code

Numeric Format	Verbose Format
1100	MQTT Initialization Fail
1101	MQTT De-initialization Fail
1102	MQTT Configuration Fail
1103	MQTT Connect Fail
1104	MQTT Disconnect Fail
1105	MQTT Publish Fail
1106	MQTT Subscribe Fail
1107	MQTT Unsubscribe Fail
1108	MQTT SSL Configuration Fail
1109	MQTT Command Timeout Fail
1110	MQTT PUBSEXT Fail
1111	MQTT Protocol Unsupported
1112	MQTT Identifier Rejected
1113	MQTT Server Unavailable
1114	MQTT Bad Username or Password
1115	MQTT Not Authorised

1200	MQTT-SN Initialization Fail
1201	MQTT-SN De-initialization Fail
1202	MQTT-SN Configuration Fail
1203	MQTT-SN Socket Fail
1204	MQTT-SN Connect Fail
1205	MQTT-SN Disconnect Fail
1206	MQTT-SN Register Fail
1207	MQTT-SN Publish Fail
1208	MQTT-SN Subscribe Fail
1209	MQTT-SN Unsubscribe Fail
1210	MQTT-SN Will Topic Update Fail
1211	MQTT-SN Will Message Update Fail
1212	MQTT-SN State Change Fail
1213	MQTT-SN Invalid Client Status

Message Service Failure Result Code - +CMS ERROR: <err>

This is NOT a command; it is the error response to +Cxxx 3GPP TS 27.005 commands.

Syntax: +CMS ERROR: <err>

Parameter: <err> - numeric error code.

The <err> values are reported in the table:

Table 4: Message Service Failure Result Code

Numeric Format	Meaning

According to 3GPP TS 24.011 section 8.2.5.4	
0...127	
According to 3GPP TS 23.040 sub clause 9.2.3.22 values	
128...255	
According to 3GPP TS 27.005 section 3.2.5 - Message Service Failure Result Code +CMS ERROR	
300	ME failure
301	SMS service of ME reserved
302	operation not allowed
303	operation not supported
304	invalid PDU mode parameter
305	invalid text mode parameter
310	SIM not inserted
311	SIM PIN required
312	PH-SIM PIN required
313	SIM failure
314	SIM busy
315	SIM wrong
316	SIM PUK required
317	SIM PIN2 required
318	SIM PUK2 required
320	memory failure
321	invalid memory index

322	memory full
330	SMSC address unknown
331	no network service
332	network time-out
340	no +CNMA acknowledgement expected
500	unknown error
510	msg blocked
<err> 512 and on are manufacturer specific	
512	No SM resources
513	TR1M timeout
514	LL error
515	No response from network

Carriage Returns, Line Feeds and Log

Generally, the number of carriage returns (<CR>) and line feeds (<LF>) at the end of command responses may vary. This scenario may also vary from software version to software version. We do not have evidence of this behavior in URC lines.

Command responses examples:

AT#LWM2MMON?<CR>

<CR><LF>

#LWM2MMON: "4"<LF>#LWM2MMON: "3"

Warning:

missing <CR>

<CR><LF>

<CR><LF>

OK

<CR><LF>

AT#LWM2MFYACKURI=0,2<CR>

<CR><LF>

#LWM2MNFYACKURI: "/3/0/9"<LF>#LWM2MNFYACKURI: "/3/0/8"

Warning:

missing <CR>

<CR><LF>

<CR><LF>

OK

<CR><LF>

The user must be aware of this aspect before writing a script to parse the command lines and the relative response lines to generate a log.

Information Responses and Result Codes

The TA response, in case of verbose response format enabled, for the previous examples command line could be as shown below:

- information response to +CMD1?
  <CR><LF>+CMD1:2,1,10<CR><LF>
- information response to +CMD1=?
  <CR><LF>+CMD1(0-2),(0,1),(0-15)<CR><LF>
- result code
  <CR><LF>OK<CR><LF>

Moreover, there are two other types of result codes:

- result codes that inform about the progress of the TA operation (e.g. connection establishment CONNECT)
- result codes that indicate the occurrence of an event not directly associated with the issuance of a command from TE (e.g. ring indication RING).

Here the basic result codes according to ITU-T V25Ter recommendation

Table 5: Basic result codes

Numeric form	Verbose form
0	OK

1	CONNECT or CONNECT <text>
2	RING
3	NO CARRIER
4	ERROR
6	NO DIALTONE
7	BUSY
8	NO ANSWER
10	CONNECT 2400
11	CONNECT 4800
12	CONNECT 9600
15	CONNECT 14400
23	CONNECT 1200/75

Note:

<text> can be"300", "1200", "2400", "4800", "9600", "14400" or "1200/75".

Command Response Time-Out

Every command issued to the Telit modules returns a result response if response codes are enabled (default). The time required to process the given command and return the response varies, depending on the command type. Commands that do not interact with the SIM or the network, and only involve internal setups or readings, have an immediate response. Commands that interact with the SIM or the network may take several seconds to send a response, depending on the SIM configuration (for example, the number of contacts stored in the phonebook, the number of stored SMS), or on the network the command may interact with.

Command Issuing Timing

The chain Command -> Response must always be respected, and a new command must not be issued before the module has finished all the ending of the response result code (whatever it is).

This applies especially to applications that "sense" the OK text and therefore may send the next command before the complete code <CR><LF>OK<CR><LF> is sent by the module.

However, It is advisable to wait for at least 20ms between the end of the reception of the response and the issuance of the next AT command.

If the response codes are disabled and therefore the module does not report any response to the command, then at least the 20ms pause time must be respected.

Command Response after Module Boot

Please note that all LWM2M commands reply with ERROR if sent just after module boot.

There is a short delay between the completion of AT interface initialization (when the module starts to respond to AT commands) and the completion of LWM2M task initialization (when the module starts to respond to LWM2M commands).

Module Reboot

Reboot of the module can be triggered by AT command. Specific AT commands, to apply the changes requested, do include an implicit reboot. These AT commands are identified by the "This AT command causes a reboot" or "This AT command could cause a reboot" warning box statements in this Guide.

Warning:

The boot-up time of the module, after issuing any of the AT commands that implicitly trigger a Reboot procedure, could be considerably longer than expected.

Please allow up to 20 seconds for boot-up before executing an HW_SHUTDOWN

Storage

Factory Profile and User Profiles

The Telit wireless modules store the values set by several commands in the internal non-volatile memory (NVM), allowing them to remember these settings even after power off. In NVM, these values are set either as factory or user profiles. There are two customizable user profiles and one factory profile in the NVM of the device: by default, the device will start with user profile 0, which is equal to the factory profile.

For backward compatibility, each profile is divided into two sections: one base section, which was historically the one saved and restored in early releases of code, and the extended section, which includes all remaining values.

The &W command is used to save the current values of both profile sections into the NVM user profile. Commands &Y and &P are both used to set the profile to be loaded at startup. &Y instructs the device to load at startup only the base section. &P instructs the device to load at startup the full profile: base + extended sections.

The &F command resets to factory profile values only the commands of the base section of the profile, while &F1 resets to factory profile values the full set of base + extended section commands.

The values set by other commands are stored in NVM outside the profile: some of them are always stored without issuing any &W, while others are stored by issuing specific commands (+CSAS, #SLEDSAV, #SKTSAV, #ESAV); all these values are read at power-up.

In this document, each AT command description begins with an "AT Command short overview table" in the following format:

Table 6: AT Command short overview table

SIM Presence	Setting saved	Can be aborted
/	see below	/

This chapter focuses on the values that the Saved Setting field can have and their meaning. The meaning of the other fields will be described in the next chapter. The Saved Setting field can have one of the values listed below (for information about the AT instance introduced hereafter, see the reference section of the #PORTCFG command):

Specific profile: The parameter values set by the command are stored in the profile base section. The stored values set is associated with the specific AT instance used to enter the command. It is a profile used by the specific AT instances.

Examples of AT commands: +IPR, E, Q, V, X, &Y, etc.

Extended profile: The parameter values set by the command are stored in the profile extended section. The stored values set is associated with the specific AT instance used to enter the command. It is a profile used by the specific AT instance.

Examples of AT commands: +FCLASS, +CREG, +CLIP, #STIA, etc.

Common profile: The parameter values set by the command are stored in the profile extended section. The stored values set is not associated with the specific AT instance used to enter the command. It is a profile shared between the AT instances.

Examples of AT commands: +CALM, #E2SLRI, #DVI, etc.

Auto: The parameter values set by the command are automatically stored in NVM, without issuing any storing AT command, and regardless of the profile (unique values). The values are automatically restored at startup.

Examples of AT commands: +COPS, +CGQREQ, #SCFG, etc.

In some cases, the parameter values are stored in the file system.

Examples of AT commands: #TEMPCFG, #TEMPMON, etc.

Other: The parameter values set by the command are stored in NVM by issuing a specific command and regardless of the profile.

Examples of AT commands:

#SLED setting is saved by #SLEDSAV

#BIQUADINEX setting is saved by #PSAV

etc.

AT Command Short Overview Table

As stated before, each AT command description begins with an "AT Command short overview table" having the following format:

Table 7: AT Command short overview table

SIM Presence	Setting saved	Can be aborted
Not required	/	No

Here are the table fields meanings:

SIM Presence: indicates if the AT command to be executed requires the presence of a SIM card.

Can be aborted: indicates if the AT command can be aborted during its execution.

AT COMMANDS

Call & DTMF

AT+CHUP - Hang Up Call

This command cancels all active and held calls

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CHUP

Execution command cancels all active and held calls, also if a multi-party session is running

AT+CHUP=?

Test command returns the OK result code

AT+CRLP - Radio Link Protocol

Set Radio Link Protocol (RLP) parameters.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CRLP=[<iws>[,<mws>[,<T1>[,<N2>[,<ver>]]]]]

Set command sets Radio Link Protocol (RLP) parameters used when non-transparent data calls originated.

Parameters:

Name	Type	Default	Description
<iws>	integer	61	IWF window dimension
	Value:		
	1÷61	:	dimension - factory default value is 61 (ver 0 or 1)
<mws>	integer	61	MS window dimension
	Value:		
	1÷61	:	dimension - factory default value is 61 (ver 0 or 1)
<T1>	integer	128	acknowledge timer in 10 milliseconds units

Value:

39÷255	:	10 milliseconds units - default value is 128 (ver 0 or 1)

<N2>	integer	255	retransmission attempts

Value:

1÷255	:	number of attempts

<ver>	integer	N/A	protocol version

Value:

0,1	:	version

AT+CRLP?

Read command returns current settings for each supported RLP version <ver>.

+CRLP: <iws>,<mws>,<T1>,<N2>,<ver>
OK

AT+CRLP=?

Test command returns the range of setting value for each supported RLP version <ver>.

Versions 0 and 1 share the same parameter set. Read and Test commands shall return only one line for this set (where <ver> is not present)

AT+CRC - Cellular Result Codes

This command controls whether the extended format of incoming call indication is used.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CRC=[<mode>]

Set command controls whether or not the extended format of incoming call indication is used.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	disables/enables extended format reporting. When enabled, an incoming call is indicated to the TE with unsolicited result code +CRING: <type> instead of the normal RING

Values:

0 : disables extended format reporting (factory default)

1 : enables extended format reporting

Unsolicited field:

Name	Type	Description
<type>	string	call type Values: ASYNC : asynchronous transparent data SYNC : synchronous transparent data REL ASYNC : asynchronous non-transparent data REL SYNC : synchronous non-transparent data VOICE : normal voice (TS 11)

AT+CRC?

Read command returns current value of the parameter <mode>

AT+CRC=?

Test command returns supported values of the parameter <mode>

AT+CVHU - Voice Hung Up Control

This command is used in order to set how to disconnect a voice connection.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CVHU=[<mode>]

Set command selects whether ATH or "drop DTR" shall cause a voice connection to be disconnected or not.

Parameter:

Name	Type	Default	Description
<mode>	integer	2	Selects how to disconnect a voice connection.

Values:

0 : "Drop DTR" ignored but OK result code given. ATH disconnects.

1 : "Drop DTR" and ATH ignored but OK result code given.

2 : "Drop DTR" behavior according to &D setting. ATH disconnects.

AT+CVHU?

Read command reports the current value of the <mode> parameter, in the format:

+CVHU: <mode>

AT+CVHU=?

Test command reports the range of supported values for parameter <mode>.

AT+CSTA - Select Type of Address

Select type of address.

3GPP TS 24.008

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSTA=[<type>]

Set command selects the type of number for further dialing commands (D) according to 3GPP specifications.

Parameter:

Name	Type	Default	Description
<type>	integer	129	type of address octet in integer format (refer to 3GPP TS 24.008, sub clause 10.5.4.7); default 145 when dialing string includes international access code character "+", otherwise 129

Values:

129 : National address type.

145 : International number. Dialing string includes international access code character "+".

AT+CSTA?

Read command returns the current value of <type> parameter

AT+CSTA=?

Test command returns supported <type> parameter values.

AT+FCLASS - Select Active Service Class

This command sets the wireless module in specified connection mode (data, voice), hence all the calls done afterwards will be data or voice.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+FCLASS=<n>

Parameter:

Name	Type	Default	Description
<n>	integer	0	type of call

Values:

0	:	data type
8	:	voice type

AT+FCLASS?

Read command returns the current configuration value of the parameter <n>.

AT+FCLASS=?

Test command returns all supported values of the parameters <n>.

AT#ECAM - Extended Call Monitoring

This command enables/disables the call monitoring function in the ME.

SIM Presence	Setting saved	Can be aborted
Required	Common profile	No

AT#ECAM=[<onoff>]

Set command enables/disables the call monitoring function in the ME.

Parameter:

Name	Type	Default	Description
<onoff>	integer	0	Enables/disables the call monitoring function in the ME, that informs the user about call events (such as incoming call, connected, hang up, etc.) using the following unsolicited indication: #ECAM: <ccid>,<ccstatus>,<calltype>,,,[<number>,<type>]

Values:

0	:	disables call monitoring function
1	:	enables call monitoring function

Unsolicited fields:

Name	Type	Description

<ccid>	integer	call ID number
<ccstatus>	integer	call status Values: 0 : idle 1 : calling (MO) 2 : connecting (MO) 3 : active 4 : hold 5 : waiting (MT) 6 : alerting (MT) 7 : Busy(not supported) 8 : Retrieved (not supported) 9 : CNAP (Calling Name Presentation) information (MT)
<calltype>	integer	call type indicator Values: 1 : voice 2 : data
<number>	string	called number (valid only if <ccstatus> is 1)

<type>	string	type of <number> Values: 129 : national number 145 : international number

The unsolicited indication is sent along with usual codes (OK, NO CARRIER, BUSY ..)

AT#ECAM?

Read command reports whether the extended call monitoring function is currently enabled or not, in the format:

#ECAM: <onoff>

AT#ECAM=?

Test command returns the list of supported values for <onoff>

AT+VTS - DTMF Tones Transmission

This command handles the transmission of DTMF tones.

3GPP TS 27.007
TIA IS-101

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+VTS=<dtmfString>[,<duration>]

Execution command allows the transmission of DTMF tones.

Parameters:

Name	Type	Default	Description
<dtmfString>	string	-	string of <dtmf>s, i.e. ASCII characters in the set (0-9), #,*,(A-D),P; the string can be at most 255 <dtmf>s long; it allows the user to send a sequence of DTMF tones, each of them with a duration that was defined through +VTD command
<duration>	integer	N/A	duration of a tone in 1/100 sec. Values: 0÷99 : not supported 100÷1000 : a single DTMF tone will be transmitted for a time <duration> (in 10 ms

multiples), no matter what the current AT+VTD setting is

- The character P does not correspond to any DTMF tone, but it is interpreted as a pause of 3 seconds between the preceding and succeeding DTMF string elements.
- This command operates in voice mode only (see +FCLASS).

AT+VTS=?

Test command provides the list of supported <dtmf>s and the list of supported <duration>s in the format:

(list of supported <dtmf>s)[,(list of supported <duration>s)]

AT+VTD - Tone Duration

This command sets the length of tones transmitted with +VTS command.

3GPP TS 27.007
TIA IS-101

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+VTD=<duration>

Set command sets the length of tones transmitted with +VTS command.

Parameter:

Name	Type	Default	Description
<duration>	integer	2	duration of a tone

Values:

0	:	the duration of every single tone is dependent on the network
1÷1000	:	duration of every single tone in 1/10 sec

This command is currently not supported; to send a DTMF with duration different from default please use directly +VTS

AT+VTD?

Read command reports the current Tone Duration, in the format:

<duration>

AT+VTD=?

Test command provides the list of supported <duration>

General Control and Config

Command Line Prefixes

AT - Starting a Command Line

AT is the prefix used to start a command line.

ITU-T Recommendation V.25 ter

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT

The prefix AT is a two-character abbreviation ("ATtention"), always used to start a command line to be sent from TE to TA, with the only exception of AT#/ prefix. As a command, it can be issued just to test if the device is responding to AT commands.

A/ - Last Command Automatic Repetition

The command immediately executes the previously issued command or commands.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

A/

If the prefix A/ is issued, the device immediately executes once again the body of the preceding command line. No editing is possible, and no termination character is necessary. A command line may be repeated multiple times through this mechanism, if desired.

If A/ is issued before any command line has been executed, the preceding command line is assumed to have been empty (that results in an OK result code).

This command works only at fixed IPR.

The custom prefix AT#/ has been defined: it causes the last command to be executed again too; but it does not need a fixed +IPR.

Generic Modem Control

AT&F - Set to Factory-Defined Configuration

Set configuration parameters to default values.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&F[<value>]

Execution command sets the configuration parameters to default values specified by manufacturer; it takes in consideration hardware configuration switches and other manufacturer-defined criteria.

Parameter:

Name	Type	Default	Description
<value>	integer	0	parameters to reset

Values:

0 : only the factory profile base section parameters are considered

1 : either the factory profile base section and the extended section are considered (full factory profile)

If parameter <value> is omitted, the command has the same behavior as AT&F0.

ATZ - Soft Reset

Soft Reset

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATZ[<n>]

Execution command loads the base section of the specified user profile and the extended section of the default factory profile

Parameter:

Name	Type	Default	Description
<n>	integer	N/A	user profile number

Value:

0,1 : user profile number

If parameter <n> is omitted, the command has the same behavior as ATZ0

Any active call is terminated.

AT&Y - Default Reset Basic Profile Designation

Define the basic profile that is loaded on startup.

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT&Y=[<n>]

Execution command defines the basic profile that will be loaded on startup.

Parameter:

Name	Type	Default	Description
<n>	integer	0	basic profile that will be loaded on startup.

Value:

0,1 : profile number: the wireless module can store 2 complete configurations (see command &W).

Differently from command Z<n>, which loads just once the desired profile, the one chosen through command &Y will be loaded on every startup.

If parameter is omitted the command has the same behavior as AT&Y0.

AT&P - Default Reset Full Profile Designation

Define which full profile is loaded at startup.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT&P[<n>]

Execution command defines which full profile will be loaded at startup.

Parameter:

Name	Type	Default	Description
<n>	integer	0	full profile that will be loaded on startup.

Value:

0,1 : profile number: the wireless module can store 2 full configurations (see command &W).

Differently from command Z<n>, which loads just once the desired profile, the one chosen through command &P will be loaded at every startup.

If parameter is omitted, the command has the same behavior as AT&P0

AT&W - Store Current Configuration

Stores the complete configuration of the device.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&W[<n>]

Execution command stores on profile <n> the complete configuration of the device.

Parameter:

Name	Type	Default	Description
<n>	integer	0	profile identifier

Value:

0,1	:	profile identifiers

If parameter is omitted, the command has the same behavior of AT&W0.

AT&N - Display Internal Phonebook Stored Numbers

The command displays telephone numbers stored in the internal phonebook.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&N[<n>]

The module has a built in nonvolatile memory where 10 telephone numbers can be stored, each one having a maximum of 24 digits.

Execution command returns the telephone number stored at the <n> position in the internal memory.

Parameter:

Name	Type	Default	Description
<n>	integer	N/A	phonebook record number

Value:

0÷9 : phonebook record number

If parameter <n> is omitted then all the internal records are shown.

AT&V - Display some Configuration and Profile

The command displays some of the basic modem configuration settings and parameters

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&V

Execution command returns some of the basic modem configuration settings and parameters, one for each row, in the format:

setting/parameter : value

Example of returned values.

- AT&V

```
DTE SPEED                    :  115200

COMMAND ECHO                 :  E0=NO

RESULT MESSAGES              :  Q0=YES

VERBOSE MESSAGES             :  V1=YES

EXTENDED MESSAGES            :  X0=NO

FLOW CONTROL OPTIONS         :  &K3=HW bidirect.

DTR (C108) OPTIONS           :  &D0=ignored

DCD (C109) OPTIONS           :  &C1=follows carrier

POWER SAVING ON DTR          :  +CFUN:1=NO
```

```
POWER SAVING ON DTR          :   +CFUN:1=NO

DEFAULT PROFILE              :   &Y0=user profile 1

OK
```

where the first POWER SAVING ON DTR line is for SIM1 slot and the second POWER SAVING ON DTR line is for SIM2 slot (see +CFUN note about Dual SIM Dual Standby configuration).

AT+GCAP - Capabilities List

This command returns the equipment supported command set list.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+GCAP

Execution command returns the equipment supported command set list:

Additional info:

Supported Command Set:

+CGSM: 3GPP TS command set
+DS: Data Service common modem command set
+ES: WCDMA data Service common modem command set
+MS: Mobile Specific command set

AT+GMI - Manufacturer Identification

This command returns the manufacturer identification.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+GMI

Execution command returns the manufacturer identification followed by an <OK> at newline.

AT+GMI=?

Test command returns OK result code.

AT+GMM - Model Identification

The command returns the model identification.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+GMM

The execution command returns the model identification followed by an <OK> at newline.

AT+GMM=?

Test command returns OK result code.

AT+GMR - Revision Identification

The command returns the software revision identification.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+GMR

Execution command returns the software revision identification followed by an <OK> at newline.

AT+GMR=?

Test command returns OK result code.

AT+GSN - Serial Number

The command reports the device board serial number.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+GSN[=<snt>]

Set command causes the TA to return IMEI (International Mobile station Equipment Identity number) and related information to identify the MT that the TE is connected to.

Parameter:

Name	Type	Default	Description
<snt>	integer	N/A	indicates the serial number type that has been requested

Values:

0 : returns <sn>

1 : returns <imei> (International Mobile station Equipment Identity)

2 : returns the <imeisv> (International Mobile station Equipment Identity and Software Version number)

3 : returns the <svn> (Software Version Number)

Additional info:

▶▶ Here below return parameters description

Name	Type	Default	Description
<sn>	string	-	Indicates the product "serial number", identified as the IMEI of the mobile, without command echo.
<imei>	string	-	string type in decimal format indicating the IMEI. IMEI is composed of Type Allocation Code (TAC) (8 digits), Serial Number (SNR) (6 digits) and the Check Digit (CD) (1 digit). Character set used in <imei> is as specified by command Select TE Character Set +CSCS.
<imeisv>	string	-	string type in decimal format indicating the IMEISV. The 16 digits of IMEISV are composed of Type Allocation Code (TAC) (8 digits), Serial Number (SNR) (6 digits) and the software version (SVN) (2 digits). Character set used in <imeisv> is as specified by command Select TE Character Set +CSCS.
<svn>	string	-	string type in decimal format indicating the current SVN which is a part of IMEISV. Character set used in <svn> is as specified by command Select TE Character Set +CSCS.

The number returned is not the IMSI, but it is the board number.

AT+GSN=?

Test command returns supported <snt> values.

AT+CGMI - Request Manufacturer Identification

The command returns device manufacturer identification code.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGMI

Execution command returns the device manufacturer identification code followed by an OK at newline.

AT+CGMI=?

Test command returns OK result code.

AT+CGMM - Request Model Identification

This command returns the device model identification.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CGMM

Execution command returns the device model identification code followed by an OK at newline.

AT+CGMM=?

Test command returns OK result code.

AT+CGMR - Request Revision Identification

The command returns device software revision number.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGMR

Execution command returns device software revision number followed by an OK at newline.

AT+CGMR=?

Test command returns OK result code.

AT+CGSN - Request Product Serial Number Identification

This command causes the TA to return IMEI (International Mobile station Equipment Identity number) and related information to identify the MT that the TE is connected to

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGSN=[<snt>]

Execution command AT+CGSN returns the product "serial number", identified as the IMEI of the mobile, without command echo.

Parameter:

Name	Type	Default	Description
<snt>	integer	N/A	the serial number type that has been requested

Values:

0 : returns <sn>

1 : returns the IMEI (International Mobile station Equipment Identity) <imei>

2 : returns the IMEISV (International Mobile station Equipment Identity and Software Version number) <imeisv>

3 : returns the SVN (Software Version Number) <svn>

Additional info:

The IMEI (15 digits) is composed of the following elements (each element shall consist of decimal digits only):

1. Type Allocation Code (TAC). Its length is 8 digits
2. Serial Number (SNR) is an individual serial number uniquely identifying each equipment within each TAC. Its length is 6 digits
3. Luhn Check Digit. Its length is 1 digit

The IMEISV (16 digits) is composed of the following elements (each element shall consist of decimal digits only):

1. Type Allocation Code (TAC). Its length is 8 digits
2. Serial Number (SNR) is an individual serial number uniquely identifying each equipment within each TAC. Its length is 6 digits
3. Software version (SVN). Its length is 2 digits

Name	Type	Default	Description
<sn>	string	-	Indicate the product "serial number", identified as the IMEI of the mobile, without command echo
<imei>	string	-	IMEI in decimal format
<imeisv>	string	-	IMEISV in decimal format
<svn>	string	-	the current SVN (which is a part of IMEISV) in decimal format

Character set used in <imei>, <imeisv> and <svn> is as specified by command Select TE Character Set +CSCS.

AT+CGSN=?

Test command returns supported <snt> values.

AT#CGMI - Request Manufacturer Identification

The command returns device manufacturer identification code.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CGMI

Execution command returns the device manufacturer identification code, with command echo.

The response is as follows

#CGMI: <code>

OK

AT#CGMI=?

Test command returns OK result code.

AT#CGMR - Request Revision Identification

The command returns device software revision number.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CGMR

Execution command returns device software revision number, with command echo.

The response is as follows

#CGMR: <num>

OK

AT#CGMR=?

Test command returns OK result code.

AT#CGSN - Product Serial Number Identification

This execution command returns the product serial number, identified as the IMEI of the mobile, with command echo

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CGSN=[<snt>]

Set command causes the TA to return IMEI (International Mobile station Equipment Identity number) and related information to identify the MT that the TE connected to.

Parameter:

Name	Type	Default	Description
<snt>	integer	N/A	serial number type that has been requested

Values:

0 : returns <sn>

1 : returns the IMEI (International Mobile station Equipment Identity)

2 : returns the IMEISV (International Mobile station Equipment Identity and Software Version number)

3 : returns the SVN (Software Version Number)

Additional info:

▶▶ <sn> Indicates the product "serial number", identified as the IMEI of the mobile, with command echo.

<imei> string type in decimal format indicating the IMEI. IMEI is composed of Type Allocation Code (TAC) (8 digits), Serial Number (SNR) (6 digits) and the Check Digit (CD) (1 digit). Character set used in <imei> is as specified by command Select TE Character Set +CSCS.

<imeisv> string type in decimal format indicating the IMEISV. The 16 digits of IMEISV are composed of Type Allocation Code (TAC) (8 digits), Serial Number (SNR) (6 digits) and the software version (SVN) (2 digits). Character set used in <imeisv> is as specified by command Select TE Character Set +CSCS.

<svn> string type in decimal format indicating the current SVN which is a part of IMEISV. Character set used in <svn> is as specified by command Select TE Character Set +CSCS.

AT#CGSN=?

Test command returns supported <snt> values.

```
AT#CGSN
#CGSN: 358677008900540
OK
```

AT#SWPKGV - Request Software Package Version

This command shows the software, modem, production, application packages versions. It provides two syntax formats: set and execution.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SWPKGV=<version>

Here is the set format.

Parameter:

Name	Type	Default	Description
<version>	string	N/A	selects the type of version to be shown; it must be entered between double quotes

Values:

M	:	shows modem version
A	:	shows App version
P	:	shows Production version
PK	:	shows the full package version

Additional info:

- The execution format returns the packages versions without #SWPKGV: command echo. The response is as follows:

  AT#SWPKGV
  <Telit Software Package Version>-<Production Parameters Version>
  <Modem Package Version>

<Production Parameters Version>
<Application Software Version>

OK

AT#SWPKGV=?

Test command returns OK result code.

AT+CPAS - Phone Activity Status

The command reports the device status in a numeric format.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CPAS

Execution command reports the device status in the format shown in Additional info section.

Additional info:

Message format returned by the execution command:

+CPAS: <pas>

Name	Type	Default	Description
<pas>	integer	0	phone activity status.

Values:

0 : ready (MT allows commands from TA/TE)

1 : unavailable (MT does not allow commands from TA/TE)

2 : unknown (MT is not guaranteed to respond to instructions)

3 : ringing (MT is ready for commands from TA/TE, but the ringer is active)

4 : call in progress (MT is ready for commands from TA/TE, but a call is in progress)

5 : asleep (MT is unable to process commands from TA/TE because it is in a low functionality state)

6 : call in active

AT+CPAS=?

Test command reports the supported range of values for <pas>.

Although +CPAS is an execution command, 3GPP TS 27.007 requires the Test command to be defined.

```
ATD03282131321;
OK

AT+CPAS
+CPAS: 4        The called phone has answered to your call
OK

ATH
OK
```

AT+CFUN - Set Phone Functionality

This command selects the level of functionality in the ME.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CFUN=[<fun>[,<rst>]]

Set command selects the level of functionality in the ME.

Parameters:

Name	Type	Default	Description
<fun>	integer	1	Power saving function mode.

Values:

0	:	disable SIM and RS232 power saving, see Additional info section
1	:	mobile full functionality with power saving disabled.
4	:	disable both TX and RX
5	:	mobile full functionality with power saving enabled
12	:	RS232 power saving, see Additional info section

<rst>	integer	0	Reset flag.

Values:

0 : do not reset the ME before setting it to <fun> functionality level.

1 : reset the device. The device is fully functional after the reset. This value is available only for <fun> = 1.

Additional info:

- <fun>=1

  the mobile is in full functionality with power saving disabled

- <fun>=0

  disable SIM
  AND
  If only UART connected and USB disconnected:
  - the mobile full functionality with power saving disabled will be active for 60 seconds after last AT command
  - 60 seconds after last AT command the mobile will enter power saving and first AT command on UART will be lost

- <fun>=4

  disable both TX and RX. Settng

  AT+CFUN=4[,0]

  actually causes the module to perform either a network deregistration and a SIM deactivation

- <fun>=5

  the mobile is in full functionality with power saving enabled.

  After set once +CFUN to 5, to activate power saving mode, plug out the USB, set the DTR (RS232) line to OFF. Once in power saving, the CTS (RS232) line switch to the OFF status to signal that the module is really in power saving condition. During the power saving condition, before sending any AT command on the serial line set the DTR line to ON and wait for the CTS line to go in ON status. With the DTR line in ON status the module will maintain the power saving mode enabled, and to to activate power saving mode again set the DTR line to OFF again.

- <fun>=12

  If only UART connected and USB disconnected:
  - the mobile full functionality with power saving disabled will be active for 60 seconds after last AT command
  - 60 seconds after last AT command the mobile will enter power saving and first AT command on UART will be lost.

- If power saving mechanism is enabled, it reduces the power consumption during the idle time, thus allowing a longer standby time with a given battery capacity.

- In AT+CFUN=5 configuration when entering in power saving with DTR line OFF, if the USB connection is active it will be deactivated; when exiting from power saving with DTR line ON, the USB connection will be reactivated.

- Right after power on or reboot AT+CFUN? can report a temporary +CFUN: 4 value for 1 to 4 seconds.

- In a Dual SIM Dual Standby configuration AT+CFUN setting is applied to the master SIM (see +SWITCHSIM).

Master SIM and slave SIM can have different AT+CFUN settings.

AT+CFUN?

Read command reports the current setting of <fun>

AT+CFUN=?

Test command returns the list of supported values for <fun> and <rst>.

AT+CIND - Indicator Control

This command is used to control the registration state of modem indicators.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CIND=[<state>[,<state[,...]]]

Set command is used to control the registration state of ME indicators, in order to automatically send the +CIEV URC, whenever the value of the associated indicator changes. The supported indicators (<descr>) and their order appear from test command AT+CIND=?

Parameter:

Name	Type	Default	Description
<state>	integer	0	registration state

Values:

0 : the indicator is deregistered; there is no unsolicited result code (+CIEV URC) automatically sent by the modem to the application, whenever the value of the associated indicator changes; the value can be directly queried with +CIND?

1 : the indicator is registered: an unsolicited result code (+CIEV URC) is automatically sent by the modem to the application, whenever the value of the

associated indicator changes; it is still possible to query the value through AT+CIND?

AT+CIND?

Read command returns the current value of ME indicators, in the format:

+CIND: <ind>,<ind>,...<ind>

The order of the values <ind>s is the one returned by test command AT+CIND=?

AT+CIND=?

Test command returns pairs, where string value <descr> is a description (max. 16 chars) of the indicator and compound value is the supported values for the indicator, in the format:

+CIND: (<descr>, (list of supported <ind>s)),(<descr>, (list of supported <ind>s)),... (<descr>, (list of supported <ind>s))

Additional info:

Test command response parameters

Name	Type	Default	Description
<descr>	string	N/A	indicator names as follows (along with their <ind> ranges).
	Values:		

"battchg"	:	dummy, it can be set but it does not have any effect; battery charge level; indicator <ind> in the range 0...5(bar levels), or 99 (not measurable)
"signal"	:	signal quality; indicator <ind> in the range 0...7(bar levels), or 99 (not measurable); same as bit error rate (<ber>) in +CSQ command
"service"	:	service availability; indicator <ind> is 0 (not registered to any network) or 1 (registered)
"sounder"	:	dummy, it can be set but it does not have any effect; sounder activity; indicator <ind> is 0 (no sound activity) or 1 (sound activity)
"message"	:	message received; indicator <ind> is 0 (no unread SMS in memory "SM") or 1 (unread SMS in memory "SM")
"call"	:	call in progress; indicator <ind> is 0 (no calls in progress) or 1 (at least a call has been established)
"roam"	:	roaming; indicator <ind> is 0 (registered to home network, or not registered) or 1 (registered to other network))

"smsfull"	:	SMS memory status; indicator <ind> is 0 (memory locations available) or 1 (an SMS storage in the modem is full)
"rssi"	:	received signal strength level; indicator <ind> values are 0 (signal strength under -112dBm), from 1 to 4 (signal strength from -97 to -66 dBm, in 15 dBm steps), 5 (signal strength greater than -51 dBm), or 99 (not measurable)
"GPRS coverage	:	there is packet service coverage; indicator <ind> is 0 (no packet service) or 1 (module attached to a packet service)
"callsetup"	:	call setup status indicator; indicator <ind> values are 0 (no active call setup), 1 (MT call is ringing), 2 (MO call was initiated), 3 (MO call ringing at called party)

"+CIEV: message,0" URC is not supported when there are no unread messages.

There will be no notification about "no unread SMS" in "SM" memory

Next command causes all the indicators to be registered

AT+CIND=1,1,1,1,1,1,1,1,1,1

Next command causes all the indicators to be de-registered

AT+CIND=0,0,0,0,0,0,0,0,0,0

Next command to query the current value of all indicators

AT+CIND?

CIND: 4,0,1,0,0,0,0,0,2,0,0

OK

AT+CMER - Mobile Equipment Event Reporting

This command configures sending of unsolicited result codes from TA to TE.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT+CMER=[<mode>[,<keyp>[,<disp>[,<ind>[,<bfr>]]]]]

Set command enables/disables sending of unsolicited result codes from TA to TE in the case of indicator state changes.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	controls the processing of unsolicited result codes

Values:

0 : buffer +CIEV Unsolicited Result Codes

1 : discard +CIEV Unsolicited Result Codes when TA-TE link is reserved (e.g. on-line data mode); otherwise forward them directly to the TE

2 : buffer +CIEV Unsolicited Result Codes in the TA when TA-TE link is reserved (e.g. on-line data mode) and flush them to the TE after reservation; otherwise forward them directly to the TE

3 : forward +CIEV Unsolicited Result Codes directly to the TE; when TA is in on-line data mode each +CIEV URC is stored in a buffer; once the ME goes into

command mode (after +++ was entered), all URCs stored in the buffer will be output

<keyp>	integer	0	keypad event reporting

Value:

0 : No keypad event reporting

<disp>	integer	0	display event reporting

Value:

0 : no display event reporting

<ind>	integer	0	indicator event reporting

Values:

0 : no indicator event reporting

2 : indicator event reporting

<bfr>	integer	0	TA buffer clearing

Values:

0 : TA buffer of unsolicited result codes is cleared when <mode> 1..3 is entered

1 : TA buffer of unsolicited result codes is flushed to the TE when <mode> 1...3 is entered (OK response shall be given before flushing the codes)

After AT+CMER has been switched on with e.g. AT+CMER=2,0,0,2 command (i.e. <bfr> is 0), URCs for all registered indicators will be issued only first time, if previous <mode> was 0, for backward compatibility. Values shown by the indicators will be current indicators values, not buffered ones.

Subsequent AT+CMER commands with <mode> different from 0 and <bfr> equal to 0 will not flush the codes, even if <mode> was set again to 0 before.

To flush the codes, <bfr> must be set to 1.

Although it is possible to issue the command when SIM PIN is pending, it will answer ERROR if "message" or "smsfull" indicators are enabled in AT+CIND, because with pending PIN it is not possible to give a correct indication about SMS status.
To issue the command when SIM PIN is pending you have to disable "message" and "smsfull" indicators in AT+CIND first

AT+CMER?

Read command returns the current setting of parameters, in the format:

+CMER: <mode>,<keyp>,<disp>,<ind>,<bfr>

AT+CMER=?

Test command returns the range of supported values for parameters <mode>, <keyp>, <disp>, <ind>, <bfr>, in the format:

+CMER: (list of supported <mode>s),(list of supported <keyp>s), (list of supported <disp>s),(list of supported <ind>s),(list of supported <bfr>s)

AT+CACM - Accumulated Call Meter

The command purpose is to reset the Advice of Charge accordingly to the Accumulated Call Meter stored in SIM (ACM): it contains the total number of home units for both the current and preceding calls.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CACM=[<pwd>]

The set command resets the Advice of Charge accordingly to the Accumulated Call Meter stored in SIM (ACM): it contains the total number of home units for both the current and preceding calls.

Parameter:

Name	Type	Default	Description
<pwd>	string	-	to access this command PIN2 is required, if after the startup the PIN2 has already been input, it is no longer required to issue the command.

AT+CACM?

Read command reports the current value of the SIM ACM in the format:

+CACM: <acm>

Additional info:

▶▶ Read command response parameter

Name	Type	Default	Description

<acm>	string	-	accumulated call meter in home units: three bytes of the ACM value in hexadecimal format (e.g. "00001E" indicates decimal value 30).

The value <acm> is in home units; price per unit and currency are defined with command +CPUC

AT+CACM=?

Test command returns the OK result code

AT+CAMM - Accumulated Call Meter Maximum

This command sets the Advice of Charge related Accumulated Call Meter Maximum Value stored in SIM (ACMmax).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CAMM=[<acmmax>[,<pwd>]]

Set command sets the Advice of Charge related Accumulated Call Meter Maximum Value stored in SIM (ACMmax). This value represents the maximum number of home units allowed to be consume by the subscriber. When ACM reaches <acmmax> value, further calls are prohibited.

Parameters:

Name	Type	Default	Description
<acmmax>	string	-	ACMmax value, integer type: it is the maximum number of home units allowed to be consumed by the subscriber.
<pwd>	string	-	PIN2; notice that if PIN2 has been already input once after startup, it is no more required

Setting <acmmax> to 0 disables the feature

AT+CAMM?

Read command reports the ACMmax value stored in SIM in the format:

+CAMM : <acmm>

Additional info:

▶▶ Read command response parameter

Name	Type	Default	Description
<acmm>	string	-	accumulated call meter in home units: three bytes of the ACMmax value in hexadecimal format (e.g. "00001E" indicates decimal value 30)

?

AT+CAMM=?

Test command returns the OK result code.

AT+CCWE - Call Meter Maximum Event

This command is used to enable/disable sending of an unsolicited result code +CCWV.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CCWE=<mode>

Set command is used to enable/disable sending of an unsolicited result code +CCWV shortly before the ACM (Accumulated Call Meter) maximum value is reached. The warning is issued approximately when 30 seconds call time remains. It is also issued when starting a call if less than 30 seconds call time remains.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	enable mode

Values:

0 : Disable the call meter warning event

1 : Enable the call meter warning event

The set command will respond with an error if the Accumulated Call Meter service is not active in SIM.

AT+CCWE?

Read command reports the currently selected <mode> in the format:
+CCWE: <mode>

AT+CCWE=?

Test command reports the supported range of values for parameter <mode>.

AT+CLAC - Available AT Commands

This command shows the available AT commands list.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CLAC

The execution command causes the ME to return one or more lines reporting the AT commands that are available to the user. The format is:

<ATcmd1>[<CR><LF><ATcmd2>[...]]

<ATcmdn> is the AT command.

AT+CLAC=?

Test command returns the OK result code.

AT+CMEE - Report Mobile Equipment Error

The command enables the use of result code.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CMEE=[<n>]

Set command enables/disables the report of result code:
+CME ERROR: <err>
as an indication of an error relating to the +Cxxx commands issued. When enabled, device related errors cause the +CME ERROR <err> result code instead of the default ERROR result code. ERROR anyway returned normally when the error message related to syntax, invalid parameters, or DTE functionality.

Parameter:

Name	Type	Default	Description
<n>	integer	0	enable flag

Values:

0 : disable +CME ERROR:<err> reports, use only ERROR report

1 : enable +CME ERROR:<err> reports, with <err> in numeric format

2 : enable +CME ERROR: <err> reports, with <err> in verbose format

- default value after powerup is "0"
- +CMEE has no effect on the final result code +CMS.
- The detailed description of <err> is available in section "ME Error Result Code - +CME ERROR: <err>".

AT+CMEE?

Read command returns the current value of subparameter <n>:

+CMEE: <n>

AT+CMEE=?

Test command returns the range of values for subparameter <n>

AT+CEER - Extended Error Report

Reports extended error related to the last unsuccessful call.

3GPP TS 27.007

GSM 04.08

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CEER

Execution command returns two lines of information text <report> offering the TA user an extended error report, in the format:

+CEER: <report>

+CEER: <report>

This report regards some error condition that may occur:

the failure in the last unsuccessful call setup (originating or answering)

the last call release

the last unsuccessful GPRS attach or unsuccessful PDP context activation,

the last GPRS detach or PDP context deactivation.

- the first line for the voice and the second line for data
- if none of these conditions have occurred since power up then "Normal, unspecified" condition is reported

AT+CEER=?

Test command returns OK result code.

AT#CEER - Extended Numeric Error Report

The command is related to extended numeric error report.

3GPP TS 24.008

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CEER

Execution command causes the TA to return a numeric code in the intermediate response format:

#CEER: <code>

which offers the user of the TA a report of the reason for

- the failure in the last unsuccessful call setup (originating or answering);

- the last call release;

- the last unsuccessful GPRS attach or unsuccessful PDP context activation;

- the last GPRS detach or PDP context deactivation.

Additional info:

Intermediate response parameters:

Name	Type	Default	Description
<code>	integer	N/A	error code with the following meaning:

Values:

0	:	No error
1	:	Unassigned (unallocated) number
3	:	No route to destination
6	:	Channel unacceptable
8	:	Operator determined barring
16	:	Normal call clearing
17	:	User busy
18	:	No user responding
19	:	User alerting, no answer
21	:	Call rejected
22	:	Number changed
26	:	Non selected user clearing
27	:	Destination out of order
28	:	Invalid number format (incomplete number)
29	:	Facility rejected
30	:	Response to STATUS ENQUIRY

31 : Normal, unspecified

34 : No circuit/channel available

38 : Network out of order

41 : Temporary failure

42 : Switching equipment congestion

43 : Access information discarded

44 : Requested circuit/channel not available

47 : Resources unavailable, unspecified

49 : Quality of service unavailable

50 : Requested facility not subscribed

55 : Incoming calls barred with in the CUG

57 : Bearer capability not authorized

58 : Bearer capability not presently available

63 : Service or option not available, unspecified

65 : Bearer service not implemented

68 : ACM equal to or greater than ACMmax

69 : Requested facility not implemented

70 : Only restricted digital information bearer capability is available

79 : Service or option not implemented, unspecified

81 : Invalid transaction identifier value

87 : User not member of CUG

88 : Incompatible destination

91 : Invalid transit network selection

95 : Semantically incorrect message

96 : Invalid mandatory information

97 : Invalid mandatory information

98 : Message type not compatible with protocol state

99 : Information element non-existent or not implemented

100 : Conditional IE error

101 : Message not compatible with protocol state

102 : Recovery on timer expiry

111 : Protocol error, unspecified

127 : Interworking, unspecified

224 : MS requested detach

225 : NWK requested detach

226 : Unsuccessful attach cause NO SERVICE

227 : Unsuccessful attach cause NO ACCESS

228 : Unsuccessful attach cause GPRS SERVICE REFUSED

229 : PDP deactivation requested by NWK

230 : PDP deactivation cause LLC link activation Failed

231 : PDP deactivation cause NWK reactivation with same TI

232 : PDP deactivation cause GMM abort

233 : PDP deactivation cause LLC or SNDCP failure

234 : PDP unsuccessful activation cause GMM error

235 : PDP unsuccessful activation cause NWK reject

236 : PDP unsuccessful activation cause NO NSAPI available

237 : PDP unsuccessful activation cause SM refuse

238 : PDP unsuccessful activation cause MMI ignore

239 : PDP unsuccessful activation cause Nb Max Session Reach

256 : PDP unsuccessful activation cause wrong APN

257 : PDP unsuccessful activation cause unknown PDP address or type

258 : PDP unsuccessful activation cause service not supported

259 : PDP unsuccessful activation cause QOS not accepted

260 : PDP unsuccessful activation cause socket error

240 : FDN is active and number is not in FDN

241 : Call operation not allowed

252 : Call barring on outgoing calls

253 : Call barring on incoming calls

254 : Call impossible

255 : Lower layer failure

If none of the described conditions has occurred since power up then 0 is reported (i.e. No error).

AT#CEER=?

Test command returns OK result code.

AT#HWREV - Hardware Identification

This command returns the device Hardware revision identification code.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#HWREV

Execution command returns the device Hardware revision identification code without command echo.

AT#HWREV=?

Test command returns the OK result code.

AT+CSCS - Select TE Character Set

The command purpose is to set different character sets that are used by the device.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSCS=[<chset>]

Set command sets the current character set used by the device.

Parameter:

Name	Type	Default	Description
<chset>	string	N/A	character set to be used by the device.

Values:

IRA : International Reference Alphabet (ITU-T T.50). Quoted string; e.g. "AB" equals two 8-bit characters with decimal values 65, 66

UCS2 : 16-bit universal multiple-octet coded character set (ISO/IEC10646).

GSM : GSM default alphabet (3GPP TS 23.038).

AT+CSCS?

Read command returns the current value of the active character set.

AT+CSCS=?

Test command returns the supported values for parameter <chset>.

To send an SMS in text mode

AT+CMCF=1

with text in UCS2, set

AT+CSMP=17,167,0,8

before

AT+CMGS

```
AT+CSCS=?
+CSCS: ("IRA","UCS2","GSM")
OK

AT+CSCS?
+CSCS: "IRA"
OK

AT+CPBW=1,"8475763000",129,"Lin Zhao"
OK

AT+CSCS="UCS2"
OK

AT+CPBR=1
```

```
+CPBR: 1,"8475763000",129,004C006E006E0020005A00680061006F
OK

AT+CSCS="IRA"
OK

AT+CPBR=1
+CPBR: 1,"8475763000",129,"Lin Zhao"
OK
```

AT+CMUX - Multiplexing Mode

This command is used to enable/disable the multiplexing protocol control channel.

3GPP TS 27.007
3GPP TS 27.010

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CMUX=<mode>[,<subset>[,<port_speed>[,<N1>[,<T1>[,<N2>[,<T2>[,<T3>[,<K>]]]]]]]]

Set command is used to enable/disable the multiplexing protocol control channel.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	multiplexer transparency mechanism. Value: 0 : basic option mode; it is currently the only supported value
<subset>	integer	0	the way in which the multiplexer control channel is set up Value: 0 : UIH frames used only; it is currently the only supported value
<port_speed>	integer	5	transmission rate

Value:

5	: 115200 bps; it is currently the only supported value

<N1>	integer	N/A	Maximum frame size.

Value:

1÷1509	: expressed in bytes

<T1>	integer	N/A	Acknowledgement timer in units of ten milliseconds

Value:

1÷255	: in units of ten milliseconds

<N2>	integer	N/A	Maximum number of re-transmissions

Value:

1÷100	: range of re-transmissions number

<T2>	integer	N/A	Response timer for the multiplexer control channel. T2 must be longer than T1.

Value:

2÷255	: in units of ten milliseconds

<T3>	integer	N/A	Wake up response timer

	Value: 1÷255 : in seconds		
<K>	integer	N/A	Window size, for Advanced option with Error-Recovery Mode Value: 1÷7 : window size range

AT+CMUX?

Read command returns the current value of the parameters, in the format:

+CMUX: <mode>,<subset>,<port_speed>,<N1>,<T1>,<N2>,<T2>,<T3>,<K>

AT+CMUX=?

Test command returns the range of supported values for all parameters.

AT#CMUXMODE - CMUX Mode Set

The module is equipped with the CMUX standard protocol to provide multiplexing features. The #CMUXMODE command configures the CMUX behavior concerning the DTR control line.

3GPP TS 27.010

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CMUXMODE=<mode>

Set command sets the CMUX mode.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	multiplexer transparency mechanism.

Values:

0 : Ignore DTR feature is disabled, a transmission of the physical DTR line instructs the DCE to disable the CMUX and switches to the normal command mode

1 : Ignore DTR feature is disabled, a transmission of the physical DTR line instructs the DCE to disable the CMUX and switches to the normal command mode

4 : Ignore DTR feature is enabled, the DCE doesn't care the physical DTR line transitions

5 : Ignore DTR feature is enabled, the DCE doesn't care the physical DTR line transitions

- DLC establishment on Virtual Channel between mode 0 and mode 1 is different.
- A software or hardware reset restores the default value
- During CMUX session the set command will return ERROR, only the read and test command can be used.

AT#CMUXMODE?

Read command reports the current parameters values in the format:

#CMUXMODE: <mode>

AT#CMUXMODE=?

Test command reports the range of parameter <mode>.

AT#USBCFG - USB Configuration

USB Composition Configuration

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#USBCFG=<composition>

Set the USB composition

Parameter:

Name	Type	Default	Description
<composition>	integer	0	USB composition configuration.

Values:

0 : VID PID 0x1BC7 0x701A: RNDIS + Diag + AT + Modem

1 : VID PID 0x1BC7 0x701B: ECM + Diag + AT + Modem

New USB composition configuration will be applied at the next reboot or complete power cycle.

AT#USBCFG?

Read command returns the current <composition> value in the following format:

#USBCFG: <composition>

AT#USBCFG=?

Test command returns all supported values of the parameter <composition>.

AT#PORTCFG - Connect Physical Ports to Service Access Points

This command allows to connect Service Access Points (software anchorage points) to the external physical ports.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#PORTCFG=<Variant>

Set command allows to connect Service Access Points to the external physical ports giving a great flexibility. Examples of Service Access Points: AT Parser Instance #1, #2, #3, etc..

Parameter:

Name	Type	Default	Description
<Variant>	integer	0	set port configuration. A short description, for each <Variant> value, is reported in test command section. The range depends on the product.

Value:

0÷max : see test command section

To enable the set port configuration, the module must be rebooted.

AT#PORTCFG?

Read command returns the requested and the active port configuration in the format:

#PORTCFG: <requested>,<active>

Additional info:

▶▶ Parameters returned by the read command, in format:

#PORTCFG: <requested>,<active>

Name	Type	Default	Description
<requested>	integer	-	value showing the requested configuration that will be activated on the next power ON.
<active>	integer	-	value showing the actual configuration.

AT#PORTCFG=?

Test command returns a brief description of the supported ports arrangement solutions.

For each <Variant> are reported, on one row, the logical connections between a physical port (USIF0, USB0, etc.) and a Service Access Point (AT#1, AT#2, etc.).

The test command returns, for example, the following message:

```
AT#PORTCFG=?
#PORTCFG: Variant=0: AT= USIF0 USB0 USB1

OK
```

AT&Z - Store Telephone Number in the Internal Phonebook

The command stores a telephone number in the internal phonebook.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&Z<n>=<nr>

The module has a built in non-volatile memory where 10 telephone numbers can be stored, each one having a maximum of 24 digits. Execution command stores the telephone number <nr> in the record <n>. The records cannot be overwritten; they must be cleared before rewriting.

Parameters:

Name	Type	Default	Description
<n>	integer	N/A	phonebook record
	Value:		
	0÷9	:	record number
<nr>	string	-	telephone number (maximum length 24 digits)

To delete the record <n> issue the command AT&Z<n>=<CR>.

The records in the module memory can be viewed with the command &N.

AT+CPUC - Price per Unit and Currency Table

Set the values of Price per Unit and Currency Table.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CPUC=<currency>,<ppu>[,<pwd>]

Set command sets the values of Advice of Charge related Price per Unit and Currency Table stored in SIM (PUCT). The PUCT information can be used to convert the home units (as used in commands +CAOC, +CACM and +CAMM) into currency units.

Parameters:

Name	Type	Default	Description
<currency>	string	-	three characters currency code (e.g. "LIT", "L. ", "USD", "DEM" etc..); used character set should be the one selected with command +CSCS
<ppu>	string	-	price per unit (dot is used as decimal separator) e.g. "1989.27"
<pwd>	string	-	SIM PIN2; if PIN2 has been already input once after startup, it is required no more

AT+CPUC?

Read command reports the current values of <currency> and <ppu> parameters in the format:

+CPUC : <currency>,<ppu>

AT+CPUC=?

Test command returns OK result code.

AT#CGMM - Request Model Identification

This command returns the device model identification.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CGMM

Execution command returns the device model identification code, with command echo.

AT#CGMM
#CGMM: <code>

OK

AT#CGMM=?

Test command returns OK result code.

AT#PDPAUTH - PDP Authentication Parameters

This set command specifies PDP authentication parameters values for a PDP context identified by the (local) context identification parameter <cid>.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#PDPAUTH=<cid>,<authType>[,<username>[,<password>]]

Set command specifies PDP authentication parameters values for a PDP context identified by the (local) context identification parameter <cid>

Parameters:

Name	Type	Default	Description
<cid>	integer	N/A	context identifier

Value:

1÷max	:	specifies a particular PDP context definition. The value of max is returned by Test command

Name	Type	Default	Description
<authType>	integer	0	authentication type

Values:

0	:	no authentication
1	:	PAP authentication
2	:	CHAP authentication

<username>	string	-	supplied by network provider. Required for <auth_type> = 1 and 2
<password>	string	-	supplied by network provider. Required for <auth_type> = 1 and 2.

AT#PDPAUTH?

Read command returns the PDP authentication parameters, excluding <password>, set for every PDP, in the format:

#PDPAUTH: <cid1>,< auth_type1 >,<username1><CR><LF>

...

#PDPAUTH:<cidmax>,<auth_typemax >,<usernamemax><CR><LF>

AT#PDPAUTH=?

Test command reports the supported range of values for parameters <cid> and <authType> and the maximum allowed length of the string parameters <password> and <username>.

S Parameters

ATS0 - Number of Rings to Auto Answer

The command controls the automatic answering feature of the DCE.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATS0=[<n>]

Set command sets the number of rings required before device automatically answers an incoming call.

Parameter:

Name	Type	Default	Description
<n>	integer	0	Number of rings

Values:

0	:	auto answer disabled
1÷255	:	number of rings required before automatic answer.

ATS0?

Read command returns the current value of S0 parameter.

ATS2 - Escape Character

The command manages the ASCII character used as escape character.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATS2=[<char>]

Set command sets the ASCII character used as escape character.

Parameter:

Name	Type	Default	Description
<char>	integer	43	escape character decimal ASCII

Value:

43 : factory default value is '+'

The escape sequence consists of three escape characters preceded and followed by n ms of idle (see S12 to set n).

ATS2?

Read command returns the current value of S2 parameter.

The format of the numbers in output is always 3 digits, left-filled with 0s.

ATS3 - Command Line Termination Character

The command manages the character configured as command line terminator.

ITU-T Recommendation V.25 ter

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATS3=[<char>]

Set command sets the value of the character either recognized by the device as command line terminator and generated by the device as part of the header, trailer, and terminator for result codes and information text, along with S4 parameter.

Parameter:

Name	Type	Default	Description
<char>	integer	13	command line termination character (decimal ASCII)

Value:

0÷127	:	command line termination character

The "previous" value of S3 is used to determine the command line termination character for entering the command line containing the S3 setting command. However, the result code issued shall use the "new" value of S3 (as set during the processing of the command line)

ATS3?

Read command returns the current value of S3 parameter.

The format of the numbers in output is always 3 digits, left-filled with 0s

ATS4 - Response Formatting Character

The command manages the character generated by the device as part of the header, trailer, and terminator for result codes and information text.

ITU-T Recommendation V.25 ter

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATS4=[<char>]

Set command sets the value of the character generated by the device as part of the header, trailer, and terminator for result codes and information text, along with the S3 parameter.

Parameter:

Name	Type	Default	Description
<char>	integer	10	response formatting character (decimal ASCII)

Value:

0÷127	:	response formatting character

If the value of S4 is changed in a command line the result code issued in response of that command line will use the new value of S4.

ATS4?

Read command returns the current value of S4 parameter.

The format of the numbers in output is always 3 digits, left-filled with 0s.

ATS5 - Command Line Editing Character

The command manages the value of the character recognized by the DCE as a request to delete from the command line the immediately preceding character.

ITU-T Recommendation V.25 ter

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATS5=[<char>]

Set command sets the value of the character recognized by the device as a request to delete from the command line the immediately preceding character.

Parameter:

Name	Type	Default	Description
<char>	integer	8	command line editing character (decimal ASCII)

Value:

0÷127	:	command line editing character

ATS5?

Read command returns the current value of S5 parameter.

The format of the numbers in output is always 3 digits, left-filled with 0s.

DTE - Modem Interface Control

ATE - Command Echo

This command allows to enable or disable the command echo.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

ATE=[<n>]

The execution command allows to enable/disable the command echo.

Parameter:

Name	Type	Default	Description
<n>	integer	0	Configuration value

Values:

0 : disables command echo

1 : enables command echo, hence command sent to the device are echoed back to the DTE before the response is given.

If parameter is omitted, the command has the same behavior of ATE0

ATE?

Read command returns the current value of <n>.

ATQ - Quiet Result Codes

This command allows to enable or disable the result code.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

ATQ[<n>]

Set command enables or disables the result codes.

Parameter:

Name	Type	Default	Description
<n>	integer	0	enables/disables result codes

Values:

0 : enables result codes

1 : disables result codes.

2 : disables result codes (only for backward compatibility).

After issuing either ATQ1 or ATQ2 every information text transmitted in response to commands is not affected

If parameter is omitted, the command has the same behavior of ATQ0.

After issuing ATQ1 or ATQ2

AT+CGACT=?

+CGACT: (0-1)

nothing is appended to the response

ATV - Response Format

Set command determines the contents of the header and trailer transmitted with result codes and information responses. It also determines if result codes are transmitted in a numeric form or an alphanumeric form (according to [1]).

[1] ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

ATV[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	1	format of information responses and result codes. See Additional info section.

Values:

0 : limited headers and trailers and numeric format of result codes

1 : full headers and trailers and verbose format of result codes

Additional info:

<n>=0	
information responses	<text><CR><LF>
result codes	<numericCode><CR>
<n>=1	
information responses	<CR><LF> <text><CR><LF>
result codes	<CR><LF> <verboseCode><CR><LF>

- the <text> portion of information responses is not affected by this setting.
- if parameter is omitted, the command has the same behavior of ATV0

ATI - Identification Information

This command returns identification information.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATI[<n>]

Execution command returns one or more lines of information text followed by a result code.

Parameter:

Name	Type	Default	Description
<n>	integer	0	information request

Values:

0 : numerical identifier

1 : module checksum

2 : checksum check result

3 : manufacturer

4 : product name

5 : DOB version

If parameter is omitted, the command has the same behavior of ATI0

AT&C - Data Carrier Detect (DCD) Control

This set command controls the DCD output behavior of the AT commands serial port.

ITU-T Recommendation V25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT&C[<n>]

Set command controls the DCD output behavior of the serial port.

Parameter:

Name	Type	Default	Description
<n>	integer	1	DCD output behavior

Values:

0 : DCD remains always High

1 : DCD follows the Carrier detect status: if carrier is detected DCD goes High, otherwise DCD is Low

2 : DCD is always High except for 1 sec "wink" when a data call is disconnected

If parameter is omitted, the command has the same behavior of AT&C0

AT&D - Data Terminal Ready (DTR) Control

This set command configures the behavior of the module according to the DTR control line transitions (RS232).

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT&D[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	0	defines the module behavior according to the DTR control line transitions

Values:

0 : module ignores DTR control line transitions

1 : when the module is connected, the high to low transition of DTR line sets the module in command mode, the current connection is not closed

2 : when the module is connected, the high to low transition of DTR line sets the module in command mode and the current connection is closed

3 : C108/1 operation enabled

4 : C108/1 operation disabled

- If AT&D2 has been issued, and the DTR line has been tied Low, auto-answering is inhibited and it is possible to answer only issuing command ATA.
- If parameter is omitted, the command has the same behavior of AT&D0.
- &D3, &D4 are used for the backward compatibility; the command has the same behavior of AT&D0.

AT&K - Flow Control

Flow Control settings.

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT&K[<n>]

Set command controls the serial port flow control behavior.

Parameter:

Name	Type	Default	Description
<n>	integer	3	flow control behavior

Values:

0 : no flow control

3 : hardware bi-directional flow control (both RTS/CTS active)

If parameter is omitted, the command has the same behavior as AT&K0

&K has no Read Command. To verify the current setting of &K, simply check the settings of the active profile issuing AT&V.

Selecting flow control with AT&K3 has no effect in command mode.

In command mode mode HW flow control is always active and this setting cannot be changed for command mode.

This command is equivalent to +IFC command.

AT+IPR - UART DCE Interface Data Rate Speed

This set command specifies the DTE speed at which the device accepts commands during command mode operations; it may be used to fix the DTE-DCE interface speed.

ITU-T Recommendation V25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+IPR=<rate>

Parameter:

Name	Type	Default	Description
<rate>	integer	115200	speed of the serial port expressed in bit per second.

Values:

300	:	bps
1200	:	bps
4800	:	bps
9600	:	bps
14400	:	bps
19200	:	bps

28800	:	bps
38400	:	bps
56000	:	bps
57600	:	bps
115200	:	bps
128000	:	bps
230400	:	bps
460800	:	bps
921600	:	bps

- DTE speed of USB does not change
- If <rate> is specified DTE-DCE speed fixed to that speed, hence no speed auto-detection (autobauding) is enabled.

AT+IPR?

Read command returns the current value of +IPR parameter.

```
AT+IPR?
+IPR: 115200
OK
```

AT+IPR=?

Test command returns the list of supported auto detectable <rate> values in the format:

+IPR:(list of supported auto detectable <rate> values)

AT+IPR=?

+IPR:
(300,1200,4800,9600,14400,19200,28800,38400,56000,57600,115200,128000,230400,460800,921

OK

AT+IFC - DTE-Modem Local Flow Control

This set command selects the flow control of the serial port in both directions, from DTE to modem and from modem to DTE.

ITU-T Recommendation V25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+IFC=<by_te>,<by_ta>

Parameters:

Name	Type	Default	Description
<by_te>	integer	2	flow control option for the data received by DTE Values: 0 : flow control None 2 : C105 (RTS)
<by_ta>	integer	2	flow control option for the data sent by modem Values:

0	:	flow control None
2	:	C106 cfo (CTS)

- The only possible commands are AT+IFC=0,0 and AT+IFC=2,2.
- Selecting flow control with AT+IFC has no effect in command mode.

  In command mode HW flow control is always active and this setting cannot be changed for command mode.
- This command is equivalent to &K command.

AT+IFC?

Read command returns active flow control settings.

AT+IFC=?

Test command returns all supported values of the parameters <by_te> and <by_ta>.

AT#CFLO - Command Flow Control

This set command enables/disables the hardware flow control in command mode. If enabled, the flow control is applied to both data mode and command mode

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT#CFLO=<enable>

Set command enables/disables the flow control in command mode. If enabled, the current flow control configured by +IFC, &K is applied to both data mode and command mode.

Parameter:

Name	Type	Default	Description
<enable>	integer	0	enable/disable flow control in the command mode

Values:

0 : disable flow control in command mode

1 : enable flow control in command mode

Setting value is saved in the profile

This behaviour is valid only for Main UART port.

In case of USB port, flow control always enabled independent of this setting.

AT#CFLO?

Read command returns current setting value in the format:

#CFLO: <enable>

AT#CFLO=?

Test command returns the range of supported values for parameter <enable>

ATX - Extended Result Codes

Set command selects the subset of result code messages the modem uses to reply to the DTE upon AT commands execution.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Required	Specific profile	No

ATX[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	0	configuration value

Values:

0 : send only OK, CONNECT, RING, NO CARRIER, ERROR, NO ANSWER results. Busy tones reporting is disabled

1÷4 : reports all messages

Additional info:

<n>=0

EXTENDED MESSAGES: X0=NO

<n>=1 .. 4

EXTENDED MESSAGES : X1=YES

When the <n> parameter is omitted, the command acts like ATX0.

AT#DTR - Data Terminal Ready (DTR) flow control

The command configures how the outbound flow on serial ports is controlled by the DTR signal, see document [1]

[1] Hardware Design Guide of the used module

[2] ITU-T Recommendation V25 ter

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT#DTR=<n>

This set command configures the behavior of the outbound flow according to the DTR control line level transitions.

Parameter:

Name	Type	Default	Description
<n>	integer	N/A	DTR outbound flow control behavior

Values:

0 : DTR level transitions are ignored and cannot control the outbound data flow. (Supported only by UART ports).

1 : DTR level transitions control the outbound data flow. If DTR is asserted, the output data is sent; if DTR is

not asserted, the output data is blocked. (Supported only by ports with DTR signal)

2 : Like setting 1, but receiving input data has the same effect of a low to high level transition. Therefore, if the DTR is not asserted, or if it is asserted but its level has not been signaled by the DTE to the DCE, after receiving input data the DTR is considered asserted. A new DTR level transition from asserted to not asserted will block again the outbound flow. (Supported only by USB ports)

Additional info:

- For USIF ports default value of <n> is 0.
- For USB ports default value of <n> is 2.
- The range depends on the port.

AT#DTR?

Read command returns the current value of <n> parameter.

AT#DTR=?

Test command returns the list of <n> values in the format:

#DTR: (list of <n> values)

This command returns only the <n> values supported by the port on which it is given;
i.e. USB ports do not support <n>=0; USIF ports without DTR pin only support <n>=0.

Test command example

AT#DTR=?

#DTR: (0,1)

OK

Call (Voice and Data) Control

ATD - Dialup Connection

This execution command starts a voice call to the phone number given as parameter or establishes a GPRS communication between the TE and the external PDN.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Required	No	Yes

ATD

The ATD execution command can assume different formats, which are described in the following Additional info section. It causes the MT to perform whatever actions are necessary to start a voice call to a given phone number or to establish communication between the TE and the external PDN.

Additional info:

ATD<number>;

Name	Type	Default	Description
<number>	string	-	phone number to be dialed Accepted characters are 0-9 and *,#,"A", "B", "C", "D","+". For backwards compatibility with landline modems, modifiers "T", "P", "R", ",", "W", "!", "@" are accepted, but have no effect.

ATD*<gprs_sc>[*[<addr>][*[<L2P>][*[<cid>]]]]#

Name	Type	Default	Description

<gprs_sc>	integer	N/A	is the GPRS Service Code, which identifies a request to use the GPRS communication
	Value:		
	99	:	GPRS Service Code
<addr>	string	-	identifies the called party in the address space applicable to the PDP
<L2P>	string	-	indicates the layer 2 protocol to be used (see +CGDATA). For communications software that does not support arbitrary characters in the dial string, the following numeric equivalents shall be used: 1 is equivalent to PPP
<cid>	integer	-	PDP context definition, see +CGDCONT command

Circuit Switched Data Calls are not supported.

The following custom result codes are received, monitoring step by step the call status:

Scenario 1:

RINGBACK: MO call in progress

CONNECT: remote call accepted

NO CARRIER: remote hang-up

Scenario 2:

RINGBACK: MO call in progress

BUSY: remote hang-up without remote call accept

ATA - Answer Incoming call

The command is used to answer to an incoming call if automatic answer is disabled.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATA

Execution command informs the DCE that answer sequence must be started if automatic answer is disabled.

This command must be the last in the command line and must be followed immediately by a <CR> character.

ATH - Hang Up/Disconnect the Current Call

This execution command is used to close the current conversation (voice or data).

.

ITU-T Recommendation V.25 ter

SIM Presence	Setting saved	Can be aborted
Not required	No	No

ATH

- This command can be issued only in command mode
- When a data conversation is active the device is in on-line mode (commands are not sensed and characters are sent to the other party), hence escape sequence (see register S2) is required before issuing this command, otherwise if &D1 option is active, DTR pin has to be tied Low to return in command mode.

ATO - Return to ON-Line Mode

This execution command is used to return to on-line mode from command mode. If there is no active connection, it returns NO CARRIER.

ITU-T Recommendation V. 25 ter

SIM Presence	Setting saved	Can be aborted
Required	No	No

ATO

After issuing this command, if the device is in conversation, to send other commands to the device you must return to command mode by issuing the escape sequence (see register S2) or tying low DTR pin if &D1 option is active.

AT#DIALMODE - Set Dialing Mode

This command sets dialing modality.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#DIALMODE=[<mode>]

Set command sets dialing modality.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	sets dialing modality

Values:

0 : voice call only, see Additional info

1 : voice call only, see Additional info

2 : voice call only, see Additional info

Additional info:

- <mode>=0

  Voice call only, OK result code is received as soon as it starts remotely ringing.

- <mode>=1

  Voice call only, OK result code is received only after the called party answers. Any character typed aborts the call and OK result code is received.

<mode>=2

Voice call only, the following custom result codes are received, monitoring step by step the call status:

DIALING (MO in progress)

RINGING (remote ring)

CONNECTED (remote call accepted)

RELEASED (after ATH)

DISCONNECTED (remote hang-up).

Any character typed before the CONNECTED message aborts the call.

The setting is saved in NVM and available on following reboot

AT#DIALMODE?

Read command returns current ATD dialling mode in the format:

#DIALMODE: <mode>

AT#DIALMODE=?

Test command returns the supported range of values for parameter <mode>.

```
AT#DIALMODE=0
OK
ATD*********;
OK
RINGBACK
OK
***answer
CONNECT
***close
NO CARRIER
```

AT#DIALMODE=1

OK

ATD*********;

OK

RINGBACK

***answer

CONNECT

OK

***close

NO CARRIER

AT#DIALMODE =2

OK

ATD*********;

OK

DIALING

RINGING

**answer

CONNECTED

**close

DISCONNECTED

Network

AT+CNUM - Subscriber Number

This command returns the MSISDN (if the phone number of the device has been stored in the SIM card).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CNUM

Execution command returns the MSISDN (if the phone number of the device has been stored in the SIM card) using the following format:

+CNUM: <alpha>,<number>,<type>[<CR><LF>
+CNUM: <alpha>,<number>,<type>[...]]

The parameters are described in the Additional info section.

Additional info:

List of the parameters meaning.

Name	Type	Default	Description
<alpha>	string	-	alphanumeric string associated to <number>; the character set depends on the value set with +CSCS.
<number>	string	-	numeric string containing the phone number in the format <type>.
<type>	integer	N/A	type of number.

Values:

129	:	national numbering scheme
145	:	international numbering scheme (contains the character "+")

AT+CNUM=?

Test command returns the OK result code.

AT+COPN - Read Operator Names

This command read operator names.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+COPN

Execution command returns the list of operator names from the ME in the format:

+COPN: <numeric1>,<alpha1>[<CR>

+COPN: <numeric2>,<alpha2>[...]]

The parameters are described in the Additional info section.

Additional info:

List of the parameters meaning.

Name	Type	Default	Description
<numericn>	string	-	operator in numeric format, see +COPS.
<alphan>	string	-	operator in long alphanumeric format, see +COPS.

Each operator code <numericn> that has an alphanumeric equivalent <alphan> in the ME memory is returned.

AT+COPN=?

Test command returns the OK result code.

AT+CREG - Network Registration Status

The command enables/disables the network registration unsolicited result code (URC) and selects its presentation format.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CREG=[<mode>]

Set command enables/disables the network registration unsolicited result code and selects one of the two available formats:

short format: +CREG: <stat>
long format: +CREG: <stat>[,<lac>,<ci>[,<AcT>]]

The parameter meanings are shown in Unsolicited code value section.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	enables/disables the network registration unsolicited result code and selects one of the two formats. The URC short format is displayed every time there is a change in the network registration status. The URC long format is displayed every time there is a change of the network cell

Values:

0 : disable the network registration unsolicited result code

1 : enable the network registration unsolicited result code, and selects the short format

2 : enable the network registration unsolicited result code, and selects the long format (includes the network cell identification data)

Unsolicited fields:

Name	Type	Description
<stat>	integer	network registration status of the module Values: 0 : not registered, terminal is not currently searching a new operator to register to 1 : registered, home network 2 : not registered, but terminal is currently searching a new operator to register to 3 : registration denied 4 : unknown 5 : registered, roaming

6 : registered for "SMS only", home network (applicable only when <AcT> indicates E-UTRAN)

7 : registered for "SMS only", roaming (applicable only when <AcT>indicates E-UTRAN)

8 : attached for emergency bearer services (not applicable)

9 : registered for "CSFB not preferred", home network (applicable only when <AcT> indicates E-UTRAN)

10 : registered for "CSFB not preferred", roaming (applicable only when <AcT> indicates E-UTRAN)

11 : only emergency services are available

<lac> string the parameter reports:

Local Area Code when <AcT>=0

Tracking Area Code when <AcT>=7

<ci> string Cell Id for the currently registered on cell

<AcT> integer access technology of the registered network.

Values:

0 : GSM

2 : UTRAN

3 : GSM w/EGPRS

4 : UTRAN w/HSDPA

5 : UTRAN w/HSUPA

6 : UTRAN w/HSDPA and HSUPA

7 : E-UTRAN

8 : EC-GSM-IoT (A/Gb mode)

9 : E-UTRAN (NB-S1 mode)

<lac>, <ci> and <AcT> network information is reported by URC only if <mode>=2, and the module is registered on some network cell.

AT+CREG?

Read command returns the current value of <mode>, the registration status <stat>, and the network information (<lac>, <ci> and <AcT>) according to the used <mode> parameter value.

+CREG: <mode>,<stat>[,<lac>,<ci>[,<AcT>]]

<lac>, <ci>, and <AcT> network information is reported only if <mode>=2 and the module is registered on some network cell.

AT+CREG=?

Test command returns supported values for parameter <mode>.

Check the registration status of the module.

AT+CREG?

+CREG: 0,2

OK

The module is in network searching state

...

...

Check again module status

AT+CREG?

+CREG: 0,1

OK

The module is registered

AT+COPS - Operator Selection

The command selects a network operator, and registers the module.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+COPS=[<mode>[,<format>[,<oper>[,<AcT>]]]]

Set command forces an attempt to select and register the network operator.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	defines the operator selection: automatic or manual.

Values:

0 : automatic selection, the parameter <oper> is ignored

1 : manual selection, the parameter <oper> must be present

2 : deregister from network. The module is unregistered until a +COPS with <mode>=0, 1 or 4 is issued

3 : set only <format> parameter, the parameter <oper> is ignored

4 : manual/automatic, <oper> parameter must be present. If manual selection fails, automatic mode (<mode>=0) is entered

5 : enable the reporting of searched network operators, *PLMNLIST indication will be reported

<format>	integer	N/A	specifies the operator name format, see <oper> parameter

Values:

0 : alphanumeric long form (max length 16 digits)

1 : alphanumeric short form

2 : numeric <oper>

<oper>	mixed	-	network operator in format defined by <format> parameter: long alphanumeric format can be up to 16 characters long and short format up to 8 characters (refer GSM MoU SE.13 [9]). numeric format is the GSM Location Area Identification number (refer 3GPP TS 24.008 [8] subclause 10.5.1.3) which consists of a three BCD digit country code coded as in ITU-T E.212 Annex A [10], plus a two BCD digit network code, which is administration specific. Returned <oper> shall not be in BCD format, but in IRA characters converted from BCD. Hence, the number has the structure: (country code digit 3) (country code digit 2) (country code digit1) (network

			code digit 3) (network code digit 2) (network code digit 1).
<AcT>	integer	N/A	selects access technology.

Values:

0	:	GSM
1	:	GSM Compact
2	:	UTRAN
3	:	GSM w/EGPRS
4	:	UTRAN w/HSDPA
5	:	UTRAN w/HSUPA
6	:	UTRAN w/HSDPA and HSUPA
7	:	E-UTRAN
8	:	UTRAN w/HSPA+
9	:	E-UTRAN CA

- 'UNKNOWN' is a value of 'read command' and not of 'set command'. This is a transient state that occur while modem is initialized.
- <mode> parameter setting is stored in NVM and available at next reboot, if it is not 3 (i.e.: set only <format> parameter).
- If <mode>=1 or 4, the selected network is stored in NVM too and is available at next reboot (this will happen even with a new SIM inserted)

- <format> parameter setting is never stored in NVM
- The network name can only be used among the list of network names displayed by the +COPN command when using the manual choice mode with alphanumeric format
- 3GPP TS 44.060 specifies the System Information messages which give the information about whether the serving cell supports EGPRS
- 3GPP TS 25.331 specifies the System Information blocks which give the information about whether the serving cell supports HSDPA or HSUPA

AT+COPS?

Read command returns current value of <mode>, <format>, <oper> and <AcT> in format <format>. If no operator is selected, <format>, <oper> and <AcT> are omitted

+COPS: <mode>[, <format>, <oper>,<AcT>]

- A network name can be displayed with different name from showed network name list by the +COPN command if the UE could get a camped network name from USIM or network.

AT+COPS=?

Test command returns a list of quintuplets, each representing an operator present in the network.
The quintuplets in the list are separated by commas:
+COPS: [list of supported (<stat>, long alphanumeric <oper>,short alphanumeric <oper>,numeric <oper>,< AcT>)s][,,(list of supported <mode>s), (list of supported <format>s)]

<stat> and <AcT> parameters are described in the Additional info section.

Additional info:

▶▶ Meaning of the <stat> and <AcT> parameters.

Name	Type	Default	Description
<stat>	integer	N/A	operator availability

Values:

0	:	unknown
1	:	available
2	:	current
3	:	forbidden

<AcT>	integer	N/A	access technology selected

Values:

0	:	GSM
1	:	GSM Compact
2	:	UTRAN
3	:	GSM w/EGPRS
4	:	UTRAN w/HSDPA
5	:	UTRAN w/HSUPA
6	:	UTRAN w/HSDPA and HSUPA

7 : E-UTRAN

Since with this command a network scan is done, this command may require some seconds before the output is given.

```
AT+COPS?
+COPS: 0,0,"Test PLMN 1-1",0
OK
AT+COPS=?
+COPS: (2,"","","45008",2),(1,"Test PLMN 1-1","Test1-1","00101",0),
(3,"","","45005",2),,(0-4),(0-2)
OK
```

AT+CLCK - Facility Lock/Unlock

This command is used to lock or unlock a ME on a network facility.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CLCK=<fac>,<mode>[,<password>[,<class>]]

Set command is used to lock or unlock a modem or a network facility.

Parameters:

Name	Type	Default	Description
<fac>	string	N/A	facility

Values:

"SC"	:	SIM (PIN request) (device asks SIM password at power-up and when this lock command issued)
"AO"	:	BAOC (Barr All Outgoing Calls)
"OI"	:	BOIC (Barr Outgoing International Calls)
"OX"	:	BOIC-exHC (Barr Outgoing International Calls except to Home Country)
"AI"	:	BAIC (Barr All Incoming Calls)
"IR"	:	BIC-Roam (Barr Incoming Calls when Roaming outside the home country)

"AB" : All Barring services (applicable only for <mode>=0)

"AG" : All outGoing barring services (applicable only for <mode>=0)

"AC" : All inComing barring services (applicable only for <mode>=0)

"FD" : SIM fixed dialling memory feature (if PIN2 authentication has not been done during the current session, PIN2 is required as <passwd>)

"PN" : network Personalisation

"PU" : network subset Personalisation

"PP" : service Provider Personalization (refer 3GPP TS 22.022 [33])

"PC" : Corporate Personalization (refer 3GPP TS 22.022 [33])

"PF" : lock Phone to the very First inserted SIM/ UICC card (also referred in the present document as PH-FSIM) (MT asks password when other than the first SIM/UICC card is inserted)

<mode>	integer	N/A	defines the operation to be done on the facility

Values:

0 : unlock facility

	1	:	lock facility
	2	:	query status

<password>	string	-	shall be the same as password specified for the facility from the DTE user interface or with command Change Password +CPWD
<class>	integer	N/A	is a sum of integers, each representing an information class of which the command refers to; default is 7 (voice + data + fax).

Values:

1	:	voice (telephony)
2	:	data (refers to all bearer services)
4	:	fax (facsimile services) (not supported by LTE)
8	:	short message service
16	:	data circuit sync
32	:	data circuit async
64	:	dedicated packet access
128	:	dedicated PAD access

Additional info:

▶▶ When <mode>=2 and command successful, it returns:

+CLCK: <status>[,<class1>[<CR><LF>+CLCK:
<status>,<class2>
[...]]

Name	Type	Default	Description
<status>	integer	N/A	the current status of the facility
	Values:		
	0	:	not active
	1	:	active
<class n>	integer	-	class of information of the facility

AT+CLCK=?

Test command reports all the facilities supported by the device.

Querying such a facility returns an output on three rows, the first for voice, the second for data, the third for fax:

AT+CLCK ="AO",2

+CLCK: <status>,1

+CLCK: <status>,2

+CLCK: <status>,4

AT+CPWD - Change Facility Password

This command changes the password for the facility lock function defined by command Facility Lock +CLCK command.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Other	Yes

AT+CPWD=<fac>,<oldpwd>,<newpwd>

Execution command changes the password for the facility lock function defined by command Facility Lock +CLCK.

Parameters:

Name	Type	Default	Description
<fac>	string	N/A	facility lock function.

Values:

"SC" : SIM (PIN request)

"AB" : All barring services

"P2" : SIM PIN2

"AC" : All inComing barring services

"AG" : All outGoing barring services

"AI" : BAIC (Barr All Incoming Calls)

"AO"	:	BAOC (Barr All Outgoing Calls)
"IR"	:	BIC-Roam (Barr Incoming Calls when Roaming outside the home country)
"OI"	:	BOIC (Barr Outgoing International Calls)
"OX"	:	BOIC-exHC (Barr Outgoing International Calls except to Home Country)

<oldpwd>	string	- old password. it shall be the same as password specified for the facility from the ME user interface or with command +CLCK	-
<newpwd>	string	- new password	-

parameter <oldpwd> is the old password while <newpwd> is the new one.

AT+CPWD=?

Test command returns a list of pairs (<fac>,<pwdlength>) which represents the available facilities and the maximum length of their password (<pwdlength>).

MAX timeout indicates the time within which the command must be executed. MAX timeout for this command is 180 s.

```
AT+CPWD=?
+CPWD:

(""CS"", 16)

(""PS"", 16)

(""PF"", 16)

(""SC"", 16)

(""AO"", 16)

(""OI"", 16)

(""OX"", 16)

(""AI"", 16)

(""IR"", 16)

(""AB"", 16)

(""AG"", 16)

(""AC"", 16)

(""FD"", 16)

(""NT"", 16)

(""NM"", 16)

(""NS"", 16)

(""NA"", 16)

(""PN"", 16)

(""PU"", 16)

(""PP"", 16)
```

(""PC"", 16)

(""P2"", 16)

OK

AT+CLIR - Calling Line Identification Restriction

The command manages the CLIR service.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CLIR=[<n>]

Set command overrides the CLIR subscription when temporary mode is provisioned as a default adjustment for all following outgoing calls. This adjustment can be revoked by using the opposite command. This command refers to CLIR service (see 3GPP TS 22.081), that allows a calling subscriber to enable or disable the presentation of the Calling Line Identification (CLI, i.e., the phone number of the caller) to the called party when originating a call.

Parameter:

Name	Type	Default	Description
<n>	integer	N/A	facility status on the Mobile

Values:

0 : CLIR facility according to CLIR service network status

1 : CLIR facility active (CLI not sent)

2 : CLIR facility not active (CLI sent)

This command sets the default behaviour of the device in outgoing calls.

AT+CLIR?

Read command gives the default adjustment for all outgoing calls (<n>) and also triggers an interrogation of the provision status of the CLIR service (<m>), in the form

+CLIR: <n>,<m>

Additional info:

Name	Type	Default	Description
<n>	integer	N/A	facility status in the Mobile

Values:

0 : CLIR facility according to CLIR service network status

1 : CLIR facility active (CLI not sent)

2 : CLIR facility not active (CLI sent)

<m>	integer	N/A	facility status in the Network

Values:

0 : CLIR service not provisioned

1 : CLIR service provisioned permanently

2 : unknown (e.g. no network present)

3 : CLI temporary mode presentation restricted

4 : CLI temporary mode presentation allowed

AT+CLIR=?

Test command reports the supported values of parameter <n>

AT+COLP - Connected Line Identification Presentation

This command enables/disables the presentation of the COL at the TE.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+COLP=[<n>]

Set command enables/disables the presentation of the COL at the TE.
When enabled (and called subscriber allows), the following intermediate result code is returned from TA to TE before any +CR or ITU T Recommendation V.25ter responses:

+COLP: <number>,<type>[,<subaddr>,<satype> [,<alpha>]]

It is manufacturer specific if this response is used when normal voice call is established.

Parameter:

Name	Type	Default	Description
<n>	integer	0	enable/disable COL indication

Values:

0	:	disable COL indication
1	:	enable COL indication

AT+COLP?

Read command gives the status of <n>, and also triggers an interrogation of the provision status of the COLP service according 3GPP TS 22.081 (given in <m>) in the format:

+COLP: <n>,<m>

Additional info:

where

Name	Type	Default	Description
<n>	integer	N/A	COL presentation enabled/disabled
	Values:		
	0 : COL presentation disabled		
	1 : COL presentation enabled		
<m>	integer	N/A	status of the COLP service on the network
	Values:		
	0 : COLP not provisioned		
	1 : COLP provisioned		
	2 : unknown (e.g. no network is present)		

AT+COLP=?

Test command returns the range for the parameter <n>.

AT+CCFC - Call Forwarding Number And Condition

This command controls the call forwarding supplementary service.

- 3GPP TS 27.007
- GSM 03.82

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CCFC=<reason>,<cmd>[,<number>[,<type>[,<class>[,<time>]]]]

The execution command controls the call forwarding supplementary service. Registration, erasure, activation, deactivation, and status query are supported.

Parameters:

Name	Type	Default	Description
<reason>	integer	0	reason of call forwarding

Values:

0 : unconditional

1 : mobile busy

2 : no reply

3 : not reachable

4 : all calls (not with query command)

5 : all conditional calls (not with query command)

<cmd>	integer	0	command parameter

Values:

0 : disable

1 : enable

2 : query status

3 : registration

4 : erasure

<number>	string	-	string type phone number of forwarding address in format specified by <type> parameter
<type>	integer	129	type of address octet in integer format

Values:

129 : National numbering scheme

145 : International numbering scheme (contains the character "+")

<class>	integer	-	sum of integers each representing a class of information which the command refers to; default 7 (voice + data + fax) 1 voice(telephony)

			2 data 4 fax (facsimile services) (not supported by LTE) 8 short message service 16 data circuit sync 32 data circuit async 64 dedicated packet access 128 dedicated PAD access
<time>	integer	20	time in seconds to wait before call is forwarded; it is valid only when <reason> "no reply" is enabled (<cmd>=1) or queried (<cmd>=2)

Value:

1÷30	:	automatically rounded to a multiple of 5 seconds

Additional info:

▶▶ when <cmd>=2 and command successful, it returns:

+CCFC: <status>,<class$_1$>[,<number>,<type>[,,,<time>]][<CR><LF>
+CCFC: <status>,<class$_2$>[,<number>,<type>[,,,<time>]][...]]

Name	Type	Default	Description
<status>	integer	0	current status of the network service

Values:

0	:	not active
1	:	active

<time>	string	-	it is returned only when <reason>=2 ("no reply") and <cmd>=2.

Please see GSM 03.82 for more info.

When querying the status of a network service (<cmd>=2) the response line for 'not active' case (<status>=0) should be returned only if service is not active for any <class>.

AT+CCFC=?

Test command reports supported values for the parameter <reason>.

- To check if the unconditional call forwarding is on or off.

  AT+CCFC=0,2

- To register the unconditional call forwarding to the network.

  AT+CCFC=0,3,"Number"

- To activate the unconditional call forwarding.

  AT+CCFC=0,1,"Number"

- To deactivate the unconditional call forwarding.

  AT+CCFC=0,0

- Example for Registration & activation:

  For registration

  AT+CCFC=0,3,"+972575684414"

  For activation

  AT+CCFC=0,1,"+ 972575684414"

AT+CCWA - Call Waiting

This command allows control of the supplementary service Call Waiting. Activation, deactivation and status query are supported.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CCWA=[<n>[,<cmd>[,<class>]]]

Set command allows the control of the call waiting supplementary service. Activation, deactivation, and status query supported

The URC has the following format:

+CCWA: <number>,<type>

Parameters:

Name	Type	Default	Description
<n>	integer	0	Enables/disables the presentation of an unsolicited result code

Values:

0	:	disable
1	:	enable

<cmd>	integer	0	Enables/disables or queries the service at network level

Values:

0	:	disable
1	:	enable
2	:	query status

<class>	integer	-	sum of integers each representing a class of information which the command refers to; default is 7 (voice + data + fax) 1 voice(telephony) 2 data 4 fax (facsimile services) (not supported by LTE) 8 short message service 16 data circuit sync 32 data circuit async 64 dedicated packet access 128 dedicated PAD access

Unsolicited fields:

Name	Type	Description
<number>	string	Phone number of calling address in format specified by <type>
<type>	integer	Type of address in integer format Values:

128	:	both the type of number and the numbering plan are unknown
129	:	Unknown type of number and ISDN/ Telephony numbering plan
145	:	International type of number and ISDN/Telephony numbering plan (contains the character "+")
161	:	National type of number and ISDN/ Telephony numbering plan

- The response to the query command is in the format:

  +CCWA: <status>,<class$_1$>[<CR><LF>
  +CCWA: <status>,<class$_2$>[...]]

  where

  <status> represents the status of the service:
  0 - inactive
  1 - active

  <class$_n$> - same as <class>

- If parameter <cmd> is omitted then network is not interrogated.

- In the query command the <class> parameter must not be issued.

- The difference between call waiting report disabling (AT+CCWA = 0,1,7) and call waiting service disabling (AT+CCWA = 0,0,7) is that in the first case the call waiting indication is sent to the device by network but this last one does not report it to the modem; instead in the second case the call waiting indication is not generated by the network. Hence the device results busy to the third party in the second case, while in the first case a ringing indication is sent to the third party.

- The command AT+CCWA=1,0 has no effect and is a non sense, then must not be issued.

AT+CCWA?

Read command reports the current value of the parameter <n>

AT+CCWA=?

Test command reports the supported values for the parameter <n>.

AT+CHLD - Call Holding Services

The command controls the network call hold service.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CHLD=[<n>]

Execution command controls the network call hold by the service. With this service, it is possible to disconnect temporarily a call and keep it suspended while it is retained by the network; at the same time it is possible to connect another party or make a multiparty connection.

Parameter:

Name	Type	Default	Description
<n>	integer	N/A	the parameter is used to release, activate or add an held call

Values:

0 : releases all held calls, or sets the UDUB (User Determined User Busy) indication for a waiting call.

1 : releases all active calls (if any exist), and accepts the other (held or waiting) call

1X : releases a specific active call X

2 : places all active calls (if any exist) on hold and accepts the other (held or waiting) call.

2X : places all active calls on hold except call X with which communication shall be supported

3 : adds an held call to the conversation

4 : connects the two calls and disconnects the subscriber from both calls (Explicit Call Transfer (ECT))

- "X" is the numbering (starting with 1) of the call given by the sequence of setting up or receiving the calls (active, held or waiting) as seen by the served subscriber. Calls hold their number until they are released. New calls take the lowest available number.
- Where both a held and a waiting call exist, the above procedures apply to the waiting call (i.e. not to the held call) in conflicting situation.
- For VOLTE conference call <n>=2X and <n>=4 parameters are not supported
- For VOLTE while no active or held calls option <n>=3 starts conference call to conference server without participants

AT+CHLD=?

Test command returns the list of supported values of parameter <n>. in the format:

+CHLD: (0,1,1X,2,2X,3,4)

The command is only applicable to voice calls.

AT+CDIP - Called line identification presentation

This command is related to a network service that provides "multiple called numbers (called line identifications) service" to an MT.

This command enables a called subscriber to get the called line identification of the called party when receiving a mobile terminated call

3GPP TS 24.008

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CDIP=[<n>]

Set command enables or disables the presentation of the called line identifications at the TE.

Parameter:

Name	Type	Default	Description
<n>	integer	N/A	sets/shows the result code presentation status to the TE

Values:

0	:	disable
1	:	enable

Additional info:

- When the presentation of the called line identification at the TE is enabled,

  +CDIP:<number>,<type>[,<subaddr>,<satype>]

response is returned after every RING (or +CRING: <type>: refer subclause "Cellular result codes +CRC") result code sent from TA to TE. It is manufacturer specific if this response used when normal voice call answered

Name	Type	Default	Description
<number>	string	-	phone number of format specified by <type>
<type>	integer	-	type of address octet in integer format (refer 3GPP TS 24.008 [8] subclause 10.5.4.7)
<subaddr>	string	-	subaddress of format specified by <satype>
<satype>	string	-	type of subaddress octet in integer format (refer 3GPP TS 24.008 [8] subclause 10.5.4.8)

AT+CDIP?

Read command gives the status of <n> and also triggers an interrogation of the provision status of the "multiple called numbers" service.

Additional info:

- Read command returns values supported as a compound value.

  +CDIP: <n>,<m>

Name	Type	Default	Description
<n>	integer	N/A	sets/shows the result code presentation status to the TE

Values:

0	:	disable
1	:	enable

<m>	integer	N/A	shows the subscriber "multiple called numbers" service status in the network

Values:

0	:	"multiple called numbers service" is not provisioned
1	:	"multiple called numbers service" is provisioned
2	:	unknown (e.g. no network, etc.)

AT+CDIP=?

Test command returns the list of supported values of parameter <n>

AT+CUSD - Unstructured Supplementary Service Data

Set command allows control of the Unstructured Supplementary Service Data (USSD 3GPP TS 22.090).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CUSD=[<n>[,<str>[,<dcs>]]]

The unsolicited result code enabled by parameter <n> is in the format:

+CUSD: <m>[,<str>,<dcs>]

to the TE

Parameters:

Name	Type	Default	Description
<n>	integer	0	disable/enable the presentation of an unsolicited result code

Values:

0 : disable the result code presentation in the DTA

1 : enable the result code presentation in the DTA

2 : cancel an ongoing USSD session (not applicable to read command response)

<str>	string	-	USSD-string (when <str> parameter is not given, network is not interrogated) If <dcs> indicates that 3GPP TS 3.38/23.038 default alphabet is used ME/TA converts GSM alphabet into current TE character set (see +CSCS). If <dcs> indicates that 8-bit data coding scheme is used: ME/TA converts each 8-bit octet into two IRA character long hexadecimal number; e.g. octet with integer value 42 is presented to TE as two characters 2A (IRA 50 and 65).
<dcs>	integer	-	3GPP TS 23.038 Cell Broadcast Data Coding Scheme in integer format (default is 0).

Unsolicited field:

Name	Type	Description
<m>	integer	Status service value Values: 0 : no further user action required (network initiated USSD-Notify, or no further information needed after mobile initiated operation)

1 : further user action required (network initiated USSD-Request, or further information needed after mobile initiated operation)

2 : USSD terminated by the network

3 : other local client has responded

4 : operation not supported

5 : network time out

AT+CUSD?

Read command reports the current value of the parameter <n>

AT+CUSD=?

Test command reports the supported values for the parameter <n>

Only mobile initiated operations are supported

AT+CAOC - Advice of Charge

This command allows the user to get information about the cost of calls and to enable an unsolicited event reporting of the Current Call Meter (CCM) information.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CAOC=<mode>

Set command refers to the Advice of Charge supplementary services that enable subscriber to get information about the cost of calls; the command also includes the possibility to enable an unsolicited event reporting of the Current Call Meter (CCM) information.

Parameter:

Name	Type	Default	Description
<mode>	integer	N/A	mode of presentation of CCM information

Values:

0 : query CCM value

1 : disables unsolicited CCM reporting

2 : enables unsolicited CCM reporting

Unsolicited field:

Name	Type	Description
<ccm>	hex	The unsolicited result code enabled by parameter <mode> is in the format: +CCCM: <ccm> where: <ccm> - current call meter in home units, string type: three bytes of the CCM value in hexadecimal format (e.g. "00001E" indicates decimal value 30)

The unsolicited result code +CCCM is sent when the CCM value changes, but not more than every 10 seconds.

AT+CAOC?

Read command reports the value of parameter <mode> in the format:
+CAOC: <mode>

AT+CAOC=?

Test command reports the supported values for <mode> parameter.

+CAOC command returns an estimate of the cost of the current call only, produced by the MS and based on the information provided by either AoCI or AOCC supplementary services; it is not stored in the SIM.

AT+CLCC - List Current Calls

This command returns the list of current calls and their characteristics

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CLCC

Execution command returns the list of current calls and their characteristics in the format:

[+CLCC:<id_1>,<dir>,<stat>,<mode>,<mpty>,<number>,<type>,<alpha>[,<priority>[,<CLI_validity>]] [<CR><LF>

+CLCC:<id_2>,<dir>,<stat>,<mode>,<mpty>,<number>,<type>,<alpha>[,<priority>[,<CLI_validity>]] [...]]]

The parameters are described in the Additional info section.

Additional info:

List of the parameters meaning.

Name	Type	Default	Description
<idn>	integer	-	call identification number
<dir>	integer	N/A	call direction

Values:

0 : mobile originated call

1 : mobile terminated call

<stat>	integer	N/A	state of the call

Values:

0 : active

1 : held

2 : dialing (MO call)

3 : alerting (MO call)

4 : incoming (MT call)

5 : waiting (MT call)

<mode>	integer	N/A	call type

Values:

0 : voice

1 : data

2 : fax (not supported by LTE)

9 : unknown

<mpty>	integer	N/A	multiparty call flag

Values:

0 :	call is not one of multiparty (conference) call parties
1 :	call is one of multiparty (conference) call parties

<number>	string	-	phone number in format specified by <type>
<type>	integer	N/A	type of phone number octet in integer format

Values:

129 :	national numbering scheme
145 :	international numbering scheme (contains the character "+")

<alpha>	string	-	alphanumeric representation of <number> corresponding to the entry found in phonebook; used character set should be the one selected with +CSCS
<priority>	integer	N/A	eMLPP priority level of the call

Value:

0÷4 :	priority

<CLI_validity>	integer	N/A	cause of <number> not containing a calling party BCD number

Values:

0 : CLI valid

1 : CLI has been withheld by the originator

2 : CLI is not available due to interworking problems or limitations of originating network

3 : CLI is not available due to calling party being of type payphone

4 : CLI is not available due to other reasons

- If no call is active then only OK message is sent.
- This command is useful in conjunction with command +CHLD to know the various call status for call holding.
- In VOLTE conference call participant's numbers start with "sip:" or "tel:", for example: sip:+12125551212
- Parameter <alpha> is dummy value.

AT+CLCC=?

Test command returns OK result code

AT+CSSN - SS Notification

This command refers to supplementary service related network initiated notifications.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSSN=[<n>[,<m>]]

Set command enables/disables the presentation of notification result codes from TA to TE.

Parameters:

Name	Type	Default	Description
<n>	integer	1	Set +CSSI result code presentation status When <n>=1 and a supplementary service notification is received after a mobile originated call setup, an unsolicited code is sent to TE before any other MO call setup result codes +CSSI: <code1>

Values:

0	:	disable
1	:	enable

<m>	integer	1	Sets the +CSSU result code presentation status When <m>=1 and a supplementary service notification is received during a mobile terminated call setup or during a call, an unsolicited result code is sent to TE +CSSU: <code2>

Values:

0	:	disable
1	:	enable

Unsolicited fields:

Name	Type	Description
<code1>	integer	+CSSI supplementary service notification. Values: 0 : unconditional call forwarding is active 1 : some of the conditional call forwardings are active 2 : call has been forwarded 3 : call is waiting 5 : outgoing calls are barred 6 : incoming calls are barred

<code2>	integer	+CSSU supplementary service notification. Values: 0 : this is a forwarded call (MT call setup) 2 : call has been put on hold (during a voice call) 3 : call has been retrieved (during a voice call) 4 : multiparty call entered (during a voice call) 5 : call on hold has been released (this is not a SS notification) (during a voice call)

AT+CSSN?

Read command reports the current value of the parameters.

AT+CSSN=?

Test command reports the supported range of values for parameters <n>, <m>.

AT+CPOL - Preferred Operator List

This command is used to edit or update the SIM preferred list of networks.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CPOL=[<index>] [,<format>[,<oper>[,<GSM_AcT>,<GSM_Compact_AcT>,<UTRAN_AcT>,<E_UTRAN_AcTn>]]]

Execution command writes an entry in the SIM list of preferred operators

Parameters:

Name	Type	Default	Description
<index>	integer	N/A	the order number of operator in the SIM preferred operator list. Value: 1÷n : order number in the list
<format>	integer	N/A	format for <oper> parameter. Values: 0 : long format alphanumeric <oper> 1 : short format alphanumeric <oper>

	2 : numeric		
<oper>	string	-	Operator Identifier.
<GSM_AcT>	integer	N/A	GSM access technology.
	Values:		
	0 : access technology not selected		
	1 : access technology selected		
<GSM_Compact_AcT>	integer	N/A	GSM compact access technology.
	Values:		
	0 : access technology not selected		
	1 : access technology selected		
<UTRAN_AcT>	integer	N/A	UTRAN access technology.
	Values:		
	0 : access technology not selected		
	1 : access technology selected		
<E_UTRAN_AcTn>	integer	N/A	E-UTRAN access technology
	Values:		

0	:	access technology not selected
1	:	access technology selected

- If <index> is used, and <oper> is not entered, the entry is deleted from the list of preferred operators.
- If <oper> is used, and <index> is not used, <oper> is put in the next free location.
- If only <format> is entered, the format of the <oper> in the read command is changed.
- Currently <GSM_Compact_AcT> not supported but set value is acceptable.

AT+CPOL?

Read command returns all used entries from the SIM list of preferred operators.

AT+CPOL=?

Test command returns the <index> range supported by the SIM and the range for the <format> parameter.

AT+CPLS - Selection of Preferred PLMN List

This command is used to select a list of preferred PLMNs in the SIM/USIM card.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CPLS=<list>

Set command select one PLMN selector with Access Technology list in the SIM card or active application in the UICC (GSM or USIM), that is used by +CPOL command.

Parameter:

Name	Type	Default	Description
<list>	integer	0	PLMNs list selector

Values:

0 : User controlled PLMN selector with Access Technology EFPLMNwAcT, if not found in the SIM/UICC then PLMN preferred list EFPLMNsel (this file is only available in SIM card or GSM application selected in UICC)

1 : Operator controlled PLMN selector with Access Technology EFOPLMNwAcT

2 : HPLMN selector with Access Technology EFHPLMNwAcT

If trying to set + CPLS which is not in supported range, an error would be returned.

AT+CPLS?

Read command returns the selected PLMN selector list from the SIM/USIM

+CPLS: <list>

AT+CPLS=?

Test command returns the whole index range supported lists by the SIM /USIM

AT+CSQ - Signal Quality

This command reports received signal quality indicators.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSQ

Execution command reports received signal quality indicators in the form:
+CSQ: <rssi>,<sq>

See Additional info section.

Additional info:

3GPP

Name	Type	Default	Description
<rssi>	integer	N/A	received signal strength indication

Values:

0	:	-113 dBm or less
1	:	-111 dBm
2÷30	:	-109 dBm ... -53 dBm; 2 dBm per step
31	:	-51 dBm or greater
99	:	not known or not detectable

TDSCDMA

Name	Type	Default	Description
<rssi>	integer	N/A	received signal strength indication

Values:

100	:	-116 dBm or less
101	:	-115 dBm
102÷191	:	from -114 dBm to -26 dBm
191	:	-25 dBm or greater
199	:	not known or not detectable

2G (GSM)

Name	Type	Default	Description
<sq>	integer	N/A	signal quality - RXQUAL [bit error rate (in percent)]

Values:

0	:	less than 0.2%
1	:	0.2% to 0.4%
2	:	0.4% to 0.8%

3 : 0.8% to 1.6%

4 : 1.6% to 3.2%

5 : 3.2% to 6.4%

6 : 6.4% to 12.8%

7 : more than 12.8%

99 : not known or not detectable

▶▶ 3G (UTRAN)

Name	Type	Default	Description
<sq>	integer	N/A	signal quality - ECIO [in dBm]

Values:

0 : -1 to 0

1 : -5 to -2

2 : -8 to -6

3 : -11 to -9

4 : -15 to -12

5 : -18 to -16

6 : -22 to -19

7 : -24 to -23

99 : not known or not detectable

▶▶ 4G (LTE)

Name	Type	Default	Description
<sq>	integer	N/A	signal quality - RSRQ [in dBm]

Values:

0 : -4 to -3

1 : -6 to -5

2 : -8 to -7

3 : -10 to -9

4 : -13 to -11

5 : -15 to -14

6 : -17 to-16

7 : -19 to -18

99 : not known or not detectable

This command should be used instead of the %Q and %L commands, since GSM/WCDMA relevant parameters are the radio link ones and no line is present, hence %Q and %L have no meaning

AT+CSQ=?

Test command returns the supported range of parameters values according to the network on which the module is registered.

Although +CSQ is an execution command without parameters, 3GPP TS 27.007 requires the Test command to be defined.

AT#SERVINFO - Serving Cell Information

This command reports information about the serving cell.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SERVINFO

Execution command reports information about serving cell. The information and the format of the returned message depends on the network type.

GSM network

#SERVINFO:<BARFCN>,<dBM>,
[<NetNameAsc>],<NetCode>,<BSIC>,<LAC>,<TA>,<GPRS>[,[<PBARFCN>],
[<NOM>],<RAC>,[PAT]]

WCDMA network

#SERVINFO:<UARFCN>,<dBM>,
[<NetNameAsc>],<NetCode>,<PSC>,<LAC>,<DRX>,<SD>,<RSCP>,<NOM>,<RAC>,<URA>

LTE network

#SERVINFO:<EARFCN>,<dBM>,
[<NetNameAsc>],<NetCode>,<PhysicalCellId>,<TAC>,<DRX>,<SD>,<RSRP>

The parameters are described in the Additional info sections.

Additional info:

▶▶ Parameters meaning.

Name	Type	Default	Description
<BARFCN>	integer	-	BCCH ARFCN of the serving cell
<dBM>	integer	-	received signal strength in dBm.

<NetNameAsc>	string	-	operator name, quoted string or "" if network name is unknown.
<NetCode>	hex	-	country code and operator code.
<BSIC>	string	-	Base Station Identification Code
<LAC>	integer	-	Localization Area Code
<TA>	integer	-	Time Advance: it is available only if a GSM or GPRS is running.
<GPRS>	integer	0	GPRS supported in the cell

Values:

0	:	not supported
1	:	supported

▶▶ The following informations will be present only if GPRS is supported in the cell.

Name	Type	Default	Description
<PBARFCN>	integer	-	Not supported by 3GPP. PBCCH ARFCN of the serving cell; it is printed only if PBCCH is supported by the cell, otherwise the label "hopping" will be printed

<NOM>	string	N/A	Network Operation Mode

Values:

"I"	:	Network Mode I
"II"	:	Network Mode II
"III"	:	Network Mode III

<PAT>	integer	N/A	Priority Access Threshold.

Values:

0	:	Priority Access Threshold
3÷6	:	Priority Access Threshold

▶▶ Parameters meaning.

Name	Type	Default	Description
<UARFCN>	integer	-	UMTS ARFCN of the serving cell
<PSC>	integer	-	Primary Scrambling Code

▶▶ Parameters meaning.

Name	Type	Default	Description

<DRX>	integer	-	Discontinuous reception cycle length.
<SD>	integer	N/A	Service Domain

Values:

0	:	No Service
1	:	CS only
2	:	PS only
3	:	CS & PS

<RSCP>	integer	-	Received Signal Code Power in dBm.
<EARFCN>	integer	-	E-UTRA Assigned Radio Channel
<PhysicalCellId>	integer	-	Physical Cell ID
<TAC>	integer	-	Tracking Area Code
<RSRP>	integer	-	Reference Signal Received Power
<URA>	integer	-	UTRAN Registration Area Identity

AT#SERVINFO=?

Test command returns OK result code.

AT#BCCHLOCK - Lock to Single BCCH ARFCN

This command enables/disables the single BCCH ARFCN locking.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#BCCHLOCK=<LockedBcch>[,<LockedUarfcn>[,<LockedPsc>[,<LockedEarfcn>[,<LockedPci>]]]]

This command allows to set the single BCCH ARFCN (also UARFCN and EARFCN) the device must be locked to, selectable within those allowed for the specific product.

Parameters:

Name	Type	Default	Description
<LockedBcch>	integer	1024	enable/disable 2G BCCH locking

Values:

1024	:	disable
0÷124	:	enable on GSM 900MHz
975÷1023	:	enable on GSM 900MHz
512÷885	:	enable on DCS 1800MHz
128÷251	:	enable on GSM 850MHz
512÷810	:	enable on PCS 1900MHz

<LockedUarfcn>	integer	0	enable/disable 3G BCCH locking

Value:

	0	:	disable

<LockedPsc>	integer	65535	enable/disable 3G BCCH locking Primary Scrambling Code selection

Value:

65535	:	disable

<LockedEarfcn>	integer	0	a number in the range 0 - 68935 representing the EARFCN to search. A value of 0 will remove the EARFCN restriction and any associated Physical Cell ID lock.

Values:

0	:	disable
1÷68935	:	enable on downlink EARFCN in LTE supported bands (some values in range 0-68935 are not supported according to product band configuration)

<LockedPci>	hex	0	E-UTRAN physical cell ID in hexadecimal format. Valid range 0 - 1F7. If parameter <LockedEarfcn> is 0 then this value is forced to 0.

Value:

0÷1F7	:	Physical cell id

- The values set by command are directly stored in NVM. Reboot is required for lock and unlock function.
- If selected locked 2G BCCH is not in power scan list, the locking function does not work and normal cell selection is performed.
- If selected locked BCCH is available but the module is not allowed to register to the corresponding PLMN, the module will be able to perform only emergency calls and will not select an alternative BCCH.
- If selected locked 2G BCCH is available, the module, in idle and in GPRS data transfer, will not perform reselection to another cell/ ARFCN.
- If selected locked 2G BCCH is available, the module, in GSM data transfer (voice call, data call, SMS), will not perform handover to another cell.
- #BCCHLOCK setting implies a RAT selection, hence it is not recommended to use this command together with +WS46.
- #BCCHLOCK setting has higher priority than PLMN selection, hence it is not recommended to use this command together with manual PLMN selection +COPS=1
- In the situation that LTE locked function is used so certain frequency and PCI should be fixed, if the handover event comes down from the network or it is in poor signal strength environment, the RLF may occur continuously. Also, even if reselection event has happened on itself, it cannot move another cells or frequencies. So, this function should be used carefully.

AT#BCCHLOCK?

Read command reports the currently stored parameters in the format:

#BCCHLOCK:
<LockedBcch>,<LockedUarfcn>,<LockedPsc>,<LockedEarfcn>,<LockedPci>

AT#BCCHLOCK=?

Test command reports the supported range of values for parameter:

<LockedBcch>,<LockedUarfcn>,<LockedPsc>,<LockedEarfcn> and <LockedPci>

AT#BND - Select Band

This command selects RF bands

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#BND=<band>[,<UMTS_band>[,<LTE_band>]]

Set command selects the current GSM and LTE bands.

Parameters:

Name	Type	Default	Description
<band>	integer	N/A	GSM band selection. The default value depends on the product. Values: 0 : GSM 900 MHz + DCS 1800 MHz 1 : GSM 900 MHz + PCS 1900 MHz 2 : GSM 850 MHz + DCS 1800 MHz 3 : GSM 850 MHz + PCS 1900 MHz 4 : GSM 900 MHz + DCS 1800 MHz + PCS 1900 MHz 5 : GSM 900 MHz + DCS 1800 MHz + PCS 1900 MHz + GSM 850 MHz
<UMTS_band>	integer	127	UMTS band selection.

			The default value depends on the product

Value:

127	:	No bands allowed

<LTE_band>	hex	N/A	indicates the LTE supported bands expressed as the sum of Band number (1+2+8 ...) calculated as shown in the table (mask of 32 bits):

Values:

0x00000	:	No bands allowed
0x00001	:	EUTRAN BAND1
0x00002	:	EUTRAN BAND2
0x00004	:	EUTRAN BAND3
0x00008	:	EUTRAN BAND4
0x00010	:	EUTRAN BAND5
0x00040	:	EUTRAN BAND7
0x00080	:	EUTRAN BAND8
0x00100	:	EUTRAN BAND9
0x00800	:	EUTRAN BAND12
0x01000	:	EUTRAN BAND13

0x02000	:	EUTRAN BAND14
0x10000	:	EUTRAN BAND17
0x20000	:	EUTRAN BAND18
0x40000	:	EUTRAN BAND19
0x80000	:	EUTRAN BAND20
0x1000000	:	EUTRAN BAND25
0x2000000	:	EUTRAN BAND26
0x8000000	:	EUTRAN BAND28
0x00200000000	:	EUTRAN BAND34
0x02000000000	:	EUTRAN BAND38
0x04000000000	:	EUTRAN BAND39
0x08000000000	:	EUTRAN BAND40
0x10000000000	:	EUTRAN BAND41
0x8000000000000	:	EUTRAN BAND66
0x0800000000000000	:	EUTRAN BAND71

127 - GSM/WCDMA invalid value

In Set command LTE band mask should be entered in HEX format without "0x".

In Read and Test commands it also appears without "0x".

In Set command, "NULL" input value is acceptable except last input parameter.

And "NULL" means that previous value is remained.

Example:

AT#BND=5,15,800C5

OK

AT#BND=5,,800C5

OK

AT#BND?

Read command returns the current selected bands in the format:

#BND: <band>,<UMTS_band>,<LTE_band>

AT#BND=?

Test command returns the supported range of values of parameters <band>, <UMTS_band> and <LTE_band>

Set command

AT#BND=0,127,800C5

OK

AT#BND=0,,800C5

OK

Test command

AT#BND=?

#BND: (0),(127),(80800C5)

OK

Read command

AT#BND?

#BND: 0,127,800C5

OK

AT#AUTOBND - Automatic Band Selection

This command has no effect and is included only for backward compatibility.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#AUTOBND=[<value>]

Set command returns the OK result code.

Parameter:

Name	Type	Default	Description
<value>	integer	2	dummy values (It has no effect and is included only for backward compatibility)

Values:

0 : only for backward compatibility.

1 : only for backward compatibility.

2 : only for backward compatibility.

AT#AUTOBND?

Read command returns the OK result code.

AT#AUTOBND=?

Test command returns the supported values for parameter <value>.

AT#CEERNET - Extended Numeric Error Report for Network Reject Cause

The command is related to extended numeric error report.

3GPP TS 24.008
3GPP TS 24.301

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#CEERNET

Execution command causes the TA to return a numeric code in the intermediate response format:

#CEERNET: <code>

which should offer the user of the TA a report for the last mobility management (MMGMM/EMM) or session management (SM/ESM) procedure not accepted by the network.

Additional info:

The following error codes are valid for mobility management (MM/GMM) or session management (SM), i.e. for 2G and 3G networks.

In 4G network the <code>s meanings are included in tables 9.9.4.4.1 (for ESM causes) and 9.9.3.9.1 (for EMM cause) of 3GPP TS 24.301 Release 9.

Name	Type	Default	Description
<code>	integer	N/A	error code

Values:

1	:	SM unknown cause
2	:	IMSI unknown in HLR
3	:	illegal MS
4	:	IMSI unknown in VISITOR LR
5	:	IMEI not accepted
6	:	illegal ME
7	:	GPRS not allowed
8	:	Operator determined barring (SM cause failure)/ GPRS and not GPRS not allowed (GMM cause failure)
9	:	MS identity cannot be derived by network
10	:	implicitly detached
11	:	PLMN not allowed
12	:	LA not allowed
13	:	roaming not allowed

14 : GPRS not allowed in this PLMN

15 : no suitable cells in LA

16 : MSC TEMP not reachable

17 : network failure

20 : MAC failure

21 : SYNCH failure

22 : congestion

23 : GSM authentication unacceptable

24 : MBMS bearer capabilities insufficient for the service

25 : LLC or SNDCP failure

26 : insufficient resources

27 : missing or unknown APN

28 : unknown PDP address or PDP type

29 : user authentication failed

30 : activation rejected by GGSN

31 : activation rejected unspecified

32 : service option not supported

33 : req. service option not subscribed

34 : serv. option temporarily out of order

35 : NSAPI already used

36 : regular deactivation

37 : QOS not accepted

38 : call cannot be identified (MM cause failure) / SMN network failure (SM cause failure)

39 : reactivation required

40 : no PDP context activated (GMM cause failure) / feature not supported (SM cause failure)

41 : semantic error in TFT operation

42 : syntactical error in TFT operation

43 : unknown PDP context

44 : semantic err in PKT filter

45 : syntactical err in PKT filter

46 : PDP context without TFT activated

47 : multicast group membership timeout

48 : retry on new cell begin (if MM cause failure) / activation rejected BCM violation (if SM cause failure)

50 : PDP type IPV4 only allowed

51 : PDP type IPV6 only allowed

52 : single address bearers only allowed

63 : retry on new cell end

81 : invalid transaction identifier

95 : semantically incorrect message

96 : invalid mandatory information

97 : MSG type non-existent or not implemented

98 : MSG type not compatible with protocol state

99 : IE non-existent or not implemented

100 : conditional IE error

101 : MSG not compatible with protocol state

111 : protocol error unspecified

112 : APN restriction value incompatible with active PDP context

- If none of this condition has occurred since power up then <code> 0: "Normal, unspecified" condition is reported.
- The <code> 1 cannot be used in OTA.
- Detach request reject cause from the network are also reported.

AT#CEERNET=?

Test command returns OK result code.

AT#PSNT - Packet Service Network Type

The command enables/disables unsolicited result code for packet service network type (PSNT)

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#PSNT=[<mode>]

Set command enables/disables unsolicited result code for packet service network type (PSNT) having the following format:

#PSNT:<nt>

Parameter:

Name	Type	Default	Description
<mode>	integer	0	enables/disables PSNT unsolicited result code.

Values:

0 : disable PSNT unsolicited result code

1 : enable PSNT unsolicited result code

2 : PSNT unsolicited result code is enabled, and read command returns the message shown in the read section.

Unsolicited field:

Name	Type	Description
<nt>	integer	network type Values:

0	:	GPRS network
1	:	EGPRS network
2	:	WCDMA network
3	:	HSDPA network
4	:	LTE network
5	:	unknown or not registered

When the type of network is HSPA, the indication is certainly valid during traffic, while it could be not valid in idle because it depends on network broadcast parameters.

AT#PSNT?

If <mode> is set to 0 or 1, read command returns the current values of the <mode> and <nt> parameters in the format:

#PSNT: <mode>,<nt>

If <mode> is set to 2, read command returns the current values of <mode> and <nt> parameters followed by four dummy parameters set to 0.

#PSNT:<mode>,<nt>,0,0,0,0

AT#PSNT=?

Test command reports the range for the parameter <mode>

AT+CEMODE - Set Mode of Operation for EPS

This set command configures the mode of operation for EPS.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CEMODE=<mode>

Parameter:

Name	Type	Default	Description
<mode>	integer	1	mode of operation.

Values:

0 : PS mode 2 of operation

1 : CS/PS mode 1 of operation

2 : CS/PS mode 2 of operation

3 : PS mode 1 of operation

The definition for UE modes of operation can be found in 3GPP TS 24.301. Other values are reserved and will result in an ERROR response to the set command.

AT+CEMODE=6

returns ERROR

AT+CEMODE?

Read command returns the current value of parameter <mode> in the format:

+CEMODE: < mode >

The read command will return right values after set command, but effectively the mode of operation changes after power cycle.

AT+CEMODE=?

Test command returns the supported range of values of parameters <mode>.

+CEMODE: (0-3)

Set EPS mode
AT+CEMODE=1
OK

Check EPS mode
AT+CEMODE?
+CEMODE: 1
OK

AT+CESQ - Extended Signal Quality

This command reports received signal quality indicators.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CESQ

Execution command reports the signal quality indicators. See Additional info section.

Additional info:

The execution command returns the following message:

+CESQ: <rxlev>,<ber>,<rscp>,<ecno>,<rsrq>,<rsrp>

Name	Type	Default	Description
<rxlev>	integer	N/A	received signal strength level, see 3GPP document, subclause 8.1.4

Values:

0 : rssi < -110 dBm

1 : - 110 ≤ rssi < -109 dBm

2 : - 109 ≤ rssi < -108 dBm

... : ...

61 : - 50 ≤ rssi < -49 dBm

62 : - 49 ≤ rssi < -48 dBm

63 : rssi ≥ -48 dBm

99 : not known or not detectable or if the current serving cell is not a GERAN cell

<ber>	integer	N/A	bit error rate [in percent]

Values:

0÷7 : as RXQUAL values in the table in 3GPP TS 45.008 subclause 8.2.4

99 : not known or not detectable or if the current serving cell is not a GERAN cell

<rscp>	integer	N/A	received signal code power. See 3GPP TS 25.133 subclause 9.1.1.3 and 3GPP TS 25.123 subclause 9.1.1.1.3.

Values:

0 : rscp < -120 dBm

1 : -120 ≤ rscp < -119 dBm

2 : -119 ≤ rscp < -118 dBm

... : ...

94	:	-27 ≤ rscp < -26 dBm
95	:	-26 ≤ rscp < -25 dBm
96	:	rscp ≥ -25 dBm
255	:	not known or not detectable or if the current serving cell is not a UTRA cell

<ecno>	integer	N/A	ratio of the received energy per PN chip to the total received power spectral density. See 3GPP TS 25.133 subclause.

Values:

0	:	Ec/Io < -24 dB
1	:	-24 ≤ Ec/Io < -23.5 dB
2	:	-23,5 ≤ Ec/Io < -23 dB
...	:	...
47	:	-1 ≤ Ec/Io < -0,5 dB
48	:	-0,5 ≤ Ec/Io < 0 dB
49	:	Ec/Io ≥ 0 dB
255	:	not known or not detectable detectable or if the current serving cell is not a UTRA cell

<rsrq>	integer	N/A	reference signal received quality, see 3GPP TS 36.133 subclause 9.1.7.

Values:

0	:	rsrq < -19.5 dB
1	:	-19.5 ≤ rsrq < -19.0 dB
2	:	-19.0 ≤ rsrq < -18.5 dB
3÷33	:	-18.5 ≤ rsrq < -3.0 dB; 0.5 dB per step
34	:	rsrq ≥ -3.0 dB
255	:	not known or not detectable or if the current serving cell is not a EUTRA cell

<rsrp>	integer	N/A	reference signal received power, see 3GPP TS36.133 subclause 9.1.4.

Values:

0	:	rsrp < -140 dBm
1	:	-140 ≤ rsrp < -139 dBm
2	:	-139 ≤ rsrp < -138 dBm
3÷96	:	-138 ≤ rsrp < -44; 1 dBm per step
97	:	rsrp ≥ -44 dBm

255	:	not known or not detectable or if the current serving cell is not a EUTRA cell

4G/3G only products do not support GSM access technology

AT+CESQ=?

Test command returns the supported range of values of the parameters <rxlev>, <ber>, <rscp>, <ecno>, <rsrq>, <rsrp>.

AT+WS46 - PCCA STD-101 Select Wireless Network

This command selects the cellular network (Wireless Data Service, WDS).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+WS46=[<n>]

Set command selects the cellular network (Wireless Data Service, WDS) to operate with the TA (WDS-Side Stack Selection).

Parameter:

Name	Type	Default	Description
<n>	integer	N/A	WDS-Side Stack to be used by the TA. The values in <n> for Query are mutually exclusive. If one value (e.g. "25") is returned, other values shall not be returned. The factory default value depends on each variant

Values:

12 : GSM Digital Cellular Systems, GERAN only

22 : UTRAN only

25 : 3GPP Systems (GERAN and UTRAN and E-UTRAN)

28 : E-UTRAN only

29 : GERAN and UTRAN

30 : GERAN and E-UTRAN

31 : UTRAN and E-UTRAN

32 : TDSCDMA only

33 : GERAN and TDSCDMA

34 : TDSCDMA and E-UTRAN

35 : GERAN and TDSCDMA and E-UTRAN

36 : GERAN and TDSCDMA and UTRAN and E-UTRAN

37 : GERAN and TDSCDMA and UTRAN

38 : TDSCDMA and UTRAN

39 : TDSCDMA and UTRAN and E-UTRAN

<n> parameter setting is stored in NVM, and available at next reboot.

AT+WS46?

Read command reports the currently selected cellular network, in the format:

+ WS46: <n>

AT+WS46=?

Test command reports the range for the parameter <n>.

AT+CEREG - EPS Network Registration Status

This command monitors the Evolved Packet System (EPS) network registration status

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CEREG=[<n>]

Set command enables/disables the EPS network registration unsolicited result code (URC), and selects one of the available formats:

short format: +CEREG: <stat>
long format: +CEREG: <stat>[,[<tac>],[<ci>],[<AcT>]]

Parameter:

Name	Type	Default	Description
<n>	integer	0	enables/disables the network registration unsolicited result code (URC), and selects one of the available formats.

Values:

0 : disable the network registration unsolicited result code

1 : enable network registration unsolicited result code; if there is a change in the terminal EPS network registration status, it is issued the URC short format

2 : enable network registration and location information unsolicited result code; if there is a change of the network cell, it is issued the URC long format (includes the network cell identification data)

Unsolicited fields:

Name	Type	Description
<stat>	integer	registration status Values: 0 : not registered, terminal is not currently searching a new operator to register to 1 : registered, home network 2 : not registered, but terminal is currently searching a new operator to register to 3 : registration denied 4 : unknown. Example, out of E-UTRAN coverage 5 : registered, roaming
<tac>	string	tracking area code (two bytes) in hexadecimal format (e.g. "00C3" equals 195 in decimal)
<ci>	string	four byte E-UTRAN cell ID in hexadecimal format

<AcT>	integer	indicates the access technology of the serving cell. Values: 0 : GSM (not applicable) 1 : GSM Compact (not applicable) 2 : UTRAN (not applicable) 3 : GSM w/EGPRS (not applicable) 4 : UTRAN w/HSDPA (not applicable) 5 : UTRAN w/HSUPA (not applicable) 6 : UTRAN w/HSDPA and HSUPA (not applicable) 7 : E-UTRAN

- <tac>, <ci> and <AcT> are reported only if <n>=2 and the mobile is registered on some network cell.
- 3GPP TS 44.060 specifies the System Information messages which give the information about whether the serving cell supports EGPRS.
- 3GPP TS 25.331 specifies the System Information blocks which give the information about whether the serving cell supports HSDPA or HSUPA.

AT+CEREG?

Read command returns the status of result code presentation mode <n> and the integer <stat> which shows whether the network has currently indicated the registration of the terminal in the format:

+CEREG: <n>,<stat>[,<tac>,<ci>[,<AcT>]]

<tac>, <ci> and <AcT> are reported only if <n>=2 and the mobile is registered on some network cell.

AT+CEREG=?

Test command returns supported values for parameter <n>.

AT#RFSTS - Read Current Network Status

Command reads current network status.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#RFSTS

Execution command returns the current network status. The format of the returned message is according to the network on which the module is registered.

GSM network

#RFSTS:<PLMN>,<ARFCN>,<RSSI>,<LAC>,<RAC>,<TXPWR>,<MM>,<RR>,<NOM>,<CID>,<IMSI>,<NetNameAsc>,<SD>,<ABND>

Parameters	Description
<PLMN>	Country code and operator code (MCC, MNC)
<ARFCN>	GSM Assigned Radio Channel
<RSSI>	Received Signal Strength Indication
<LAC>	Localization Area Code
<RAC>	Routing Area Code
<TXPWR>	Tx Power

Parameter/values	Description
<MM>	Mobility Management state (for debug purpose only)
-	Ignore these values

Parameter/values	Description
<RR>	Radio Resource state (for debug purpose only)
-	Ignore these values

Parameters	Descriptions
<NOM>	Network Operator Mode
<CID>	Cell ID
<IMSI>	International Mobile Subscriber Identity
<NetNameAsc>	Operator name

Parameter/values	Description
<SD>	Service Domain
0	No Service
1	CS only
2	PS only
3	CS+PS

Parameter/values	Description
<ABND>	Active Band
1	GSM 850

2	GSM 900
3	DCS 1800
4	PCS 1900

LTE network

#RFSTS:<PLMN>,<EARFCN>,<RSRP>,<RSSI>,<RSRQ>,<TAC>,<RAC>,[<TXPWR>],<DRX>,<MM>,<RRC>,<CID>,<IMSI>,[<NetNameAsc>],<SD>,<ABND>,<T3402>,<T3412>,<SINR>

Parameters	Description
<PLMN>	Country code and operator code(MCC, MNC)
<EARFCN>	E-UTRA Assigned Radio Channel
<RSRP>	Reference Signal Received Power
<RSSI>	Received Signal Strength Indication
<RSRQ>	Reference Signal Received Quality
<TAC>	Tracking Area Code
<RAC>	Routing Area Code
<TXPWR>	Tx Power (In traffic only)
<DRX>	Discontinuous reception cycle Length (cycle length in ms)

Parameter/values	Description
<MM>	Mobility Management state (for debug purpose only)

-	Ignore these values

Parameters	Description
<RRC>	Radio Resource state (for debug purpose only)
-	Ignore these values

Parameters	Description
<CID>	Cell ID
<IMSI>	International Mobile Station ID
<NetNameAsc>	Operator name

Parameter/values	Description
<SD>	Service Domain
0	No Service
1	CS only
2	PS only
3	CS+PS

Parameters/values	Description
<ABND>	Active Band
1..63	According to 3GPP TS 36.101

Parameters	Description
<T3402>	Timer T3402 in seconds
<T3412>	Timer T3412 in seconds
<SINR>	Signal-to-Interface plus Noise Ratio

AT#RFSTS=?

Test command tests for command existence.

AT#MONI - Cell Monitor

This command is both a set and an execution command.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MONI[=[<number>]]

Set command sets one cell out of seven, in a neighbor list of the serving cell including it, from which extract GSM/LTE related information.
After entering the set command, enter the execution command AT#MONI<CR> to get the GSM/LTE related information for the selected cell and dedicated channel (if exists) in the format shown, for each network, in the Additional info section.

Parameter:

Name	Type	Default	Description
<number>	integer	-	the parameter meaning depends on the network, see Additional info section.

Additional info:

GSM network

Name	Type	Default	Description
<number>	integer	0	GSM network

Values:

0÷6 : it is the ordinal number of the cell, in the neighbor list of the serving cell.

7	:	it is a special request to obtain GSM-related information from the whole set of seven cells in the neighbor list of the serving cell

- LTE network

Name	Type	Default	Description
<number>	integer	0	LTE network

Values:

0	:	it is the serving cell
1	:	it is the intra-frequency cells
2	:	it is the inter-frequency cells
3	:	it is the W-CDMA neighbor cells, the report message is empty.
4	:	it is the GSM neighbor cells
5,6	:	it is not available
7	:	it is a special request to obtain LTE-related information from the all available neighbor cells.

- Execution command AT#MONI<CR> reports GSM/LTE related information for selected cell and dedicated channel (if exists) in the following formats:

a) When extracting data for the serving cell and the network name is known the format is:

GSM network

#MONI: <netname> BSIC:<bsic> RxQual:<qual> LAC:<lac> Id:<id> ARFCN:<arfcn> PWR:<dBm> dBm TA: <timadv>

LTE network

#MONI: <netmame> RSRP:<rsrp> RSRQ:<rsrq> TAC:<tac> Id:<id> EARFCN:<earfcn> PWR:<dBm> DRX:<drx> pci:<physicalCellId> QRxLevMin:<QRxLevMin>

b) When the network name is unknown, the format is:

GSM network

#MONI: <cc> <nc> BSIC:<bsic> RxQual:<qual> LAC:<lac> Id:<id> ARFCN:<arfcn> PWR:<dBm> dBm TA: <timadv>

LTE network

#MONI: Cc:<cc> Nc:<nc> RSRP:<rsrp> RSRQ:<rsrq> TAC:<tac> Id:<id> EARFCN:<earfcn> PWR:<dBm> DRX:<drx> pci:<physicalCellId> QRxLevMin:<QRxLevMin>

c) When extracting data for an adjacent cell (or active set cell), the format is:

GSM network

#MONI: Adj Cell<n> [LAC:<lac> Id:<id>] ARFCN:<arfcn> PWR:<dBm>

LTE network

LTE intra-frequency and inter-frequency cells

#MONI: RSRP:<rsrp> RSRQ:<rsrq> PhysCellId:<physicalCellId> EARFCN:<earfcn> PWR:<dBm>

LTE GSM neighbor cells

#MONI: Adj Cell<n> BSIC:<bsic> ARFCN:<arfcn> PWR:<dBm>

Name	Type	Default	Description
<netname>	string	-	name of network operator

<cc>	string	-	country code
<nc>	string	-	network operator code
<n>	integer	-	progressive number of adjacent cell
<bsic>	string	-	base station identification code
<qual>	integer	-	quality of reception: 0..7
<lac>	string	-	localization area code
<id>	integer	-	cell identifier
<arfcn>	integer	-	assigned radio channel
<dBm>	integer	-	received signal strength in dBm.
<timadv>	integer	-	timing advance
<rscp>	integer	-	Received Signal Code Power in dBm.
<drx>	string	-	Discontinuous reception cycle length
<physicalCellId>	integer	-	physical cell identifier
<rsrp>	integer	-	Reference Signal Received Power
<rsrq>	integer	-	Reference Signal Received Quality

<tac>	integer	-	Tracking Area Code
<earfcn>	integer	-	E-UTRA Assigned Radio Channel
<QRxLevMin>	integer	-	minimum required RX level in the cell

- TA: <timadv> reported only for the serving cell.
- When AT#MONI=7 is the last setting entered, the execution command AT#MONI<CR> reports the information previously listed for each of the cells in the neighbor of the serving cell. The information is formatting in a sequence of <CR><LF>-terminated strings.
- The timing advance value is meaningful only during calls or GPRS transfers active.

AT#MONI=?

Test command reports the maximum number of cells, in a neighbor list of the serving cell excluding it, from which we can extract GSM/LTE related information, along with the ordinal number of the current selected cell, in the format:

#MONI: (<MaxCellNo>,<CellSet>)

Additional info:

- Parameters meaning.

Name	Type	Default	Description

<MaxCellNo>	integer	-	maximum number of cells in a neighbor of the serving cell and excluding it from which we can extract GSM/ LTE related information. This value is always 6.
<CellSet>	integer	-	last setting done with command #MONI.

The module supports GSM network

```
AT+WS46?
+WS46: 30
OK
AT+CREG?
+CREG: 0,1
OK
AT#MONI=1
OK
AT#MONI=?
#MONI: (6,1)
OK
AT#MONI
#MONI: I TIM BSIC:25 RxQual:0 LAC:D5BD Id:3A27 ARFCN:1018 PWR:-72dbm
TA:-1
OK
AT#MONI=7
OK
AT#MONI=?
#MONI: (6,7)
```

```
OK
AT#MONI
#MONI: Cell BSIC LAC CellId ARFCN Power C1 C2 TA RxQual PLMN
#MONI: S 25 D5BD 3A27 1018 -74dbm 31 31 4 7 I TIM
#MONI: N1 26 D5BD 3A26 1023 -79dbm -1 -1
#MONI: N2 21 D5BD 5265 1009 -78dbm -1 -1
#MONI: N3 27 D5BD 5266 13 -87dbm -1 -1
#MONI: N4 25 D5BD 5251 1020 -88dbm -1 -1
#MONI: N5 27 D5BD 5286 1011 -95dbm -1 -1
#MONI: N6 30 00D2 C5A0 16 -99dbm -1 -1
OK
```

AT#RPMCFG - RPM function enable

This command enables/disables RPM (Radio Policy Management) function.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#RPMCFG=<rpm_ena>

Set command enables/disable the RPM function.

Parameter:

Name	Type	Default	Description
<rpm_ena>	integer	N/A	Indicates whether RPM functionality is to be enabled or disabled at power up.

Values:

0	:	Disabled
1	:	Enabled

AT#RPMCFG?

Read command reports RPM function status

#RPMCFG: <rpm_ena>

AT#RPMCFG=?

Test command returns OK result.

AT#RPMCFGEXT - RPM function configuration

This command configures RPM (Radio Policy Management) function.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#RPMCFGEXT=<app_max_num_reset>,<app_reset_counter>,<app_reset_timer>,
<app_reset_blk_leak_rate_timer>,
<avg_reset_time>,<leak_rate_time>,<backoff_cnt_0>,<backoff_cnt_1>,
<backoff_cnt_2>,<backoff_cnt_3>,
<cpdp_cnt_0>,<cpdp_cnt_1>,<cpdp_cnt_2>,<cpdp_cnt_3>,<lr3_timer>

Set command sets the RPM configuration parameters.

Parameters:

Name	Type	Default	Description
<app_max_num_reset>	integer	N/A	N1. Max number of SW resets per Hour allowed by RPM following "permanent" MM/GMM/EMM reject

Value:

0÷255	:	Max number of SW resets

<app_reset_counter>	integer	N/A	APP-initiated reset per hour Value: 0÷255 : reset counter
<app_reset_timer>	integer	N/A	1 hour Timer for tracking APP initiated reset Value: 0÷65535 : reset timer
<app_reset_blk_leak_rate_timer>	integer	N/A	LR-1. Leak rate for C-BR-1 (Counter related to N1) Value: 0÷65535 : leak rate timer
<avg_reset_time>	integer	N/A	T1. Average time before RPM resets modem following "permanent" MM/GMM/EMM reject Value:

			0÷255 : reset time
<leak_rate_time>	integer	N/A	LR-2. Leak rate for C-R-1 (Counter related to T1) Value: 0÷65535 : leak rate time
<backoff_cnt_0>	integer	N/A	F1. Max number of PDP Activation Requests / PDN Connectivity Requests per Hour allowed by RPM following a PDP Activation ignore Scenario. Value: 0÷255 : backoff_cnt_0
<backoff_cnt_1>	integer	N/A	F2. Max number of PDP Activation Requests / PDN Connectivity Requests per Hour allowed by RPM following a "permanent" PDP Activation Reject.

	Value:		
	0÷255	:	backoff_cnt_1
<backoff_cnt_2>	integer	N/A	F3. Max number of PDP Activation Requests / PDN Connectivity Requests per Hour allowed by RPM following a "temporary" PDP Activation Reject.
	Value:		
	0÷255	:	backoff_cnt_2
<backoff_cnt_3>	integer	N/A	F4. Max number of PDP context or PDN Connectivity Activation/ Deactivation Requests per Hour allowed by RPM
	Value:		
	0÷255	:	backoff_cnt_3
<cpdp_cnt_0>	integer	N/A	CPDP-1
	Value:		

	0÷255	:	cpdp_cnt_0

<cpdp_cnt_1>	integer	N/A	CPDP-2

Value:

	0÷255	:	cpdp_cnt_1

<cpdp_cnt_2>	integer	N/A	CPDP-3

Value:

	0÷255	:	cpdp_cnt_2

<cpdp_cnt_3>	integer	N/A	CPDP-4

Value:

	0÷255	:	cpdp_cnt_3

<lr3_timer>	integer	N/A	LR-3. Leak Rate3 timer (hour). Leak rate for CPDP-1 to C-PDP-4 (Counter related to F1)

Value:

	0÷255	:	lr3_timer

AT#RPMCFGEXT?

Read command reports the RPM configuration parameters.

#RPMCFGEXT: <app_max_num_reset>,<app_reset_counter>,<app_reset_timer>,
<app_reset_blk_leak_rate_timer>,
<avg_reset_time>,<leak_rate_time>,<backoff_cnt_0>,<backoff_cnt_1>,
<backoff_cnt_2>,<backoff_cnt_3>,
<cpdp_cnt_0>,<cpdp_cnt_1>,<cpdp_cnt_2>,<cpdp_cnt_3>,<lr3_timer>

AT#RPMCFGEXT=?

Test command returns OK result.

AT+CLIP - Calling Line Identification Presentation

This command enables/disables the presentation of the CLI (Calling Line Identity).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CLIP=[<n>]

Set command refers to the supplementary service CLIP (Calling Line Identification Presentation) that enables a called subscriber to get the calling line identity (CLI) of the calling party when receiving a mobile terminated call. If enabled the device reports after each RING the response:

+CLIP: <number>,<type>

Parameter:

Name	Type	Default	Description
<n>	integer	0	enables/disables CLI indication.

Values:

0 : disable

1 : enable

Unsolicited fields:

Name	Type	Description
<number>	string	phone number of format specified by <type>
<type>	integer	type of address octet Values: 128 : both the type of number and the numbering plan are unknown 129 : unknown type of number and ISDN/ Telephony numbering plan 145 : international type of number and ISDN/Telephony numbering plan (contains the character "+") 161 : National type of number and ISDN/ Telephony numbering plan

in the +CLIP: response they are currently not reported either the subaddress information (it's always "" after the 2nd comma) and the subaddress type information (it's always 128 after the 3rd comma)

AT+CLIP?

Read command returns the presentation status of the CLI in the format:

+CLIP: <n>,<m>

Additional info:

Parameters:

Name	Type	Default	Description
<n>	integer	N/A	status of the local setting
	Values:		
	0 :	CLI presentation disabled	
	1 :	CLI presentation enabled	
<m>	integer	N/A	status of the CLIP service on the network
	Values:		
	0 :	CLIP not provisioned	
	1 :	CLIP provisioned	
	2 :	unknown (e.g. no network is present)	

Read command issues a status request to the network, hence it may take a few seconds to give the answer due to the time needed to exchange data with it.

AT+CLIP=?

Test command returns the supported values of parameter <n>

The command changes only the report behavior of the device, it does not change CLI supplementary service setting on the network.

AT#CODEC - GSM and UMTS Audio Codec

GSM and UMTS audio codec mode settings

3GPP TS 24.008

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#CODEC=[<codec>]

Set command sets the GSM and UMTS audio codec mode.

Parameter:

Name	Type	Default	Description
<codec>	integer	63	1-255: sum of integers each representing a specific codec mode as follows: - 1 = FR, full rate mode enabled (always enabled and cannot be disabled) - 2 = EFR, enhanced full rate mode enabled (always enabled and cannot be disabled) - 4 = HR, half rate mode enabled (always enabled and cannot be disabled) - 8 = AMR-FR, AMR full rate mode enabled (always enabled and cannot be disabled) - 16 = AMR-HR, AMR half rate mode enabled (always enabled and cannot be disabled) - 32 = FAWB, full rate AMR wide band - 64 = UAMR2, UMTS AMR version 2 (always disabled and cannot be enabled) - 128 = UAWB, UMTS AMR wide band (always disabled and cannot be enabled)

Value:

0÷255	:	codec modes setting

AT#CODEC?

Read command returns current audio codec mode in the format:

#CODEC: <codec>

AT#CODEC=?

Test command returns the range of available values for parameter <codec>.

```
AT#CODEC=31
OK
```

SIM

AT+SWITCHSIM - Set master SIM

This command sets the master SIM. Only applied for dual-SIM projects

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+SWITCHSIM=<simID>

Set command is used to set the master SIM.

Parameter:

Name	Type	Default	Description
<simID>	integer	0	SIM identifier

Values:

0	:	SIM 1
1	:	SIM2

AT+SWITCHSIM?

Read command returns current value of the parameter <simID> in the format:

+SWITCHSIM: <simID>

OK

AT+SWITCHSIM=?

Test command returns the supported range of values for the parameter <simID>.

This command can be used for dual SIM card switching only through UART port.

USB port not supported.

AT+DUALSIM - Set Dual SIM Mode

This command sets the Dual SIM mode. Only applied for dual-SIM projects.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+DUALSIM=<mode>

Set command is used to set the Dual SIM mode.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	SIM identifier

Values:

0 : Dual-SIM Dual Standby mode (DSDS)

1 : Dual-SIM Single Standby mode (DSSS)

AT+DUALSIM?

Read command returns current value of the parameter <mode> in the format:

+DUALSIM: <mode>

OK

AT+DUALSIM=?

Test command returns the supported range of values for the parameter <mode>.

Due to Network Module interoperability potential limitations it is recommended to configure Dual-SIM Single Standby (DSSS) mode (AT+DUALSIM=1) when SIM1 and SIM2 are present at the same time and DSDS functionality is not needed.

AT#PCT - Display PIN Counter

This command reports the PIN/PUK or PIN2/PUK2 input remaining attempts, if +CPIN password is required.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#PCT

Execution command reports the PIN/PUK or PIN2/PUK2 input remaining attempts, depending on +CPIN requested password, in the format:

#PCT: <n>

Additional info:

▶▶ Here is shown the parameter meaning.

Name	Type	Default	Description
<n>	integer	N/A	remaining attempts.

Values:

0	:	the SIM is blocked
1÷3	:	if the device is waiting either SIM PIN or SIM PIN2 to be given.
1÷10	:	if the device is waiting either SIM PUK or SIM PUK2 to be given.

AT#PCT=?

Test command returns the OK result code.

```
AT+CPIN?
+CPIN: SIM PIN
OK

AT#PCT                    Check PIN remained counter
#PCT: 3
OK

AT+CPIN=1111              Input incorrect PIN number
+CME ERROR: incorrect password

AT#PCT
#PCT: 2
OK
```

AT+CCID - Read ICCID

Execution command reads on SIM the Integrated Circuit Card Identification (ICCID). It is the card identification number that provides a unique identification number for the SIM.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CCID

The command returns the following message:

+CCID: <ICCID>
OK

AT+CCID?

Read command returns the following message:

+CCID: <ICCID>
OK

AT+CCID=?

Test command returns the OK result code.

```
AT+CCID
+CCID: 89861109091740011006
OK
```

AT+CIMI - International Mobile Subscriber Identity (IMSI)

This command returns the International Mobile Subscriber Identity (IMSI number).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CIMI

Execution command returns the value of the International Mobile Subscriber Identity stored in the SIM, the returned message has the following format (with command no echoed):

<IMSI value>
OK

If the SIM is not inserted, the command returns ERROR.

AT+CIMI=?

Test command returns OK result code

```
AT+CIMI
22201701202507
OK
```

AT#CIMI - International Mobile Subscriber Identity (IMSI)

This command returns the International Mobile Subscriber Identity (IMSI).

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#CIMI

Execution command returns the value of the International Mobile Subscriber Identity stored in the SIM with command echo

If the SIM is not inserted, the command returns ERROR.

AT#CIMI=?

Test command returns the OK result code.

```
AT#CIMI
#CIMI: 450050209516643
OK
```

AT#SIMDET - SIM Detection Mode

The command manages the SIM Detection mode.

[1] Telit Hardware Design Guide of the used module.

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT#SIMDET=<mode>

Set command simulates the SIM status, or selects the automatic SIM Detection status. This command is used by modules providing the dedicated SIMIN pin. Refer to document [1] to have information on dedicated SIMIN pin.

Parameter:

Name	Type	Default	Description
<mode>	integer	2	SIM detection mode

Values:

0 : ignore dedicated SIMIN pin and simulate the status "SIM Not Inserted"

1 : ignore dedicated SIMIN pin and simulate the status "SIM Inserted"

2 : automatic SIM detection using dedicated SIMIN Pin

When #SIMDET=1 (that simulates "SIM Inserted") is issued, a query to detect the presence of the SIM is forced, regardless of SIMIN pin status.
If SIM is not responding (for example, because it is not present) then, after a timeout, the modem gives up and sets the SIM status

to SIM NOT INSERTED.
If SIM is correctly answering to queries then SIM status becomes SIM READY.

AT#SIMDET?

Read command returns the currently selected Sim Detection Mode in the format:

#SIMDET: <mode>,<simIn>

Additional info:

The values for <simIn> are:

Name	Type	Default	Description
<simIn>	integer	N/A	SIMIN pin status.
	Values:		
	0 : SIM not inserted		
	1 : SIM inserted		

AT#SIMDET=?

Test command reports the supported range of values for parameter <mode>.

AT#CCID - Read ICCID

Execution command reads on SIM the Integrated Circuit Card Identification (ICCID). It is the card identification number that provides a unique identification number for the SIM.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#CCID

The command returns the following message:

#CCID: <ICCID>
OK

AT#CCID=?

Test command returns the OK result code.

```
AT#CCID
#CCID: 89861109091740011006
OK
```

AT#QSS - Query SIM Status

Query SIM Status.

SIM Presence	Setting saved	Can be aborted
Not required	Specific profile	No

AT#QSS=[<mode>]

Set command enables/disables the Query SIM Status unsolicited indication in the ME. The format of the unsolicited indication is the following:

#QSS: <status>

The parameter is described in the Unsolicited field section.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	Type of notification

Values:

0 : disabled. It is only possible to query the current SIM status through read command AT#QSS?

1 : enabled. The ME informs at every SIM status change through the unsolicited indication where <status> range is 0...1

2 : enabled. The ME informs at every SIM status change through the unsolicited indication where <status> range is 0...3

Unsolicited field:

Name	Type	Description

<status>	integer	current SIM status Values: 0 : SIM not inserted 1 : SIM inserted 2 : SIM inserted and PIN unlocked 3 : SIM inserted and READY (SMS and Phonebook access are possible)

The command reports the SIM status change after the <mode> has been set to 2.

It is strongly suggested to set <mode>=2 and save the value in the user profile, then power off the module. The proper SIM status will be available at the next power on.

AT#QSS?

Read command reports whether the unsolicited indication #QSS is currently enabled or not, along with the SIM status, in the format:

#QSS: <mode>,<status>

The parameters are described in the previous sections.

AT#QSS=?

Test command returns the supported range of values for parameter <mode>.

AT+CRSM - Restricted SIM access

The command transmits to the UICC some specific commands and their required parameters.

3GPP TS 11.11
3GPP TS 27.007
3GPP TS 51.011

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CRSM=<command>[,<fileId>[,<P1>[,<P2>[,<P3>[,<data>]]]]]

Set command transmits to the UICC the specific command and its required parameters. The command response shows the status words and response data returned by the UICC in the format:

+CRSM: <sw1>,<sw2>[,<response>]

For parameters meanings see Additional info section.

Parameters:

Name	Type	Default	Description
<command>	integer	176	command passed on to the UICC

Values:

176	:	READ BINARY
178	:	READ RECORD
192	:	GET RESPONSE
214	:	UPDATE BINARY

220	:	UPDATE RECORD
242	:	STATUS

<fileId>	integer	-	identifier of an elementary data file on UICC; mandatory for every command except STATUS
<P1>	integer	0	parameter passed on to the UICC; it is mandatory for every command except GET RESPONSE and STATUS

Value:

0÷255	:	parameter P1 passed on to the UICC in a command APDU

<P2>	integer	0	parameter passed on to the UICC; it is mandatory for every command except GET RESPONSE and STATUS

Value:

0÷255	:	parameter P2 passed on to the UICC in a command APDU

<P3>	integer	0	parameter passed on to the UICC; it is mandatory for every command except GET RESPONSE and STATUS

Value:

	0÷255	:	parameter P3 passed on to the UICC in a command APDU
<data>	string	-	information to be read from UICC or written to the UICC (hexadecimal character format).

Additional info:

▶▶ Response data fields:

Name	Type	Default	Description
<sw1>	integer	-	information from the UICC about the execution of the actual command (successful or failed)
<sw2>	integer	-	information from the UICC about the execution of the actual command (successful or failed)
<response>	hex	-	on a successful completion of the previously issued command it shows the response data. It is not returned after a successful UPDATE BINARY or UPDATE RECORD command

This command requires PIN authentication. However commands READ BINARY and READ RECORD can be issued before PIN authentication and if the SIM is blocked (after three failed PIN

authentication attempts) to access the contents of the Elementary Files.

Use only decimal numbers for parameters <command>,<fileid>, <P1>, <P2> and <P3>.

AT+CRSM=?

Test command returns the OK result code

Read binary, ICCID(2FE2)

```
AT+CRSM=176,12258,0,0,10
+CRSM: 144,0,982850702001107686F4
OK
```

Read record, ADN(6F3A)

```
AT+CRSM=178,28474,1,4,40
+CRSM: 144,0,42434A554EFFFFFFFFFFFFFFFFFFFFFFFF06811056789282FFFFFFFFFFFFFF
OK
```

Update Binary, KcGPRS(6F52)

```
AT+CRSM=214,28539,0,0,8,C69018C7958C87
+CRSM: 144,0
OK
```

Update Record, ADN(6F3A)

```
AT+CRSM=220,28474,9,4,30,657469FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
FFFFFFFFFFF
+CRSM: 144,0
OK
```

Status, FPLMN(6F7B)

```
AT+CRSM=242,28539
+CRSM:144,0,623C820238218410A0000000871002FFFFFFFF89040300FFA51180013181030
10A3282011E8304000030E08A01058B032F0609C6099001C0830101830181
OK
```

AT+CSIM - Generic SIM Access

This command sends a <command> to the SIM/UICC.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSIM=<lenght>,<command>

Set command transmits to the MT the <command>; it then shall send as it is to the SIM. In the same manner the SIM <response> shall be sent back by the MT to the TA as it is. Refer subclause 9.2 for <err> values.
This command allows a direct control of the SIM by an distant application on the TE. The TE shall then take care of processing SIM information within the frame specified by GSM/UMTS.

The response message of the command is in the format:

+CSIM: <length>,<response>

Error case:
+CME ERROR: <err>

The response message parameters are described in the Additional info section.

Parameters:

Name	Type	Default	Description
<lenght>	integer	-	Number of the characters that are sent to TE in <command> (two times the actual length of the command)
<command>	string	-	Command passed on by the ME to the SIM/UICC in the format as described in standard GSM 51.011 (hexadecimal character format; refer +CSCS)

Additional info:

List of the meaning of the response message parameters.

Name	Type	Default	Description
<lenght>	string	-	see Parameter section.
<response>	string	-	response to the command passed on by the SIM to the ME in the format as described in standard GSM 51.011 (hexadecimal character format; refer +CSCS)
<err>	string	N/A	error values (numeric format followed by verbose format).

Values:

3 : operation not allowed (operation mode is not allowed by the ME, wrong interface lock/unlock status)

4 : operation not supported (wrong format or parameters of the command)

13 : SIM failure (SIM no response)

Compared to Restricted SIM Access command +CRSM, the definition of +CSIM allow TE to take more control over the SIM-MT interface. The locking and unlocking of the interface may be by a special <command> value or automatically by TA/MT (by interpreting <command> parameter). In case that TE application does not use the unlock command (or does not send a <command> causing

automatic unlock) in a certain timeout value, MT may release the locking.

AT+CSIM=?

Test command returns the OK result code.

AT+CCHO - Open Logical Channel

Open Logical Channel

3GPP TS 31.101

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CCHO=<dfname>

Execution of the command causes the MT to return <sessionId> to allow the TE to identify a channel that is being allocated by the currently selected UICC, which is attached to ME.

The currently selected UICC will open a new logical channel; select the application identified by the <dfname> received with this command and return a <sessionId> as the response.

The ME shall restrict the communication between the TE and the UICC to this logical channel.

The response message of the command is in the format:
+CCHO: < sessionId >

The <sessionId> is described in the Additional info section.

Error case:
+CME ERROR: <err>

Parameter:

Name	Type	Default	Description
<dfname>	string	-	all selectable applications in the UICC are referenced by a DF name coded on 1 to 16 bytes (hexadecimal character format; refer +CSCS).

Additional info:

<sessionId> returned by the +CCHO command.

Name	Type	Default	Description
<sessionId>	integer	-	session Id to be used to target a specific application on the smart card (e.g. (U)SIM, WIM, ISIM) using logical channels mechanism. Session Id is used when sending commands with Restricted UICC Logical Channel access +CRLA, or Generic UICC Logical Channel access +CGLA commands.

The logical channel number is contained in the CLASS byte of an APDU command, thus implicitly contained in all APDU commands sent to a UICC.

In this case it will be up to the MT to manage the logical channel part of the APDU CLASS byte and to ensure that the chosen logical channel is relevant to the <sessionId> indicated in the AT command. Refer to 3GPP TS 31.101.

AT+CCHO=?

Returns the OK result code.

AT+CCHC - Close Logical Channel

Close a communication session.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CCHC=<sessionId>

Set command asks the ME to close a communication session with the active UICC. The ME shall close the previously opened logical channel. The TE will no longer be able to send commands on this logical channel. The UICC will close the logical channel when receiving this command.

Parameter:

Name	Type	Default	Description
<sessionId>	integer	-	a session ID to be used in order to target a specific application on the smart card (e.g. (U)SIM, WIM, ISIM) using logical channels mechanism

Additional info:

Error case:

+CME ERROR: <err>

Name	Type	Default	Description
<err>	integer	N/A	possible <err> values (numeric format followed by verbose format)

Values:

3	:	operation not allowed (operation mode is not allowed by the ME)
4	:	operation not supported (wrong format or parameters of the command)
13	:	SIM failure (SIM response SW1 SW2 status byte Error)
15	:	SIM wrong (SIM response SW1 SW2 status byte Error)
21	:	invalid index (not correspond to an opened channel)
100	:	unknowns (generic error)

AT+CCHC=?

Test command returns the OK result code.

AT+CGLA - Generic UICC Logical Channel Access

This command is used to control the currently selected UICC on the TE.

[1] 3GPP TS 11.11
[2] 3GPP TS 31.101
[3] 3GPP TS 31.102

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGLA=<sessionId>,<length>,<command>

Set command transmits to the MT the <command> it then shall send as it is to the selected UICC. In the same manner the UICC <response> shall be sent back by the MT to the TA as it is. This command allows a direct control of the currently selected UICC by a distant application on the TE. The TE shall then take care of processing the received UICC information. The response of the command is in the format:

+CGLA: <length>,<response>

Error case:

+CME ERROR: <err>

The response messages parameters are described in the Additional info section.

Parameters:

Name	Type	Default	Description
<sessionId>	integer	-	this parameter is the identifier of the session to be used in order to send the APDU commands to the UICC. It is mandatory to send commands to the UICC when targeting applications on the smart card using a logical channel other than the default channel (channel "0").

			<sessionId> is returned by the +CCHO command.
<length>	integer	-	characters number of the <command> sent to UICC (two times the actual length of the command)
<command>	string	-	command passed on by the MT to the UICC in the format as described in standard [1] or [2] (hexadecimal character format; refer to +CSCS).

Additional info:

▶▶ List of the meaning of the response messages parameters.

Name	Type	Default	Description
<length>	integer	-	characters number of the <response> sent to TE (two times the actual length of the response)
<response>	string	-	response to the command passed on by the UICC to the TE in the format as described in standard [1] or [2] (hexadecimal character format) See standard [2] for more information about defined values
<err>	string	N/A	error values (numeric format followed by verbose format)
	Values:		

3	:	operation not allowed (operation mode is not allowed by the ME)
4	:	operation not supported (wrong format or parameters of the command)
13	:	SIM failure (SIM response SW1 SW2 status byte Error)
15	:	SIM wrong (SIM response SW1 SW2 status byte Error)
21	:	invalid index (not correspond to an opened channel)
100	:	unknowns (generic error)

When the SW1 SW2 bytes received from UICC in response to <command> are "61 XX", MT automatically send to UICC a GET RESPONSE command with length "XX" and the +CGLA <response> is that retuned by GET RESPONSE command.

AT+CGLA=?

Test command returns the OK result code.

AT+ICCID - Read ICCID

Execution command reads on SIM the ICCID (card identification number that provides a unique identification number for the SIM)

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+ICCID

The command returns the following message:

+ICCID: <ICCID>

AT+ICCID=?

Test command returns the OK result code.

AT+ICCID

+ICCID: 89014104243627585306

OK

AT#SIMINCFG - SIMIN Pin Configuration

This command allows to set the SIMIN pin status for SIM detection.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SIMINCFG=<GPIOPin>,<siminDetMode>

Set command sets SIMIN pin status for SIM detection.

Parameters:

Name	Type	Default	Description
<GPIOPin>	integer	-	number of the GPIO pin used as SIMIN pin. Actually, it is a dummy parameter that is not used.
<siminDetMode>	integer	0	status of SIMIN pin for SIM detection

Values:

0 : SIMIN pin LOW means SIM inserted, HIGH means SIM removed

1 : SIMIN pin LOW means SIM removed, HIGH means SIM inserted

AT#SIMINCFG?

Read command reports the selected SIMIN pin status in the format:

#SIMINCFG: 0,<siminDetMode>

AT#SIMINCFG=?

Test command reports supported range of values for parameters <GPIOPin> and <siminDetMode>.

If SIMIN pin is not connected, the hot-swap function cannot be supported.

AT+CPIN - Enter the PIN

The command sends to the device a password which is necessary before it can be operated.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CPIN=<pin>[,<newpin>]

Set command sends the device a necessary password before it can be operated (SIM PIN, SIM PUK, PH-SIM PIN, etc.).
If the PIN required is SIM PUK or SIM PUK2, the <newpin> is required. This second pin, <newpin> will replace the old pin in the SIM.
The command may be used to change the SIM PIN by sending it with both parameters <pin> and <newpin> when PIN request is pending; if no PIN request is pending the command will return an error code and to change the PIN the command +CPWD must be used instead.

Parameters:

Name	Type	Default	Description
<pin>	string	-	PIN required or old PIN if the command is used to change the SIM PIN
<newpin>	string	-	new PIN that will replace old pin

To check the status of the PIN request use the command

AT+CPIN?

If all parameters omitted then the behavior of Set command is the same as Read command.

AT+CPIN?

Read command reports the PIN/PUK/PUK2 request status of the device in the form:

+CPIN: <code>

Additional info:

▶▶ Parameters:

Name	Type	Default	Description
<code>	string	N/A	PIN/PUK/PUK2 request status code

Values:

READY	:	ME is not pending for any password
SIM PIN	:	ME is waiting SIM PIN to be given
SIM PUK	:	ME is waiting SIM PUK to be given
PH-SIM PIN	:	ME is waiting phone-to-SIM card password to be given
PH-FSIM PIN	:	ME is waiting phone-to-very first SIM card password to be given
PH-FSIM PUK	:	ME is waiting phone-to-very first SIM card unblocking password to be given
SIM PIN2	:	ME is waiting SIM PIN2 to be given; this <code> is returned only when the last executed command

		resulted in PIN2 authentication failure (i.e. +CME ERROR: 17)
SIM PUK2	:	ME is waiting SIM PUK2 to be given; this <code> is returned only when the last executed command resulted in PUK2 authentication failure (i.e. +CME ERROR: 18)
PH-NET PIN	:	ME is waiting network personalization password to be given
PH-NET PUK	:	ME is waiting network personalization unblocking password to be given
PH-NETSUB PIN	:	ME is waiting network subset personalization password to be given
PH-NETSUB PUK	:	ME is waiting network subset personalization unblocking password to be given
PH-SP PIN	:	ME is waiting service provider personalization password to be given
PH-SP PUK	:	ME is waiting service provider personalization unblocking password to be given
PH-CORP PIN	:	ME is waiting corporate personalization password to be given

PH-CORP PUK	:	ME is waiting corporate personalization unblocking password to be given

Pin pending status at start-up depends on PIN facility setting, to change or query the default power up setting use the command

AT+CLCK=SC,<mode>,<pin>

After reboot SIM in state "READY" (always) if SIM inserted and active

AT+CPIN=?

Test command returns OK result code.

What follows is a list of the commands which are accepted when ME is pending SIM PIN or SIM PUK

A	+COPS	+CHUP	+GMR
D	+CLIP	+FCLASS	+CGSN
H	+CPAS	+IPR	+GSN
O	+CFUN	+IFC	+CPIN
E	+CGMI	+CRC	+CSQ
I	+CGMM	+CRLP	+CCLK
Q	+CGMR	+GMI	+CREG
V	+CRSM	+GMM	+CMEE

Z	+CLAC	+CGREG
X	+CNMI	+CSDH

All the above commands, but +CSDH and +CNMI, can be issued even if ME is waiting for phone-to-SIM card password to be given

```
AT+CMEE=1
OK

AT+CPIN?
+CME ERROR: 10
error: you have to insert the SIM

AT+CPIN?
+CPIN: READY
OK
you inserted the SIM and module is not waiting for PIN
```

SIM Toolkit

AT#STIA - SIM/USIM Toolkit Interface Action

The SIM/USIM Application Toolkit (SAT/USAT) provides an interface to the applications existing in the SIM/USIM device. The module must support the mechanisms required by the SIM/USIM applications.

[1] 3GPP TS 23.038
[2] 3GPP TS 31.111
[3] Telit SIM/USIM Application Toolkit Application Note

SIM Presence	Setting saved	Can be aborted
Required	Common profile	No

AT#STIA=[<mode>[,<timeout>]]

Set command is used to enable/disable the SIM/USIM Application Toolkit (SAT/USAT). In addition, the command can enable the URCs sending.

Parameters:

Name	Type	Default	Description
<mode>	integer	1	enables/disables SAT/USAT. In addition, <mode> parameter enables the: #STN URCs notifying the user that the SIM/USIM application has issued a proactive command. Some proactive commands require a user response #STN URCs that are the SIM/USIM device responses concerning actions initiated by the user, refer to Additional info section.

If <mode>=2, the URC format depends on the <cmdType> as described in the Additional info sections. For <cmdType>, and all other URC parameters refer to #STGI command. The <mode> parameter values are listed below.

Values:

0	:	disable SAT/USAT
1	:	enable SAT/USAT without #STN URC
2	:	enable SAT/USAT and extended #STN URC
3	:	enable SAT/USAT and reduced #STN URC
17	:	enable SAT/USAT without #STN URC and the alphabet used
18	:	enable SAT/USAT, extended #STN URC, and the alphabet used
19	:	enable SAT/USAT, reduced #STN URC, and the alphabet used
33	:	enable SAT/USAT without #STN URC and the UCS2 alphabet used
34	:	enable SAT/USAT, extended #STN URC, and the UCS2 alphabet used
35	:	enable SAT/USAT, reduced #STN URC, and the UCS2 alphabet used

<timeout>	integer	1	When an ongoing proactive command, requiring a user response, is not answered before <timeout> minutes, it is automatically aborted. In this case, the terminal response from the module is either "ME currently unable to process command" or, if applicable, "No response from user". In addition, the following URC is sent on the AT interface. For parameter meaning of the URC refer to Unsolicited fields section. #STN:<cmdTerminateValue> Follows the <timeout> range.

Value:

1,2	:	timeout expressed in minutes

Additional info:

<mode>=3, the URC has the following reduced format: #STN: <cmdType>

<mode>=2, and <cmdType>=1 (REFRESH proactive command), the URC has the following extended format:

#STN: 1,<refreshType>

<mode>=2, and <cmdType>=33 (DISPLAY TEXT proactive command), the URC is sent if allowed by SIM/USIM, the extended format is:

#STN: 33[,<cmdDetails>[,<alphaIdentifier>]]

If bit 7 of <cmdDetails>=1, the response with the #STSR command is required.

<mode>=2, and <cmdType>=34 (GET INKEY proactive command), the URC has the following extended format:

#STN: 34,<cmdDetails>,<text>

<mode>=2, and <cmdType>=35 (GET INPUT proactive command), the URC has the following extended format:

#STN: 35,<cmdDetails>,<text>,<responseMin>,<responseMax>[,<defaultText>]

<mode>=2, and <cmdType>=36 (SELECT ITEM proactive command), the URC has the following extended format:

the first line of output is:

#STN: 36,<cmdDetails>,<numOfItem>[,<titleText>]<CR><LF>

one line follows for every item, repeated <numOfItems> times:

#STN: 36,<itemId>,<itemText>[,<nextActionId>]

<mode>=2, and <cmdType>=37 (SET UP MENU proactive command), the URC has the following extended format:

the first line of output is:

#STN: 37,<cmdDetails>,<numOfItem>,<titleText><CR><LF>

one line follows for every item, repeated for <numOfItems>:

#STN: 37,<itemId>,<itemText>[,<nextActionId>]

<mode>=2, and <cmdType>=40 (SET UP IDLE MODE TEXT proactive command), the URC has the following extended format:

#STN: 40[,<idleModeTextString>]

- The settings are saved on user profile and available on following reboot. SIM/USIM Toolkit activation/deactivation is only performed at power on according to the saved setting.

- A typical SAT/USAT session on AT interface starts after an #STN: 37 unsolicited code is received, if enabled (see above). At that point usually an AT#STGI=37 command is issued (see #STGI), and after the SAT main menu has been displayed on TE an AT#STSR=37,0,x command is issued to select an item in the menu (see #STSR).

AT#STIA?

Read command can be used to get information about the SAT/USAT interface. The message format is:

#STIA: <state>,<mode>,<timeout>,<SatProfile>

Additional info:

- Returned parameters.

Name	Type	Default	Description
<state>	integer	N/A	state of the sending of the SET UP MENU proactive command (37)

Values:

0 : SIM/USIM has not sent the SET UP MENU proactive command (37)

	1	:	SIM/USIM has sent the SET UP MENU proactive command (37)
<mode>	integer	-	refer to Set section
<timeout>	integer	-	refer to Set section
<SatProfile>	string	-	SAT/USAT Terminal Profile. Is the list of SIM/USIM Application Toolkit facilities supported by the ME. The profile cannot be changed by the TA.

In SAT/USAT applications an SMS message is usually sent to the network provider containing service requests, e.g. to send the latest news. The provider returns a message with the requested information. Before activating SAT/USAT, it is recommended to set the SMS text mode with the AT+CMGF=1 command and enable URC for incoming SMS messages with +CNMI command.

AT#STIA=?

Test command returns the range of available values for the parameters <mode> and <timeout>.

AT#STGI - SIM Toolkit Get Information

The #STGI command interfaces to the SIM/USIM Application Toolkit to get information on the ongoing proactive command.

3GPP TS 31.111
Telit SIM/USIM Application Toolkit Application Note

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#STGI=[<cmdType>]

Set command gets parameters of the ongoing proactive command. The command can be used after the reception of the #STN: <cmdType> URC. If no proactive command is ongoing, #STGI returns an ERROR message.

Parameter:

Name	Type	Default	Description
<cmdType>	integer	N/A	proactive command code. For each proactive command listed below, its #STGI response format is described in the Additional info sections.

Values:

1	:	REFRESH
33	:	DISPLAY TEXT
34	:	GET INKEY
35	:	GET INPUT

36 : SELECT ITEM

37 : SET UP MENU

40 : SET UP IDLE MODE TEXT

Additional info:

<cmdType>=1 (REFRESH proactive command)

#STGI response format:

#STGI: 1,<refreshType>

Name	Type	Default	Description
<refreshType>	integer	N/A	identifies the refresh type

Values:

0 : SIM Initialization and Full File Change Notification

1 : File Change Notification

2 : SIM Initialization and File Change Notification

3 : SIM Initialization

4 : SIM Reset

5 : NAA Application Reset

6 : NAA Session Reset

7 : Steering of Roaming

8 : Steering of Roaming WLAN

<cmdType>=33 (DISPLAY TEXT proactive command)

#STGI response format:

#STGI: 33,<cmdDetails>[,<text>]

Name	Type	Default	Description
<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - normal priority 1 - high priority bits 1-6: reserved for future use bit 7: 0 - clear message after a delay 1 - wait for user to clear message Value: 0x00÷0xFF : mask
<text>	string	-	text provided by the SIM/USIM device in ASCII format

<cmdType>=34 (GET INKEY proactive command)

#STGI response format:

#STGI: 34,<cmdDetails>,<text>

Name	Type	Default	Description
<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - digits only (0-9, *, # and +) 1 - alphabet set bit 1: 0 - SMS default alphabet (GSM character set) 1 - UCS2 alphabet bit 2: 0 - character sets defined by bit 0 and bit 1 are enabled 1 - character sets defined by bit 0 and bit 1 are disabled and the "Yes/No" response is requested bits 3-6: 0 bit 7: 0 - no help information available 1 - help information available Value: 0x00÷0x87 : mask
<text>	string	-	string as prompt for test

<cmdType>=35 (GET INPUT proactive command)

#STGI response format:

#STGI:
35,<cmdDetails>,<text>,<responseMin>,<responseMax>[,<defaultText>]

Name	Type	Default	Description
<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - digits only (0-9, *, #, and +) 1 - alphabet set bit 1: 0 - SMS default alphabet (GSM character set) 1 - UCS2 alphabet bit 2: 0 - ME may echo user input on the display 1 - user input shall not be revealed in any way. Hidden entry mode is only available when using digit input. In hidden entry mode only characters ('0'-'9', '*' and '#') are allowed. bit 3: 0 - user input to be in unpacked format 1 - user input to be in SMS packed format bits 4-6: 0 bit 7: 0 - no help information

			available 1 - help information available
	Value:		
	0x00÷0x8F	:	mask
<text>	string	-	string as prompt for text
<responseMin>	integer	N/A	minimum number of characters of the user input
	Value:		
	0÷255	:	minimum length of user input.
<responseMax>	integer	N/A	maximum number of characters of the user input.
	Value:		
	0÷255	:	maximum length of user input
<defaultText>	string	-	string supplied as default response text

<cmdType>=36 (SELECT ITEM proactive command)

#STGI response format:

the first line of output is:

#STGI: 36,<cmdDetails>,<numOfItem>[,<titleText>]<CR><LF>

one line follows for every item, repeated <numOfItems> times:

#STGI: 36,<itemId>,<itemText>[,<nextActionId>]

Name	Type	Default	Description
<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - presentation type is not specified 1 - presentation type is specified in bit 1 bit 1: 0 - presentation as a choice of data values if bit 0 = 1 1 - presentation as a choice of navigation options if bit 0 is 1 bit 2: 0 - no selection preference 1 - selection using soft key preferred bits 3-6: 0 bit 7: 0 - no help information available 1 - help information available
	Value:		
	0x00÷0x87		: mask
<numOfItems>	integer	-	number of items in the list
<titleText>	string	-	menu title string

<itemId>	integer	N/A	item identifier
	Value:		
	1÷numOfItems	:	item identifier range
<itemText>	string	-	item title string
<nextActionId>	integer	-	is the code of next proactive command to be issued upon execution of the menu item. If <nextActionId>=0, no next action information available.

<cmdType>=37 (SET UP MENU proactive command)

#STGI response format:

the first line of output is:

#STGI: 37,<cmdDetails>,<numOfItem>,<titleText><CR><LF>

one line follows for every item, repeated for <numOfItems>:

#STGI: 37,<itemId>,<itemText>[,<nextActionId>]

Name	Type	Default	Description
<cmdDetails>	hex	N/A	a bit mask where each bit position, according to its value, has a specific meaning: bit 0: 0 - no selection preference 1 - selection using soft key preferred bit 1-6: 0

			bit 7: 0 - no help information available 1 - help information available
	Value:		
	0x00÷0x81	:	mask
<numOfItems>	integer	-	number of items in the list
<titleText>	string	-	menu title string
<itemId>	integer	N/A	item identifier
	Value:		
	1÷numOfItems	:	item identifier range
<itemText>	string	-	item title
<nextActionId>	integer	-	numerical code of next proactive command type to be issued upon execution of the menu item. If <nextActionId>=0, no next action information available.

<cmdType>=40 (SET UP IDLE MODE TEXT proactive command)

#STGI response format:

#STGI: 40,<idleModeTextString>

Name	Type	Default	Description
<idleModeTextString>	string	-	text provided by the SIM/USIM device in ASCII format

The proactive commands are only those command types that use the AT interface. SAT/USAT commands which are not using the AT interface (not MMI related SAT commands, e.g. PROVIDE LOCAL INFORMATION) are executed without sending any indication to the user.

AT#STGI?

The read command returns the ongoing proactive command and the SAT/USAT state. The message format is:

#STGI: <state>,<cmdType>

Additional info:

Returned parameters:

Name	Type	Default	Description
<state>	integer	-	state of the sending of the SET UP MENU proactive command (37), refer to AT#STIA? command
<cmdType>	integer	-	ongoing proactive command code

AT#STGI=?

Test command returns the supported values of parameters <state> and <cmdType>.

A typical SAT/USAT session, running on AT interface, starts when the user receives the #STN: 37 URC. The unsolicited result code must be previously enabled by the #STIA command. The #STN: 37 notifies the user that the main menu of the SIM/USIM Application has been sent to TA, and TA has stored the just received menu. Later, at any time, you can type in the AT#STGI=37 command to display the main menu of the SIM/USIM Application on TE.

Upon receiving the #STGI response, you must enter the #STSR command to confirm the execution of the proactive command and provide any required user response. In this case, you must enter the AT#STSR=37,0,x command to select the x item of the SIM/USIM Application menu.

The #STN: 237 URC indicates that the main menu of the SIM/USIM Application has been removed from TA, and it is no longer available. In this case, AT#STGI=37 command returns ERROR message.

AT#STSR - SIM Toolkit Send Response

This command allows the user to provide a response to confirm the execution of the ongoing proactive command.

3GPP TS 31.111
Telit SIM/USIM Toolkit Application Note

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#STSR=[<cmdType>[,<userAction>[,<data>]]]

The set command allows the user to provide a response action to the ongoing proactive command when the action is required by the command itself.

Parameters:

Name	Type	Default	Description
<cmdType>	integer	-	proactive command code, refer to #STGI command to have information on the <cmdType>
<userAction>	integer	0	identify the user action

Values:

0 : the user accepts the ongoing proactive command

16 : proactive SIM/USIM session terminated by user

17 : backward move in the proactive SIM/USIM session requested by the user

18	:	no response from user
19	:	help information required by the user
32	:	TA currently unable to process command

<data>	string	-	data entered by user, see Additional info section

Additional info:

- <data> parameter is used according to <cmdType>, and when <userAction>=0:

- <cmdType>=34 (GET INKEY proactive command)

  <data> contains the key pressed by the user. The character set is selected by +CSCS command.

  If the ongoing proactive command requires to the user a binary choice (yes/no), the valid content of <data> is:

  - "Y" or "y" (positive answer) and "N" or "n" (negative answer) for "IRA", "8859-1", "PCCP437" character sets

  - "0079" or "0059" (positive answer) and "006E" or "004E" (negative answer) for UCS2 alphabet

  The ongoing proactive command to require a binary choice sets bit 2 of the <cmdDetails> parameter to 1, see #STGI command.

- <cmdType>=35 (GET INPUT proactive command).

  <data> contains the string of characters entered by the user.

- <cmdType>=36 (SELECT ITEM proactive command).

  <data> contains the item identifier selected by the user.

Use of icons is not supported. All icon related actions will respond with no icon available.

AT#STSR?

The read command returns the ongoing proactive command and the SAT/USAT interface state. The format message is:

#STSR: <state>,<cmdType>

If there is no ongoing proactive command, an ERROR message is returned.

Additional info:

Returned parameters.

Name	Type	Default	Description
<state>	integer	-	state of the sending of the SET UP MENU proactive command (37), refer to AT#STIA? command
<cmdType>	integer	-	proactive command code, refer to #STGI command to have information on the <cmdType>

AT#STSR=?

The test command returns the range for the parameters <state> and <cmdType>.

SMS & CB

AT+CSMS - Select Message Service

Set command selects messaging service <service>

3GPP TS 27.005
3GPP TS 23.040
3GPP TS 23.041

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CSMS=<service>

Set command selects messaging service <service>. It returns the types of messages supported by the ME:

+CSMS: <mt>,<mo>,<bm>

For parameters meaning see Additional info section.

Parameter:

Name	Type	Default	Description
<service>	integer	0	Select Message Service

Values:

0 : The syntax of SMS AT commands is compatible with 3GPP TS 27.005 Phase 2 version 4.7.0

1 : The syntax of SMS AT commands is compatible with 3GPP TS 27.005 Phase 2+ version

Additional info:

▶▶ Parameters meaning of the returned message.

Name	Type	Default	Description
<mt>	integer	0	mobile terminated messages support: Values: 0 : type not supported 1 : type supported
<mo>	integer	0	mobile originated messages support Values: 0 : type not supported 1 : type supported
<bm>	integer	0	broadcast type messages support Values: 0 : type not supported 1 : type supported

AT+CSMS?

Read command reports current service setting along with supported message types in the format:

+CSMS: <service>, <mt>, <mo>, <bm>

AT+CSMS=?

Test command reports the supported value of the parameter <service>

```
AT+CSMS=1
+CSMS: 1,1,1
OK
AT+CSMS?
+CSMS: 1,1,1,1
OK
```

AT+CPMS - Preferred Message Storage

The command selects the memory storage used by SMs (Short Messages).

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CPMS=<memr>[,<memw>[,<mems>]]

Set command selects memory storages <memr>, <memw> and <mems> to be used for reading, writing, sending and storing SMs

The command returns the memory storage status in the format:

+CPMS: <usedr>,<totalr>,<usedw>,<totalw>,<useds>,<totals>

The parameters are described in the Additional info section.

Parameters:

Name	Type	Default	Description
<memr>	string	SM	memory from which messages are read and deleted

Values:

SM : SIM SMS memory storage

ME : SMS memory storage in Flash

SR : Status Report message storage (in SIM EF-SMSR file exists otherwise in the RAM volatile memory). NOT SUPPORTED. Do not use this value

<memw>	string	SM	memory to which writing and sending operations are made

Values:

SM	:	SIM SMS memory storage
ME	:	SMS memory storage in Flash

<mems>	string	SM	memory to which received SMs are preferred to be stored

Values:

SM	:	SIM SMS memory storage
ME	:	SMS memory storage in Flash

Additional info:

▶▶ Here is the meaning of the parameters returned by the command.

Name	Type	Default	Description
<usedr>	integer	-	number of SMs stored in <memr>
<totalr>	integer	-	max number of SMs that <memr> can contain
<usedw>	integer	-	number of SMs stored in <memw>

<totalw>	integer	-	max number of SMs that <memw> can contain
<useds>	integer	-	number of SMs stored in <mems>
<totals>	integer	-	max number of SMs that <memw> can contain

It is only possible to set the three memory storages <memr>, <memw> and <mems> to the same value.

If <memr> is set to a specific memory storage value, also <memw> and <mems> will be set to the same value.

AT+CPMS?

Read command reports the message storage status.

+CPMS:<memr>,<usedr>,<totalr>,<memw>,<usedw>,<totalw>,<mems>,<useds>,<totals>

The parameters are described in previous sections.

AT+CPMS=?

Test command reports the supported values for parameters <memr>, <memw> and <mems>.

```
AT+CPMS?
+CPMS: "ME",27, 50,"ME",27, 50,"ME",27,50
OK
AT+CPMS="SM","SM","SM"
+CPMS: 1,20,1,20,1,20
OK
AT+CPMS?
+CPMS: "SM",1,20,"SM",1, 20,"SM",1,20
```

OK

(You have 1 out of 20 SMS SIM positions occupied)

AT+CMGF - Message Format

Selects the format of SMS messages to be used in following SMS commands.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Common profile	No

AT+CMGF=[<mode>]

Set command selects the format of SMS messages used with send, list, read and write commands.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	format to use for SMS operations

Values:

0	:	PDU mode
1	:	text mode

AT+CMGF?

Read command reports the current value of the parameter <mode> in the format:

+CMGF: <mode>

AT+CMGF=?

Test command returns the supported values of parameter <mode>.

AT+CSCA - Service Center Address

This command allows to set the Service Center Address for SMS transmissions.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CSCA=<number>[,<type>]

Set command sets the Service Center Address to be used for mobile originated SMS transmissions

Parameters:

Name	Type	Default	Description
<number>	string	-	String type phone number of forwarding address in format specified by <type> parameter
<type>	integer	145	The type of number

Values:

129 : National numbering scheme

145 : International numbering scheme (contains the character "+")

To use the SM service, is mandatory to set a Service Centre Address at which service requests directed

- In Text mode the settings is used by send & write commands; in PDU mode, setting is used by the same commands, but only when the length of the SMSC address coded into the <pdu> parameter equals zero.
- The current settings are stored as default in SIM profile. Destination can be changed through +CSAS

AT+CSCA?

Read command reports the current value of the SCA in the format:

+CSCA: <number>,<type>

- If SCA is not present the device reports an error message.

AT+CSCA=?

Test command returns the OK result code.

- To use the SM service, is mandatory to set a Service Center Address at which service requests will be directed.
- In Text mode, this setting is used by send and write commands; in PDU mode, setting is used by the same commands, but only when the length of the SMSC address coded into the <pdu> parameter equals zero.
- The current settings are stored through +CSAS

```
AT+CSCA="821029190903",145
OK
```

```
AT+CSCA?
+CSCA: "+821029190903",145
OK
```

AT+CSMP - Set Text Mode Parameters

This command is used to select values for additional parameters for storing and sending SMS when the text mode is used (AT+CMGF=1).

3GPP TS 27.005

3GPP TS 23.40

3GPP TS 23.038

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CSMP=[<fo>[,<vp>[,<pid>[,<dcs>]]]]

Set command is used to select values for additional parameters for storing and sending SMS when the text mode is used (AT+CMGF=1).

Parameters:

Name	Type	Default	Description
<fo>	integer	17	first octet - depending on the command or result code: first octet of 3GPP TS 23.040 SMS-SUBMIT or SMS-DELIVER, in integer format (default 17, i.e. SMS-SUBMIT with validity period in relative format). As first octet of a PDU has the following bit field description (bit[7]bit[6]bit[5]bit[4]bit[3]bit[2]bit[1]bit[0]): bit[1]bit[0]: Message Type Indicator, 2-bit field describing the message type; [00] - SMS-DELIVER; [01] - SMS-SUBMIT (default);

bit[2]: Reject Duplicates, 1-bit field: user is not responsible for setting this bit and, if any set, it will have no meaning (default is [0]);

bit[4]bit[3]: Validity Period Format, 2-bit field indicating whether or not the Validity Period field is present (default is [10]):

[00] - Validity Period field not present

[01] - Validity Period field present in enhanced format(i.e. quoted time-string type, see below)

[10] - Validity Period field present in relative format, (i.e. integer type, see below)

[11] - Validity Period field present in absolute format (i.e. quoted time-string type, see below)

bit[5]: Status Report Request, 1-bit field indicating the MS is requesting a status report (default is [0]);

[0] - MS is not requesting a status report

[1] - MS is requesting a status report

bit[6]: User Data Header Indicator, 1-bit field: user is not responsible for setting this bit and, if any set, it will have no meaning (default is [0]);

bit[7]: Reply Path, 1-bit field indicating the request for Reply Path (default is [0]);

[0] - Reply Path not requested

[1] - Reply Path requested

Value:

0÷255	: mask

<vp>	mixed	167	validity period - depending on <fo> setting: a) if <fo> asks for a Not Present Validity Period, <vp> can be any type and it will be not

considered;
b) if <fo> asks for a Validity Period in relative format, <vp> shall be integer type (default 167, i.e. 24 hours);
0..143 - (<vp> + 1) x 5 minutes
144..167 - 12 hours + ((<vp> - 143) x 30 minutes)
168..196 - (<vp> - 166) x 1 day
197..255 - (<vp> - 192) x 1 week
c) if <fo> asks for a Validity Period in absolute format, <vp> shall be quoted time-string type (see +CCLK)
d) if <fo> asks for a Validity Period in enhanced format, <vp> shall be the quoted hexadecimal representation (string type) of 7 octets, as follows:
- the first octet is the Validity Period Functionality Indicator, indicating the way in which the other 6 octets are used; following bit field description:
bit[7]: extension bit
[0] - there are no more VP Fuctionality Indicator extension octets to follow
bit[6]: Single Shot SM;
[0] - the SC is not required to make up to one delivery attempt
[1] - the SC is required to make up to one delivery attempt
bit[5]bit[4]bit[3]: reserved
[000]
bit[2]bit[1]bit[0]: Validity Period Format
[000] - No Validity Period specified
[001] - Validity Period specified as for the relative format. The following octet contains the VP value as described before; all the other octets are 0s.
[010] - Validity Period is relative in integer representation. The following octet contains the VP value in the range 0 to 255, representing 0 to 255 seconds; all the other octets are 0s.

[011] - Validity Period is relative in semi-octet representation. The following 3 octets contain the relative time in Hours, Minutes and Seconds, giving the length of the validity period counted from when the SMS-SUBMIT is received by the SC; all the other octets are 0s.

Value:

0÷255	:	range for Validity Period in relative format

<pid>	integer	0	TP-Protocol-Identifier, as described in 3GPP TS 23.40

Value:

0÷255	:	described in 3GPP TS 23.40

<dcs>	integer	0	SMS Data Coding Scheme, as described in 3GPP TS 23.038

Value:

0÷255	:	described in 3GPP TS 23.038

The current settings are stored through +CSAS.

AT+CSMP?

Read command returns the current setting in the format:

+CSMP: <fo>,<vp>,<pid>,<dcs>

If the Validity Period Format (<fo>'s bit[4]bit[3]) is [00] (i.e. Not Present), <vp> is represented just as a quoted empty string ("").

AT+CSMP=?

Test command returns the OK result code.

Set the parameters for an outgoing message with 24 hours of validity period and default properties:

AT+CSMP=17,167,0,0

OK

- Set the parameters for an outgoing message with 24 hours of validity period and default properties:
  AT+CSMP=17,167,0,0
  OK

  Set the parameters for an outgoing message with validity period in enhanced format: the <vp> string actually codes 24 hours of validity period.
  AT+CSMP=9,"01A80000000000"
  OK

  Set the parameters for an outgoing message with validity period in enhanced format: the <vp> string actually codes 60 seconds of validity period.
  AT+CSMP=9,"023C0000000000"
  OK

  Set the parameters for an outgoing message with validity period in enhanced format: the <vp> string actually codes 29 hours 85 minutes 30 seconds of validity period.
  AT+CSMP=9,"03925803000000"
  OK

AT+CSDH - Show Text Mode Parameters

This command controls whether detailed header information is shown in text mode.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSDH=[<show>]

Set command controls whether detailed header information is shown in text mode (AT+CMGF=1) result codes.

Parameter:

Name	Type	Default	Description
<show>	integer	0	control the display of the result codes.

Values:

0 : see Additional info section

1 : show the values in result codes

Additional info:

- If <show>=0

do not show header values defined in commands +CSCA and +CSMP (<sca>,<tosca>, <fo>, <vp>, <pid> and <dcs>) nor <length>, <toda> or <tooa> in +CMT, +CMGL, +CMGR result codes for SMS-DELIVERs and SMS-SUBMITs in text mode.

For SMS-COMMANDs in +CMGR result code do not show <pid>, <mn>, <da>, <toda>, <length> or <cdata>

AT+CSDH?

Read command reports the current setting in the format:

+CSDH: <show>

AT+CSDH=?

Test command reports the supported range of values for parameter <show>.

AT+CSCB - Select Cell Broadcast

The command selects which types of Cell Broadcast Messages are to be received by the device.

3GPP TS 27.005
3GPP TS 23.041
3GPP TS 23.038

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSCB=[<mode>[,<mids>[,<dcss>]]]

Set command selects which types of Cell Broadcast Messages are to be received by the device

Parameters:

Name	Type	Default	Description
<mode>	integer	0	select which types of Cell Broadcast messages are to be received

Values:

0 : the message types defined by <mids> and <dcss> are accepted

1 : the message types defined by <mids> and <dcss> are rejected

<mids>	string	-	message Identifiers: all different possible combinations of the CBM message identifiers;
<dcss>	string	-	Data Coding Schemes: all different possible combinations of CBM data coding schemes;

The current settings are stored also by +CSAS command

AT+CSCB?

Read command reports the current value of parameters <mode>, <mids> and <dcss>

AT+CSCB=?

Test command returns the range of values for parameter <mode>

```
AT+CSCB?
+CSCB: 1,"","""

OK      (all CBMs are accepted, none is rejected, only in UMTS NW)
AT+CSCB=0,"0,1,300-315,450","0-3"
OK
```

AT+CSAS - Save Settings

This command saves settings which have been made by the +CSCA, +CSMP and +CSCB commands in local non volatile memory

- 3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CSAS[=<profile>]

Execution command saves settings which have been made by the +CSCA, +CSMP and +CSCB commands in local non volatile memory

Parameter:

Name	Type	Default	Description
<profile>	integer	0	Index of the profile where the settings are saved

Values:

0	:	it saves the settings to NVM
1÷n	:	SIM profile number; the value of <n> depends on the SIM and its max is 3

Certain settings may not be supported by the SIM and therefore they are always saved to NVM, regardless the value of <profile>.

If parameter is omitted the settings are saved in the non volatile memory.

AT+CSAS=?

Test command returns the possible range of values for the parameter <profile>.

It is allowed to store up to 5 profiles.

AT+CRES - Restore Settings

Execution command restores message service settings saved by +CSAS command from either NVM or SIM.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CRES=[<profile>]

Parameter:

Name	Type	Default	Description
<profile>	integer	N/A	Defines which message service profiles to restore.

Values:

0 : restores message service settings from NVM

1÷n : restores message service settings from SIM. The n value depends on the SIM.

Certain settings may not be supported by the SIM and therefore they are always restored from NVM, regardless the value of <profile>

If parameter is omitted the command restores message service settings from NVM.

AT+CRES?

Read command reports the current value of the parameter <profile>.

AT+CRES=?

Test command returns the possible range of values for the parameter <profile>.

It is allowed to store up to 5 profiles.

AT+CMMS - More Message to Send

Set command controls the continuity of SMS relay protocol link. When feature is enabled (and supported by network) multiple messages can be sent much faster as link is kept open.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMMS=[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	0	enables/disables the relay protocol link continuity.

Values:

0 : disable

1 : keep link opened while messages are sent. If the delay between two messages exceeds 3 seconds, the link is closed and the parameter <n> is automatically reset to 0

2 : keep link opened while messages are sent. If the delay between two messages exceeds 3 seconds, the link is closed but the parameter <n> remains set to 2

AT+CMMS?

Read command reports the current value of the parameter <n>

AT+CMMS=?

Test command reports the supported value of <n> parameter.

AT+CNMI - New Message Indications to Terminal Equipment

This command sets the parameters for receiving SMS messages.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CNMI=[<mode>[,<mt>[,<bm>[,<ds>[,<bfr>]]]]]

Set command selects the behavior of the device on how the receiving of new messages from the network is indicated to the DTE.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	unsolicited result codes buffering option.

Values:

0 : Buffer unsolicited result codes in the TA. If TA result code buffer is full, indications can be buffered in some other place or the oldest indications may be discarded and replaced with the new received indications.

1 : Discard indication and reject new received message unsolicited result codes when TA-TE link is reserved, otherwise forward them directly to the TE.

2 : Buffer unsolicited result codes in the TA in case the DTE is busy and flush them to the TE after reservation. Otherwise forward them directly to the TE.

3 : if <mt> is set to 1, it enables the hardware ring line for 1 sec. Not supported

<mt>	integer	0	result code indication reporting for SMS-DELIVER.

Values:

0 : No SMS-DELIVER indications are routed to the TE and message is stored.

1 : If SMS-DELIVER is stored into ME/TA, indication of the memory location is routed to the TE using the URC described in Additional info section.

2 : SMS-DELIVERs (except class 2 messages and messages in the message waiting indication group) are routed directly to the TE using the URC described in the Additional info section.

3 : Class 3 SMS-DELIVERs are routed directly to TE using unsolicited result codes defined in <mt>=2. Messages of other data coding schemes result in indication as defined in <mt>=1.

<bm>	integer	0	broadcast reporting option

Values:

0 : Cell Broadcast Messages are not sent to the DTE

2 : New Cell Broadcast Messages are sent to the DTE with the URC described in Additional info section.

<ds> integer 0 SMS-STATUS-REPORTs reporting option

Values:

0 : status report receiving is not reported to the DTE and is not stored

1 : the status report is sent to the DTE with the URC described in the Additional info section.

2 : if a status report is stored, then the unsolicited result code, described in Additional info section, is sent.

<bfr> integer 0 buffered result codes handling method

Values:

0 : TA buffer of unsolicited result codes defined within this command is flushed to the TE when <mode>=1..3 is entered (OK response shall be given before flushing the codes)

1 : TA buffer of unsolicited result codes defined within this command is cleared when <mode>=1..3 is entered.

Additional info:

<mt>=1:

+CMTI: <memr>,<index>

<mt>=2:

PDU mode

+CMT: <alpha>,<length><CR><LF><PDU>

TEXT mode

+CMT:<oa>,<alpha>,<scts>[,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]<CR><LF><data>

The parameters written in italics will be present depending on +CSDH last setting.

Class 2 messages and messages in the message waiting indication group (stored message) result in indication as defined in <mt>=1.

Acknowledge for the received SMS-DELIVER SM is sent to network immediately when +CSMS <service> is set to '0' or when +CSMS <service> is set to '1', acknowledge is sent via +CNMA command during predefine time-out, an error is sent to network in case timeout expire, next +CMT response is depend on acknowledge of current received +CMT response in case +CSMS <service> parameter set to '1'.

<bm>=2:

PDU mode

+CBM: <length><CR><LF><PDU>

Text mode

+CBM:<sn>,<mid>,<dcs>,<pag>,<pags><CR><LF><data>

<ds>=1:

PDU mode

+CDS: <length><CR><LF><PDU>

TEXT mode

+CDS: <fo>,<mr>,<ra>,<tora>,<scts>,<dt>,<st>

Acknowledge for the received SMS-STATUS-REPORT SM is sent to network immediately when +CSMS <service> is set to '0' or when +CSMS <service> is set to '1', acknowledge is sent via +CNMA command during pre-defined timeout, an error is sent to network in case timeout expire, next +CDS response is depend on acknowledge of current received +CDS response in case +CSMS <service> parameter set to '1'.

<ds>=2:

+CDSI: <mems>,<index>

Unsolicited fields:

Name	Type	Description
<memr>	string	memory storage where the new message is stored: "SM", "ME".
<index>	integer	location on the memory where SMS is stored.
<alpha>	string	alphanumeric representation of originator/destination number corresponding to the entry found in MT phonebook; used character set should be the one selected with command +CSCS.
<lenght>	integer	PDU length
<PDU>	string	PDU message

<oa>	string	originating address, string type converted in the currently selected character set (see +CSCS)
<alpha>	string	alphanumeric representation of <oa>; used character set should be the one selected with command +CSCS.
<scts>	string	arrival time of the message to the SC
<tooa>	integer	type of number <oa>: 129 - number in national format 145 - number in international format (contains the "+")
<fo>	string	first octet of message PDU, see 3GPP TS 03.40/23.040
<pid>	string	Protocol Identifier
<dcs>	string	Data Coding Scheme
<sca>	string	Service Centre address, string type, converted in the currently selected character set (see +CSCS)
<tosca>	integer	type of number <sca>: 129 - number in national format 145 - number in international format (contains the "+")
<length>	integer	text length
<data>	string	TP-User-Data If <dcs> indicates that GSM 03.38 default alphabet is used and <fo> indicates that GSM 03.40 TP-User-Data-Header-Indication is not set (bit 6 of <fo> is 0), each character of GSM alphabet will be converted into current TE character set (see +CSCS). If <dcs> indicates that 8-bit or UCS2 data coding scheme is used or <fo> indicates that GSM 03.40 TP-User-Data-Header-Indication is set (bit 6 of <fo> is 1), each 8-bit octet will be converted into

		two IRA character long hexadecimal number (e.g. octet 0x2A will be converted as two characters 0x32 0x41). Class 2 messages and messages in the "store" message waiting indication group result in indication as defined in <mt>=1.
<sn>	integer	message serial number
<mid>	integer	message ID
<dcs>	string	Data Coding Scheme
<pag>	integer	page number
<pags>	integer	total number of pages of the message
<data>	string	CBM Content of Message If <dcs> indicates that GSM 03.38 default alphabet is used, each character of GSM alphabet will be converted into current TE character set (see +CSCS) If <dcs> indicates that 8-bit or UCS2 data coding scheme is used, each 8-bit octet will be converted into two IRA character long hexadecimal number (e.g. octet 0x2A will be converted as two characters 0x32 0x41)
<mr>	integer	message reference number; 3GPP TS 23.040 TP-Message-Reference in integer format
<ra>	string	recipient address, string type, represented in the currently selected character set (see +CSCS)
<tora>	integer	type of number <ra>: 129 - number in national format 145 - number in international format (contains the "+")

<scts>	string	arrival time of the message to the SC
<dt>	string	sending time of the message
<st>	string	message status as coded in the PDU

DTR signal is ignored, hence the indication is sent even if the DTE is inactive (DTR signal is Low). In this case the unsolicited result code may be lost so if MODULE remains active while DTE is not, at DTE startup is suggested to check whether new messages have reached the device meanwhile with command AT+CMGL=0 that lists the new messages received.

<ds>=2 has the same behavior as <ds>=1 if the SMS memory storage is in the "SM".

<ds>=2 generates the expected URC only if the SMS memory storage is "ME".

AT+CNMI?

Read command returns the current parameter settings for +CNMI command in the form:

+CNMI: <mode>,<mt>,<bm>,<ds>,<bfr>

AT+CNMI=?

Test command reports the supported range of values for the +CNMI command parameters

Concatenated SMS is not supported in text mode

It is the application responsibility to decode the HEX format of the Concatenated SMS and assemble the pieces into text.

```
AT+CMGF=1
OK

AT+CNMI=1,2,0,1,0
OK


Received message from network

+CMT: "+821020955219",,"07/07/26,20:09:07+36"

TEST MESSAGE
```

AT+CNMA - New Message Acknowledgement

This command is used to confirm the correct reception of a new message.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CNMA=<n>

Parameter:

Name	Type	Default	Description
<n>	integer	N/A	type of acknowledgement in PDU mode

Values:

0 : send RP-ACK without PDU

1 : send RP-ACK with optional PDU message

2 : send RP-ERROR with optional PDU message

AT+CNMA=?

Test command returns the possible range of values for the parameter <n>.

AT+CMGL - List Messages

This command is used to list the messages.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMGL[=<stat>]

Execution command reports the list of all the messages with status value <stat> stored into <memr> message storage (<memr> is the message storage for read and delete SMs as last settings of command +CPMS).

Parameter:

Name	Type	Default	Description
<stat>	mixed	-	<stat> parameter type and the command output depend on the last settings of the +CMGF command (message format to be used). There are two modes: PDU mode Text mode See the following Additional info sections.

Additional info:

When message format is PDU mode, the <stat> parameter is:

Name	Type	Default	Description
<stat>	integer	N/A	status value

Values:

0	:	new message
1	:	read message
2	:	stored message not sent yet
3	:	stored message already sent
4	:	all messages

▶▶ In case of PDU mode the representation format (see +CMGF) is:

+CMGL: <index>,<stat>,<alpha>,<length><CR><LF><pdu>[<CR><LF>
+CMGL: <index>,<stat>,<alpha>,<length><CR><LF><pdu>[...]]

Name	Type	Default	Description
<index>	integer	-	message position in the storage list.
<stat>	integer	-	message status. See the above <stat> parameter description.
<alpha>	string	-	String type alphanumeric representation of <da> or <oa>, corresponding to an entry found in the phonebook; used character set is the one selected with command +CSCS.
<length>	integer	-	PDU length in bytes

<pdu>	string	-	message in PDU format, according to 3GPP TS 23.040

▶▶ When message format is TEXT mode, the <stat> parameter is:

Name	Type	Default	Description
<stat>	string	N/A	status value

Values:

"REC UNREAD"	:	new message
"REC READ"	:	read message
"STO UNSENT"	:	stored message not sent yet
"STO SENT"	:	stored message already sent
"ALL"	:	all messages

▶▶ In case of TEXT mode, the representation format for stored messages (either sent or unsent) or received messages (either read or unread, not message delivery confirm) is:

+CMGL: <index>,<stat>,<oa/da>,<alpha>,<scts>[,<tooa/toda>,<length>]<CR><LF> <data>[<CR><LF>
+CMGL: <index>,<stat>,<oa/da>,<alpha>,<scts>[,<tooa/toda>,<length>]<CR><LF> <data>[...]]

The information written in italics will be present depending on +CSDH last setting.

Name	Type	Default	Description
<index>	integer	-	message position in the storage list.
<stat>	string	-	message status. See the above <stat> parameter description.
<oa/da>	string	-	originator/destination address, represented in the currently selected character set (see +CSCS).
<alpha>	string	-	The alphanumeric representation of <da> or <oa>, corresponding to an entry found in the phonebook; used character set is the one selected with command +CSCS.
<scts>	string	-	TP-Service Centre Time Stamp in Time String Format.
<tooa/ toda>	integer	N/A	type of number <oa/da>
	Values:		
	129 :		number in national format
	145 :		number in international format (contains the "+")
<length>	integer	-	text length
<data>	string	-	TP-User-Data

If <dcs> indicates that 3GPP TS 23.038 default alphabet is used, each character of GSM alphabet will be converted into current TE character set (see +CSCS)

If <dcs> indicates that 8-bit or UCS2 data coding scheme is used, each 8-bit octet will be converted into two IRA character long hexadecimal number (e.g. octet 0x2A will be converted as two characters 0x32 0x41)

If <fo> indicates that a UDH is present each 8-bit octet will be converted into two IRA character long hexadecimal number. The <length> indicates text length in characters without UDH length.

▶▶ In case of TEXT mode, the representation format for delivery confirm messages is:

+CMGL: <index>,<stat>,<fo>,<mr>,<ra>,<tora>,<scts>,<dt>,<st>[<CR><LF>
+CMGL: <index>,<stat>,<fo>,<mr>,<ra>,<tora>,<scts>,<dt>,<st>[...]]

Name	Type	Default	Description
<index>	integer	-	message position in the storage list.
<stat>	string	-	Message status. See the last <stat> parameter description.
<fo>	integer	-	first octet of the message PDU

<mr>	integer	-	message reference number; 3GPP TS 23.040 TP-Message-Reference in integer format
<ra>	string	-	recipient address, represented in the currently selected character set (see +CSCS)
<tora>	string	-	type of number <ra>
<scts>	string	-	arrival time of the message to the SC
<dt>	string	-	sending time of the message
<st>	integer	-	message status as coded in the PDU

If parameter is omitted the command returns the list of SMS with "REC UNREAD" status.

AT+CMGL=?

Test command returns a list of supported <stat>s

Concatenated SMS is not supported in text mode

It is the application responsibility to decode the HEX format of the Concatenated SMS and assemble the pieces into text.

AT+CMGF=1 Set Text mode

OK

AT+CMGL

+CMGL: 1,"REC UNREAD","+821020955219",,"07/07/26,20:05:11+36"

SMS Test message

+CMGL: 2,"REC UNREAD","+821020955219",,"07/07/26,20:05:58+36"

SMS Test message

+CMGL: 3,"REC UNREAD","+821020955219",,"07/07/26,20:06:37+36"

SMS Test Message.

+CMGL: 4,"REC UNREAD","+821020955219",,"07/07/26,20:07:43+36"

TEST MESSAGE..
+CMGL: 5,"REC UNREAD","+821020955219",,"07/07/26,20:09:07+36"

TEST MESSAGE

OK

AT+CMGF=0 Set PDU mode
OK

AT+CMGL=2
+CMGL: 0,2,,24
0791280192919030116 40A8110567892820000A70CF4F29C0E6A97E7F3F0B90C
+CMGL: 1,2,,21
0791280192919030116 40A8110516529700000A709027A794E77B95C2E
+CMGL: 26,2,,17
0881280100990102591 1640A8110567892820014A704C7D1B1DB
OK

AT+CMGR - Read Message

This command is used to read a message.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMGR=<index>

Execution command reports the message with location value <index> from <memr> message storage (<memr> is the message storage for read and delete SMs as last settings of command +CPMS).

Parameter:

Name	Type	Default	Description
<index>	integer	-	message index. The command output depends on the last settings of command +CMGF (message format to be used). There are two modes: PDU mode Text mode See the following Additional info sections.

Additional info:

- In case of PDU mode, if there is a message in location <index>, the output has the following format:

  +CMGR: <stat>,<alpha>,<length><CR><LF><pdu>

Name	Type	Default	Description
<stat>	integer	N/A	status of the message Values: 0 : new message 1 : read message 2 : stored message not yet sent 3 : stored message already sent
<alpha>	string	-	string type alphanumeric representation of <da> or <oa>, corresponding to an entry found in the phonebook; used character set is the one selected with command +CSCS
<length>	integer	-	PDU length in bytes
<pdu>	string	-	message in PDU format, according to 3GPP TS 23.040

▶▶ In case of Text mode, if there is a received message in location <index>, the output has the following format (the information written in italics will be present depending on +CSDH last setting):

+CMGR:<stat>,<oa>,<alpha>,<scts>
[,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]<CR><LF><data>

If there is either a sent or an unsent message in location <index> the output format is:

+CMGR:<stat>,<da>,<alpha> [,<toda>,<fo>,<pid>,<dcs>, [<vp>],<sca>,<tosca>,<length>]<CR><LF><data>

If there is a Message Delivery Confirm message in location <index> the output format is:

+CMGR: <stat>,<fo>,<mr>,,,<scts>,<dt>,<st>

Name	Type	Default	Description
<stat>	string	N/A	status of the message
	Values:		
	"REC UNREAD"	:	new received message
	"REC READ"	:	received message read
	"STO UNSENT"	:	message stored not yet sent
	"STO SENT"	:	message stored already sent
<fo>	integer	-	first octet of the message PDU
<mr>	integer	-	message reference number; 3GPP TS 23.040 TP-Message-Reference in integer format
<scts>	string	-	arrival time of the message to the SC
<dt>	string	-	sending time of the message

<st>	integer	-	message status as coded in the PDU
<pid>	integer	-	Protocol Identifier
<dcs>	integer	-	Data Coding Scheme
<vp>	mixed	-	Validity Period; its format depends on SMS-SUBMIT <fo> setting (see +CPMS): 1. Not present: if <fo> tells that Validity Period Format is not present 2. Integer: if <fo> tells that Validity Period Format is relative 3. Quoted time-string type: if <fo> tells that Validity Period Format is absolute 4. Quoted hexadecimal representation of 7 octets: if <fo> tells that Validity Period Format is enhanced
<oa>	string	-	Originator address, represented in the currently selected character set (see +CSCS).
<da>	string	-	Destination address, represented in the currently selected character set (see +CSCS).
<alpha>	string	-	The alphanumeric representation of <da> or <oa>, corresponding to an entry found in the phonebook; used character set is the one selected with command +CSCS.

<sca>	string	-	Service Centre Address
<tooa>	integer	N/A	type of number of <oa>

Values:

129 : number in national format

145 : number in international format (contains the "+")

<toda>	integer	N/A	type of number of <da>

Values:

129 : number in national format

145 : number in international format (contains the "+")

<tosca>	integer	N/A	type of number of <sca>

Values:

129 : number in national format

145 : number in international format (contains the "+")

<length>	integer	-	text length
<data>	string	-	TP-User-Data

If <dcs> indicates that 3GPP TS 23.038 default alphabet is used, each character of GSM alphabet will be converted into current TE character set (see +CSCS)

If <dcs> indicates that 8-bit data coding scheme is used, each 8-bit octet will be converted into two IRA character long hexadecimal number (e.g. octet 0x2A will be converted as two characters 0x32 0x41)

In both cases (in PDU or Text mode) if status of the message is 'received unread', status in the storage changes to 'received read'.

AT+CMGR=?

Test command returns the OK result code

Concatenated SMS is not supported in text mode

It is the application responsibility to decode the HEX format of the Concatenated SMS and assemble the pieces into text.

```
AT+CMGF=0
OK

AT+CMGR=1
+CMGR:2,,21
07912801929190301164 0A8110516529700000A709027A794E77B95C2E
OK

AT+CMGF=1
OK

AT+CMGR=3
+CMGR:"REC READ","+821020955219",,"07/07/19,10:06:34+36" test
```

message/.....
OK

AT+CMGS - Send Short Message

The command is related to sending short messages.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CMGS

Execution command sends a short message to the network. It can have two syntax formats according to the SMS format: PDU or Text mode (see +CMGF command). If short message is successfully sent to the network, the result is shown with the following URC:

+CMGS: <mr>[,<scts>]

Additional info:

In PDU mode the +CMGS command has the following syntax:

AT+CMGS=<length>

After command line is terminated with <CR>, the module responds sending a four-character sequence prompt:

<CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

and waits for the specified number of bytes. the PDU shall be hexadecimal format (each octet of the PDU is given as two IRA character long hexadecimal number) and given in one line.

To send the message issue Ctrl-Z char (0x1A hex). To exit without sending the message issue ESC char (0x1B hex).

Name	Type	Default	Description

<length>	integer	N/A	length in bytes of the PDU to be sent (excluding the SMSC address octets)

Value:

7÷164 : number of bytes

In Text mode the +CMGS command has the following syntax:

AT+CMGS=<da>[,<toda>]

After command line is terminated with <CR>, the module responds sending a four-character sequence prompt:

<CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

After this prompt, you can enter text that should be formatted as follows:

if current <dcs> (see +CSMP) indicates that GSM03.38 default alphabet is used and current <fo> (see +CSMP) indicates that 3GPP TS 23.040 TP-User-Data-Header-Indication is not set, then ME/TA converts the entered text into GSM alphabet, according to 3GPP TS 27.005, Annex A; backspace can be used to delete last character and carriage returns can be used; after every <CR> entered by the user the sequence <CR><LF><greater_than><space> is sent to the TE.

if current <dcs> (see +CSMP) indicates that 8-bit or UCS2 data coding scheme is used or current <fo> (see +CSMP) indicates that 3GPP TS 23.040 TP-User-Data-Header-Indication is set, the entered text should consist of two IRA character long hexadecimal numbers which ME/TA converts into 8-bit octet (e.g. the 'asterisk' will be entered as 2A (IRA50 and IRA65) and this will be converted to an octet with integer value 0x2A)

To send the message issue Ctrl-Z char (0x1A hex). To exit without sending the message issue ESC char (0x1B hex).

Name	Type	Default	Description
<da>	string	-	destination address, string type represented in the currently selected character set (see +CSCS).
<toda>	string	129	type of destination address Values: 129 : number in national format 145 : number in international format (contains the "+")

Unsolicited fields:

Name	Type	Description
<mr>	integer	TP-Message-Reference number as per 3GPP TS 23.040
<scts>	string	TP-Service Centre Time Stamp in Time String Format. <scts> is returned when +CSMS <service> value is 1 and network supports.

- The DCD signal shall be in ON state while PDU is given. The echoing of given characters back from the TA is controlled by echo command E.
- in PDU mode: when the length of the SMSC address equals 0, then the SMSC address set with command +CSCA is used; in this case the SMSC Type of Address octet shall not be present in the PDU.

- To ensure that during the command execution, which may take several seconds, no other SIM interacting commands issued, care must take.

- It is possible to send a concatenation of at most 16 SMs; the maximum number of chars depends on the <dcs>:

  2432 chars

  2128 chars if 8-bit is used

  1056 chars if 16-bit is used

- If message sending fails for some reason, then an error code is reported.

AT+CMGS=?

Test command returns the OK result code.

- To avoid malfunctions is suggested to wait for the +CMGS: <mr> or +CMS ERROR: <err> response before issuing further commands.

Concatenated SMS is not supported in text mode

It is the application responsibility to decode the HEX format of the Concatenated SMS and assemble the pieces into text.

Set PDU mode
AT+CMGF=0

AT+CMGS=18
> 0881280100990102591155078110209055 12F90000A704F4F29C0E

+CMGS: 124

OK

Set text mode
AT+CMGF=1

AT+CSMP=17,167,0,0

AT+CMGS="01090255219",129

>TEST MESSAGE

+CMGS:125

OK

AT+CMSS - Send Message from Storage

This command sends to the network a message which is already stored in the <memw> storage.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMSS=<index>[,<da>[,<toda>]]

Execution command sends to the network a message which is already stored in the <memw> storage (see +CPMS) at the location <index>.

Parameters:

Name	Type	Default	Description
<index>	string	-	location value in the message storage <memw> of the message to send
<da>	string	-	destination address, string type represented in the currently selected character set (see +CSCS); if it is given it shall be used instead of the one stored with the message.
<toda>	integer	N/A	type of destination address

Values:

129 : number in national format

145 : number in international format (contains the "+")

Additional info:

If message is successfully sent to the network then the result is sent in the format:

+CMSS: <mr>[, <scts>]

where:

<mr> - message reference number.

<scts> - TP-Service Centre Time Stamp in Time String Format.

If message sending fails for some reason, an error code is reported:

+CMS ERROR:<err>

To store a message in the <memw> storage see command +CMGW.

Care must be taken to ensure that during the command execution, which may take several seconds, no other SIM interacting commands are issued.

AT+CMSS=?

Test command returns the OK result code.

To avoid malfunctions is suggested to wait for the +CMSS: <mr> or +CMS ERROR: <err> response before issuing further commands.

Concatenated SMS is not supported in text mode

It is the application responsibility to decode the HEX format of the Concatenated SMS and assemble the pieces into text.

```
AT+CMGF=1
OK
AT+CMGW="0165872928"
> test message...
+CMGW: 28
```

OK

AT+CMSS=28

+CMSS: 136

OK

AT+CMGW - Write Short Message to Memory

The command is related to writing short messages.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CMGW

Execution command writes a new short message in the <memw> memory storage (see +CPMS). It can have two syntax formats according to the SMS format: PDU or Text mode (see +CMGF command). If short message is successfully written the following URC is displayed:

+CMGW: <index>

Additional info:

In PDU mode the +CMGW command has the following syntax:

AT+CMGW=<length>[,<stat>]

After command line is terminated with <CR>, the module responds sending a four-character sequence prompt:

<CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

and waits for the specified number of bytes.

To write the message issue Ctrl-Z char (0x1A hex). To exit without sending the message issue ESC char (0x1B hex).

Name	Type	Default	Description
<length>	integer	N/A	length in bytes of the PDU to be written
	Value:		

	7÷164	:	number of bytes

<stat>	integer	2	message status

Values:

0	:	new message
1	:	read message
2	:	stored message not sent yet
3	:	stored message already sent

<data>	hex	-	PDU bytes, given in online mode

In Text mode the +CMGW command has the following syntax:

AT+CMGW[=<da>[,<toda>[,<stat>]]]

After command line is terminated with <CR>, the module responds sending a four-character sequence prompt:

<CR><LF><greater_than><space> (IRA 13, 10, 62, 32)

After this prompt, you can enter text that should be formatted as follows:

if current <dcs> (see +CSMP) indicates that GSM03.38/23.038 default alphabet is used and current <fo> (see +CSMP) indicates that 3GPP TS 03.40/23.040 TP-User-Data-Header-Indication is not set, then ME/TA converts the entered text into GSM alphabet, according to 3GPP TS 27.005, Annex A; backspace can be used to delete last character and carriage returns can be used..

if current <dcs> (see +CSMP) indicates that 8-bit or UCS2 data coding scheme is used or current <fo> (see +CSMP) indicates that 3GPP TS 23.040 TP-User-Data-Header-Indication is set, the entered text should consist of two IRA character long hexadecimal numbers which ME/TA converts into 8-bit octet (e.g. the 'asterisk' will be entered as 2A (IRA50 and IRA65) and this will be converted to an octet with integer value 0x2A)

The command waits for the specified number of bytes.

To write the message issue Ctrl-Z char (0x1A hex). To exit without writing the message issue ESC char (0x1B hex).

Name	Type	Default	Description
<da>	string	-	destination address, string type represented in the currently selected character set (see +CSCS).
<toda>	integer	N/A	type of destination address
	Values:		

129	:	number in national format
145	:	number in international format (contains the character "+")

<stat>	string	"STO UNSENT"	message status

Values:

"REC UNREAD"	:	new received message unread
"REC READ"	:	received message read
"STO UNSENT"	:	message stored not yet sent (default)
"STO SENT"	:	message stored already sent

Unsolicited field:

Name	Type	Description
<index>	integer	message location index in the memory <memw> (see +CPMS). If message storing fails for some reason, an error code is reported.

The DCD signal shall be in ON state while <data> is entered. The echoing of <data> is controlled by echo command E.

- Care must be taken to ensure that during the command execution, which may take several seconds, no other SIM interacting commands are issued.

- It is possible to save a concatenation of at most 16 SMs; the maximum number of chars depends on <dcs>:

  2432 chars

  2128 chars if 8-bit is used

  1056 chars if 16-bit is used

- In text mode, not only SUBMIT messages can be stored in SIM, but also DELIVER messages.

  The type of saved message depends upon the current <fo> parameter (see +CSMP). For a DELIVER message, current <vp> parameter (see +CSMP) is used to set the message Service Centre Time Stamp <scts>, so it must be an absolute time string, e.g. "09/01/12,11:15:00+04".

  SUBMIT messages can only be stored with status "STO UNSENT" or "STO SENT"; DELIVER messages can only be stored with status "REC UNREAD" or "REC READ".

- If message writing fails for some reason, then an error code is reported.

AT+CMGW=?

Test command returns the OK result code.

- To avoid malfunctions it is suggested to wait for the +CMGW: <index> or +CMS ERROR: <err> response before issuing further commands.

Concatenated SMS is not supported in text mode

It is the application responsibility to decode the HEX format of the Concatenated SMS and assemble the pieces into text.

```
AT+CMGF=0          set PDU mode
OK

AT+CMGW=18

> 0881280100990102591155078110209055 12F90000A704F4F29C0E

+CMGW: 29

OK

AT+CMGF=1          set text mode
OK

AT+CSMP=17,167,0,0
OK

AT+CSCA="821029190903",145
OK

AT+CMGW="0165872928"

> test message...

+CMGW: 28

OK
```

AT+CMGD - Delete Message

This command allows to delete from memory messages.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CMGD=<index>[,<delflag>]

Execution command deletes from memory <memr> the message(s).

Parameters:

Name	Type	Default	Description
<index>	integer	-	message index in the selected storage <memr>
<delflag>	integer	0	Type of multiple message deletion

Values:

0 : delete message specified in <index>

1 : delete all read messages from <memr> storage, leaving unread messages and stored mobile originated messages (whether sent or not) untouched

2 : delete all read messages from <memr> storage and sent mobile originated messages, leaving unread messages and unsent mobile originated messages untouched

3 : delete all read messages from <memr> storage, sent and unsent mobile originated messages, leaving unread messages untouched

4 : Delete all messages from <memr> storage

if <delflag> is present and not set to 0 then <index> is ignored and ME shall follow the rules for <delflag> shown above.

AT+CMGD=?

Test command shows the valid memory locations <index> and the supported values of <delflag>.

+CMGD: (supported <index>s list)[,(supported <delflag>s list)]

AT+CMGD=?

+CMGD: (0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,
(0-4)

OK

AT+CMGD=11 Delete message in 10th record

OK

AT+CMGD=1,4 Delete all messages

OK

AT+CGSMS - Select Service for MO SMS Messages

Set command is used to specify the service or service preference that the MT will use to send MO SMS messages.

3GPP TS 27.005

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CGSMS=[<service>]

Parameter:

Name	Type	Default	Description
<service>	integer	3	indicates the service or service preference to be used

Values:

0 : GPRS

1 : circuit switched

2 : GPRS preferred. Use circuit switched if SMS via GPRS service not available or GPRS not registered.

3 : circuit switched preferred. Use GPRS if SMS via circuit switched not available.

If SMS transfer via Packet Domain fails, <service> parameter automatically reset to Circuit switched.

AT+CGSMS?

The read command returns the currently selected service or service preference in the format:

+CGSMS: <service>

AT+CGSMS=?

Test command reports the supported list of currently available <service>.

AT#SMSMODE - SMS Commands Operation Mode

SMS Commands Operation Mode.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SMSMODE=<mode>

Set command enables/disables the check for presence of SMS Service Centre Address (SCA) in the Fixed Dialing Number (FDN) phonebook.

Parameter:

Name	Type	Default	Description
<mode>	integer	1	Enables/disables the check for presence of SCA in FDN phonebook.

Values:

1 : Disables the check for presence of SCA in FDN phonebook.

2 : Enables the check for presence of SMS SCA in the FDN phonebook when FDN are enabled. If the SMS SCA is not present a SMS cannot be sent.

In AT#SMSMODE=1 with FDN active (AT+CLCK="FD",1,<pin>) it is still enabled the check for the SCA presence in the FDN phonebook.

AT#SMSMODE?

Read command reports whether the check of SMS SCA in FDN phonebook is enabled or not, in the format:

#SMSMODE: <mode>

AT#SMSMODE=?

Test command reports the range of <mode> parameter values.

AT#E2SMSRI - SMS Ring Indicator

This set command enables/disables the Ring Indicator pin response to an incoming SMS message. If enabled, a negative going pulse is generated when receiving an incoming SMS message.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#E2SMSRI=[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	0	RI enabling

Values:

0	:	disables RI pin response for incoming SMS messages
50÷1150	:	enables RI pin response for incoming SMS. The value of <n> is the duration in ms of the pulse generated on receipt of an incoming SMS.

If +CNMI=3,1 command is issued, and the module is in a GPRS connection, a 100 ms break signal is sent and a 1 sec. pulse is generated on RI pin, no matter if the RI pin response is either enabled or not.

If +CNMI=0,0,0,0,0 command is issued, no ring pulse will be generated, regardless of #E2SMSRI value

AT#E2SMSRI?

Read command reports the duration in ms of the pulse generated on receipt of an SMS, in the format:

#E2SMSRI: <n>

<n>=0 means that the RI pin response to an incoming SMS is disabled

AT#E2SMSRI=?

Reports the range of supported values for parameter <n>

Phonebook

AT+CPBS - Select Phonebook Memory Storage

This set command selects phonebook memory storage, which will be used by other phonebook commands.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CPBS=<storage>

Parameter:

Name	Type	Default	Description
<storage>	string	"SM"	specify the phonebook memory storage

Values:

"SM"	:	SIM phonebook
"FD"	:	SIM fixed dialing-phonebook (FDN)(only phase 2/2+ SIM)
"LD"	:	SIM last-dialing-phonebook (+CPBF is not applicable for this storage). Not supported.

"MC"	:	device missed (unanswered received) calls list (+CPBF is not applicable for this storage). Not supported.
"RC"	:	ME received calls list (+CPBF is not applicable for this storage). Not supported.
"MB"	:	mailbox numbers stored on SIM; it is possible to select this storage only if the mailbox service is provided by the SIM (see #MBN). Not supported.
"DC"	:	ME last-dialing-phonebook (+CPBF is not applicable for this storage). Not supported.
"ME"	:	ME phonebook. Not supported.
"EN"	:	SIM emergency numbers phonebook (+CPBW and +CPBF not applicable for this storage). Not supported.
"ON"	:	SIM own numbers (MSISDNs) phonebook (+CPBF is not applicable for this storage)
"SDN"	:	SIM Service Dialing Numbers (SDN) phonebook (+CPBW is not applicable for this storage)
"AP"	:	SIM application phonebook (DFPHONEBOOK under ADFUSIM)
"MN"	:	SIM card CPHS MN
"MBDN"	:	SIM card CPHS MBDN

AT+CPBS?

Read command returns the actual values of the parameter <storage>, the number of occupied records <used> and the maximum index number <total>, in the format:

+CPBS: <storage>,<used>,<total>

AT+CPBS=?

Test command returns the supported range of values for the parameter <storage>

Current phonebook storage is SIM

AT+CPBS="SM"

OK

AT+CPBR=1

+CPBR: 1,"0105872928",129,"James"

OK

AT+CPBR - Read Phonebook Entries

The command reads phonebook entries.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CPBR=<index1>[,<index2>]

Execution command returns phonebook entries in location number range <index1>..<index2> from the current phonebook memory storage selected with +CPBS. If <index2> is omitted, only location <index1> is returned.

The intermediate response format is:

[+CPBR:<index1>,<number>,<type>,<text>[,<hidden>][,<group>] [,<adnumber>][,<adtype>][,<secondtext>][,<email>]]

...

[<CR><LF> +CPBR:<index2>,<number>,<type>,<text>[,<hidden>] [,<group>][,<adnumber>] [,<adtype>][,<secondtext>][,<email>]]]

or
+CME ERROR: <err>

Parameters:

Name	Type	Default	Description
<index1>	integer	-	value in the range of location numbers of the currently selected phonebook memory storage (see +CPBS).
<index2>	integer	-	value in the range of location numbers of the currently selected phonebook memory storage (see +CPBS).

Additional info:

▶▶ Intermediate response parameters

Name	Type	Default	Description
<indexn>	integer	-	the location number of the phonebook entry
<number>	string	-	phone number of format <type>
<type>	integer	N/A	type of phone number octet
	Values:		
	129 : national numbering scheme		
	145 : international numbering scheme (contains the character "+")		
<test>	string	-	the alphanumeric text associated to the number; character set as specified by command Select TE Character Set +CSCS
<hidden>	integer	0	indicates if the entry is hidden or not
	Values:		
	0 : phonebook entry not hidden		
	1 : phonebook entry hidden		

<group>	string	-	the group the entry may belong to; character set as specified by command Select TE Character Set +CSCS
<adnumber>	string	-	additional phone number of format <adtype>
<adtype>	integer	-	type of address octet
<secondtext>	string	-	second text field associated with the number; character set as specified by command Select TE Character Set +CSCS
<email>	string	-	email address; character set as specified by command Select TE Character Set +CSCS

AT+CPBR=?

Test command returns the supported range of values for parameters <indexn> and the maximum lengths of <number>, <text>, <group>, <secondtext> and <email> string parameters, in the format:

+CPBR: (<minIndex> - <maxIndex>),<nlength>,<tlength>,<glength>,<slength>,<elength>

Additional info:

▶▶ Test command response parameters

Name	Type	Default	Description

<minIndex>	integer	-	the minimum <index> number
<maxIndex>	integer	-	the maximum <index> number
<nlength>	integer	-	maximum <number> field length
<tlength>	integer	-	maximum <name> field length
<glength>	integer	-	maximum <group> field length
<slength>	integer	-	maximum <secondtext> field length
<elength>	integer	-	maximum <email> field length

The value of <nlength> could vary, depending on the availability of Extension service, in the following situations:

1. if "SM" memory storage has been selected (see +CPBS) and the SIM supports the Extension1 service
2. if "FD" memory storage has been selected (see +CPBS) and the SIM supports the Extension2 service
3. if "MB" memory storage has been selected (see +CPBS) and the SIM supports the Extension6 service

Remember to select the PB storage with +CPBS command before issuing PB commands.

```
AT+CPBS="ME"
OK
AT+CPBS?
+CPBS: "ME",1,100
OK
AT+CPBR=?
+CPBR: (1-500),40,20,2,20,20 OK
AT+CPBR=1
+CPBR: 1,"01048771234",129,"James","","",0,"","
OK
```

AT+CPBF - Find Phonebook Entries

This command returns phonebook entries.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CPBF=<findtext>

Execution command returns phonebook entries (from the current phonebook memory storage selected with +CPBS) which alphanumeric field start with string <findtext>.

Parameter:

Name	Type	Default	Description
<findtext>	string	-	string to be searched among the phonebook entries; used character set should be the one selected with command +CSCS.

Additional info:

The command returns a report in the form:

[+CPBF:<index1>,<number>,<type>,<text>[,<hidden>][,<group>] [,<adnumber>] [,<adtype>][,<secondtext>][,<email>]<CR><LF>

+CPBF:<index2>,<number>,<type>,<text>[,<hidden>][,<group>] [,<adnumber>] [,<adtype>][,<secondtext>][,<email>][...]]]

Name	Type	Default	Description

<indexn>	integer	-	The location number of the phonebook entry
<number>	string	-	Phone number of format <type>
<type>	integer	N/A	type of phone number octet

Values:

129	:	national numbering scheme
145	:	international numbering scheme (contains the character "+")

<text>	string	-	The alphanumeric text associated to the number; the character set used should be the one selected with command +CSCS
<group>	string	-	Field of maximum length <glength> indicating a group the entry may belong to; character set as specified by command Select TE Character Set +CSCS
<adnumber>	string	-	additional number; phone number of format <adtype>
<adtype>	integer	-	type of address octet
<secondtext>	string	-	Field of maximum length <slength> indicating a second text field associated

			with the number; character set as specified by command Select TE Character Set +CSCS
<email>	string	-	field of maximum length <elength> indicating an email address; character set as specified by command Select TE Character Set +CSCS
<hidden>	string	N/A	indicates if the entry is hidden or not

Values:

0	:	phonebook entry not hidden
1	:	phonebook entry hidden

- +CPBF is not applicable if the current selected storage (see +CPBS) is either "MC", "RC" or "LD" or "DC".
- If <findtext>="" the command returns all the phonebook records.
- If no PB records satisfy the search criteria then an ERROR message is reported.

?

AT+CPBF=?

Test command reports the maximum lengths of <number> and <text> fields, in the format:

+CPBF: <nlength>,<tlength>,<glength>,<slength>,<elength>

Additional info:

Test command response fields

Name	Type	Default	Description
<nlength>	integer	-	Maximum length of field <number>
<tlength>	integer	-	Maximum length of field <text>
<glength>	integer	-	Maximum length of field <group>
<slength>	integer	-	Maximum length of field <secondtext>
<elength>	integer	-	Maximum length of field <email>

The value of <nlength> could vary, depending on the availability of Extension service, in the following situations:

1. if "SM" memory storage has been selected (see +CPBS) and the SIM supports the Extension1 service
2. if "FD" memory storage has been selected (see +CPBS) and the SIM supports the Extension2 service
3. if "MB" memory storage has been selected (see +CPBS) and the SIM supports the Extension6 service

Remember to select the PB storage with +CPBS command before issuing PB commands.

AT+CPBS="ME" Selecting phonebook

OK

AT+CPBF="J" Searching for string "J"

+CPBF: 1,"01048771234",129,"James","","",0,"","",""

+CPBF: 2,"0169998888",129,"Jane","","",0,"","","" OK

Searching for everything in phone book, and finding all entries

AT+CPBF=""

+CPBF: 1,"01048771234",129,"James","","",0,"","",""

+CPBF: 2,"0169998888",129,"Jane","","",0,"","",""

+CPBF: 7,"0115556666",129,"Juliet","","",0,"","",""

+CPBF: 5,"0181111234",129,"Kevin","","",0,"","",""

OK

AT+CPBW - Write Phonebook Entry

This command writes phonebook entry in the current phonebook memory.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT+CPBW=[<index>[,<number>[,<type>[,<text>]]]]

Execution command writes phonebook entry in location number <index> in the current phonebook memory storage selected with +CPBS.

Parameters:

Name	Type	Default	Description
<index>	integer	-	value in the range of location numbers of the currently selected phonebook memory storage (see +CPBS)
<number>	string	-	phone number in the format <type>
<type>	integer	129	type of number Values: 129 : national numbering scheme 145 : international numbering scheme (contains the character "+")
<text>	string	-	text associated to the number; used character set should be the one selected with command +CSCS

- If record number <index> already exists, it will be overwritten.
- If either <number>, <type> and <text> are omitted, the phonebook entry in location <index> is deleted.
- If either "LD", "MC" or "RC" memory storage has been selected (see +CPBS) it is possible just to delete the phonebook entry in location <index>.
- If either <number>, <type> and <text> omitted, the phonebook entry in location <index> deleted.
- If <index> is omitted or <index>=0, the number <number> is stored in the first free phonebook location.

AT+CPBW=?

Test command returns location range supported by the current storage as a compound value, the maximum length of <number> field, supported number format of the storage and maximum length of <text> field. The format is:

+CPBW: (list of supported <index>s),<nlength>,(list of supported <type>s),<tlength>

Additional info:

- Test command response fields

Name	Type	Default	Description
<nlength>	integer	-	Maximum length of field <number>
<tlength>	integer	-	Maximum length of field <text>

the value of <nlength> could vary, depending on the availability of Extension service, in the following situations:

1. if "SM" memory storage has been selected (see +CPBS) and the SIM supports the Extension1 service
2. if "FD" memory storage has been selected (see +CPBS) and the SIM supports the Extension2 service
3. if "MB" memory storage has been selected (see +CPBS) and the SIM supports the Extension6 service

Remember to select the PB storage with +CPBS command before issuing PB commands.

```
AT+CPBW=?
+CPBW: (1-500),40,(129,145),255
OK
AT+CPBW=6,"18651896699",129,"John"
OK
```

Time & Alarm

AT+CCLK - Clock Management

The command is related to real time clock management.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CCLK=<time>

Set command sets the real-time clock of the module.

Parameter:

Name	Type	Default	Description
<time>	string	N/A	Current time as quoted string in the format: "yy/MM/dd,hh:mm:ss±zz"

Values:

yy : year (two last digits are mandatory), range is 00..99

MM : month (two digits are mandatory), range is 01..12

dd : day (two digits are mandatory) The range for dd(day) depends either on the month and on the year it refers to. Available ranges are: (01..28) (01..29) (01..30) (01..31). Trying to enter an out of range value will raise an ERROR message.

hh : hour (two digits are mandatory), range is 00..23

mm : minute (two digits are mandatory), range is 00..59

ss : seconds (two digits are mandatory), range is 00..59

±zz : time zone (indicates the difference, expressed in quarter of an hour, between the local time and GMT; two digits are mandatory), range is -96...+96

AT+CCLK?

Read command returns the current setting <time> of the real-time clock, in the format:

+CCLK: <time>

AT+CCLK=?

Test command returns the OK result code.

Set date and time:
AT+CCLK="02/09/07,22:30:00+00"
OK

Read date and time:
AT+CCLK?
+CCLK: "02/09/07,22:30:25"
OK

AT+CALA - Alarm Management

This command is related to the alarm management.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CALA=<time>[,<n>[,<type>[,<text>[,<recurr>[,<silent>]]]]]

Set command stores in the internal Real Time Clock of the module an alarm time with respective settings.

It is possible to set up a recurrent alarm for one or more days in the week.

Currently just one alarm can be set.

Alarms are not supported after disconnecting from power. Coin cell are supported. In case of a power cut, alarm will be deleted and needs to be re-set.

When the RTC time reaches the alarm time then the alarm starts, the behavior of the module depends on the setting <type> and if the module was already ON at the moment when the alarm time had come.

Parameters:

Name	Type	Default	Description
<time>	string	-	current alarm time as quoted string in the format: "yy/MM/dd,hh:mm:ssÂ±zz" Refer to +CCLK for the string meaning. Empty string (+CALA="") deletes the current alarm and resets all the +CALA parameters to the factory default configuration.

			"hh:mm:ssÂ±zz" string must be used only when issuing +CALA with parameter <recurr>.
<n>	integer	0	index of the alarm

Value:

0	:	the only value supported

<type>	integer	1	alarm behavior type

Values:

0	:	reserved
1	:	the module wakes up fully operative as if the ON/OFF button has been pressed. If the module is already ON when the alarm times out, then it does nothing.
2,3	:	see Additional info section.
4	:	Not Supported
5	:	see Additional info section.
6÷8	:	Not Supported

<text>	string	-	alarm code text string used in the URC +CALA. It has meaning only if <type> is equal to 2 or 5.
<recurr>	string	N/A	sets a recurrent alarm for one or more days in the week in the following format:

"X[,Y[,...]]"

where X, Y, ... can assume the following values:

Values:

0	:	all days in the week
1	:	Monday
2	:	Tuesday
3	:	Wednesday
4	:	Thursday
5	:	Friday
6	:	Saturday
7	:	Sunday

<silent>	integer	N/A	indicates if the alarm is silent or not

Values:

0	:	the alarm is not silent
1	:	the alarm is silent

Additional info:

▶▶ <type>=2

The module wakes up in "alarm mode" if at the alarm time it was powered OFF, otherwise it remains fully operative. In both cases the module issues an unsolicited code every 3 s:

+CALA: <text>

Where <text> is the +CALA optional parameter previously set.

The module keeps on sending the unsolicited code every 3 s until a #WAKE or #SHDN command is received or a 90 seconds timer expires. If the module is in "alarm mode" and it does not receive the #WAKE command within 90s then it shuts down.

- <type>=3

  The module wakes up in "alarm mode" if at the alarm time it was powered OFF, otherwise it remains fully operative. In both cases the module starts playing the alarm tone on the selected path for the ringer (see command #SRP).

  The module keeps on playing the alarm tone until a #WAKE or #SHDN command is received or a 90 s time-out occurs. If the device is in "alarm mode" and it does not receive the #WAKE command within 90s then it shuts down.

  If alarm expires during a call alarm sound will stop when the call is disconnected.

- <type>=5

  The module will make both the actions as for <type>=2 and <type>=3.

The "alarm mode" is indicated by hardware pin CTS to the ON status and DSR to the OFF status, while the "power saving" status is indicated by a CTS - OFF, DSR - OFF and USB_VBUS - OFF status. The normal operating status is indicated by DSR - ON or USB_VBUS - ON status.

During the "alarm mode" the device will not make any network scan and will not register to any network and therefore is not able to dial or receive any call or SMS.

The only commands that can be issued to the module in this state are the #WAKE and #SHDN, every other command must not be issued during this state.

AT+CALA?

Read command returns the list of current active alarm settings in the device, in the following format:

[+CALA: <time>,<n>,<type>,[<text>],<recurr>,<silent>]

On READ command <time> does not include the time zone.

AT+CALA=?

Test command returns the list of supported index values, alarm types, maximum length of the text to be displayed in the URC +CALA, maximum length of <recurr> and supported <silent>s, in the format:

+CALA: (list of supported <n>s),(list of supported <type>s),<tlength>,<rlength>, (list of supported <silent>s)

Additional info:

Parameters:

Name	Type	Default	Description
<tlength>	string	-	maximum length of <text> parameter
<rlength>	string	-	maximum length of <recurr> parameter

```
AT+CALA="02/09/07,23:30:00+00"
OK
```

AT+CTZR - Time Zone Reporting

This command enables and disables the time zone change event reporting.

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CTZR=<onoff>

Set command permits to enable/disable the time zone change event reporting.

If the reporting is enabled and whenever the time zone is changed, the MT returns the unsolicited result code:

+CTZV: <tz>

Parameter:

Name	Type	Default	Description
<onoff>	string	0	enable/disable the time zone change event reporting. Values: 0 : Disable time zone change event reporting 1 : Enable time zone change event reporting

Unsolicited field:

Name	Type	Description

<tz>	string	New time zone.

AT+CTZR?

Read command reports the currently selected <onoff> in the format:

+CTZR: <onoff>

AT+CTZR=?

Test command reports the supported range of values for parameter <onoff>

AT+CTZU - Automatic Time Zone Update

Set command enables/disables the automatic time zone update via NITZ.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CTZU=<onoff>

Parameter:

Name	Type	Default	Description
<onoff>	integer	0	enables/disables the automatic time zone update

Values:

0	:	disable
1	:	enable

Despite of the name, the command +CTZU enables automatic update of the date and time set by +CCLK command (not only time zone).

AT+CTZU?

Read command reports the current setting of <onoff> in the format:
+CTZU: <onoff>

AT+CTZU=?

Test command returns the supported values of parameter <onoff>.

AT#NITZ - Network Identity and Time Zone

This command handles Network Identity and Time Zone.

SIM Presence	Setting saved	Can be aborted
Required	Common profile	No

AT#NITZ=[<val>[,<mode>]]

Set command enables/disables the automatic date/time updating and the Full Network Name applying. It enables also the #NITZ URC in the format:

#NITZ: <datetime>

and permits to change its format.

Parameters:

Name	Type	Default	Description
<val>	integer	7	identifies the functionalities to enable. The <val> parameter is a sum of integer values, where every value corresponds to a functionality: 1 - enables automatic date/time updating 2 - enables Full Network Name applying 4 - sets the #NITZ URC 'extended' format (see <datetime> below) 8 - sets the #NITZ URC 'extended' format with Daylight Saving Time (DST) support (see <datetime> below)

Values:

0	:	disables every functionality
1÷15	:	sum of integer values

<mode> integer 0 enables/disables the #NITZ URC

Values:

0	:	disables the URC
1	:	enables the URC

Unsolicited field:

Name	Type	Description
<datetime>	string	string format depends on parameter <val> "yy/MM/dd,hh:mm:ss" - 'basic' format, if <val> is in (0..3) "yy/MM/dd,hh:mm:ss±zz" - 'extended' format, if <val> is in (4..7) "yy/MM/dd,hh:mm:ss±zz,d" - 'extended' format with DST support, if <val> is in (8..15) For the meaning of the <datetime> subfields, please check +CCLK and #CCLK commands

- If the DST information is not sent by the network, then the <datetime> parameter will have the format "yy/MM/dd,hh:mm:ss±zz".
- Date and time information can be sent by the network after GSM registration or after PS attach.

AT#NITZ?

Read command reports whether

- automatic date/time updating

- Full Network Name applying

- #NITZ URC (as well as its format)

are currently enabled or not in the format:

#NITZ: <val>,<mode>

AT#NITZ=?

Test command returns supported values of parameters <val> and <mode>.

AT#CCLK - Clock Management

The command is related to real time clock management.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#CCLK=<time>

Set command sets the real-time clock of the module.

Parameter:

Name	Type	Default	Description
<time>	string	N/A	Current time as quoted string in the format: "yy/MM/dd,hh:mm:ss±zz,d"

Values:

yy : year (two last digits are mandatory), range is 00..99

MM : month (two digits are mandatory), range is 01..12

dd : day (two digits are mandatory) The range for dd(day) depends either on the month and on the year, it refers to. Available ranges are: (01..28) (01..29) (01..30) (01..31). Trying to enter an out of range value will raise an ERROR message.

hh : hour (two digits are mandatory), range is 00..23

mm	:	minute (two digits are mandatory), range is 00..59
ss	:	seconds (two digits are mandatory), range is 00..59
±zz	:	time zone (indicates the difference, expressed in quarter of an hour, between the local time and GMT; two digits are mandatory), range is: -96..+96
d	:	number of hours added to the local TZ because of Daylight Saving Time (summertime) adjustment; range is 0-2.

AT#CCLK?

Read command returns the current setting of the real-time clock, in the format <time>.

If the time is set by the network but the Daylight-Saving Time (DST) information is missing, or the time is set by +CCLK command, then the <time> format is:

"yy/MM/dd,hh:mm:ss±zz"

AT#CCLK=?

Test command returns the OK result code.

The way of writing the <time>:

"yy/mm/dd,hh:mm:ss±zz,d"

AT#CCLK="02/09/07,22:30:00+00,1"

- Set command:

AT#CCLK="14/02/18,08:00:00+12,1"

OK

- Read command:

AT#CCLK?
#CCLK: "14/02/18,08:00:02+12,1"

OK

AT#CCLKMODE - Clock Mode

This command allows to enable the local time or the UTC time.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#CCLKMODE=<mode>

Set command enables the local time or the UTC time in +CCLK and #CCLK commands

Parameter:

Name	Type	Default	Description
<mode>	integer	0	Time and date mode

Values:

0 : Local time + local time zone offset

1 : UTC time + local time zone offset

The setting is saved automatically in NVM.

AT#CCLKMODE?

Read command reports whether the local time or the UTC time is enabled, in the format:

#CCLKMODE: <mode>

AT#CCLKMODE=?

Test command reports the supported range of values for parameter <mode>

AT#WAKE - Wake from Alarm Mode

Stop any alarm activity

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#WAKE=<opmode>

Execution command stops any eventually present alarm activity and, if the module is in alarm mode, it exits the alarm mode and enters the normal operating mode.

Parameter:

Name	Type	Default	Description
<opmode>	integer	0	operating mode

Value:

0 : normal operating mode; the module exits the alarm mode and enters the normal operating mode, any alarm activity is stopped (e.g. alarm tone playing) and an OK result code is returned.

If #WAKE=0 command is issued after an alarm has been set with +CALA command, but before the alarm has expired, it will answer OK but have no effect.

AT#WAKE?

Read command returns the operating status of the device in the format:

#WAKE: <status>

where:
<status>

0 - normal operating mode
1 - alarm mode or normal operating mode with some alarm activity.

AT#WAKE=?

Test command returns OK result code.

The alarm mode is indicated by status ON of hardware pin CTS and by status ON of pin DSR; the power saving status is indicated by a CTS - OFF and DSR - OFF status; the normal operating status is indicated by DSR - ON.

During the alarm mode the device will not make any network scan and will not register to any network and therefore it is not able to dial or receive any call or SM, the only commands that can be issued to the MODULE in this state are the #WAKE and #SHDN, every other command must not be issued during this state.

Audio

Audio Basic Configuration

AT+CLVL - Loudspeaker Volume Level

This command allows to set the volume of internal loudspeaker audio output.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CLVL=<level>

Set command is used to select the volume of the internal loudspeaker audio output of the device.

Parameter:

Name	Type	Default	Description
<level>	integer	12	Loudspeaker volume (+2dB/step)

Value:

0÷max : max value can be read by issuing the Test command.

AT+CLVL?

Read command reports the current <level> setting of the loudspeaker volume in the format:

+CLVL: <level>

AT+CLVL=?

Test command returns the supported values of parameter <level> in the format:
+CLVL: (0-max)

AT+CMUT - Microphone Mute Control

This command enables/disables the muting of the microphone audio line during a voice call.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CMUT=<n>

Set command enables/disables the uplink voice muting during a voice call.

Parameter:

Name	Type	Default	Description
<n>	integer	0	Controls the muting of the microphone audio line

Values:

0 : mute off, microphone active

1 : mute on, microphone muted

This command mutes/activates both microphone audio paths, internal mic and external mic.

AT+CMUT?

Read command reports whether the muting of the microphone audio line during a voice call is enabled or not, in the format:

+CMUT:<n>

AT+CMUT=?

Test command reports the supported values for <n> parameter.

AT+CSIL - Silence Command

This command enables the silent mode.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CSIL=[<n>]

Set command enables/disables the silent mode. When the phone is in silent mode, all signaling tones from MT are suppressed.

Parameter:

Name	Type	Default	Description
<n>	integer	0	enables/disables the silent mode

Values:

0	:	disable
1	:	enable

AT+CSIL?

Read command returns the current value of the parameter <n> in the format:

+CSIL: <n>

AT+CSIL=?

Test command returns the supported values of parameter <n>.

AT#CAP - Change Audio Path

This command has no effect and is included only for backward compatibility.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CAP=[<n>]

Parameter:

Name	Type	Default	Description
<n>	integer	1	change physical audio path

Values:

0 : Not supported

1 : Handsfree MIC1 – SPKR1 SPKR2 (automatic DSP profile 1 Hands Free)

2 : Headset – MIC2 – SPKR1 SPKR2 (automatic DSP profile 2 Headset)

The audio path is mutually exclusive, enabling one disables the other.

When changing the audio path, the volume level is set at the previously stored value for that audio path (see +CLVL).

AT#CAP?

Read command reports the set value of the parameter <n> in the format:

#CAP: <n>

AT#CAP=?

Test command reports the supported values for the parameter <n>.

AT#DIGMICG - Digital Microphone Gain

This command manages the gain of the handset microphone.

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT#DIGMICG=<gainLevel>

Set command sets the handset microphone input gain through 46 levels by 1 dB steps.

Parameter:

Name	Type	Default	Description
<gainLevel>	integer	33	handset microphone input gain

Value:

0÷45	:	input gain (+1dB/step)

AT#DIGMICG?

Read command returns the current handset microphone input gain in the format:

#DIGMICG: <gainLevel>

AT#DIGMICG=?

Test command returns the supported range of values of parameter <gainLevel>.

Set the functionality in a certain way

AT#DIGMICG=30

OK

Check parameters value

AT#DIGMICG?

#DIGMICG=30

OK

AT#OAP - Open Audio Loop

This set command enables/disables the Open Audio Path.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#OAP=[<mode>]

Parameter:

Name	Type	Default	Description
<mode>	integer	0	enables/disables the Open Audio Path

Values:

0	:	disable
1	:	enable

AT#OAP?

Read command reports the current value of the parameter <mode> in the format:

#OAP: <mode>

AT#OAP=?

Test command returns the supported values of parameter <mode>.

The audio loop is established between microphone and speaker using sidetone scaling value.
#OAP is intended for testing purposes only. Thus, care must be

taken to ensure that during the command execution no other audio interacting commands are issued.

AT#HFMICG - Handsfree Microphone Gain

Handsfree Microphone Gain

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT#HFMICG=[<level>]

Set the handsfree analogue microphone input gain

Parameter:

Name	Type	Default	Description
<level>	integer	1	handsfree microphone input gain

Value:

0÷7 : handsfree microphone gain (+6dB/step)

Parameter is saved in NVM issuing AT&W command

AT#HFMICG?

Returns the current handsfree microphone input gain, in the format:

#HFMICG: <level>

AT#HFMICG=?

Test command returns the supported range of values for parameter <level>

Tones Configuration

AT#TSVOL - Tone Classes Volume

This command is used to manage the tone classes volume selection.

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT#TSVOL=<class>,<mode>[,<volume>]

Set command is used to select the volume mode for one or more tone classes.

Parameters:

Name	Type	Default	Description
<class>	integer	-	the <class> is a bit mask (1 Byte long) where each bit, if set, indicates the volume corresponding to a class of tones: bit 0 = GSM tones bit 1 = ringer tones bit 2 = alarm tones bit 3 = signaling tones bit 4 = DTMF tones bit 5 = SIM Toolkit tones bit 6 = user defined tones bit 7 = dial tones
<mode>	integer	0	it indicates which volume is used for the classes of tones represented by <class>

Values:

0	:	default volume is used
1	:	the volume <volume> is used

<volume>	integer	N/A	volume to be applied to the set of classes of tones represented by <class>; mandatory if <mode> is 1

Value:

0÷max	:	the 'max' value can be read issuing the Test Command AT#TSVOL=?

The only volume that changes with the TSVOL configuration is the ringer tones class volume (bit 1). The other are dummy values.

AT#TSVOL?

Read command returns for each class of tones the last setting of <mode> in the format:

#TSVOL: 1,<mode1><CR><LF>
..
#TSVOL:128,<mode128>

Additional info:

If <mode> is not 0, also the <volume> is reported in the format:

#TSVOL: 1,<mode1>,<volume1><CR><LF>
#TSVOL: 2,<mode2>,<volume2><CR><LF>
#TSVOL: 4,<mode4>,<volume4><CR><LF>
#TSVOL: 8,<mode8>,<volume8><CR><LF>
#TSVOL: 16,<mode16>,<volume16><CR><LF>
#TSVOL: 32,<mode32>,<volume32><CR><LF>
#TSVOL: 64,<mode64>,<volume64><CR><LF>
#TSVOL:128,<mode128>,<volume128>

AT#TSVOL=?

Test command returns the supported range of values of parameters <class>, <mode> and <volume>.

- Set tone class volume

  AT#TSVOL=64,1,5
  OK

  Read current <mode> setting values

  AT#TSVOL?
  #TSVOL:1,0
  #TSVOL:2,0
  #TSVOL:4,1,5
  #TSVOL:8,0
  #TSVOL:16,1,5
  #TSVOL:32,0
  #TSVOL:64,1,5
  #TSVOL:128,0
  OK

Embedded DTMF Decoder & TTY

AT#DTMF - Embedded DTMF Decoder Enabling

This command enables/disables the embedded DTMF decoder.

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT#DTMF=<mode>

Set command enables/disables the embedded DTMF decoder.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	enable/disable the embedded DTMF decoder

Values:

0 : disable DTMF decoder

1 : enable the embedded DTMF decoder with URC activation

2 : same as <mode>=1 - enable the embedded DTMF decoder with URC activation

Additional info:

The DTMF tone receiving is notified with an unsolicited message through AT interface in the following format:

#DTMFEV: x

Unsolicited field:

Name	Type	Description
<x>	string	DTMF Digit Detected. Values: (0-9, A-D, *, #)

#DTMF = 2 has been added only for backward compatibility, hence refer to #DTMF = 1

AT#DTMF?

Read command returns the currently selected <mode> in the format:

#DTMF: <mode>

AT#DTMF=?

Test command returns the supported values of the parameter <mode>.

If #DTMF=2 is set, the read command will still give #DTMF=1 since the configuration is the same.

Digital Voice Interface

AT#DVI - Digital Voiceband Interface

Digital Voiceband Interface handling.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#DVI=<mode>[,<dviport>,<clockmode>]

Set command enables/disables the Digital Voiceband Interface. To have information about the pins used by the interface, refer to document [1].

Parameters:

Name	Type	Default	Description
<mode>	integer	2	enables/disables the DVI
	Values:		
	0 :	disable DVI	
	1 :	enable DVI: audio is forwarded to the DVI block	
	2 :	Legacy : DVI settings are managed by "audio_config.nvm" configuration file. AT#DVIEXT and AT#DVICLK settings are ignored	
<dviport>	integer	2	select DVI port

Value:

2	:	DVI port, the only available.

<clockmode>	integer	1	select DVI clock mode.

Values:

0	:	NOT SUPPORTED
1	:	DVI master

- <clockmode> parameter 0 DVI SLAVE is not supported.
- System reboot is required for the changes to take effect.

AT#DVI?

Read command reports last setting, in the format:

#DVI: <mode>,<dviport>,<clockmode>

AT#DVI=?

Test command reports the range of supported values for parameters <mode>,<dviport> and <clockmode>.

Modules Supporting the command
LE910R1-EU

Configure DVI as master using the DVI Port #2 (the only available)

```
AT#DVI=1,2,1
OK
```

AT#DVIEXT - Digital Voiceband Interface Extension

Digital Voiceband Interface Extension.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#DVIEXT=<config>[,<samplerate>[,<samplewidth>[,<audiomode>[,<edge>]]]]

Set command configures the Digital Voiceband Interface.

Parameters:

Name	Type	Default	Description
<config>	integer	0	Transmission mode
	Values:		
	0 : Burst Mode (PCM)		
	1 : Normal Mode (I2S)		
<samplerate>	integer	1	Sample rate
	Values:		
	0 : audio scheduler sample rate 8 kHz		
	1 : audio scheduler sample rate 16 kHz		
<samplewidth>	integer	0	Number of bit per sample
	Values:		

0	:	16 bits per sample
1	:	NOT SUPPORTED
2	:	NOT SUPPORTED
3	:	NOT SUPPORTED
4	:	NOT SUPPORTED

<audiomode> integer 1 Mode of audio

Values:

0	:	Mono Mode
1	:	Dual Mono

<edge> integer 1 Edge on which bit is transmitted

Values:

0	:	Falling edge
1	:	Rising edge

System reboot is required for the changes to take effect.

Only Dual Mono is supported for <audiomode> : the set command does not affect the <audiomode> value which is constant at 1 (Dual Mono).

AT#DVIEXT?

Read command reports last setting, in the format:

#DVIEXT: <config>,<samplerate>,<samplewidth>,<audiomode>,<edge>

AT#DVIEXT=?

Test command reports the range of supported values for parameters:

<config>,<samplerate>,<samplewidth>,<audiomode>,<edge>

Modules Supporting the command
LE910R1-EU

Configure DVI as I2S, mono 8 kHz, rising edge using

AT#DVIEXT=1,0,0,0,1

OK

AT#DVICLK - Digital Voiceband Interface Clock

This set command configures and activates the DVICLK clock signal.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#DVICLK=<clk>

Parameter:

Name	Type	Default	Description
<clk>	integer	0	Set DVI bit clock frequency

Values:

0 : DVI clock activated at 32 fs (default)

1 : DVI clock activated at 64 fs

2 : DVI clock activated at 128 fs

3 : DVI clock activated at 256 fs

AT#DVICLK?

Read command reports last setting, in the format:

#DVICLK: <clk>

AT#DVICLK=?

Test command reports the range of parameter <clk>.

sampling frequency [fs] is 16 kHz

AT#DVICLK=1

setting means that bit clock is 64*[fs] = 64*16 = 1024 kHz

Audio File Management

AT#ASEND - Save Audio File

This command allows user to send an audio file to serial port and store it in the modem file system.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#ASEND=<filename>,<filesize>

Execution command allows user to send an audio file to serial port and store it in the modem file system

Parameters:

Name	Type	Default	Description
<filename>	string	-	file name Allowed extensions for <filename>: *.pcm or no extension - the audio supported format is PCM raw data *,wav - the audio supported format is RIFF/WAVE File Format
<filesize>	integer	-	file size in bytes

Additional info:

- When the sending is stopped or an error occurs, an URC is provided with the following format:

  #ASENDEV: <result>

  Where:

  <result>

  0 - file storing done

  1 - file storing error

  - In case Storing is completed or an error occurs, URC is received.

- Filename has a maximum of 32 characters

- The total size of all audio files must not be over <total_size> in #ASIZE

- The file should be sent using RAW ASCII file transfer. the flow control is set to hardware and baudrate is set to 115200 bps in the UART port settings.

- Ii is not allowed for TE to use two or more serial ports as DATA service (DUN and asend) simultaneously.

AT#ASEND=?

Test command returns the OK result code.

After the CONNECT, an audio file has to be sent to serial port

AT#ASEND="test.pcm",159182

CONNECT

AT#APLAY - Play an Audio File

This command plays an audio file.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#APLAY<mode>[,<dir>,<filename>[,<volume>]]

This command plays audio file on the speaker or uplink path.

When the playing is ended, the following URC is displayed showing the playing result code.

#APLAYEV: <result>

For <result> code values see unsolicited code value section.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	Playing mode

Values:

0 : Stop playing, optional parameters are ignored

1 : Start playing, optional parameters are mandatory

<dir>	integer	0	audio path selection

Values:

0 : Send to the speaker

1 : Send to the uplink path

<filename>	string	-	File name, maximum length is 128. It is mandatory to specify the file extension and write file name between double quotes.
<volume>	integer	36	Reproduction volume. The default value is 36 until the volume setting is not changed by means of the #APLAY command. If this parameters is not set by #APLAY command, the reproduction will use default value.

Value:

0÷48	:	One dB per step

Unsolicited field:

Name	Type	Description
<result>	integer	Playing result Values: 0 : File playing done 1 : File playing error

The format of mono audio file is 8k/16k samples/sec and 16 bits/sample.

.wav file formats are the only supported.

AT#APLAY?

Read command reports the currently selected <mode>, <dir>, and <volume> in the format:

#APLAY: <mode>,<dir>,[<volume>]

The <volume> is shown only if its setting has been changed by means of the #APLAY command.

AT#APLAY=?

Test command reports the supported range of values for the parameters <mode>, <dir>, the maximum length of file name, and <volume> in the format:

#APLAY: (0,1),(0,1),128,(0-48)

- Start the playing on the speaker channel with the default value of <volume>.

  The value is 36 until the volume setting is not changed by means of the #APLAY command.

  AT#APLAY = 1,0,"1K_0DB_3S.wav"

  OK

  The playing is successfully ended.

  #APLAYEV: 0

- Start the playing on the speaker channel with <volume> = 44.

  AT#APLAY = 1,0,"1K_0DB_3S.wav",44

  OK

  The playing is successfully ended.

  #APLAYEV: 0

AT#ALIST - List Audio Files

The command lists all audio files stored in file system.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#ALIST

Execution command reports the list of the audio files stored in file system, in the format:

#ALIST: <filename>,<filesize>,<crc><CR><LF>

Additional info:

Intermediate response parameters.

Name	Type	Default	Description
<filename>	string	-	file name
<filesize>	integer	-	file size in bytes
<crc>	hex	-	file CRC16 poly (x^16+x^12+x^5+1) in hex format.

CRC16 is calculated using the standard reversed CRC16-CCITT x^16+x^12+x^5+1 polynomial (0x1021 representation, reversed) with initial value FFFF.

CRC calculation time depends on file size.

AT#ALIST=?

Test command returns the OK result code.

AT#AFIND - Find a Specific Audio File

This command finds a specific audio file.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#AFIND=<fileName>

Execution command finds a specific audio file stored in the file system.
Audio files are .wav files.
Command returns OK result code if the file is present in the file system.

Parameter:

Name	Type	Default	Description
<fileName>	string	-	file name

<fileName> has a maximum of 16 characters (filename extension included).

Quotation marks around <fileName> can also be used.

AT#AFIND=?

Test command returns the OK result code.

```
AT#ALIST
#ALIST: "test.wav",159,498E
OK
AT#AFIND="test.wav"
OK
AT#AFIND=test.wav
OK
```

AT#AFIND="abc.wav"

ERROR

AT#ASIZE - Audio Available Size

This command shows residual space in bytes available to store audio files.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#ASIZE

Execution command shows residual space in bytes available to store audio files, in the format:

#ASIZE: <totalSize>,<usedSize>,<freeSize>

Additional info:

Intermediate response parameters.

Name	Type	Default	Description
<totalSize>	integer	-	size of total NVM file system memory expressed in bytes
<usedSize>	integer	-	size of used NVM file system memory expressed in bytes
<freeSize>	integer	-	size of available free NVM file system memory expressed in bytes

AT#ASIZE=?

Test command returns the OK result code.

AT#ADELF - Delete Audio File

This command deletes a specific audio file.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#ADELF=<filename>

Set command deletes a specific audio file.

Parameter:

Name	Type	Default	Description
<filename>	string	-	File name to be deleted.

filename has a maximum of 32 characters.

AT#ADELF=?

Test command returns the OK result code

AT#ADELA - Delete all Audio Files

This command deletes all audio files stored on the File system.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#ADELA

Set command deletes all audio files stored on the File system.

AT#ADELA=?

Test command returns the OK result code.

HW and Radio Control

AT#TEMPCFG - Temperature Monitor Configuration

Set command sets the temperature zones shown in Note section. Use #TEMPMON command to get the current internal temperature of the module.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#TEMPCFG=<etlz_clr>,<etlz>,<etlz_act_in>,<otlz_clr>,<otlz>,<otlz_act_in >, <otnz_clr>,<otnz>,<otnz_act_in>,<otuz_clr>,<otuz>,<otuz_act_in>, <etuz_clr>,<etuz>,<etuz_act_in>

Parameters:

Name	Type	Default	Description
<etlz_clr>	integer	-273	extreme low zone temperature threshold clear. It has only one valid value: -273 Celsius degrees. See note.
	Value:		
	-273 : Celsius degrees. See note		
<etlz>	integer	-33	extreme low zone temperature threshold. Default value -33 Celsius degrees.
	Value:		
	-40÷105 : Celsius degrees		
<etlz_act_in>	integer	0	extreme low zone action info. Default value 0.

Value:

0÷5	: action

<otlz_clr> integer -35 operate low zone temperature threshold clear. Default value -35 Celsius degrees.

Value:

-40÷105	: Celsius degrees

<otlz> integer -28 operate low zone temperature threshold. Default value -28 Celsius degrees.

Value:

-40÷105	: Celsius degrees

<otlz_act_in> integer 0 operate low zone action info. Default value 0.

Value:

0÷5	: action

<otnz_clr> integer -30 operate normal zone temperature threshold clear. Default value -30 Celsius degrees.

Value:

	-40÷105	:	Celsius degrees

<otnz>	integer	94	operate normal zone temperature threshold. Default value 94 Celsius degrees.

Value:

-40÷105	:	Celsius degrees

<otnz_act_in>	integer	0	operate normal zone action info. Default value 0.

Value:

0÷5	:	action

<otuz_clr>	integer	80	operate up zone temperature threshold clear. Default value 80 Celsius degrees.

Value:

-40÷105	:	Celsius degrees

<otuz>	integer	99	operate up zone temperature threshold. Default value 99 Celsius degrees.

Value:

-40÷105	:	Celsius degrees

<otuz_act_in>	integer	2	operate up zone action info. Default value 2.

Value:

0÷5	: action

<etuz_clr>	integer	82	extreme up zone temperature threshold clear. Default value 82 Celsius degrees.

Value:

-40÷105	: Celsius degrees

<etuz>	integer	105	extreme up zone temperature threshold. It has only one valid value: 105 Celsius degrees. See note.

Value:

105	: Celsius degrees. See note

<etuz_act_in>	integer	3	extreme up zone action info. Default value 3.

Value:

0÷5	: action

<etlz_clr> extreme low zone threshold clear is enforced to have value of '-273'.

Module does not operate in such temperature, but this value is logically set in order to define clearly 'thermal state' to temperatures below -40 Celsius degrees.

- <etuz> extreme up zone threshold is enforced to have value of '105'.

  Module does not operate in such temperature, but this value is logically set to define clearly 'thermal state' to temperatures above 105 Celsius degrees.

- The module is shutdown (<action> 5) at temperature above the Extreme up zone threshold (105 Celsius degrees).

- #TEMPMON set command, changes field "actions" to "mitigate" or "none" to all zones.

- The temperature correctly set are saved in a configuration file in the module file system.

AT#TEMPCFG?

Read command reports the current parameter setting for #TEMPCFG command in the format:

#TEMPCFG:<etlz_clr>,<etlz>,<etlz_act_in>,<otlz_clr>,<otlz>,<otlz_act_in>,<otnz_clr>,<otnz>,<otnz_act_in>,<otuz_clr>,<otuz>,<otuz_act_in>,<etuz_clr>,<etuz>,<etuz_act_in>

AT#TEMPCFG=?

Test command reports the supported range values for parameters <x_clr>, <x>, <x_act_in>, where "x" is substitute for etlz, otlz, otnz, otuz, etuz.

Values are:

#TEMPCFG: (-40-105),(-40-105),(0-5)

Thermal mitigation mechanism

After setting new values, the mitigation algorithm operates with them if a power cycle or a #REBOOT command is executed. Thermal mitigation mechanism works like this:

the whole temperature scale is divided into 5 states (zones).

each measured temperature should belong to a state called the current state.

Each state is defined by the following fields:

threshold	upper temperature boundary of the state. Values are in Celsius degrees.
threshold_clr	lower temperature boundary of the state. Values are in Celsius degrees.
action_in	indicator that indicates if an action should be taken or not in the current state.

action_in can be none or mitigate, their values are shown in the table below.

none	mitigate	mitigation action
0		no mitigation
	1	data throttling (reducing uplink baud rate)
	2	TX back off (reducing MTPL - Max Tx Power Limit)
	3	emergency calls only
	4	RF OFF. RX and TX circuits automatically disabled (using +CFUN=4)
	5	automatic shutdown. Module is powered OFF

There are five limitations on setting temperature and actions, in-order to keep module safety.
- User is prohibited to set an action of "automatic shutdown" to 'operate normal zone'.
- User is prohibited to set an action of "no mitigation" or "data throttling" to "operate up zone".
- User is prohibited to set an action of "no mitigation" or "data throttling" or "tx backoff" to "extreme up zone".
- User is prohibited to set "normal zone" above 97 Celsius degrees.
- If the module enters into a state of "emergency only" calls, registration again to a regular call, happens just when the module returns to "no mitigation" state only.
- A "+CME ERROR: operation not supported" error will be received as a response.

When temperature exceeds the current state threshold, the thermal mitigation algorithm searches the next state when this temperature is lower than threshold. After it finds it, the current state is updated to that state and then it checks whether action is mitigate. If yes, then it activates the mitigation according to the action_in of the current state.

When temperature decreases below threshold_clr then it does the same algorithm as above, but in the opposite direction. It searches the next state

when this temperature is greater than threshold_clr, updates the current state to that state, and activates mitigation as described above.

In the state definitions there are the following two rules:

rule 1: overlap between 2 adjacent states of at least 2 Celsius degrees, i.e.:

thr of state(x) - thr_clr of state(x+1) >= 2

rule 2: every state shall have free temperature range which has no part in any overlap range. This range should be at least 2 Celsius degrees, i.e.:

thre_clr state(x+2) - thre state(x) >= 2

Rule 1 comes to ensure hysteresis in the transition between two states.

Rule 2 comes to ensure a minimum range for a stable state.

Referring to the figure below:

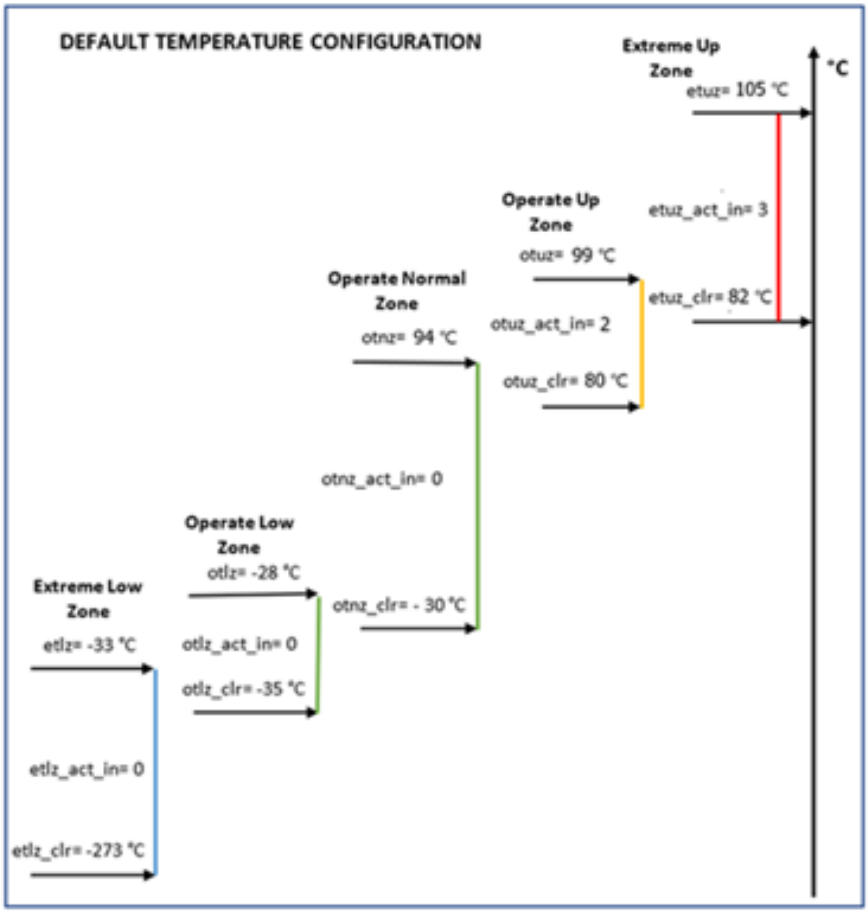
State 0 is Extreme low zone

State 1 is Operate low zone

State 2 is Operate normal zone

State 3 is Operate up zone

State 4 is Extreme up zone



After moving to zone with activity 3 (emergency call only), only when moving to zone with activity 0 (no mitigation) the device will register to the network.

<action> for high-zone cannot be 'no action' or 'data throttling'.

<action> for extreme high zone cannot be 'no action', 'data throttling' or 'tx backoff'.

- AT#TEMPCFG= -273,-33,3,-35,-28,2,-30,80,0,78,90,3,88,528,3
  OK

  Rules 1 and 2 are respected.

zone	thr_clr	thr	action_in
Extreme low zone	-273	-33	3: emergency call only
Operate low zone	-35	-28	2: TX back off
Operate normal zone	-30	80	0: no mitigation
Operate up zone	78	90	3: emergency call only
Extreme up zone	88	105	3: emergency call only

- AT#TEMPCFG=-273,-33,3,-35,-28,2,-30,80,0,79,90,3,88,105,3
  +CME ERROR: operation not supported

  Rule 1 is not respected: hysteresis less than 2 Celsius degrees

  (thr of Operate normal zone) - (thr_clr of Operate up zone) = 1 < 2

zone	thr_clr	thr	action_in
Extreme low zone	-273	-33	3: emergency call only
Operate low zone	-35	-28	2: TX back off

Operate normal zone	-30	80	0: no mitigation
Operate up zone	79	90	3: emergency call only
Extreme up zone	88	105	3: emergency call only

- AT#TEMPCFG=-273,-33,3,-35,-28,2,-30,80,0,78,90,3,81,105,3

  +CME ERROR: operation not supported

  Rule 2 is not respected: free temperature range less than 2 Celsius degrees

  (thr_clr of Extreme up zone) - (thr of Operate normal zone) = 1 < 2

zone	thr_clr	thr	action_in
Extreme low zone	-273	-33	3: emergency call only
Operate low zone	-35	-28	2: TX back off
Operate normal zone	-30	80	0: no mitigation
Operate up zone	78	90	3: emergency call only
Extreme up zone	81	105	3: emergency call only

AT#GPIO - General Purpose Input/Output Pin Control

Set the value of the general-purpose input/output GPIO pins.

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT#GPIO=[<pin>,<mode>[,<dir>[,<save>]]]

Execution command sets the value of the general purpose GPIO pin.

Parameters:

Name	Type	Default	Description
<pin>	integer	N/A	GPIO pin number. The supported range goes from 1 to Max value that is hardware dependent. Use AT#GPIO=? test command to know Max value. Value: 1÷Max : GPIO pin identifier
<mode>	integer	N/A	sets GPIO pin configuration, its action depends on <dir> value. Refer to Additional info sections. Value: 0÷4 : mode identifier
<dir>	integer	N/A	sets the GPIO pin in input, output, or alternate functions. If <dir> = 2, 3, 4, 5, 6, 7, 17, 18

pin direction is Alternate Function ALT1, ALT2, ALT3, ALT4, ALT5, ALT6, ALT16, ALT17 respectively this direction.

ALT6 cannot be set via #GPIO

Values:

0	:	pin set as input
1	:	pin set as output
2÷14	:	pin set in alternate functions (some values not supported)
16÷18	:	pin set in alternate functions (some values not supported)

<save>	integer	0	GPIO pin save configuration. If <save> is omitted, the configuration is stored in NVM only if user set or reset ALTx function on <dir> parameter.

Values:

0	:	GPIO pin configuration is not saved
1	:	GPIO pin configuration is saved

Additional info:

▶▶ meaning of <mode>

0 - remove any Pull-up/Pull-down if <dir>=0 - INPUT
- output pin cleared to 0 (Low) if <dir>=1 - OUTPUT

1 - remove any Pull-up/Pull-down if <dir>=0 - INPUT
- output pin set to 1 (High) if <dir>=1 - OUTPUT

2 - Reports the read value from the input pin if <dir>=0 - INPUT
- Reports the read value from the input pin if <dir>=1 - OUTPUT
- Reports a no meaning value if ALTERNATE FUNCTION

3 - enable Pull-Up if <dir>=0 - INPUT

4 - enable Pull-Down if <dir>=0 - INPUT

▶▶ While GPIO<pin> in used by ALT function, it cannot be set as GPIO output or any other ALT function before the original ALT released or change it to GPIO input (default state).

Name	Type	Default	Description
<dir>	integer	-	current direction setting for the GPIO<pin>
<stat>	integer	-	the parameter can be: logic value read from pin GPIO<pin> in the case the pin <dir> is set to input; logic value present in output of the pin GPIO<pin> in the case the pin <dir> is currently set to output; No meaning value for the pin GPIO<pin> in the case the pin <dir> is set to alternate function or Tristate pull down.

▶▶ ALT6 is available for all GPIOs and this value is only possible to set by #FASTSHDN

"ALT1" value is valid for GPIO1 as "SLED"

"ALT2" value is valid for all GPIOs: alternate function is "Alarm Pin"

"ALT3" value is valid for all GPIOs as "TempMon Pin"

"ALT4" value is valid for all GPIOs as "AD_Det Pin" (not supported)

"ALT5" value is valid for all GPIOs as "AD_Rep Pin" (not supported)

"ALT6" value is valid for all GPIOs as "FASTSHDN"

"ALT16" value is valid for all GPIOs as "Antenna Control 0" (not supported)

"ALT17" value is valid for all GPIOs as "Antenna Control 1" (not supported)

Not all configurations for the three parameters are valid.

While using the pins in the alternate function, the GPIO read/write access to that pin is not accessible and must be avoided.

SLED will work depending on HW version.

To activate SLED by default, #SLED command should be set to <mode>=2 after setting ALT1 as follows. (See #SLED,#SLEDSAV command)

```
AT#GPIO=1,0,2 OK
AT#SLED=2
OK
AT#SLEDSAV
OK
```

AT#GPIO?

Read command reports the read direction and value of all GPIO pins, in the format:

#GPIO: <dir>, <stat>[<CR><LF>#GPIO: <dir>, <stat> [...]]

where:

<dir> - as seen before

<stat> - as seen before

If <mode> = 3,4 the output format is

#GPIO: <dir>, <stat>, <mode>[<CR><LF>

#GPIO: <dir>, <stat>, <mode> [...]]

AT#GPIO=?

Test command reports the supported range of values of the command parameters <pin>, <mode>, <dir>, and <save>.

AT#GPIO=?

#GPIO: (1-10),(0-4),(0-14,16-18),(0-1)

OK

AT#GPIO=3,1,1 // setting GPIO3 as output and value is HIGH

OK

AT#GPIO=4,1,1,1 // setting GPIO4 as output and value is HIGH and GPIO pin save configuration

OK

AT#GPIO=3,2 // report GPIO3 state

#GPIO: 1,1

OK

AT#GPIO? // read command

#GPIO: 1,1 // GPIO1 is output and output value is HIGH

#GPIO: 0,0

#GPIO: 1,1

#GPIO: 0,0

#GPIO: 0,1 // GPIO5 is input and input value is HIGH

#GPIO: 1,0

OK

AT#SLED - STAT_LED GPIO Setting

The command configures the behavior of the STAT_LED GPIO.

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT#SLED=<mode>[,<onDuration>[,<offDuration>]]

Set command sets the behaviour of the STAT_LED GPIO.

Parameters:

Name	Type	Default	Description
<mode>	integer	2	defines the STAT_LED GPIO behavior.

Values:

0 : GPIO tied low

1 : GPIO tied high

2 : GPIO is handled with specific timings. See Additional info section

3 : GPIO is turned ON/OFF alternatively, with period defined by the sum <onDuration> + <offDuration>

4 : GPIO is handled with specific timings. See Additional info section

5 : status led disabled

<onDuration>	integer	10	duration of period in which STAT_LED GPIO is tied high while <mode>=3

Value:

1÷100	:	in tenth of seconds

<offDuration>	integer	10	duration of period in which STAT_LED GPIO is tied low while <mode>=3

Value:

1÷100	:	in tenth of seconds

Additional info:

- <mode>=2, the timings of STAT_LED GPIO are:

  not registered: always ON

  registered in idle: blinking 1 s ON and 2 s OFF

  registered in idle with power saving: blinking time depends on network condition to minimize power consumption

- <mode>=4, the timings of STAT_LED GPIO are:

  not registered: blinking 0,5 s ON and 0,5 s OFF

  registered in idle: blinking 300 ms ON and 2,7 s OFF

  registered in idle with power saving: blinking time depends on network condition to minimize power consumption

When module boot the STAT_LED GPIO always tied High and holds this value until the first NVM reading.

SLED function of a GPIO corresponds to ALT1 function of the GPIO, So ALT1 function should be set through AT#GPIO command.

AT#SLED?

Read command returns the STAT_LED GPIO current setting, in the format:

#SLED: <mode>,<onDuration>,<offDuration>

AT#SLED=?

Test command returns the range of available values for parameters <mode>, <onDuration> and <offDuration>.

The setting is saved using the #SLEDSAV command.

AT#SLEDSAV - Save STAT_LED GPIO Setting

This command allows to save the current STAT_LED GPIO setting.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SLEDSAV

Execution command saves the STAT_LED GPIO setting in NVM.

AT#SLEDSAV=?

Test command returns OK result code.

AT#ADC - Read Analog/Digital Converter Input

This command returns the current voltage value of the specified ADC inputs, expressed in mV.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#ADC=[<adc>,<mode>[,<dir>]]

Execution command reads selected <adc> pin voltage, converted by ADC, and prints out the result as shown in Additional info section.

Parameters:

Name	Type	Default	Description
<adc>	integer	N/A	index of input pin
	Values:		
	1 : available		
	2 : available		
<mode>	integer	2	required action
	Value:		
	2 : query ADC value		

<dir>	integer	0	direction. Its interpretation is currently not implemented.

Value:

0	:	no effect

Additional info:

Format of the message printed out by the execution command:

#ADC:<value>

Name	Type	Default	Description
<adc>	integer	-	pin <adc> voltage expressed in mV.

The command returns the last valid measure.

AT#ADC?

Read command reports all pins voltage, converted by ADC, in the format:

#ADC: <value>[<CR><LF>#ADC: <value>[...]]

AT#ADC=?

Test command reports the supported range of values of the command parameters <adc>, <mode> and <dir>.

AT#V24CFG - V24 Output Pins Configuration

This command sets the AT commands serial port interface output pins mode.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#V24CFG=<pin>,<mode>[,<save>]

Set command sets the AT commands serial port interface output pins mode.

Parameters:

Name	Type	Default	Description
<pin>	integer	N/A	AT commands serial port interface hardware pin

Values:

0 : DCD (Data Carrier Detect)

1 : Not supported. CTS (Clear To Send)

2 : RI (Ring Indicator)

3 : DSR (Data Set Ready)

4 : Not supported. DTR (Data Terminal Ready). This is not an output pin, so its state cannot be set through the AT#V24 command

5 : Not supported. RTS (Request To Send). This is not an output pin, so its state cannot be set through the AT#V24 command

6 : Not supported. RXD (Receive Data). This is not an output pin, so its state cannot be set through the AT#V24 command

7 : Not supported. TXD (Transmit Data)

<mode>	integer	0	AT commands serial port interface hardware pins mode

Values:

0 : AT commands serial port mode: the V24 pins are controlled by the serial port device driver

1 : GPIO mode: the V24 output pins can be managed through the AT#V24 command

<save>	integer	1	Save V24 pin configuration

Values:

0 : Not supported. Pin configuration is not saved

1 : Pin configuration is saved

If optional <save> parameter is omitted, the pin configuration will be stored in NVM (default <save> parameter is 1).
If optional <save> parameter is present, it must be 1 (only supported <save> parameter is 1) and the pin configuration will be stored in NVM.

AT#V24CFG pin configuration takes effect only after reboot.

Changing V24 pins configuration may affect the cellular module functionality set through +CFUN.

AT#V24CFG?

Read command returns the current configuration for all the pins (both output and input) in the format:

#V24CFG: <pin1>,<mode1>[<CR><LF><CR><LF>
#V24CFG: <pin2>,<mode2>[...]]

AT#V24CFG=?

Test command reports supported range of values for parameters <pin>,<mode> and <save>.

AT#V24 - V24 Output Pins Control

This command sets the state of the output pins of the AT commands serial port interface.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#V24=<pin>[,<state>]

Set command sets the AT commands serial port interface output pins state, when pins are in GPIO mode (see #V24CFG).

Parameters:

Name	Type	Default	Description
<pin>	integer	N/A	AT commands serial port interface hardware pin: Values: 0 : DCD (Data Carrier Detect) 1 : Not supported. CTS (Clear To Send) 2 : RI (Ring Indicator) 3 : DSR (Data Set Ready) 4 : Not supported. DTR (Data Terminal Ready). This is not an output pin: trying to set its state raises the result code "ERROR" 5 : Not supported. RTS (Request To Send). This is not an output pin: trying to set its state raises the result code "ERROR"

6 : Not supported. RXD (Receive Data). This is not an output pin: trying to set its state raises the result code "ERROR"

7 : Not supported. TXD (Transmit Data)

<state>	integer	N/A	State of AT commands serial port interface output hardware pins (0, 2, 3) when pins are in GPIO mode (see #V24CFG):

Values:

0 : Low state

1 : High state

If <state> is omitted the command returns the actual state of the pin <pin>.

AT#V24?

Read command returns actual state for all the available pins in the format:

#V24: <pin1>,<state1>[<CR><LF>
#V24: <pin2>,<state2>[...]]

AT#V24=?

Test command returns the supported values of parameters <pin> and <state>.

AT#I2CWR - Write to I2C

This command is used to send data to an I2C peripheral connected to module.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#I2CWR=<sdaPin>,<sclPin>,<deviceId>,<registerId>,<len>

Execution command sends data to an I2C peripheral connected to module GPIOs. After the writing activity has been accomplished, the GPIOs will not be restored to the original setting. Use #GPIO command to see the status of the used GPIOs. To have information on GPIO pins refer to document [1].

Parameters:

Name	Type	Default	Description
<sdaPin>	integer	-	GPIO number for SDA. To know the range use #I2CWR test command.
<sclPin>	integer	-	GPIO number for SCL. To know the range use #I2CWR test command.
<deviceId>	hex	N/A	address of the I2C device (7 bits). The Least Significant Bit is used for read/ write command, but in this #I2CWR implementation, it does not matter if the LSB is set to 0 or 1. Address must be written in hexadecimal form without 0x. 10 bit address is also supported.

Value:

	0÷3FF	:	addressing range extended to 10 bit

<registerId>	hex	N/A	register to write data to

Value:

0÷FF	:	value must be written in hexadecimal form without 0x

<len>	integer	N/A	number of data to send

Value:

1÷254	:	number of data to send

Additional info:

After entering the command, the module returns the prompt ">" and waits for the data to send. To complete the operation, send Ctrl-Z char (0x1A hex); to exit without writing the message send ESC char (0x1B hex). Data must be written in hexadecimal form.

If data are successfully sent, the response is OK, otherwise an error code is reported.

AT#I2CWR=?

Test command returns the range of available values for parameters <sdaPin>, <sclPin>, <deviceId>, <registerId>, <len>.

Set GPIO_2 as SDA, and GPIO_3 as SCL. Device I2C address is 0x20; 0x10 is the address of the first register where to write I2C data; 14 data bytes will be written starting from register 0x10.

```
AT#I2CWR=2,3,20,10,14
> 00112233445566778899AABBCCDD<ctrl-z>
OK
```

AT#I2CRD - Read from I2C

This command is used to read data from an I2C peripheral connected to module.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#I2CRD=<sdaPin>,<sclPin>,<deviceId>,<registerId>,<len>

Execution command reads data from an I2C peripheral connected to module GPIOs. After the reading activity has been accomplished, the GPIOs will not be restored to the original setting. Use #GPIO command to see the status of the used GPIOs. To have information on GPIO pins refer to document [1].

Parameters:

Name	Type	Default	Description
<sdaPin>	integer	-	GPIO number for SDA. To know the range use #I2CRD test command.
<sclPin>	integer	-	GPIO number for SCL. To know the range use #I2CRD test command.
<deviceId>	hex	N/A	address of the I2C device (7 bits). The Least Significant Bit is used for read/ write command, but in this #I2CRD implementation, it does not matter if the LSB is set to 0 or 1. Address must be written in hexadecimal form without 0x. 10 bit address is also supported

Value:

	0÷3FF : addressing range extended to 10 bit		
<registerId>	hex	N/A	Register to read data from Value: 0÷FE : value must be written in hexadecimal form without 0x
<len>	integer	N/A	Number of data to receive - Data Read from I2C will be dumped in hexadecimal format - If data requested are more than data available in the device, dummy data (normally 0x00 or 0xff) will be dumped Value: 1÷254 : number of data to receive

AT#I2CRD=?

Test command returns the range of available values for parameters <sdaPin>, <sclPin>, <deviceId>, <registerId>, <len>.

Read 12 bytes from I2C device with address 0x20, starting from register address 0x10. SDA is mapped on GPIO_02, SCL is mapped on GPIO_03.

```
AT#I2CRD=2,3,20,10,12
#I2CRD: 00112233445566778899AABBCC
OK
```

AT#I2CCF - Combined Format for I2C Writing and Reading

This command is used to write and read data to/from an I2C device using the I2C Combined Format. The module acts as an I2C master.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#I2CCF=<sdaPin>,<sclPin>,<deviceId>,<lenwr>,<lenrd>

The module, as master, transmits data to the slave and then, reads data from it through two GPIOs. Transfer direction is changed when writing section is ended. After the write/read activity has been accomplished, the GPIOs will not be restored to the original setting. Use #GPIO command to see the status of the used GPIOs. To have information on GPIO pins refer to document [1].

Parameters:

Name	Type	Default	Description
<sdaPin>	integer	-	GPIO number for SDA. To know the range use #I2CCF test command.
<sclPin>	integer	-	GPIO number for SCL. To know the range use #I2CCF test command.
<deviceId>	hex	N/A	address of the I2C device (7 bits). The Least Significant Bit is used for read/write command, but in this #I2CCF implementation, it does not matter if the LSB is set to 0 or to 1. Address must be written in hexadecimal form without 0x. 10 bit address is also supported.

Value:

0÷3FF	: addressing range extended to 10 bit

<lenwr>	integer	N/A	number of data to write.

Value:

0÷254	: number of data to write.

<lenrd>	integer	N/A	number of data to read.

Value:

0÷254	: number of data to read.

Additional info:

▶▶ After entering the command, and if <lenwr> > 0, the module returns the prompt ">" and waits for the data to send. To complete the operation enter Ctrl-Z char (0x1A hex); to exit without writing the message enter ESC char (0x1B hex).

Data must be written in hexadecimal form without 0x.

If data are successfully sent, the response is OK, otherwise an error code is reported.

AT#I2CCF=?

Test command returns the range of available values for parameters <sdaPin>, <sclPin>, <deviceId>, <lenwr>, <lenrd>.

- Set GPIO_2 as SDA, GPIO_3 as SCL; Device I2C address is 0x20; First is send data 0x0a; after a "RESTART", 4 data bytes are read.

  AT#I2CCF=2,3,20,1,4
  >0a<ctrl-z>
  #I2CCF: abcdef12

  OK

  The sequence is the following:
  START - 0x20- 0x0a -RESTART - 0X21 - data read 1 -...- data read 4 - STOP

- Set GPIO_2 as SDA, GPIO_3 as SCL; Device I2C address is 0x20; read data:

  AT#I2CCF=2,3,20,0,2
  #I2CCF: abcd

  OK

  The sequence is the following:
  START - 0x21- - data read 1 - data read 2 - STOP

AT#TEMPMON - Temperature Monitor

This command is used to retrieve internal temperature information of the module.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#TEMPMON=<mod>[,<urcMode>[,<action>[,<GPIO>]]]

Set command sets the behavior of the module internal temperature monitor.

#TEMPMEAS: <level>,<value>

The temperature monitoring message parameters are described in the Unsolicited fields section.

Parameters:

Name	Type	Default	Description
<mod>	integer	N/A	select the temperature monitoring mode. Values: 0 : sets the command parameters 1 : triggers the measurement of the module internal temperature, reporting the result using the format shown above
<urcMode>	integer	0	URC presentation mode Values: 0 : disables the presentation of the temperature monitoring URC.

1 : enables the presentation of the temperature monitoring URC, whenever the module internal temperature reaches either operating or extreme levels.

<action>	integer	0	sum of integers, each representing the action to be done whenever the module internal temperature reaches either operating or extreme levels.

Values:

0 : no action

1 : (01) Activating of thermal mitigation according to thermal configuration file

2 : (10) Output pin <GPIO> is tied HIGH when operating temperature bounds are reached; when the temperature is back to normal the output pin <GPIO> is tied LOW. If this <action> is required, it is mandatory to set the <GPIO> parameter too

3 : (11) This value contains <action=1> and <action=2> i.e. activate thermal mitigation and a GPIO indication. If this <action> is required, it is mandatory to set the <GPIO> parameter too

<GPIO>	integer	-	GPIO number. Valid range is any GPIO pin as described in #GPIO command. This parameter is needed and required only if <action> 2 or 3 is enabled.

Unsolicited fields:

Name	Type	Description
<level>	integer	threshold level (see Notes section) Values: -2 : Extreme temperature lower bound -1 : Operating temperature lower bound 0 : normal temperature 1 : Operating temperature upper bound 2 : Extreme temperature upper bound
<value>	integer	current temperature expressed in Celsius degrees.

AT#TEMPMON?

Read command reports the current parameter settings for the command in the format:

#TEMPMON: <urcMode>,<action>[,<GPIO>]

AT#TEMPMON=?

Test command reports the supported range of values for parameters <mod>, <urcMode>, <action> and <GPIO>.

- Thresholds levels are defined in #TEMPCFG command. See there for detailed description on thermal mitigation configuration.
- Last <GPIO> is saved in the NVM.

Thermal mitigation is disabled automatically when using laboratory test SIM.

AT#TESTMODE - Test Mode Configuration

Set module in Test Mode for configuring and testing the POWER level in 4G technology (not signaling mode).

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#TESTMODE=<band>,<apc>

The Test Mode is entered using the following command sequence:

AT+CFUN=0

AT#TESTMODE=<band>,<apc>

Now it is possible to perform tests on instrument.

To exit from Test Mode and go back to Operative Mode perform a reboot.

Parameters:

Name	Type	Default	Description
<band>	integer	N/A	Integer that identifies the 4G band. See Additional info section for list of 4G bands.
	Value:		
	1÷41	:	4G band number
<apc>	integer	N/A	Integer that identifies the power.
	Value:		

1550÷1900	:	Power levels. See Additional info section for list

Additional info:

▶▶ 4G commands: <band> parameter

Each <band> value corresponds to a center-band UL channel according to the following table:

EARFCN UL	Band
18300	1
18900	2
19500	3
20175	4
20525	5
20700	6
21100	7
21625	8
23790	17
24300	20
27435	28
36275	34

38000	38
39150	40
40620	41

4G commands: <apc> parameter

Changes the 4G transmission power.

<apc> has the range 1550 - 1900, see table below:

<apc>	dBm
1500 ..	-1.7
1550 ..	1.3
1600 ..	4.4
1650 ..	7.4
1700 ..	10.6
1750 ..	13.1
1800 ..	16.2
1850 ..	19.2
1900 ..	22.5

In Test Mode the other AT commands do not work.

AT#TESTMODE is intended for RF functional verification or connection testing.

The reported output power level (dBm) is not calibrated and therefore should not be used as a reference for accurate power measurement.

AT#TESTMODE?

Read command reports the current Test Mode Status.

AT#TESTMODE=?

Test command returns the OK result code

This AT command must be used only in a controlled environment for prototype or production test purpose with RF pin connected to a 50 Ohm termination in order to avoid RF radiation in the field.

During this usage a large amount of heat may be generated during the test, we suggest user do not use this command for long time.

We decline any responsibility related to malicious usage of this AT command.

Set mode 0

AT+CFUN=0

OK

Set Test Mode parameters

AT#TESTMODE:1, 1700

AT#TESTMODE2G - Test Mode 2G Configuration

Set module in Test Mode for configuring and testing the POWER level in 2G technology (not signaling mode).

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#TESTMODE2G=<op>[,<band>,<ch>,<afcDAC>[,<txRampScale>,<rxGainCode>]]

The Test Mode is entered using the following command sequence:

AT+CFUN=0

AT#TESTMODE2G=<op>[,<band>,<ch>,<afcDAC>[,<txRampScale>,<rxGainCode>]]

Now it is possible to perform tests on instrument.

Possible answers from the module, depending on the chosen operation, are:

#TESTMODE2G:<rssiDbmValue>

OK

#TESTMODE2G:0

OK

To exit from Test Mode and go back to Operative Mode perform a reboot.

Parameters:

Name	Type	Default	Description
<op>	integer	N/A	Integer that identifies the type of test. See Additional info section for the list of available operations.
	Value:		

	0÷3 : Operation to be performed in test mode		
<band>	integer	N/A	Integer that identifies the 2G band. See Additional info section for the list of 2G bands. Value: 2÷64 : 2G band number
<ch>	integer	N/A	Integer that identifies the 2G channel. See Additional info section for the list of 2G channel. Value: 0÷1023 : 2G channel number
<afcDAC>	integer	N/A	Integer that identifies the AFC DAC value. Value: 5000÷32000 : AFC DAC value
<txRampScale>	integer	N/A	Integer that identifies the Tx ramp scale value. Value:

	0÷1200	:	Tx ramp scale

<rxGainCode>	integer	N/A	Integer for the Rx ramp scale value. Only valid for Rx mode

Value:

0÷49	:	Rx ramp scale

Additional info:

▶▶ Output parameter:

Name	Type	Default	Description
<rssiDbmValue>	integer	N/A	received signal strength indication in dBm

Value:

110÷20	:	received signal strength indication

▶▶ 2G commands: <op> parameter.

Defines the type of operation to be performed in test mode.

Admitted <op> values are defined in table below:

0	start GSM Tx

1	start GSM Rx
3	stop test

- 2G commands: <band> parameter.

  Change the 2G transmission band.

  <band> has the range 2 - 64, see table below:

Band	Name
2	DCS 1800
4	PCS 1900
8	E-GSM 900
64	GSM 850

- 2G commands: <ch> parameter.

  Change the 2G transmission channel.

  <ch> has the range 0 - 1023, see table below:

2G channels	Band
512 .. 885	DCS 1800 (2)
512 .. 810	PCS 1900 (4)
975 .. 1023	E-GSM 900 (8)
0 .. 124	E-GSM 900 (8)
128 .. 251	GSM 850 (64)

In Test Mode the other AT commands do not work.

Before changing operation send the Stop Test Mode command, otherwise the module returns ERROR.

AT#TESTMODE2G is intended for RF functional verification or connection testing.

The reported output power level (dBm) is not calibrated and therefore should not be used as a reference for accurate power measurement.

AT#TESTMODE2G=?

Test command returns the expected syntax:

#TESTMODE2G=<op>[,<band>,<arfcn>,<afcDac>[,<txRampScale>,<rxGainCode>]]

This AT command must be used only in a controlled environment for prototype or production test purpose with RF pin connected to a 50 Ohm termination in order to avoid RF radiation in the field.

During this usage a large amount of heat may be generated during the test, we suggest user do not use this command for long time.

We decline any responsibility related to malicious usage of this AT command.

Set mode 0

AT+CFUN=0

OK

Set GSM Rx: band 900 ch 62

AT#TESTMODE2G=1,8,62,6000,700,30

expected answer (with received power):

#TESTMODE2G:-85

OK

Set GSM Tx: band 1800 ch 690

AT#TESTMODE2G=0,2,690,6000,700,20

expected answer:

#TESTMODE2G:0

OK

Stop Test Mode

AT#TESTMODE2G=3

expected answer:

#TESTMODE2G:0

OK

Power Down

AT#REBOOT - Module Reboot

Immediate module reboot.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#REBOOT

Execution command reboots immediately the unit.
It can be used to reboot the system after a remote update of the script in order to have the new one running.

If #REBOOT follows an AT command that stores some parameters in NVM, it is recommended to insert a delay of at least 5 seconds before to issue #REBOOT, to permit the complete NVM storing.

#REBOOT is an obsolete AT command; please refer to #ENHRST to perform a module reboot.

AT#REBOOT=?

Test command returns OK result code.

- Reboot the module
  AT#REBOOT
  OK
  (the module reboots)

AT#ENHRST - Periodic Reset

Enable or Disable the one shot or periodic unit reset

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#ENHRST=<mode>,<delay>

Set commands enables/disables the unit reset after the specified <delay> in minutes

Parameters:

Name	Type	Default	Description
<mode>	integer	0	Enable\Disable mode

Values:

0 : disables the unit reset

1 : enables the unit reset only one time (one shot reset)

2 : enables periodically unit reset

<delay>	integer	-	time interval in minutes after that the unit reboots

The settings are saved automatically in NVM only if old or new <mode> value is 2, i.e. unit set in periodic reset mode. Therefore, any change from 0 to 1 or conversely is not stored in NVM.

The command AT#ENHRST=1,0 causes the immediate module reboot. If it follows an AT command that stores some parameters in NVM, it is strongly recommended to insert a delay of at least 5 sec

before issuing it, to permit the complete NVM storing process.

AT#ENHRST?

Read command reports the current parameter settings in the following format:

#EHNRST: <mode>[,<delay>,<remainingTime>]

Additional info:

Read command parameter for <mode> =1 or 2.

Name	Type	Default	Description
<remainingTime>	integer	-	time in minutes remaining before next reset

AT#ENHRST=?

Test command reports supported range of values for parameters <mode> and <delay>.

Example of #ENHRST usage and expected unit behavior.

- AT#ENHRST=1,60
  ...
  Module reboots after 60 minutes
- AT#ENHRST=1,0
  Module reboots immediately
- AT#ENHRST=2,40
  ...

Module reboots after 40 minutes, and after every following power on, it will continue to reboot always after 40 minutes.

AT#SHDN - Software Shutdown

This command turns the module OFF.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SHDN

Execution command causes device to detach from the network and shut down. Before definitive shut down an OK response is returned.

When issuing the command any previous activity terminated and the device will not respond to any further command.

To turn it on again hardware pin ON/OFF must be tied low.

The maximum time to shut down the device, completely is 25 seconds.

AT#SHDN=?

Test command returns OK result code.

AT#FASTSHDN - Fast Shutdown Configuration

This command can be used as a set command to configure a GPIO pin performing a fast shutdown when on it is forced a High to Low level transition. Or can be used as an execute command to force immediately a fast shutdown.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#FASTSHDN[=<enable>,[<GPIO>]]

Parameters:

Name	Type	Default	Description
<enable>	integer	0	enable/disable the fast shutdown execution via a GPIO pin. Values: 0 : disabled 1 : enabled
<GPIO>	integer	-	selects the GPIO to execute the fast shutdown. When the selected <GPIO> pin goes from High to Low level and the <enable> is set to 1, the module execute immediately the fast shutdown.

Additional info:

- The execution command #FASTSHDN<CR><LF> forces the module to execute immediately the fast shutdown.

The GPIO pin selected by the #FASTSHDN command must be used for this purpose only. If you need to use the selected GPIO pin for different activities, it must be free with the following command:

#FASTSHDN=0,<GPIO>

AT#FASTSHDN?

Read command reports the currently selected configuration in the format:

AT#FASTSHDN: <enable>,<GPIO>

AT#FASTSHDN=?

Test command returns the supported range of values for the parameters <enable> and <GPIO>.

Enable fast shutdown on GPIO_05
AT#FASTSHDN=1,5
OK

AT#FASTSHDN?
#FASTSHDN: 1,5
OK

Force immediate fast shutdown
AT#FASTSHDN
OK

Easy Scan

AT#CSURV - Network Survey

This command performs a network survey through band channels.

SIM Presence	Setting saved	Can be aborted
Required	No	Yes

AT#CSURV[= [<s>,<e>]]

Execution command performs a quick survey on channels belonging to the band selected by last #BND command issue, starting from channel <s> to channel <e>.

Network registration is required.

Issuing AT#CSURV<CR>, a full band scan is performed. The command responds with the following string:

Network survey started...

After a while, a list of network survey information text lines, one for each received carrier, is reported.

The #CSURV output ends in two ways, depending on the last #CSURVF setting.

If AT#CSURVF=0 or AT#CSURVF=1 the output will end with the string:

Network survey ended

If AT#CSURVF=2 the output will end with the string:

Network survey ended (Carrier: <NoARFCN> BCCh: <NoBCCh>)

The network survey information text lines parameters are described in the Additional info sections.

Parameters:

Name	Type	Default	Description
<s>	integer	-	starting channel, see note section

<e>	integer	-	ending channel, see note section

Additional info:

▶▶ 2G Networks for BCCH-Carrier

Network survey information text lines:
arfcn: <arfcn> bsic: <bsic> rxLev: <rxLev> ber: <ber> mcc: <mcc> mnc: <mnc> lac: <lac> cellId: <cellId> cellStatus: <cellStat>

<CR><LF><CR><LF><CR><LF>

Name	Type	Default	Description
<arfcn>	integer	-	C0 carrier assigned radio channel (BCCH - Broadcast Control Channel).
<bsic>	integer	-	base station identification code; if #CSURVF last setting is 0, <bsic> is a decimal number, else it is a 2-digits octal number.
<rxLev>	integer	-	decimal number; it is the reception level (in dBm).
<ber>	integer	-	always 0, dummy parameter.
<mcc>	hex	-	hexadecimal 3-digits number; it is the mobile country code.

<mnc>	hex	-	hexadecimal 2 or 3 digits number; it is the mobile network code.
<lac>	mixed	-	location area code; if #CSURVF last setting is 0, <lac> is a decimal number, else it is a 4-digits hexadecimal number.
<cellId>	mixed	-	cell identifier; if #CSURVF last setting is 0, <cellId> is a decimal number, else it is a 4-digits hexadecimal number.
<cellStat>	string	N/A	cell status

Values:

CELL_SUITABLE	:	C0 is a suitable cell.
CELL_LOW_PRIORITY	:	the cell is low priority based on the received system information.
CELL_FORBIDDEN	:	the cell is forbidden.
CELL_BARRED	:	the cell is barred based on the received system information.

CELL_LOW_LEVEL	:	the cell <rxLev> is low.
CELL_OTHER	:	none of the above (e.g. exclusion timer running, no BCCH available, etc.).

4G Networks

Network survey information text lines for LTE:

earfcn: <earfcn> rxLev: <rxLev> mcc: <mcc> mnc: <mnc> phycellId: <cellId> tac: <tac> cellIdentity: <cellIdentity> rsrp: <rsrp> rsrq: <rsrq>

Name	Type	Default	Description
<earfcn>	integer	-	E-UTRA Assigned Radio Channel
<tac>	integer	-	Tracking Area Code. if #CSURVF last setting is 0, <tac> is a decimal number, else it is a 4-digits hexadecimal number
<cellId>	integer	-	Physical cell identifier; if #CSURVF last setting is 0, <cellId> is a decimal number, else it is a hexadecimal number

<cellIdentity>	integer	-	Cell identifier; if #CSURVF last setting is 0, <cellIdentity> is a decimal number, else it is a hexadecimal number
<rsrp>	integer	-	Reference Signal Received Power. If value cannot be retrieved then value is reported as 0
<rsrq>	integer	-	Reference Signal Received Quality. If value cannot be retrieved then value is reported as 0

▶▶ #CSURV end output parameters if AT#CSURVF=2:

Network survey ended (Carrier: <NoARFCN> BCCh: <NoBCCh>)

Name	Type	Default	Description
<NoARFCN>	integer	-	number of scanned frequencies
<NoBCCH>	integer	-	number of found BCCH

2G Network

<s> and <e> parameters can assume the values shown in the Channels Ranges column, according to the module in use.

Channels Ranges	GSM Bands
0,124	GSM900
975,1023	GSM900
512,885	DCS1800
128,251	GSM850
512,810	PCS1900
0,1023	all supported GSM bands

4G Network

<s> and <e> parameters can assume the values shown in the Channels Ranges column, according to the module in use.

Channels Ranges	LTE Bands
0,599	1
600,1199	2
1200,1949	3
1950,2399	4
2400,2649	5
2750,3449	7
3450,3799	8
4750,4949	11
5010,5179	12
5180,5279	13
5730,5849	17
5850,5999	18
6000,6149	19
6150,6449	20
6450,6599	21

8690,9039	26
9210,9659	28
37750,38249	38
38650,39649	40
39650,41589	41
0,65534	all supported LTE bands

2G Network, for BCCH-Carrier

AT#CSURV

Network survey started ...

arfcn: 1018 bsic: 21 rxLev: -71 ber: 0 mcc: 222 mnc: 01 lac: 54717 cellId: 14887 cellStatus: CELL_SUITABLE numArfcn: 0

arfcn:

arfcn: 1009 bsic: 10 rxLev: -85 ber: 0 mcc: 222 mnc: 01 lac: 54717 cellId: 21093 cellStatus: CELL_SUITABLE numArfcn: 0

arfcn:

...

Network survey ended

OK

4G Network

AT#CSURV

Network survey started ...

earfcn: 1300 rxLev: -65 mcc: 460 mnc: 00 phycellId: 17D tac: 4020 cellIdentity: 363284E rsrp: -102 rsrq: -30

earfcn: 1506 rxLev: -71 mcc: 460 mnc: 01 phycellId: 138 tac: 0B03 cellIdentity: 11AE007 rsrp: -107 rsrq: -28

earfcn: 1650 rxLev: -72 mcc: 460 mnc: 01 phycellId: 1A6 tac: 0B03 cellIdentity: 12DAB02 rsrp: -107 rsrq: -28

earfcn: 1850 rxLev: -74 mcc: 460 mnc: 01 phycellId: 1A9 tac: 0B03 cellIdentity: C1453BB rsrp: -93 rsrq: -11

earfcn: 1506 rxLev: -71 mcc: 460 mnc: 11 phycellId: 138 tac: 0B03 cellIdentity: 11AE007 rsrp: -107 rsrq: -28

earfcn: 1650 rxLev: -72 mcc: 460 mnc: 11 phycellId: 1A6 tac: 0B03 cellIdentity: 12DAB02 rsrp: -107 rsrq: -28

earfcn: 1850 rxLev: -74 mcc: 460 mnc: 11 phycellId: 1A9 tac: 0B03 cellIdentity: C1453BB rsrp: -93 rsrq: -11

Network survey ended (Carrier: 4 BCCH: 4)

OK

AT#CSURVF - Network Survey Format

This command configures the numbers format used in the messages related to the surveying of the network bands channels.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#CSURVF=[<format>]

Set command configures the numbers format in each information text line of the network survey message (Easy Scan (R)).

Parameter:

Name	Type	Default	Description
<format>	integer	0	format of some numbers (see #CSURV) in each network survey information text line

Values:

0 : Decimal

1 : Hexadecimal

2 : Hexadecimal and extended network survey end message format

AT#CSURVF?

Read command reports the current format of the numbers in each network survey information text line, as follows:

#CSURVF: <format>

AT#CSURVF=?

Test command reports the supported range of values for the parameter <format>.

Jamming Detection and Report

AT#JDRENH2 - Enhanced Jamming Detection and Report

This command enables/disable jamming detection, and reports the relative result to the user.

[1] Hardware Design Guide of the used module

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#JDRENH2=<mode>[,<sat2G>,<unused>,<carrNum2G>,<pRxLevT2G>,<unused>,<unused>,<unused>[,<P_RSRP_T4G>[,<P_RSRQ_T4G>[,<unused>]]]]

The set command performs the following activities:

- enables/disables the detection of the presence of jamming in the module coverage.

- selects one of the following two reports mode or both: on dedicated GPIO, by means of the URC, or GPIO + URC.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	enable/disable jamming detection and select reporting mode

Values:

0 : disable jamming detection

1 : enable jamming detection, and report its condition on a GPIO pin, see Additional info section

2 : enable jamming detection, and report its condition with an URC, see Additional info section

3 : enable jamming detection, and report its condition as <mode>=1 and <mode>=2

4 : enable jamming detection, and report its condition with an URC every 3 sec, see <mode>=2

5 : enable jamming detection, and report its condition as <mode>=1 and <mode>=4

6 : not used

<sat2G>	integer	45	is the starting absolute threshold of RSSI 2G Network. After a frequency scan in 2G bands, if the measured power of a carrier is greater than <sat2G> threshold, that carrier is counted as possible jammed carrier.

Value:

0÷63 : threshold values

<unused>	integer	N/A	unused parameter

Value:

0÷91 : dummy value

<carrNum2G>	integer	100	is the minimum number of possible jammed carriers to consider that the module is under jamming condition

Value:

	0÷200	:	number of carriers

<pRxLevT2G>	integer	15	set the threshold of RxLev in 2G Network. The RxLev_Thr threshold is calculated as shown below: RxLev_Thr = RxLev_Av * (1 + (<pRxLevT2G>/100)) where RxLev_Av is the average of the last RxLev measures.

Value:

0÷100	:	values used to compute RxLev_Thr threshold

<unused>	integer	N/A	unused parameter

Value:

0÷100	:	dummy value

<unused>	integer	N/A	unused parameter

Value:

0÷100	:	dummy value

<unused>	integer	N/A	unused parameter

Value:

			0÷100 : dummy value
<P_RSRP_T4G>	integer	6	set the threshold of RSRP. The threshold (RSRP_Thr) is calculated as: RSRP_Thr = RSRP_Av*(1-(<P_RSRP_T4G>/100)) where RSRP_Av is the average of the last RSRP measures Value: 0÷100 : threshold of RSRP
<P_RSRQ_T4G>	integer	60	Set the threshold of RSRQ. The threshold (RSRQ_Thr) is calculated as RSRQ_Thr = RSRQ_Avg*(1-(<P_RSRQ_T4G>/100)) where RSRQ_Avg is the average of the last RSRQ measures. Value: 0÷100 : RSRQ threshold values
<unused>	integer	N/A	unused parameter Value: 0÷100 : dummy value

Additional info:

- <mode>=1

  The jamming condition is reported on pin GPIO_02 (JDR):

  GPIO_02 (JDR) = Low, Normal Operating Condition

  GPIO_02 (JDR) = High, Jammed Condition

  To have information on GPIO_02 pin, refer to document [1]. GPIO_02 pin can be used also by other functionality, see #GPIO command.

- <mode>=2

  the jamming condition is reported with a single URC on serial line, in the format:

  #JDR: <status>

Unsolicited field:

Name	Type	Description
<status>	string	jamming condition status, <mode>=2 Values: JAMMED RAT : jamming condition detected, <RAT> is the Radio Access Technology: GSM or LTE. OPERATIVE RAT : Normal Operating condition restored. Status shown only after a jamming condition has occurred.

AT#JDRENH2?

Read command reports the current parameters values, in the format:
#JDRENH2:<mode>,<sat2G>,<unused>,<carrNum2G>,<pRxLevT2G>,<unused>,<unused>,<unused>,<P_RSRP_T4G>, <P_RSRQ_T4G>,<unused>

AT#JDRENH2=?

Test command reports the supported range of the parameters values.

AT#JDR4GCFG - LTE Jamming Detection Threshold Configuration

The command configures the LTE Jamming Detection thresholds.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#JDR4GCFG=<P_RSRP_T4G>,<P_RSRQ_T4G>,<unused>[,<Initial_Delay>[,<Sampling_Number>[,<P_RSSI_S4G>]]]

Set command allows to configure the LTE Jamming Detection thresholds. After configuration, use the #JDRENH2 command to enable/disable LTE jamming detection and select reporting mode.

Parameters:

Name	Type	Default	Description
<P_RSRP_T4G>	integer	6	set the threshold of RSRP. The threshold (RSRP_Thr) is calculated as: RSRP_Thr = RSRP_Av*(1+(<P_RSRP_T4G>/100)) where RSRP_Av is the average of the last10 RSRP measures Value: 0÷100 : threshold of RSRP
<P_RSRQ_T4G>	integer	60	set the threshold of RSRQ. The threshold (RSRQ_Thr) is calculated as: RSRQ_Thr= RSRQ_Av*(1-(<P_RSRQ_T4G>/100))

			where RSRQ_Av is the average of the last 10 RSRQ measures
	Value:		
	0÷100	:	threshold of RSRQ
<unused>	integer	N/A	unused parameter
	Value:		
	0÷100	:	dummy value
<Initial_Delay>	integer	0	Sets the delay in tens of ms from power ON, to the beginning of the RSSI samples collection
	Value:		
	0÷1000	:	Value of Initial Delay
<Sampling_Number>	integer	30	Number of samples required to validate the Jamming condition.
	Value:		
	1÷200	:	Number of samples
<P_RSSI_S4G>	integer	-50	RSSI threshold level in dBm, above which the Jamming condition is detected.
	Value:		

120÷20	:	Value of RSSI Threshold

If you change the <P_RSRP_T4G> and <P_RSRQ_T4G> parameters of the AT#JDR4GCFG command, it will be automatically changed the parameters of the AT#JDRENH2 command, too, without notice, and vice-versa.

All the parameter settings are saved in NVM memory

AT#JDR4GCFG?

Read command returns the current settings in the format:

#JDR4GCFG:
<P_RSRP_T4G>,<P_RSRQ_T4G>,0,<Initial_Delay>,<Sampling_Number>,<P_RSSI_S4G>

AT#JDR4GCFG=?

Test command returns the range of supported values for all the parameters.

Packet Domain

AT+CGCLASS - GPRS Mobile Station Class

This command sets the mode of operation

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CGCLASS=[<class>]

Set command sets the mode of operation according to <class> parameter.

Parameter:

Name	Type	Default	Description
<class>	string	N/A	mode of operation

Values:

B : Class-B mode of operation (A/Gb mode), or CS/PS mode of operation (Iu mode)

CG : Class-C mode of operation in PS only mode (A/Gb mode), or PS mode of operation - (Iu mode)

CC : Class-C mode of operation in CS only mode (A/Gb mode), or CS (Iu mode) - (lowest mode of operation)

The setting is saved in NVM (and available on following reboot).

AT+CGCLASS?

Read command returns the current value of the mode of operation in the format:

+CGLASS: <class>

AT+CGCLASS=?

Test command reports the range for the parameter <class>.

AT+CGQREQ - Quality of Service Profile (Requested)

This command allows to specify a Quality of Service Profile (requested) that is used when the terminal sends an Activate PDP Context Request message to the network.

3GPP TS 27.007
3GPP TS 03.060
3GPP TS 23.060

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CGQREQ=[<cid>[,<precedence>[,<delay>[,<reliability>[,<peak>[,<mean>]]]]]]

Set command allows specify Quality of Service Profile that used when the terminal sends an Activate PDP Context Request message to the network. It specifies a profile for the context identified by the (local) context identification parameter, <cid>.

Parameters:

Name	Type	Default	Description
<cid>	integer	-	PDP context identification (see +CGDCONT command).
<precedence>	integer	-	precedence class
<delay>	integer	-	delay class
<reliability>	integer	-	reliability class
<peak>	integer	-	peak throughput class
<mean>	integer	-	mean throughput class

- If a value is omitted for a particular class then this class is not checked
- A special form of the Set command, +CGQREQ=<cid> causes the requested profile for context number <cid> to become undefined.

AT+CGQREQ?

Read command returns the current settings for each defined context in the format:

+CGQREQ:<cid>,<precedence>,<delay>,<reliability>,<peak>,<mean>[<CR><LF>
+CGQREQ:<cid>,<precedence>,<delay>,<reliability>,<peak>,<mean>[...]]

- If no PDP context has been defined, it has no effect and OK result code is returned.

AT+CGQREQ=?

Test command returns as a compound value the type of the current PDP context and the supported values for the sub parameters in the format:

+CGQREQ: <PDP_Type>,
(list of supported <precedence>s),(list of supported <delay>s),
(list of supported <reliability>s),(list of supported <peak>s),
(list of supported <mean>s)

Additional info:

- PDP type meaning.

Name	Type	Default	Description
<PDP_Type>	string	-	specifies the type of packet data protocol (see +CGDCONT command)

Only the "IP" <PDP_Type> is currently supported.

```
AT+CGQREQ?
+CGQREQ: 1,0,0,3,0,0
OK
AT+CGQREQ=1,0,0,3,0,0
OK
AT+CGQREQ=?
+CGQREQ: "IP",(0-3),(0-4),(0-5),(0-9),(0-18,31)
+CGQREQ: "PPP",(0-3),(0-4),(0-5),(0-9),(0-18,31)
+CGQREQ: "IPV6",(0-3),(0-4),(0-5),(0-9),(0-18,31)
OK
```

AT+CGDCONT - Define PDP Context

Define PDP Context.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CGDCONT=[<cid>[,<PDP_type>[,<APN>[,<PDP_addr>

[,<d_comp>[,<h_comp>[,<IPv4AddrAlloc>[,<Emergency_ind>]]]]]]]]

Set command specifies PDP context parameter values for a PDP context identified by the (local) context identification parameter, <cid>.

Parameters:

Name	Type	Default	Description
<cid>	integer	-	(PDP Context Identifier) numeric parameter which specifies a particular PDP context definition 1..max - where the value of max is returned by the Test command
<PDP_type>	string	N/A	(Packet Data Protocol type) a string parameter which specifies the type of packet data protocol

Values:

"IP" : Internet Protocol

"PPP"	:	Point to Point Protocol
"IPV6"	:	Internet Protocol, Version 6
"IPV4V6"	:	Virtual <PDP_type> introduced to handle dual IP stack UE capability

<APN>	string	-	(Access Point Name) a string parameter which is a logical name that is used to select the GGSN or the external packet data network. If the value is empty ("") or omitted, then the subscription value will be requested
<PDP_addr>	string	-	A string parameter that identifies the terminal in the address space applicable to the PDP. The allocated address may be read using the +CGPADDR command
<d_comp>	integer	0	Numeric parameter that controls PDP data compression

Values:

0	:	PDP data compression off (default if value is omitted)
1	:	PDP data compression on
2	:	V.42bis

<h_comp>	integer	0	Numeric parameter that controls PDP header compression

Values:

	0 : PDP header compression off (default if value is omitted) 1 : PDP header compression on 2 : RFC1144 (applicable for SNDCP only) 3 : RFC2507		
<IPv4AddrAlloc>	integer	N/A	a numeric parameter that controls how the MT/TA requests to get the IPv4 address information
	Values: 0 : IPv4 Address Allocation through NAS Signaling 1 : IPv4 Address Allocated through DHCP		
<Emergency_ind>	integer	N/A	a numeric parameter used to indicate whether the PDP context is for emergency bearer services or not
	Values: 0 : PDP context is not for emergency bearer services 1 : PDP context is for emergency bearer services		

A special form of the Set command,

+CGDCONT=<cid>

causes the values for context number <cid> to become undefined, except cid = 1 and emergency profile.

- Emergency profile could be undefined only after setting emergency indication parameter to 0.
- Although max number of PDP profile is 16, the user can only create 6 PDP profiles (persistent profiles), the rest 10 are temporary profiles.
- Profile 1 is the default profile and the attach profile, so even if you delete all profiles, the profile 1 is always regenerated regardless rebooting.
- Predefined PDP profile file that operator requested should not recommended to delete.

AT+CGDCONT?

Read command returns the current settings for each defined context in the format:

+CGDCONT:
<cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>,<IPv4AddrAlloc>
,<Emergency_ind><CR><LF>

+CGDCONT:<cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>,<IPv4AddrAlloc>
,<Emergency_ind>

AT+CGDCONT=?

Test command returns values supported as a compound value.

```
AT+CGDCONT=1,"IPV4V6","apn","10.10.10.10"
OK

AT+CGDCONT?
+CGDCONT: 1,"IPV4V6","apn","10.10.10.10",0,0,0,0
```

```
+CGDCONT: 2,"IPV4V6","ims","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0
+CGDCONT: 3,"IPV4V6","sos","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,1
OK

AT+CGDCONT=?
+CGDCONT: (1-15),"IP",,,(0-2),(0-2),(0-1),(0-1)
+CGDCONT: (1-15),"PPP",,,(0-2),(0-2),(0-1),(0-1)
+CGDCONT: (1-15),"IPV6",,,(0-2),(0-2),(0-1),(0-1)
+CGDCONT: (1-15),"IPV4V6",,,(0-2),(0-2),(0-1),(0-1)
OK
```

AT+CGDSCONT - Define Secondary PDP Context

Define secondary PDP Context.

[1] 3GPP TS 44.065
[2] 3GPP TS 25.323
[3] RFC1144
[4] RFC2507
[5] RFC3095

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CGDSCONT=[<cid>,<p_cid>[,<d_comp>[,<h_comp>]]]

Set command specifies PDP context parameter values for a Secondary PDP context identified by the (local) context Identification parameter, <cid>.

The number of PDP contexts that may be in a defined state at the same time is given by the range returned by the test command. In EPS the command is used to define traffic flows.

Parameters:

Name	Type	Default	Description
<cid>	integer	-	specifies a particular PDP Context Identifier, see +CGDCONT command.
<p_cid>	integer	-	specifies a particular PDP context definition which has been specified by use of the +CGDCONT command. The parameter is local to the TE-MT interface.
<d_comp>	integer	0	controls PDP data compression (applicable for SNDCP only) (refer 3GPP TS 44.065)
	Values:		

0 : off (default if value is omitted)

1 : on (manufacturer preferred compression)

2 : V.42 bis

<h_comp>	integer	0	controls PDP header compression (refer 3GPP TS 44.065 and 3GPP TS 25.323).

Values:

0 : off (default if value is omitted)

1 : on (manufacturer preferred compression)

2 : RFC1144 (applicable for SNDCP only)

3 : RFC2507 - Not supported

4 : RFC3095 (applicable for PDCP only) - Not supported

A special form of the set command, +CGDSCONT= <cid> causes the values for context number <cid> to become undefined.

AT+CGDSCONT?

The read command returns the current settings for each defined context in the format:

+CGDSCONT: <cid>,<p_cid>,<d_comp>,<h_comp>[<CR><LF>
+CGDSCONT: <cid>,<p_cid>,<d_comp>,<h_comp> [...]]

AT+CGDSCONT=?

Test command returns the supported range of values of parameters.

AT+CGSCONTRDP - Secondary PDP Context Read Dynamic Parameters

This command returns parameters for a given <cid>.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGSCONTRDP[=<cid>]

The execution command returns <p_cid> and <bearer_id> for a given <cid>. If the parameter <cid> is omitted, the <cid>, <p_cid> and <bearer_id> returned for all active secondary PDP contexts.
In EPS, the Traffic Flow parameters returned.

Format of the returned message:

+CGSCONTRDP:<cid>,<p_cid>,<bearer_id>[<CR><LF>
+CGSCONTRDP: <cid>, <p_cid>,<bearer_id> [...]]

Parameter:

Name	Type	Default	Description
<cid>	integer	-	specifies a particular PDP context or Traffic Flows definition. The parameter is local to the TE-MT interface and is used in other PDP context-related commands.

Additional info:

Here is the list of the parameters meanings returned by the +CGSCONTRDP command.

Name	Type	Default	Description
<p_cid>	integer	-	specifies a particular active secondary context definition or default EPS context

			Identifier which has been specified by use of the +CGDCONT command. The parameter is local to the TE-MT interface
<bearer_id>	integer	-	identifies the bearer, EPS Bearer in EPS and NSAPI in UMTS/GPRS.

Parameters for network-initiated PDP contexts returned as well. The dynamic part of the PDP context will only exist if established by the network.

AT+CGSCONTRDP=?

The test command returns a list of <cid>s associated with active secondary PDP contexts:

AT+CGDATA - Enter Data State

Execution command causes to perform whatever actions are necessary to establish a communication with the network using one or more GPRS PDP types.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CGDATA=[<L2P>[,<cid>[,...]]]

Parameters:

Name	Type	Default	Description
<L2P>	string	N/A	indicates the layer 2 protocol to be used
	Value:		
	"PPP" : PPP Point-to-point protocol		
<cid>	integer	-	specifies a particular PDP context definition (see +CGDCONT command).

If parameter <L2P> is omitted, the layer 2 protocol is unspecified

AT+CGDATA=?

Test command reports information on the supported layer 2 protocols.

```
AT+CGDATA=?
+CGDATA: ("PPP")
```

OK

AT+CGDATA="PPP",1

OK

AT+CGTFTRDP - Traffic Flow Template Read Dynamic Parameters

This command returns the relevant information about Traffic Flow Template for an active secondary or non secondary PDP context

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CGTFTRDP=[<cid>]

The execution command returns the relevant information about Traffic Flow Template for an active secondary or non secondary PDP context specified by <cid> together with the additional network assigned values when established by the network.

If the parameter <cid> is omitted, the Traffic Flow Templates for all active secondary and non secondary PDP contexts are returned.

Parameters of both network and MT/TA initiated PDP contexts returned.
Possible Response(s):

+CGTFTRDP: <cid>, <packet filter identifier>, <evaluation precedence index>, <source address and subnet mask>, <protocol number (ipv4) / next header(ipv6)>,<destination port range>,<source port range>, <ipsec security parameter index (spi)>, <type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>,<flow label ipv6)>, <direction>, <NW packet filter Identifier><CR><LF>

+CGTFTRDP: <cid>, <packet filter identifier>, <evaluation precedence index>, <source address and subnet mask>, <protocol number (ipv4) / next header (ipv6)>, <destination port range>,<source port range>, <ipsec security parameter index (spi)>, <type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>,<flow label (ipv6)>, <direction>,<NW packet filter Identifier> [...]]

Parameter:

Name	Type	Default	Description

<cid>	integer	-	specifies a particular PDP context definition or Traffic Flows definition.

Additional info:

▶▶ The following parameters are defined in 3GPP TS 23.060

<packet filter identifier>: a numeric parameter. The value range is from 1 to 16.

<evaluation precedence index>: a numeric parameter. The value range is from 0 to 255.

<source address and subnet mask>: string type. The string is given as dot-separated numeric (0-255) parameters on the form: "a1.a2.a3.a4.m1.m2.m3.m4" for IPv4 or "a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16.m1.m2.m3.m4.m5.m6.m7.m8.m9.m10.m11.m12.m13.m14.m15.m16" for IPv6.

<protocol number (ipv4) / next header (ipv6)>: a numeric parameter, value range from 0 to 255.

<destination port range>: string type. The string is given as dot-separated numeric (0-65535) parameters on the form "f.t".

<source port range>: string type. The string is given as dot-separated numeric (0-65535) parameters on the form "f.t".

<ipsec security parameter index (spi)>: numeric value in hexadecimal format. The value range is from 00000000 to FFFFFFFF.

<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>: string type. The string given as dot-separated numeric (0-255) parameters on the form "t.m".

<flow label (ipv6)>: numeric value in hexadecimal format. The value range is from 00000 to FFFFF. Valid for IPv6 only.

<direction> a numeric parameter which specifies the transmission direction in which the Packet Filter shall be applied.
0- Pre-Release 7 TFT Filter (see 3GPP TS 24.008, table 10.5.162)
1- Uplink
2- Downlink
3- Bidirectional (Used for Uplink and Downlink)

<NW packet filter Identifier> a numeric parameter. The value range is from 1 to 16. In EPS the value is assigned by the network when established

Some of the above listed attributes can coexist in a Packet Filter while others mutually exclude each other. The possible combinations listed on 3GPP TS 23.060

AT+CGTFTRDP=?

+CGTFTRDP: (list of <cid>s associated with active secondary or non secondary contexts) in the format

+CGTFTRDP: (<cid>),(<operation code>), (<No of PKT filters>), (<packet filter identifier>), (<evaluation precedence index>), (<source address and subnet mask>), (<protocol number(ipv4)/next header(ipv6)>), (<destination port range>), (<source port range>), (<ipspec security parameter index(spi)>), (<type of service(tos)(ipv4) and mask/traffic class(ipv6) and mask>), (<flow label(ipv6)>), (<direction>)

AT+CGQMIN - Quality of Service Profile (Minimum Acceptable)

This command allows to specify a minimum acceptable profile of QoS.

3GPP TS 27.007
3GPP TS 03.060
3GPP TS 23.060

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CGQMIN=[<cid>[,<precedence>[,<delay>[,<reliability>[,<peak>[,<mean>]]]]]]

Set command allows specify a minimum acceptable profile, checked by the terminal against the negotiated profile returned in the Activate PDP Context Accept message.

Parameters:

Name	Type	Default	Description
<cid>	integer	-	PDP context identification (see +CGDCONT command)
<precedence>	integer	-	precedence class
<delay>	integer	-	delay class
<reliability>	integer	-	reliability class
<peak>	integer	-	peak throughput class
<mean>	integer	-	mean throughput class

If a value is omitted for a class, then this class is not checked.

A special form of the set command, AT+CGQMIN=<cid> causes the requested profile for context number <cid> to become undefined.

AT+CGQMIN?

Read command returns the current settings for each defined context in the format:

+CGQMIN: <cid>,<precedence>,<delay>,<reliability>,<peak>,<mean>[<CR><LF>

+CGQMIN: <cid>,<precedence>,<delay>,<reliability>,<peak>,<mean>[...]]

If no PDP context has been defined, it has no effect and OK result code is returned

AT+CGQMIN=?

Test command returns as a compound value the type of the current PDP context and the supported values for the sub parameters in the format:

+CGQMIN: <PDP_Type>,
(list of supported <precedence>s),(list of supported <delay>s),
(list of supported <reliability>s),(list of supported <peak>s),
(list of supported <mean>s)

Additional info:

PDP type meaning.

Name	Type	Default	Description
<PDP_Type>	string	-	specifies the type of packet data protocol, see +CGDCONT command Only the "IP" <PDP_Type> is currently supported.

</>

- AT+CGQMIN=1,0,0,3,0,0 OK

  AT+CGQMIN?

  +CGQMIN: 1,0,0,5,0,0

  OK

- AT+CGQMIN=?

  +CGQMIN: "IP",(0-3),(0-4),(0-5),(0-9),(0-18,31)

  +CGQMIN: "PPP",(0-3),(0-4),(0-5),(0-9),(0-18,31)

  +CGQMIN: "IPV6",(0-3),(0-4),(0-5),(0-9),(0-18,31)

  +CGQMIN: "IPV4V6",(0-3),(0-4),(0-5),(0-9),(0-18,31)

  OK

AT+CGPADDR - Show PDP Address

This command returns a list of PDP addresses for the specified context identifiers.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGPADDR=[<cid>[,...]]

Execution command returns a list of PDP addresses for the specified context identifiers in the format:

+CGPADDR: <cid>[,<PDP_addr_1>[,<PDP_addr_2>]][<CR><LF>

+CGPADDR: <cid>,[<PDP_addr_1>[,<PDP_addr_2>]][...]]

Parameter:

Name	Type	Default	Description
<cid>	integer	-	specifies a particular PDP context definition (see +CGDCONT command). If no <cid> specified, the addresses for all defined contexts are returned. 1 - 24

Additional info:

<PDP_addr_1> and <PDP_addr_2>

each is a string that identifies the MT in the address space applicable to the PDP. The address may be static or dynamic. For a static address, it will be the one set by the +CGDCONT command when the context was defined. For a dynamic address it will be the one assigned during the last PDP context activation that used the context definition referred to by <cid>. Both <PDP_addr_1> and <PDP_addr_2> are omitted if none is available. Both <PDP_addr_1> and <PDP_addr_2> are included when both IPv4 and IPv6 addresses

are assigned, with <PDP_addr_1> containing the IPv4 address and <PDP_addr_2> containing the IPv6 address.

The string is given as dot-separated numeric (0-255) parameter of the form:

a1.a2.a3.a4 for IPv4 and
a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16 for IPv6.

In dual-stack terminals (<PDP_type> IPV4V6), the IPv6 address will be provided in <PDP_addr_2>. For terminals with a single IPv6 stack (<PDP_type> IPV6) or due to backwards compatibility, the IPv6 address can be provided in parameter <PDP_addr_1>.

AT+CGPADDR=?

Test command returns a list of defined <cid>.

```
AT#GPRS=1
+IP: xxx.yyy.zzz.www
OK
AT+CGPADDR=1
+CGPADDR: 1,"xxx.yyy.zzz.www"
OK
AT+CGPADDR=?
+CGPADDR: (1)
OK
```

AT+CGCMOD - Modify PDP Context

This command has no effect and is included only for backward compatibility with landline modems

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGCMOD=[<cid_n>[,...]]

The execution command is used to modify the specified PDP context(s) with repect to QoS profiles and TFTs. After command has completed, the MT returns to V.250 online data state.If the requested modification for any specified context cannot be achieved, an ERROR or +CME ERROR response is returned. Extended error responses are enabled by the +CMEE command.

For EPS, the modification request for an EPS bearer resource will be answered by the network by an EPS bearer

Modification request. The request must be accepted by the MT before the PDP context effectively changed.

Parameter:

Name	Type	Default	Description
<cid_n>	integer	-	specifies a particular PDP context definition (see the +CGDCONT and +CGDSCONT commands).

If no parameters are specified (no <cid_n> specified), the command modifies all active contexts.

AT+CGCMOD=?

Test command returns a list of <cid_n>s associated with active contexts.

AT+CGTFT - Traffic Flow Template

This command allows the TE to specify a Packet Filter - PF for a Traffic Flow Template - TFT that is used in the GGSN in UMTS/GPRS and Packet GW in EPS for routing of packets onto different QoS flows towards the TE.

The concept is further described in the 3GPP TS 23.060. A TFT consists of from one and up to 16 Packet Filters, each identified by a unique <packet filter identifier>. A Packet Filter also has an <evaluation precedence index> that is unique within all TFTs associated with all PDP contexts that are associated with the same PDP address.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CGTFT=[<cid>,<packet filter identifier>,<evaluationprecedenceindex>

[,<source address and subnet mask> [,<protocol number(ipv4) / next header(ipv6)>

[,<destination portrange> [,<sourceport range>[,<ipsecsecurity parameter index (spi)>

[,<typeof service (tos)(ipv4) and mask /traffic class (ipv6)and mask> [,<flowlabel (ipv6)>]]]]]]]]]

This command allows the TE to specify a Packet Filter - PF for a Traffic Flow Template - TFT that is used in the GGSN in UMTS/GPRS and Packet GW in EPS for routing of packets onto different QoS flows towards the TE.

The set command specifies a Packet Filter that whom added to the TFT stored in the MT and used for the context identified by the (local) context identification parameter, <cid>. The specified TFT will be stored in the GGSN in UMTS/GPRS and Packet GW in EPS only at activation or MS-initiated modification of the related context.

Since this is the same parameter that is used in the +CGDCONT and +CGDSCONT commands, the +CGTFT command is effectively an extension to these commands. The Packet Filters consist of a number of parameters, each of which may be set to a separate value.
Possible Response(s):
OK
ERROR

Parameter:

Name	Type	Default	Description
<cid>	integer	-	specifies a particular PDP context definition (see the +CGDCONT and +CGDSCONT commands).

Additional info:

The following parameters are defined in 3GPP TS 23.060:

<packet filter identifier>: a numeric parameter, value range from 1 to 16.

<evaluation precedence index>: a numeric parameter. The value range is from 0 to 255.

<source address and subnet mask>: string type. The string is given as dot-separated numeric (0-255)

parameters on the form:

"a1.a2.a3.a4.m1.m2.m3.m4" for IPv4 or

"a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16.m1.m2.m3.m4.m5.m6.m7.

m8.m9.m10.m11.m12.m13.m14.m15.m16", for IPv6.

<protocol number (ipv4) / next header (ipv6)>: a numeric parameter, value range from 0 to 255.

<destination port range>: string type. The string is given as dot-separated numeric (0-65535) parameters on the form "f.t".

<source port range>: string type. The string is given as dot- separated numeric (0-65535) parameters on the form "f.t".

<ipsec security parameter index (spi)>: numeric value in hexadecimal format. The value range is from 0 to FFFFFFFF.

<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>: string type. The string is given as dot-separated numeric (0-255) parameters on the form "t.m".

<flow label (ipv6)>: numeric value in hexadecimal format. The value range is from 0 to FFFFF. Valid for IPv6 only.

- A special form of the set command, +CGTFT= <cid> causes all of the Packet Filters in the TFT for context number <cid> to become undefined.
- At any time there may exist only one PDP context with no associated TFT amongst all PDP contexts associated to one PDP address. At an attempt to delete a TFT, which would violate this rule, an ERROR or +CME ERROR response is returned. Extended error responses are enabled by the +CMEE command.
- Some of the above listed attributes may coexist in a Packet Filter while others mutually exclude each other, the possible combinations are shown in 3GPP TS 23.060.

AT+CGTFT?

The read command returns the current settings for all Packet Filters for each defined context.

AT+CGTFT: <cid>, <packet filter identifier>, <evaluation precedence index>, <source address and subnet mask>, <protocol number (ipv4) / next header (ipv6)>, <destination port range>, <source port range>, <ipsec security parameter index (spi)>, <type of service (tos) (ipv4) and mask /traffic class (ipv6) and mask>, <flow label (ipv6)> [<CR><LF>+CGTFT: <cid>,<packet filter identifier>, <evaluation precedence index>, <source address and subnet mask>, <protocol number (ipv4) / next header (ipv6)>, <destination port range>, <source port range>, <ipsec security parameter index (spi)>, <type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>, <flow label (ipv6)> [...]]

AT+CGTFT=?

The test command returns values supported as a compound value. If the MT supports several PDP types, the parameter value ranges for each PDP type returned on a separate line. TFTs shall be used for PDP-type IP and PPP only. For PDP-type PPP a TFT is applicable only when IP traffic is carried over PPP. If PPP carries header-compressed IP packets, then a TFT cannot be use.

AT+CGTFT: <PDP_type>, (list of supported <packet filter identifier>s), (list of supported <evaluation precedence index>s), (list of supported <source address and subnet mask>s), (list of supported <protocol number (ipv4) / next header (ipv6)>s), (list of supported <destination port range>s), (list of supported <source port range>s), (list of supported <ipsec security parameter index (spi)>s), (list of supported <type of service (tos) (ipv4) and mask / traffic class(ipv6) and mask>s), (list of supported <flow label (ipv6)>s)
[<CR><LF>+CGTFT: <PDP_type>, (list of supported <packet filter identifier>s), (list of supported <evaluation precedence index>s), (list of supported <source address and subnet mask>s),(list of supported <protocol number (ipv4) / next header (ipv6)>s), (list of supported <destination port range>s), (list of supported <source port range>s), (list of supported <ipsec security parameter index (spi)>s),(list of supported <type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>s), (list of supported <flow label (ipv6)>s)
[...]]

AT#MSCLASS - Multislot Class Control

Set command sets the GPRS/EGPRS multislot class.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#MSCLASS=[<class>[,<autoattach>]]

Parameters:

Name	Type	Default	Description
<class>	integer	12	GPRS/EGPRS multislot class. Value: 1÷12 : class
<autoattach>	integer	1	specify when the new multislot class will be enabled. Values: 0 : the new multislot class is enabled only at the next detach / attach or after a reboot. Not supported. 1 : the new multislot class is enabled immediately, automatically forcing a detach / attach procedure

It is only available for variants that support GSM.

If <autoattach> parameter value 0 is set, then AT#MSCLASS will return ERROR because this value is not supported.

The default value for <autoattach> parameter is 1

AT#MSCLASS?

Read command reports the current value of the multislot class in the format:

#MSCLASS: <class>

AT#MSCLASS=?

Test command reports the range of available values for both parameters <class> and <autoattach>.

AT+CGCONTRDP - PDP Context Read Dynamic Parameters

The execution command returns the relevant information for a PDP Context established by the network.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CGCONTRDP[=<cid>]

The execution command returns the relevant information: <bearer_id>, <apn>, <ip&subnet>,<gw_addr>, <DNS_prim_addr>, <DNS_sec_addr>, <P-CSCF_prim_addr> and <P-CSCF_sec_addr> for an active non secondary PDP context with the context identifier <cid>.

If the parameter <cid> is omitted, the relevant information for all active non secondary PDP contexts is returned.

Possible response(s):

+CGCONTRDP:<cid>,<bearer_id>,<apn>[,<ip&subnet>[,<gw_addr>[,<DNS_prim_addr>

[,<DNS_sec_addr>[,<P-CSCF_prim_addr>[,<P-CSCF_sec_addr>]]]]]][<CR><LF>
+CGCONTRDP:<cid>,<bearer_id>,<apn>[,<ip&subnet>[,<gw_addr>[,<DNS_prim_addr>

[,<DNS_sec_addr>[,<PCSCF_prim_addr>[,<PCSCF_sec_addr>]]]]]][...]]

The response message parameters are described in the Additional info section.

Parameter:

Name	Type	Default	Description

<cid>	integer	-	identifies a non secondary PDP context definition. The parameter is local to the TE-MT interface and is used in other PDP context-related commands.

Additional info:

▶▶ List of the meaning of the response message parameters.

Name	Type	Default	Description
<bearer_id>	integer	-	identifies the bearer, EPS Bearer in EPS and NSAPI in UMTS/GPRS.
<apn>	string	-	logical name used to select the GGSN or the external packet data network.
<ip&subnet>	string	-	IP address and subnet mask of the MT. The string is given as dot-separated numeric (0-255) parameters on the form. For more information, see next Additional info section.

<gw_addr>	string	-	Gateway address of the MT. The string is given as dot-separated numeric (0-255) parameters. If the MT has dual stack capabilities the parameter shows first the dot separated IPV4 Gateway address followed by the dot separated IPV6 Gateway Address. The gateway addresses are separated by space.
<DNS_prim_addr>	string	-	IP address of the primary DNS Server. If the MT has dual stack capabilities the parameter shows first the dot separated IPV4 Address, followed by the dot separated IPV6 Address of DNS Server.
<DNS_sec_addr>	string	-	IP address of the secondary DNS Server. If the MT has dual stack capabilities the parameter shows first the dot separated IPV4 Address, followed by the dot separated IPV6 Address of DNS Server.
<P_CSCF_prim_addr>	string	-	IP address of the primary P-CSCF Server.

			If the MT has dual stack capabilities the parameter shows first the dot separated IPV4 Address, followed by the dot separated IPV6 primary Address of P-CSCF Server.
<P_CSCF_sec_addr>	string	-	IP address of the secondary P-CSCF Server. If the MT has dual stack capabilities the parameter shows first the dot separated IPV4 Address, followed by the dot separated IPV6 Address of P-CSCF Server.

▶▶ Referring to <ip&subnet> parameter:

the string is given as dot-separated numeric (0-255) parameters. The format is:

for IPv4:

"a1.a2.a3.a4.m1.m2.m3.m4"

for IPv6:

"a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16. m1.m2.m3.m4.m5.m6.m7.m8.m9.m10.m11.m12. m13.m14.m15.m16"

If the MT has dual stack capabilities the string shows first the dot separated IPv4 Address followed by the dot Separated IPv6 Address. The IPv4 address and the IPv6 address parameters are separated by space: "a1.a2.a3.a4 a1.m1.m2.m3.m4 a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16.

m1.m2. m3.m4.m5.m6.m7.m8.m9.m10.m11.m12.m13.m14.m15.m16"

The dynamic part of the PDP context will only exist if established by the network. The test command returns a list of <cid>s associated with active contexts.

The execution command

AT+CGCONTRDP=9

returns

+CME ERROR: 100

AT+CGCONTRDP?

Read command returns

+CGCONTRDP: 1

OK

AT+CGCONTRDP=?

Return the list of <cid>s associated with active contexts in the following form

+CGCONTRDP: (<p_cid>), (<bearer_id>),(<apn>),(<source_addr and subnet_mask>), (<gw_addr>), (<DNS_prim_addr>), (<DNS_sec_addr>), (<P-CSCF_prim_addr>), (<P-CSCF_sec_addr>),(<IM_CN_Signalling_Flag>)

AT+CGEQOS - Define EPS Quality of Service

The command specifies the EPS Quality of Service parameters.

[1] 3GPP TS 23.203

[2] 3GPP TS 24.301

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT+CGEQOS=[<cid>[,<QCI> [,<DL_GBR>,<UL_GBR> [,<DL_MBR>,<UL_MBR>]]]]

Set command specifies the EPS Quality of Service parameters.

A special form of the set command, +CGEQOS= <cid> causes the values for context number <cid> to become undefined.

Parameters:

Name	Type	Default	Description
<cid>	integer	-	it specifies a particular EPS Traffic Flows definition in EPS
<QCI>	integer	9	it specifies a class of EPS QoS, see standard [1].

Values:

0	:	QCI is selected by network
1÷4	:	value range for guaranteed bit rate Traffic Flows

	5÷9	:	value range for non-guaranteed bit rate Traffic Flows
<DL_GBR>	integer	-	it indicates DL GBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI, see standard [2].
<UL_GBR>	integer	-	indicates UL GBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI, see standard [2].
<DL_MBR>	integer	-	it indicates DL MBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI, see standard [2].
<UL_MBR>	integer	-	it indicates UL MBR in case of GBR QCI. The value is in kbit/s. This parameter is omitted for a non-GBR QCI, see standard [2].

Additional info:

▶▶ Possible Response(s):

+CME ERROR: <err>

AT+CGEQOS?

Read command returns the current settings for each defined QoS.

+CGEQOS: <cid>, <QCI>,[<DL_GBR>,<UL_GBR>],[<DL_MBR>,<UL_MBR>]

[<CR>>LF>

+CGEQOS: <cid>, <QCI>,[<DL_GBR>,<UL_GBR>],[<DL_MBR>,<UL_MBR>][...]]

AT+CGEQOS=?

Test command returns the ranges of the supported parameters:

+CGEQOS: (list of supported <cid>s), (list of supported <QCI>s), (list of supported <DL_GBR>s), (list of supported <UL_GBR>s), (list of supported <DL_MBR>s), (list of supported <UL_MBR>s)

AT+CGEQOSRDP - EPS Quality of Service Read Dynamic Parameters

The command returns the parameters related to the Quality of Service.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CGEQOSRDP=[<cid>[,...]]

The execution command returns the Quality of Service parameters <QCI>, [<DL_GBR> and <UL_GBR>] and [<DL_MBR> and <UL_MBR>] of the active secondary or non secondary PDP context associated to the provided context identifier <cid>.

Possible Response(s):

+CGEQOSRDP:<cid>,<QCI>,[<DL_GBR>,<UL_GBR>],[<DL_MBR>,<UL_MBR>]

[<CR>>LF>

+CGEQOSRDP:<cid>,<QCI>,[<DL_GBR>,<UL_GBR>],[<DL_MBR>,<UL_MBR>] [...]]

See +CGEQOS command to have information on the meaning of the returned parameters.

Parameter:

Name	Type	Default	Description
<cid>	integer	-	It specifies a particular Traffic Flows definition in EPS and a PDN connection definition in UMTS/GPRS.

If the parameter <cid> is omitted, the Quality of Service parameters for all established PDN connections are returned.

AT+CGEQOSRDP=?

The test command returns a list of <cid>s associated with active contexts.

Parameters of both network and MT/TA initiated PDN connections will be returned.

AT+CGPIAF - Printing IP Address Format

This command selects the printout format of the IPv6 address.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT+CGPIAF=[<IPv6_AddressFormat>[,<IPv6_SubnetNotation>[,<IPv6_LeadingZeros>[,<Pv6_CompressZeros>]]]]

Set command decides what the format to print IPv6 address parameter.

Parameters:

Name	Type	Default	Description
<IPv6_AddressFormat>	integer	0	selects the IPv6 address format. Relevant for all AT command parameters that can hold an IPv6 address. Values: 0 : use IPv4-like dot-notation. IP addresses, and subnetwork mask if applicable, are dot-separated. 1 : use IPv6-like colon-notation. IP address, and subnetwork mask if applicable and when given explicitly, are separated by a space.
<IPv6_SubnetNotation>	integer	0	selects the subnet-notation for remote address and subnet mask. Setting does not apply if

			IPv6 address format <IPv6_AddressFormat>=0.

Values:

0	:	both IP address, and subnet mask are started explicitly, separated by a space.
1	:	the printout format is applying / (forward slash) subnet-prefix Classless Inter-Domain Routing (CIDR) notation.

<IPv6_LeadingZeros>	integer	0	selects whether leading zeros are omitted or not. Setting does not apply if IPv6 address format <IPv6_AddressFormat>=0.

Values:

0	:	leading zeros are omitted.
1	:	leading zeros are included.

AT+CGPIAF?

Read command returns the current parameter setting.

AT+CGPIAF=?

Test command returns supported values in the format:

+CGPIAF: (0-1),(0-1),(0-1),(0-1)

OK

```
AT+CGPIAF=0,0,0,0
OK
```

```
AT#SGACT=1,1
#SGACT: 252.1.171.171.205.205.239.224.0.0.0.0.0.0.0.1
OK

AT+CGPIAF=1,0,0,0
OK

AT#SGACT=1,1
#SGACT: FC01:ABAB:CDCD:EFE0:0:0:0:1
OK
```

AT+CEVDP - Voice Domain Preference

Set command selects the voice domain preference.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT+CEVDP=<domain>

Parameter:

Name	Type	Default	Description
<domain>	string	3	voice domain preference.

Values:

1 : CS voice only

2 : CS voice preferred, IMS PS voice as secondary

3 : IMS PS voice preferred, CS as secondary

4 : IMS PS voice only

The domain is saved into the NVM

AT+CEVDP=6

returns

+CME ERROR: 50

AT+CEVDP?

Read command returns the selected domain in the format:

+CEVDP: <domain>

AT+CEVDP=?

Test command returns the supported range of values of the parameter <domain>.

AT+CGACT - PDP Context Activate or Deactivate

This command activates or deactivates the specified PDP context(s).

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGACT=[<state>[,<cid>[,<cid>][,...]]]]

Execution command is used to activate or deactivate the specified PDP context(s)

Parameters:

Name	Type	Default	Description
<state>	integer	N/A	activate/deactivate the PDP context
	Values:		
	0	:	deactivate
	1	:	activate
<cid>	integer	-	specifies a PDP context definition (see +CGDCONT command)

If no <cid> are specified the activation/deactivation form of the command activates/deactivates all defined contexts.

AT+CGACT?

Read command returns the current activation state for all the defined PDP contexts in the format:

+CGACT: <cid>,<state>[<CR><LF>
+CGACT: <cid>,<state>[...]]

AT+CGACT=?

Test command reports information on the supported PDP context activation states parameters in the format:

+CGACT: (0,1)

AT+CGACT=1,1

OK

AT+CGACT?

+CGACT: 1,1

OK

AT+CGEREP - Packet Domain Event Reporting

This command enables or disables the presentation of unsolicited result codes.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGEREP=[<mode>[,<bfr>]]

Set command enables or disables sending of unsolicited result codes +CGEV: XXX from TA to TE in the case of certain events occurring in the TA or the network. The URC formats and related events are shown in the Additional info sections.

Parameters:

Name	Type	Default	Description
<mode>	integer	0	controls the processing of URCs specified with this command.

Values:

0 : buffer unsolicited result codes in the TA. If TA result code buffer is full, the oldest one can be discarded. No codes are forwarded to the TE.

1 : discard unsolicited result codes when TA-TE link is reserved (e.g. in on-line data mode); otherwise forward them directly to the TE.

2 : buffer unsolicited result codes in the TA when TA-TE link is reserved (e.g. in on-line data mode) and flush them to the TE when TA-TE link becomes available; otherwise forward them directly to the TE.

<bfr>	integer	0	controls the effect on buffered codes when <mode> 1 or 2 is entered.

Values:

0 : TA buffer of unsolicited result codes defined within this command is cleared when <mode> 1 or 2 is entered.

1 : TA buffer of unsolicited result codes defined within this command is flushed to the TE when <mode> 1 or 2 is entered (OK response shall be given before flushing the codes)

Additional info:

▶▶ The following unsolicited result codes and the corresponding events are defined:

+CGEV: NW DEACT <p_cid>, <cid>
The network has deactivated a context. The <cid> for this context is provided to the TE in addition to the associated primary <p_cid>.

+CGEV: ME DEACT <p_cid>, <cid>
The mobile termination has forced a context deactivation. The

associated <cid> is provided to the TE in addition to the PDN connection associated <p_cid>.

+CGEV: ME ACT <p_cid>, <cid>
The network has responded to an ME initiated Traffic Flow activation request with an EPS bearer activation or modification. The associated MT allocated context identifier <cid> is provided to the TE in addition to the PDN connection associated <p_cid>.

+CGEV: NW ACT <p_cid>, <cid>
The network has activated a context. The <cid> for this context is provided to the TE in addition to the associated primary <p_cid>.

+CGEV: NW DETACH
The network has forced a PS detach. This implies that all active contexts have been deactivated. These are not reported separately.

+CGEV: ME DETACH
The mobile termination has forced a PS detach. This implies that all active contexts have been deactivated. These are not reported separately.

+CGEV: NW PDN ACT <cid>
The network has activated a context. The context represents a Primary PDP context in GSM/UMTS. The <cid> for this context is provided to the TE.

+CGEV: ME PDN ACT <cid>[,<reason>[,<cid_other>]]
The mobile termination has activated a PDN connection.
For <reason> and <cid_other> parameters see below.

+CGEV: NW PDN DEACT <cid>
The network has deactivated a context. The context represents a PDN connection in LTE or a Primary PDP context in GSM/UMTS. The associated <cid> for this context is provided to theTE.

+CGEV: ME PDN DEACT <cid>
The mobile termination has deactivated a PDN connection. The associated <cid> is provided to the TE.

+CGEV: EPS ACT <cid>

The network has activated a PDP context. The associated <cid> is provided to the TE.

+CGEV: EPS MODIFY <cid>, <change_reason>
The network has modified EPS bearer context parameter(s). The associated <cid> is provided to the TE in addition of the change reason: TFT and/or QoS modification. For <change_reason> parameter see below

+CGEV: NW REATTACH
Internal definition. Used to notify IMS the NW reattach action

Name	Type	Default	Description
<reason>	integer	N/A	indicates the reason why the context activation request for PDP type IPv4v6 was not granted. This parameter is only included if the requested PDP type associated with <cid> is IPv4v6, and the PDP type assigned by the network for <cid> is either IPv4 or IPv6.

Values:

0 : IPv4 only allowed

1 : IPv6 only allowed

			2 : single address bearers only allowed 3 : single address bearers only allowed and MT initiated context activation for a second address type bearer was not successful
<cid_other>	integer	-	indicates the context identifier allocated by MT for an MT initiated context of a second address type. MT shall only include this parameter if <reason> parameter indicates single address bearers only allowed, and MT supports MTinitiated context activation of a second address type without additional commands from TE, and MT has activated the PDN connection or PDP context associated with <cid_other>.
<change_reason>	integer	N/A	indicates what kind of change occurred Values: 1 : TFT only changed

2 : QoS only changed

3 : Both TFT and QoS changed

Unsolicited fields:

Name	Type	Description
<PDP_type>	string	Packet Data Protocol type, which specifies the type of packet data protocol
<PDP_addr>	string	identifies the terminal in the address space applicable to the PDP
<cid>	string	PDP Context Identifier
<p_cid>	string	Primary PDP Context Identifier

AT+CGEREP?

Read command returns the current <mode> and <bfr> settings, in the format:

+CGEREP: <mode>,<bfr>

AT+CGEREP=?

Test command reports the supported range of values for the +CGEREP command parameters.

AT+CGREG - GPRS Network Registration Status

The command enables/disables the GPRS network registration unsolicited result code (URC) and selects its presentation format.

3GPP TS 27.007

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CGREG=[<n>]

Set command enables/disables the GPRS network registration unsolicited result code and selects one of the two available formats:

short format: +CGREG:<stat>
long format: +CGREG:<stat>[,<lac_tac>,<ci>[,<AcT>,<rac_mme_code>]]

Parameter:

Name	Type	Default	Description
<n>	integer	0	enables/disables the GPRS network registration unsolicited result code, and selects one of the two formats. The URC short format is displayed every time there is a change in the GPRS network registration status. The URC long format is displayed every time there is a change of the network cell.

Values:

0 : disable the network GPRS registration unsolicited result code

1 : enable the network GPRS registration unsolicited result code, and selects the short format

2 : enable the network GPRS registration unsolicited result code, and selects the long format (includes the network cell identification data)

Unsolicited fields:

Name	Type	Description
<stat>	integer	registration status of the module Values: 0 : not registered, terminal is not currently searching a new operator to register to 1 : registered, home network 2 : not registered, but terminal is currently searching a new operator to register to 3 : registration denied 4 : unknown 5 : registered, roaming

<lac_tac>	string	location area code (2G, 3G) or tracking area code (4G) in hexadecimal format (e.g. "00C3" equals 195 in decimal)
<ci>	string	cell ID in hexadecimal format
<AcT>	integer	access technology of the serving cell Values: 0 : GSM 2 : UTRAN 3 : GSM w/EGPRS 4 : UTRAN w/HSDPA 5 : UTRAN w/HSUPA 6 : UTRAN w/HSDPA and HSUPA 7 : E-UTRAN (not applicable) 8 : EC-GSM-IoT (A/Gb mode) 9 : E-UTRAN (NB-S1 mode) (not applicable)
<rac_mme_code>	string	one byte routing area code (2G, 3G) or Mobile Management Entity code (4G) in hexadecimal format

AT+CGREG?

Read command returns the status of result code presentation mode <n> and the integer <stat> which shows whether the network has currently indicated the registration of the terminal in the format:

+CGREG : <n>,<stat>[,<lac_tac>,<ci>,<Act>,<rac_mme_code>]

<lac_tac>,<ci>,<Act> and <rac_mme_code> are reported only if <n>=2 and the mobile is registered on some network cell.

AT+CGREG=?

Test command returns supported values for parameter <n>.

AT+CGATT - PS Attach or Detach

This execution command is used to register (attach) the terminal to or deregister (detach) the terminal from the Packet Domain service.

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SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+CGATT=<state>

Parameter:

Name	Type	Default	Description
<state>	integer	N/A	state of PS attachment

Values:

0 : detached

1 : attached

AT+CGATT?

Read command returns the current PS state in the format:

+CGATT: <state>

AT+CGATT=?

Test command returns the values range of the <state> parameter.

```
AT+CGATT?
+CGATT: 0
OK

AT+CGATT=?
+CGATT: (0-1)
OK

AT+CGATT=1
OK
```

IPEasy

AT#SGACT - Context Activation

This command enables/disables the context activation.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SGACT=<cid>,<stat>[,<userId>,<pwd>]

Execution command is used to activate the specified PDP context, followed by binding data application to the PS network. Also, it is used to deactivate the PDP context and unbind data application from PS network.

Parameters:

Name	Type	Default	Description
<cid>	integer	N/A	PDP context identifier PDP context identifier numeric parameter which specifies a particular PDP context definition The value of max is returned by the Test command.
	Value:		
	1÷max	:	specifies a particular PDP context definition (see +CGDCONT command)
<stat>	integer	N/A	activate/deactivate the PDP context specified
	Values:		
	0	:	deactivate the context

	1	:	activate the context

<userId>	string	-	userId, used only if the context requires it
<pwd>	string	-	password, used only if the context requires it

Additional info:

- Execution command returns a list of IP addresses for the specified context identifiers in the format:

  If IP or IPV6 PDP context:

  #SGACT: <ipAddr>

  For DUAL STACK IPV4V6 PDP context:

  #SGACT: [<ipAddrV4>],[<ipAddrV6>]

Name	Type	Default	Description
<ipAddr>	string	-	ip address ipv4 or ipv6
<ipAddrV4>	string	-	ip address ipv4 (if v4 PDP context activated)
<ipAddrV6>	string	-	ip address ipv6 (if v6 PDP context activated)

- Context activation/deactivation returns ERROR if there is not any socket associated to it (see AT#SCFG).

- In LTE network, default PDP context (<cid> = 1) is activated by piggybacking on LTE attach procedure and maintained until detached from NW. This command with <cid> = 1 is just binding or unbinding application to the default PDP context.

- If the unsolicited result code for obtaining IP address was enabled (urcmode value) using #SGACTCFG command, on start-up and due to USB enumeration timing the unsolicited may not appear, user should manually use +CGPADDR command to see the IP address.
- PDP authentication is changed by #SGACTAUTH and #USERID and #PASSW when activated using #SGACT and it is synchronized with the #PDPAUTH and you cannot sync when you are deactivated.
- PDP context deactivation returns an ERROR if the socket session of applications such as Socket, FTP and etc. is active.

AT#SGACT?

Read command returns the state of all the contexts that have been defined in the format:

#SGACT: <cid$_1$>,<stat$_1$><CR><LF>

...

#SGACT: <cid$_{max}$>,<stat$_{max}$>

Additional info:

- parameters meaning

Name	Type	Default	Description
<cidn>	integer	-	as <cid> in Set section
<statn>	integer	N/A	context status

Values:

0	:	context deactivated
1	:	context activated

AT#SGACT=?

Test command reports the range for the parameters <cid> and <stat>.

AT#SGACTCFG - PDP Automatic Context Activation-Reactivation

This command configures the automatic activation/reactivation of the specified PDP context

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SGACTCFG=<cid>,<retry>[,<delay>[,<urcmode>]]

Set command is used to enable or disable the automatic activation/reactivation of the context for the specified PDP context, to set the maximum number of attempts and to set the delay between an attempt and the next one. The context is activated automatically after every GPRS Attach or after a NW PDP CONTEXT deactivation if at least one IPEasy socket is configured to this context (see AT#SCFG).

Parameters:

Name	Type	Default	Description
<cid>	integer	N/A	numeric parameter which specifies a particular PDP context definition. The value of max is returned by the Test command
	Value:		
	1÷max	:	specifies a particular PDP context definition
<retry>	integer	0	specifies the maximum number of context activation attempts in case of activation failure
	Values:		

0	:	disable the automatic activation/reactivation of the context
1÷15	:	supported range

<delay>	integer	N/A	specifies the delay in seconds between an attempt and the next one

Value:

180÷3600	:	supported range

<urcmode>	integer	0	URC presentation mode.

Values:

0	:	disable unsolicited result code
1	:	enable unsolicited result code, after an automatic activation/reactivation, of the local IP address obtained from the network. It has meaning only if <auto>=1

Additional info:

▶▶ The unsolicited message is in the format:

If IP or IPV6 PDP context:

#SGACT: <ip_address>

For DUAL STACK IPV4V6 PDP context:

#SGACT: [<ipAddrV4>],[<ipAddrV6>]

Reporting the local IP address obtained from the network.

- The URC presentation mode <urcmode> is related to the current AT instance only. Last <urcmode> setting is saved for every instance as extended profile parameter, thus it is possible to restore it even if the multiplexer control channel is released and set up, back and forth.
- < retry > and <delay> setting are global parameters saved in NVM.
- If the automatic activation is enabled on a context, then it is not allowed to modify by the command #SCFG the association between the context itself and the socket connection identifier; all the other parameters of command #SCFG are modifiable while the socket is not connected.
- The context is also activated automatically after successful AT#SGACTCFG set command.

AT#SGACTCFG?

Read command reports the states of all configured PDP contexts, in the format:

#SGACTCFG: <cid$_1$>,<retry$_1$>,<delay$_1$>, <urcmode>CR><LF>

...

#SGACTCFG: <cid$_n$>,<retry$_n$>,<delay$_n$>,<urcmode>

Additional info:

- parameters meaning.

Name	Type	Default	Description
<cidn>	integer	-	PDP context identifier
<retryn>	integer	-	specifies the maximum number of context activation attempts in case of activation failure

<delayn>	integer	-	specifies the delay in seconds between an attempt and the next one
<urcmode>	integer	-	URC presentation mode

AT#SGACTCFG=?

Test command reports supported range of values for parameters <cid>, <retry>, <delay> and <urcmode>.

AT#SCFG - Socket Configuration

This command sets the configuration for the socket.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SCFG=<connId>,<cid>,<pktSz>,<maxTo>,<connTo>,<txTo>

Set command sets the socket configuration parameters.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	Socket connection identifier. Value: 1÷6 : Socket connection identifier value
<cid>	integer	N/A	specifies a particular PDP context definition Value: 1÷max : The value of max is returned by the Test command
<pktSz>	integer	N/A	Packet size in bytes to be used by the TCP/UDP/IP stack for data sending. Used for online data mode only. Values:

0	:	automatically chosen by the device
1÷1500	:	packet size in bytes

<maxTo>	integer	90	Exchange timeout in seconds (or socket inactivity timeout); if there's no data exchange within this timeout period the connection is closed.

Values:

0	:	no timeout
1÷65535	:	timeout

<connTo>	integer	600	Connection timeout in tenths of seconds. If we cannot establish a connection to the remote within this timeout period, an error is raised.

Value:

10÷1200	:	timeout value in hundreds of milliseconds

<txTo>	integer	50	data sending timeout; data are sent even if they are less than max packet size, after this period. Used for online data mode only.

Values:

0	:	no timeout

1÷255	:	timeout in tenths of seconds
256	:	timeout value of 10 ms
257	:	timeout value of 20 ms
258	:	timeout value of 30 ms
259	:	timeout value of 40 ms
260	:	timeout value of 50 ms
261	:	timeout value of 60 ms
262	:	timeout value of 70 ms
263	:	timeout value of 80 ms
264	:	timeout value of 90 ms

These values are automatically saved in NVM.

AT#SCFG?

Read command returns the current socket configuration parameters values for all the six sockets, in the format:

#SCFG: <connId1>,<cid1>,<pktsz1>,<maxTo1>,<connTo1>,<txTo1><CR><LF>
#SCFG: <connId2>,<cid2>,<pktsz2>,<maxTo2>,<connTo2>,<txTo2><CR><LF>
...
#SCFG: <connId6>,<cid6>,<pktsz6>,<maxTo6>,<connTo6>,<txTo6>

AT#SCFG=?

Test command returns the range of supported values for all the parameters.

Example of test command.

- AT#SCFG?
  #SCFG: 1,1,300,90,600,50
  #SCFG: 2,2,300,90,600,50
  #SCFG: 3,2,250,90,600,50
  #SCFG: 4,1,300,90,600,50
  #SCFG: 5,1,300,90,600,50
  #SCFG: 6,1,300,90,600,50
  OK

AT#SCFGEXT - Socket Configuration Extended

This command sets the socket configuration extended parameters.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SCFGEXT=<connId>,<srMode>,<recvDataMode>,<keepalive>[,<ListenAutoRsp>[,<sendDataMode>]]

Set command sets the socket configuration extended parameters.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷6 : socket connection identifier		
<srMode>	integer	0	SRING unsolicited mode, see Additional info section.
	Values:		
	0 : Normal		
	1 : Data amount		
	2 : Data view		
	3 : Data view with UDP datagram information		

<recvDataMode>	integer	0	Data view mode for received data in command mode (#SRECV or <srMode> = 2)

Values:

0	:	Text mode
1	:	Hexadecimal mode

<keepalive>	integer	0	Set the TCP Keepalive value in minutes

Values:

0	:	deactivated
1÷240	:	keepalive time in minutes

<ListenAutoRsp>	integer	0	set the listen auto-response mode, that affects the commands #SL and #SLUDP

Values:

0	:	deactivated
1	:	activated

<sendDataMode>	integer	0	Data mode for sending data in command mode (#SSEND)

Values:

0	:	data represented as text
1	:	data represented as sequence of hexadecimal numbers (from 00 to FF). Each octet of the data is given as two IRA character long.

Additional info:

▶▶ These are the SRING formats, depending on <srMode> setting:

if <srMode> = 0 (Normal):

SRING: <connId>

if <srMode> = 1 (Data amount):

SRING: <connId>,<recData>

if <srMode> = 2 (Data view):

SRING: <connId>,<recData>,<data>

if <srMode> = 3 (Data view with UDP datagram information):

SRING: <sourceIP>,<sourcePort>,<connId>,<recData>,<dataLeft>,<data>

Name	Type	Default	Description
<recData>	integer	-	amount of data received on the socket connection number <connId>
<data>	mixed	-	data received displayed following <recvDataMode> value
<sourceIP>	string	-	IP address of the source of data

<sourcePort>	string	-	IP port of the source of data
<dataLeft>	integer	-	number of bytes left in the UDP datagram

- KeepAlive Interval - Interval between two successive keepalive retransmissions, if acknowledgement to the previous keepalive transmission is not received.

  Non configurable value: 75 sec.

  KeepAlive Probes - The number of unacknowledged retransmissions to send out before closing socket.

  Non configurable value: 9 retransmissions.

- These values are automatically saved in NVM

- For the behavior of #SL and #SLUDP in case of auto-response mode or in case of no auto-response mode, see the description of the two commands.

- <sendDataMode> parameter has effect on #SSEND and #SSENDUDP

AT#SCFGEXT?

Read command returns the current socket extended configuration parameters values for all the six sockets, in the format:

#SCFGEXT: <connId$_1$>,
<srMode$_1$>,<recvDataMode$_1$>,<keepalive$_1$>,<ListenAutoRsp$_1$>,<sendDataMode$_1$><CR><LF>

. . .

#SCFGEXT: <connId$_6$>,
<srMode$_6$>,<recvDataMode$_6$>,<keepalive$_6$>,<ListenAutoRsp$_6$>,<sendDataMode$_6$>

AT#SCFGEXT=?

Test command returns the range of supported values for all the sub parameters.

Socket 1 set with data view sring, text data mode, a keepalive time of 30 minutes and listen auto-response set.

Socket 3 set with data amount sring, hex recv data mode, no keepalive and listen auto-response not set.

Socket 4 set with hex recv and send data mode

```
AT#SCFGEXT?

#SCFGEXT: 1,2,0,30,1,0

#SCFGEXT: 2,0,0,0,0,0

#SCFGEXT: 3,1,1,0,0,0

#SCFGEXT: 4,0,1,0,0,1

#SCFGEXT: 5,0,0,0,0,0

#SCFGEXT: 6,0,0,0,0,0

OK
```

AT#SCFGEXT2 - Socket Configuration Extended 2

Socket Configuration Extended.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SCFGEXT2=<connId>,<bufferStart>[,<abortConnAttempt>[,<unused_B>[,<unused_C>[,<noCarrierMode>]]]]

Set command sets the socket configuration extended parameters for features not included in #SCFGEXT command.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier Value: 1÷6 : socket connection identifier value
<bufferStart>	integer	0	Set the sending timeout method based on new data received from the serial port (<txTo> timeout value is set by #SCFG command). Restart of transmission timer done when new data received from the serial port. It is necessary to avoid overlapping of the two methods. Enabling new method, the old method for transmission timer (#SCFG) automatically disabled to avoid overlapping.

			Check if new data have been received from serial port is done with a granularity that is directly related to #SCFG<txTo> setting with a maximum period of 1 sec
	Values:		
	0 :	old behaviour for transmission timer (#SCFG command 6th parameter old behaviour, start only first time if new data are received from the serial port)	
	1 :	new behaviour for transmission timer: Restart when new data received from serial port	
<abortConnAttempt>	integer	0	Enable connection attempt (#SD / #SKTD) abort before CONNECT (online mode) or OK (command mode). Values automatically saved in NVM
	Values:		
	0 :	not possible to interrupt connection attempt	
	1 :	it is possible to interrupt the connection attempt (<connTo> set by #SCFG or DNS resolution running if required) and give back control to AT interface by reception of a character. As soon as the control given to the AT interface, the ERROR message will be received on the interface itself.	
<unused_B>	integer	-	reserved for future use

<unused_C>	integer	-	reserved for future use
<noCarrierMode>	integer	0	select the NO CARRIER message format received when the socket is closed.

Values:

0 : no additional information is attached to NO CARRIER message

1 : NO CARRIER: <connId> message

2 : NO CARRIER: <connId>, <cause> message. Refer to Additional info section

Additional info:

▶▶ <noCarrierMode>=2 selects the following NO CARRIER message format:

NO CARRIER: <connId>, <cause>

For possible <cause> values, see also #SLASTCLOSURE

Like #SLASTCLOSURE, in case of subsequent consecutive closure causes received, the original disconnection cause indicated.

In the case of command mode connection and remote closure with subsequent inactivity timeout closure without retrieval of all available data (#SRECV or SRING mode 2), it is indicated cause 1 for both possible FIN and RST from remote. parameters values for all the six sockets, in the format:

#SCFGEXT2:<connId1>,<bufferStart1>,<abortConnAttempt>,0,0, 0 <CR><LF>
...
#SCFGEXT2:<connId1>,<bufferStart1>,<abortConnAttempt>,0,0, 0

- Is necessary to avoid overlapping of the two methods. Enabling new method, the old method for transmission timer (#SCFG) is automatically disabled to avoid overlapping.
- Check if new data have been received from serial port is done with a granularity directly related to <txTo> parameter which is set by #SCFG command. The maximum period is 1 sec.
- Like #SLASTCLOSURE, in case of subsequent consecutive closure causes received, the original disconnection cause indicated.
- In the case of command mode connection and remote closure with subsequent inactivity timeout closure without retrieval of all available data (#SRECV or SRING mode 2), it is indicated cause 1 for both possible FIN and RST from remote.

AT#SCFGEXT2=?

Test command returns the range of supported values for all parameters.

```
AT#SCFGEXT2=1,1
OK
AT#SCFGEXT2=2,1
OK
AT#SCFGEXT2?
#SCFGEXT2: 1,1,0,0,0,0
#SCFGEXT2: 2,1,0,0,0,0
#SCFGEXT2: 3,0,0,0,0,0
#SCFGEXT2: 4,0,0,0,0,0
#SCFGEXT2: 5,0,0,0,0,0
#SCFGEXT2: 6,0,0,0,0,0
OK
AT#SCFG?
```

#SCFG: 1,1,300,90,600,50

#SCFG: 2,1,300,90,600,50

#SCFG: 3,1,300,90,600,50

#SCFG: 4,2,300,90,600,50

#SCFG: 5,2,300,90,600,50

#SCFG: 6,2,300,90,600,50

OK

AT#SCFG=1,1,300,90,600,30

OK

Current configuration: socket with connId 1 and 2 are configured with new transmission timer behaviour.

<txTo> corresponding value has been changed (#SCFG) for connId 1, for connId 2 has been left to default value.

AT#SCFGEXT3 - Socket Configuration Extended 3

This command sets the socket configuration extended parameters for features not included in #SCFGEXT command nor in #SCFGEXT2 command.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

- AT#SCFGEXT3=<connId>,<immRsp>[,<closureTypeCmdModeEnabling>[,<fastSRING>[,<lingerTime>[,<UDPSocketMode>[,<ssendTimeout>]]]]]

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier Value: 1÷6 : socket connection identifier value
<immRsp>	integer	0	enables #SD command mode immediate response Values: 0 : #SD in command mode (see #SD) returns after the socket is connected

		1 : #SD in command mode returns immediately; then the state of the connection can be read by the AT command #SS	
<closureTypeCmdModeEnabling>	integer	0	it has no effect and is included only for backward compatibility
	Values:		
	0 : disabled		
	1 : enabled		
<fastSRING>	integer	0	it has no effect and is included only for backward compatibility
	Values:		
	0 : disabled		
	1 : enabled		
<lingerTime>	integer	1	Defines the time (in seconds) that the connection will not return until all queued messages for the socket have been successfully sent or the linger timeout has been reached.

Value:

1÷120	:	linger time in seconds

<UDPSocketMode>	integer	1	Defines the socket mode of operation WRT the remote socket. Parameter is saved in NVM

Values:

1	:	DL will accept any remote IP and PORT, UL for specified IP and port only.
2	:	DL will accept any remote IP and PORT, UL changes the remote IP and PORT to the last used (received/ sent) IP and port.
3	:	Connected mode– DL and UL for specified IP and port only.

<ssendTimeout>	integer	0	Timeout for #SSEND. If data sending fails data sending is retried until the timer has expired. If <ssendTimeout> = 0, and data sending fails, data sending is retried until it is successful.

If <ssendTimeout> = 1, and data sending fails, data sending is not retried.

Values:

0	:	no timeout
1	:	immediate exit (error) in case of stack temporary suspended internally
100÷600	:	timeout value in hundreds of milliseconds

- AT#SCFGEXT3?

  Read command returns the current socket extended configuration parameters values for all the six sockets, in the format:

  #SCFGEXT3: <connId1>, <immRsp1>,<closureTypeCmdModeEnabling1>,

  <fastsring1>,<lingerTime1>,<UDPSocketMode1>,<ssendTimeout1><CR><LF>

  ..

  #SCFGEXT3: <connId6>, <immRsp6>,<closureTypeCmdModeEnabling6>,

  <fastsring6>,<lingerTime6>,<UDPSocketMode6>,<ssendTimeoutcon6><CR><LF>

- AT#SCFGEXT3=?

  Test command returns the range of supported values for all the parameters.

AT#SD - Socket Dial

Execution command opens a remote connection via socket.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SD=<connId>,<txProt>,<rPort>,<IPaddr>[,<closureType>[,<lPort>[,<connMode>[,<txTime>[,<userIpType>]]]]]

Execution command opens a remote connection via socket.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	Socket connection identifier.
	Value:		
	1÷6 : socket connection identifier value		
<txProt>	integer	N/A	Transmission protocol.
	Values:		
	0 : TCP		
	1 : UDP		
<rPort>	integer	N/A	Remote host port to contact.
	Value:		

			1÷65535 : remote host port number
<IPaddr>	string	-	IP address of the remote host. This parameter can be either: - any valid IP address in the format: "xxx.xxx.xxx.xxx" - any host name to be solved with a DNS query - any valid IPv6 address in the format: xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx or xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx. xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx
<closureType>	integer	0	Socket closure behavior only for TCP when remote host has closed. The parameter has no effects for UDP connections. Values: 0 : local host closes immediately when remote host has closed 255 : local host closes after an escape sequence (+++)
<lPort>	integer	N/A	UDP connections local port only for UDP connections. The parameter has no effects for TCP connections. Value:

	1÷65535	:	UDP local port number

<connMode>	integer	0	Connection mode.

Values:

0	:	online mode connection
1	:	command mode connection

<txTime>	integer	0	adjusting a time interval for series of UDP data packets will be uploaded.

Values:

0	:	Time interval is not requested
1÷1000	:	Time interval in milliseconds

<userIpType>	integer	0	ip type for socket to open

Values:

0	:	no ip type chosen
1	:	ipv4
2	:	ipv6

- <userIpType> parameter is only valid when <IPaddr> is domain name and dual stack connection is open by #SGACT.

  When <userIpType> is "no ip type chosen" ipv6 will be requested firstly. When ipv6 DNS server does not support so ipv4 will be requested.

- <closureType> parameter is valid for TCP connections only and has no effect (if used) for UDP connections.

- <lPort> parameter is valid for UDP connections only and has no effect (if used) for TCP connections

- If we set <connMode> to online mode connection and the command is successful we enter in online data mode and we see the intermediate result code CONNECT. After the CONNECT we can suspend the direct interface to the socket connection (N.B. the socket stays open) using the escape sequence (+++): the module moves back to command mode and we receive the final result code OK after the suspension.

  After such a suspension, it is possible to resume it in every moment (unless the socket inactivity timer timeouts, see #SCFG) by using the #SO command with the corresponding <connId>.

- If we set <connMode> to command mode connection and the command is successful, the socket is opened, and we remain in command mode and we see the result code OK.

- If there are input data arrived through a connected socket and not yet read because the module entered command mode before reading them (after an escape sequence or after #SD has been issued with <connMode> set to command mode connection), these data are buffered and we receive the SRING URC (SRING presentation format depends on the last #SCFGEXT setting); it is possible to read these data afterwards issuing #SRECV. Under the same hypotheses it is possible to send data while in command mode issuing #SSEND.

- <txTime> parameter is valid for UDP connections only and has no effect (if used) for TCP connections. For slow servers it is recommended to adjust the time interval for uploading series of data packets in order to do not lose data. The following data packet will be sent after the previous data packet's time interval has been expired.

If we set <connMode> to online mode connection and the command is successful we enter in online data mode and we see the intermediate result code CONNECT. After the CONNECT we can open additional online mode connection only after suspending the first socket connection (socket stays open) using the escape sequence (+++).

AT#SD=?

Test command reports the range of values for all the parameters.

Examples of socket dial in online and command mode.

- Open socket 1 in online mode
  AT#SD=1,0,80,"www.google.com",0,0,0

  CONNECT
  ...

  ...
  Open socket 1 in command mode
  AT#SD=1,0,80,"www.google.com",0,0,1
  OK

AT#SO - Socket Restore

Execution command resumes the direct interface to a socket connection which has been suspended by the escape sequence.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SO=<connId>

Parameter:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier

Value:

1÷conMax	:	socket connection identifier. conMax value is returned by test command

AT#SO=?

Test command reports the range of values for <connId> parameter

AT#SH - Socket Shutdown

This command is used to close a socket.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SH=<connId>

Parameter:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier to be closed

Value:

1÷6 : socket connection identifier

A socket connection can be closed only when it is in suspended mode (with pending data too). Trying to close an active socket connection will produce an error.

AT#SH=?

Test command reports the range for parameter <connId>

AT#SL - Socket Listen

This command opens/closes socket listening.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SL=<connId>,<listenState>,<listenPort>[,<lingerT>]

Set command opens/closes a socket listening for an incoming TCP connection on a specified port.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷6	:	socket connection identifier
<listenState>	integer	N/A	listening action
	Values:		
	0	:	close socket listening
	1	:	start socket listening
<listenPort>	integer	N/A	local listening port
	Value:		

		1÷65535	: local listening port value
<lingerT>	integer	N/A	linger time

Values:

0	:	immediate closure after remote closure
255	:	local host closes only after an escape sequence (+++)

If successful, command returns a final result code OK. If the ListenAutoRsp flag has not been set through the command #SCFGEXT (for the specific <connId>), then, when a TCP connection request comes on the input port, if the sender is not filtered by internal firewall (see #FRWL), an URC is received:

+SRING : <connId>

Afterwards we can use #SA to accept the connection or #SH to refuse it.

If the ListenAutoRsp flag has been set, then, when a TCP connection request comes on the input port, if the sender is not filtered by the internal firewall (see #FRWL), the connection is automatically accepted: the CONNECT indication is given and the modem goes into online data mode.

If the socket is closed by the network the following URC is received:

#SL: ABORTED

AT#SL?

Read command returns all the actual listening TCP sockets.

AT#SL=?

Test command returns the range of supported values of the parameters.

Open a socket listening for TCP on port 3500.

```
AT#SL=1,1,3500
OK
```

AT#SLUDP - Socket Listen UDP

This command opens/closes a socket listening for an incoming UDP connection on a specified port.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT#SLUDP=<connId>,<listenState>,<listenPort>

Execution command opens/closes a socket listening for an incoming UDP connection on a specified port.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷6	:	socket connection identifier
<listenState>	integer	0	indicates the action that will be performed
	Values:		
	0	:	closes socket listening
	1	:	starts socket listening
<listenPort>	integer	1	local listening port
	Value:		

1÷65535	:	available port numbers

If the ListenAutoRsp flag has not been set through the command #SCFGEXT (for the specific connId), then, when an UDP connection request comes on the input port, if the sender is not filtered by internal firewall (see #FRWL), an URC is received:

+SRING : <connId>

Afterwards we can use #SA to accept the connection or #SH to refuse it.

If the ListenAutoRsp flag has been set, then, when an UDP connection request comes on the input port, if the sender is not filtered by the internal firewall (see command #FRWL), the connection is automatically accepted: the CONNECT indication is given and the modem goes into online data mode.

If the socket is closed by the network the following URC is received:

#SLUDP: ABORTED

When closing the listening socket <listenPort> is a don't care parameter

AT#SLUDP?

Read command returns all the actual listening UDP sockets.

AT#SLUDP=?

Test command returns the range of supported values for all the sub parameters.

Next command opens a socket listening for UDP on port 3500.

```
AT#SLUDP=1,1,3500
OK
```

AT#SA - Socket Accept

Execution command accepts an incoming socket connection.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SA=<connId>[,<connMode>]

Execution command accepts an incoming socket connection after an URC

SRING: <connId>

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	Socket connection identifier. Value: 1÷6 : Socket connection identifier value
<connMode>	integer	0	Connection mode, as for command #SD. Values: 0 : online mode connection 1 : command mode connection

The SRING URC has to be a consequence of a #SL issue.

AT#SA=?

Test command reports the range of values for all the parameters.

AT#SSEND - Send Data in Command Mode

This command is used to send data through a connected socket.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSEND=<connId>

Execution command permits, while the module is in command mode, to send data through a connected socket.

After entering AT#SSEND=... command, and terminated the command line with <CR>, the module returns the following four characters sequence prompt, and waits for data to send:

<CR><LF><greater_than><space> (see IRA character set: 13, 10, 62, 32)

To send the entered data, enter Ctrl-Z char (0x1A hex); to abort the operation enter ESC char (0x1B hex).

If data are successfully sent, the command returns OK. If data sending fails for some reason, an error code is reported.

Parameter:

Name	Type	Default	Description
<connId>	integer	N/A	Selection on which Socket connection identifier send data. Value: 1÷6 : Socket connection identifier supported

The maximum number of bytes to send is 1500 bytes

- It is possible to use #SSEND only if the connection was opened by #SD, else the ME is raising an error.
- A byte corresponding to BS char (0x08) is treated with its corresponding meaning; therefore previous byte will be cancelled (and BS char itself will not be sent).

AT#SSEND=?

Test command returns the range of supported values for parameter <connId>

Send data through socket number 2

```
AT#SSEND=2
>Test<CTRL-Z>
OK
```

AT#SSENDEXT - Send Data in Command Mode extended

This command allows to send data through a connected socket including all possible octets (from 0x00 to 0xFF).

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSENDEXT=<connId>,<bytesToSend>

Execution command permits, while the module is in command mode, to send data through a connected socket including all possible octets (from 0x00 to 0xFF).

After entering AT#SSENDEXT=... command, and terminated the command line with <CR>, the module returns the following four characters sequence prompt, and waits for data to send:

<CR><LF><greater_than><space> (see IRA character set: 13, 10, 62, 32)

When <bytesToSend> bytes have been sent, operation is automatically completed.

If data are successfully sent, the command returns OK. If data sending fails for some reason, an error code is reported.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷6	:	socket connection identifier
<bytesToSend>	integer	N/A	number of bytes to be sent
	Value:		

1÷maxBytes	:	maxBytes is the maximum number of bytes that can be sent and it is reported by the test command

- It's possible to use #SSENDEXT only if the connection was opened by #SD, else the modem returns an error.
- All special characters are sent like a generic byte.

AT#SSENDEXT=?

Test command returns the range of supported values for parameters <connId> and <bytesToSend>.

Open the socket in command mode:

AT#SD=1,0,<port>,"IP address",0,0,1
OK

Enter the command specifying total number of bytes as second parameter:

AT#SSENDEXT=1,256

> ..... ; Terminal echo of bytes sent is displayed here
OK

All possible bytes (from 0x00 to 0xFF) are sent on the socket as generic bytes.

AT#SRECV - Socket Receive Data in Command Mode

This command permits the user to read data arrived through a connected socket when the module is in command mode.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SRECV=<connId>,<maxByte>[,<UDPInfo>]

Execution command permits the user to read data arrived through a connected socket but buffered and not yet read because the module entered command mode before reading them; the module is notified of these data by a SRING: URC, whose presentation format depends on the last #SCFGEXT setting.

Parameters:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷6	:	socket connection identifier.
<maxByte>	integer	N/A	max number of bytes to read
	Value:		
	1÷1500	:	max number of bytes to read
<UDPInfo>	integer	0	enables/disables the visualization of UDP datagram information.
	Values:		

0 : UDP information disabled

1 : UDP information enabled, see Additional info section.

Additional info:

If <UDPInfo> is set to 1 (AT#SRECV=<connId>,<maxBytes>,1), the command returns a message having the following format:

#SRECV: <remoteIP>,<remotePort><connId>,<recData>,<dataLeft>

Name	Type	Default	Description
<remoteIP>	string	-	remote ip address
<remotePort>	string	-	remote port address
<recData>	integer	-	received data
<dataLeft>	integer	-	remaining bytes in the datagram.

Issuing #SRECV when there is no buffered data raises an error.

AT#SRECV=?

Test command returns the range of supported values for parameters <connId> <maxByte> and <UDPInfo>.

SRING URC

(<srMode> be 0, <dataMode> be 0) telling data have just come through connected socket identified by <connId>=1 and are now buffered

SRING: 1

Read in text format the buffered data

AT#SRECV=1,15
#SRECV: 1,15
test string
OK

Or:
if the received datagram, received from <IPaddr> and <IPport> is of 60 bytes

AT#SRECV=1,15,1
#SRECV: <IPaddr>,<IPport>,1,15,45
test string
OK

SRING URC

(<srMode> be 1, <dataMode> be 1) telling 15 bytes data have just come through connected socket identified by <connId>=2 and are now buffered

SRING: 2,15

Read in hexadecimal format the buffered data

AT#SRECV=2,15
#SRECV: 2,15 737472696e67612064692074657374
OK

SRING URC

(<srMode> be 2, <dataMode> be 0) displaying (in text format) 15 bytes data that have just come through connected socket identified by <connId>=3; there is no need to issue #SRECV to read the data; no data remain in the buffer after this URC

SRING: 3,15, test string

AT#SSENDUDP - Send UDP Data to a Specific Remote Host

This command allows to send data over UDP to a specific remote host.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSENDUDP=<connId>,<remoteIP>,<remotePort>

This command allows, while the module is in command mode, to send data over UDP to a specific remote host. UDP connection has to be previously completed with a first remote host through #SLUDP / #SA. Then, if module receives data from this or another host, it is able to send data to it. Like command #SSEND, the device responds with ">" prompt and waits for the data to send.

Parameters:

Name	Type	Default	Description
<connId>	integer	1	socket connection identifier
	Value:		
	1÷6	:	identifier number
<remoteIP>	string	-	IP address of the remote host in dotted decimal notation, string type: "xxx.xxx.xxx.xxx"
<remotePort>	integer	1	remote host port
	Value:		
	1÷65535	:	host port number

- After SRING that indicates incoming UDP data and issuing #SRECV to receive data itself, through #SS is possible to check last remote host (IP/Port).
- If successive resume of the socket to online mode is performed (#SO), connection with first remote host is restored as it was before.
- The maximum number of bytes to send is 1500.

AT#SSENDUDP=?

Test command reports the supported range of values for parameters <connId>,<remoteIP> and <remotePort>.

- Starts listening on <LocPort> (previous setting of firewall through #FRWL has to be done)

```
AT#SLUDP=1,1,<LocPort>
OK

SRING: 1            UDP data from a remote host available

AT#SA=1,1
OK

SRING: 1

AT#SI=1
#SI: 1,0,0,23,0       23 bytes to read
OK

AT#SRECV=1,23
#SRECV:1,23
message from first host
OK

AT#SS=1
#SS: 1,2,<LocIP>,<LocPort>,<RemIP1>,<RemPort1>
OK
```

AT#SSENDUDP=1,<RemIP1>,<RemPort1>
>response to first host
OK

SRING: 1 UDP data from a remote host available

AT#SI=1
#SI: 1,22,23,24,0 24 bytes to read
OK

AT#SRECV=1,24
#SRECV:1,24
message from second host
OK

AT#SS=1
#SS: 1,2,<LocIP>,<LocPort>,<RemIP2>,<RemPort2>
OK

Remote host has changed, we want to send a response:

AT#SSENDUDP=1,<RemIP2>,<RemPort2>
>response to second host
OK

AT#SS - Socket Status

Execution command reports the current sockets status.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SS[=<connId>]

Set command reports the current socket status for the selected socket in the format::

#SS: <connId>,<state>,<locIP>,<locPort>,<remIP>,<remPort>

If optional parameter <connId> is not present, execution command reports the current socket status for all the sockets in the format:

#SS: <connId>,<state>,<locIP>,<locPort>,<remIP>,<remPort><CR><LF>

[<connId>,<state>,<locIP>,<locPort>,<remIP>,<remPort><CR><LF>[...]]

Parameter:

Name	Type	Default	Description
<connId>	integer	N/A	socket connection identifier
	Value:		
	1÷6	:	socket connection identifier

Additional info:

▶▶ Other parameters meaning.

Name	Type	Default	Description

<state>	integer	N/A	actual state of the socket

Values:

0	:	socket closed
1	:	socket with an active data transfer connection
2	:	socket suspended
3	:	socket suspended with pending data
4	:	socket listening
5	:	socket with an incoming connection. Waiting for the user accept or shutdown command
6	:	socket in opening process. The socket is not in Closed state but still not in Active or Suspended or Suspended with pending data state

<locIP>	string	-	IP address associated by the context activation to the socket
<locPort>	integer	-	two meanings: the listening port if we put the socket in listen mode the local port for the connection if we use the socket to connect to a remote machine

<remIP>	string	-	when we are connected to a remote machine this is the remote IP address
<remPort>	string	-	it is the port we are connected to on the remote machine

AT#SS=?

Test command returns the OK result code.

AT#SI - Socket Info

This command is used to get socket information.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SI[=<connId>]

Execution command returns information about all sockets data traffic.

Parameter:

Name	Type	Default	Description
<connId>	integer	NA	socket connection identifier.

Value:

1÷conMax	:	socket connection identifier; conMax value is returned by test command

Additional info:

- The response format is:

  #SI: <connId>,<sent>,<received>,<buff_in>,<ack_waiting>[,<cause>]

Name	Type	Default	Description
<connId>	integer	-	socket connection identifier
<sent>	integer	-	total amount (in bytes) of sent data since the last time the socket connection identified by <connId> has been opened

<received>	integer	-	total amount (in bytes) of received data since the last time the socket connection identified by <connId> has been opened
<buff_in>	integer	-	total amount (in bytes) of data just arrived through the socket connection identified by <connId> and currently buffered, not yet read
<ack_waiting>	integer	-	total amount (in bytes) of sent and not yet acknowledged data since the last time the socket connection identified by <connId> has been opened
<cause>	integer	-	socket disconnection cause 0 - not available (socket has not yet been closed) 1 - remote host TCP connection close due to FIN/END: normal remote disconnection decided by the remote application 2 - remote host TCP connection close due to RST, all other cases in which the socket is aborted without indication from peer (for instance because peer doesn't send ack after maximum number of retransmissions/ peer is no more alive). All these cases include all the "FATAL" errors after recv or send on the TCP socket (named as different from EWOULDBLOCK) 3 - socket inactivity timeout 4 - network deactivation (PDP context deactivation from network)

Not yet acknowledged data are available only for TCP connections.

The value <ack_waiting> is always 0 for UDP connections.

Issuing #SI<CR> causes getting information about data traffic of all the sockets, the response format is:

#SI: <connId1>,<sent1>,<received1>,<buff_in1>,<ack_waiting1><CR><LF> ...

#SI: <connIdconMax>,<sentconMax>,<receivedconMax>,<buff_inconMax>,<ack_waitingconMax>

AT#SI=?

Test command reports the range of <connId> parameter.

</>

- AT#SI

  #SI: 1,123,400,10,50

  #SI: 2,0,100,0,0

  #SI: 3,589,100,10,100

  #SI: 4,0,0,0,0

  #SI: 5,0,0,0,0

  #SI: 6,0,98,60,0

  ...

  OK

  Sockets 1,2,3,6 are opened with some data traffic.

  For example, socket 1 has 123 bytes sent, 400 bytes received, 10 bytes waiting to be read and 50 bytes waiting to be acknowledged from the remote side.

  AT#SI=1

  #SI: 1,123,400,10,50

  OK

  We have information only about socket number 1

AT#FRWL - Firewall Setup

This command controls the internal firewall settings.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FRWL=[<action>[,<ip_addr>[,<net_mask>]]]

Set command controls the internal firewall settings

Parameters:

Name	Type	Default	Description
<action>	integer	0	command action
	Values:		
	0 : remove selected chain		
	1 : add an ACCEPT chain		
	2 : remove all chains (DROP everything); <ip_addr> and <net_mask> have no meaning in this case.		
<ip_addr>	string	-	remote address to be added into the ACCEPT chain; it can be any valid IP address in the format: xxx.xxx.xxx.xxx
<net_mask>	string	-	mask to be applied on the <ip_addr>; it can be any valid IP address mask in the format: xxx.xxx.xxx.xxx

Additional info:

Firewall criterion

The firewall applies for incoming (listening) connections only. Its general policy is DROP, therefore all packets that are not included into an ACCEPT chain rule will be silently discarded.

When a packet comes from the IP address incoming_IP, the firewall chain rules will be scanned for matching with the following criteria:

incoming_IP & <net_mask> = <ip_addr> & <net_mask>

If criterion is matched, then the packet is accepted and the rule scan is finished; if criteria are not matched for any chain the packet is silently dropped

Default AT#FRWL is:

AT#FRWL?

#FRWL: "000.000.000.000","000.000.000.000"

OK

Default white list is:

- all IP address allowed

If firewall need to be set than first delete default white list with

AT#FRWL=0,"000.000.000.000","000.000.000.000"

or with

AT#FRWL=2

and then set the desired white list with

AT#FRWL=1,<ip_addr>,<net_mask>

in order to have the desired white list

AT#FRWL?

#FRWL: "<ip_addr>","<net_mask>"

- Only tre zero format 000 is accepted for both <ip_addr> and <net_mask>.

  Do not use single zero format 0 for both <ip_addr> and <net_mask>,

- The only accepted values for <net_mask> are:

  255.255.255.255

  255.255.255.000

  255.255.000.000

  255.000.000.000

AT#FRWL?

Read command reports the list of all ACCEPT chain rules registered in the Firewall settings in the format:

#FRWL: <ip_addr>,<net_mask>
#FRWL: <ip_addr>,<net_mask>
...
OK

AT#FRWL=?

Test command returns the allowed values for parameter <action>.

Let us assume that we want to accept connections only from our devices which are on the IP addresses ranging from:

197.158.1.1 to 197.158.255.255

We need to add the following chain to the firewall:

AT#FRWL=1,"197.158.1.1","255.255.000.000"

OK

AT#ICMP - Ping Support

Set command enables/disables the ICMP Ping support.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#ICMP=<mode>

Parameter:

Name	Type	Default	Description
<mode>	integer	1	ICMP mode selection.

Values:

0 : disable ICMP Ping support

1 : enable firewalled ICMP Ping support: the module is sending a proper ECHO_REPLY only to a subset of IP Addresses pinging it; this subset of IP Addresses has been previously specified through #FRWL command.

2 : enable free ICMP Ping support; the module is sending ECHO_REPLY to every IP Address pinging it.

AT#ICMP?

Read command returns whether the ICMP Ping support is currently enabled or not, in the format:

#ICMP: <mode>

AT#ICMP=?

Test command reports the supported range of values for the <mode> parameter.

AT#PING - Send PING Request

This command is used to send Ping Echo Request.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT#PING=<IPaddr>[,<retryNum>[,<len>[,<timeout>[,<ttl>[,<pdpId>]]]]]

Execution command is used to send Ping Echo Request messages and to receive the corresponding Echo Reply. Before sending PING Request the PDP context must be activated by #SGACT command.

Once the single Echo Reply message is received, a string like that is displayed:

#PING: <replyId>,<Ip Address>,<replyTime>,<ttl>

Parameters:

Name	Type	Default	Description
<IPaddr>	string	-	address of the remote host, string type. This parameter can be either: - any valid IP address in the format: "xxx.xxx.xxx.xxx" - any host name to be solved with a DNS query - any valid IPV6 address (IPV6 HEX, IPV6 decimal)
<retryNum>	integer	4	the number of Ping Echo Request to send

Value:

1÷64	:	Ping Echo Request number

<len>	integer	32	the length of Ping Echo Request message

Value:

32÷1460	:	Ping Echo Request length

<timeout>	integer	50	the timeout, in 100 ms units, waiting a single Echo Reply

Value:

1÷600	:	timeout, in 100 ms units

<ttl>	integer	128	time to live

Value:

1÷255	:	time to live / Hop Limit

<pdpId>	integer	1	PDP context identifier Numeric parameter which specifies a particular PDP context definition. The value of max is returned by the Test command.

Value:

1÷max	:	PDP context identifier

Unsolicited fields:

Name	Type	Description
<replyId>	integer	Echo Reply number
<IpAddress>	string	IP address of the remote host
<replyTime>	integer	time, in 100 ms units, required to receive the response
<ttl>	integer	time to live of the Echo Reply message

- When the Echo Request timeout expires (no reply received on time) the response will contain <replyTime> set to 600 and <ttl> set to 255.
- To receive the corresponding Echo Reply it is not required to enable separately #ICMP
- The format of IPv6 address is not represented according to +CGPIAF setting.
- If <pdpId> is not used it will be tried PING by default value.

AT#PING=?

Test command reports the supported range of values for the #PING command parameters.

```
AT#PING="www.telit.com"
#PING: 01,"81.201.117.177",6,50
#PING: 02,"81.201.117.177",5,50
#PING: 03,"81.201.117.177",6,50
#PING: 04,"81.201.117.177",5,50

OK
```

AT#QDNS - Query DNS

The command executes a DNS query

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#QDNS=<host name>[,<userIpType>]

Execution command executes a DNS query to solve the host name into an IP address. If the DNS query is successful, then the IP address will be reported in the result code as follows:

#QDNS: <host name>,<IP address>

Parameters:

Name	Type	Default	Description
<host name>	string	-	Host name string
<userIpType>	integer	0	Currently unused

Values:

0	:	IPv4v6
1	:	IPv4
2	:	IPv6

Additional info:

- IP address in the result code

Name	Type	Default	Description
<IP address>	string	-	this parameter is in the format: xxx.xxx.xxx.xxx

The command activates the PDP context if it was not previously activated. In this case the context is deactivated after the DNS query.

This command requires that the authentication parameters are correctly set and the PS network is present.

AT#QDNS=?

Test command returns the supported range of values for parameter <userIpType>.

AT#DNS - Manual DNS Selection

This command manually set primary and secondary DNS servers.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#DNS=<cid>,<primary>,<secondary>

Set command allows to manually set primary and secondary DNS servers for a PDP context defined by +CGDCONT

Parameters:

Name	Type	Default	Description
<cid>	integer	N/A	numeric parameter which specifies a particular PDP context definition. The value of max is returned by the Test command Value: 1÷max : specifies a particular PDP context definition
<primary>	string	-	Ipv4- manual primary DNS server, string type, in the format "xxx.xxx.xxx.xxx" used for the specified cid; we are using this value instead of the primary DNS server come from the network (default is "0.0.0.0") Ipv6- manual primary DNS server, string type, in the format "xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx

			.xxx.xxx" used for the specified cid; we are using this value instead of the primary DNS server come from the network (default is "0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0") Ipv6 can also be in HEX format: "xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx"
<secondary>	string	-	Ipv4- manual primary DNS server, string type, in the format "xxx.xxx.xxx.xxx" used for the specified cid; we are using this value instead of the primary DNS server come from the network (default is "0.0.0.0") Ipv6- manual primary DNS server, string type, in the format "xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx .xxx.xxx" used for the specified cid; we are using this value instead of the primary DNS server come from the network (default is "0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0") Ipv6 can also be in HEX format: "xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx"

- If <primary> is "0.0.0.0" and <secondary> is not "0.0.0.0", then issuing AT#DNS=... returns an error.
- If <primary> is "0.0.0.0" we are using the primary DNS server coming from the network as consequence of a context activation.
- if <primary> is not "0.0.0.0" and <secondary> is "0.0.0.0", then we are using only the manual primary DNS server.

- The context identified by <cid> must be previously defined, otherwise issuing AT#DNS=... returns an error.
- Issuing AT#DNS=... returns an error if the context identified by <cid> has already been activated by AT commands.

AT#DNS?

Read command returns the manual DNS servers set either for every defined PDP context and for the single GSM context (only if defined), in the format:

[#DNS: <cid>,<primary>,<secondary>[<CR><LF>

#DNS: <cid>,<primary>,<secondary>]]

In case +CGDCONT determined as ipv4v6 the format is:

[#DNS: <cid>,<primary ip4>,<primary ip6>,<secondary ip4>,<secondary ip6> [<CR><LF>

#DNS: <cid>,<primary ip4>,<primary ip6>,<secondary ip4>,<secondary ip6>]]

AT#DNS=?

Test command reports the supported range of values for the <cid> parameter only, in the format:

#DNS: (1-8)

AT#NWDNS - DNS from Network

The command allows to get the primary and secondary DNS addresses for selected GSM or PDP context identifiers

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#NWDNS=[<cid>[,<cid>[..]]]

Execution command returns either the primary and secondary DNS addresses for the GSM context (if specified) and/or a list of primary and secondary DNS addresses for the specified PDP context identifiers.

The command returns a row of information for every specified <cid> whose context has been already defined. No row is returned for a <cid> whose context has not been defined yet. The response is in the form:

#NWDNS: $<cid_n>$,$<PDNSaddress_n>$,$<SDNSaddress_n>$<CR><LF>
#NWDNS: $<cid_p>$,$<PDNSaddress_p>$,$<SDNSaddress_p>$<CR><LF>
...
#NWDNS: $<cid_q>$,$<PDNSaddress_q>$,$<SDNSaddress_q>$<CR><LF>

Parameter:

Name	Type	Default	Description
<cid>	integer	N/A	numeric parameter which specifies a particular PDP context definition. The value of max is returned by the test command.
	Value:		

1÷max	:	specifies a particular PDP context definition (see +CGDCONT command)

Additional info:

▶▶ Parameters meaning:

Name	Type	Default	Description
<cid>	integer	-	context identifier, as before
<PDNSaddress>	string	-	primary DNS addresses set through AT#DNS command. If not set, they are the primary and secondary DNS addresses assigned during the PDP (or GSM) context activation.
<SDNSaddress>	string	-	secondary DNS addresses set through AT#DNS command. If not set, they are the primary and secondary DNS addresses assigned during the PDP (or GSM) context activation.

Entering AT#NWDNS= (no <cid> specified), the DNS addresses for all defined contexts are returned.

- Issuing the command with more than 6 input parameters raises an error.
- The command returns only one row of information for every specified <cid>, even if the same <cid> is present more than once.

AT#NWDNS=?

Test command returns a list of defined <cid>s.

AT#NTP - Calculate and Update Date and Time with NTP

The command handles the date and time update using NTP protocol.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#NTP=<NTPAddr>,<NTPPort>,<updModClock>,<timeout>[,<timeZone>]

Execution command permits to calculate and update date and time through NTP protocol sending a request to a NTP server. The command returns an intermediate response having the following format:

Execution command permits to calculate and update date and time through NTP protocol sending a request to a NTP server. By default, #NTP uses the PDP context identifier <cid>=1, see +CGDCON command. Make sure that <cid>=1 is not used, otherwise the command returns an ERROR.

#NTP: <time><CR><LF>OK<CR><LF>
or
<CR><LF>ERROR<CR><LF>

The <time> parameter is described in Additional info section.

Parameters:

Name	Type	Default	Description
<NTPAddr>	string	address of the NTP server. This parameter can be either: - any valid IP address in the format: "xxx.xxx.xxx.xxx"	

		- any host name to be solved with a DNS query	
<NTPPort>	integer	N/A	NTP server port to contact
	Value:		
	1÷65535	:	port
<updModClock>	integer	N/A	update mode
	Values:		
	0	:	no update module clock
	1	:	update module clock
<timeout>	integer	N/A	waiting timeout for server response in seconds
	Value:		
	1÷10	:	waiting timeout for server response in seconds
<timeZone>	string	0	Time Zone: indicates the difference, expressed in quarter of an hour,

between the local time and GMT.

Value:

47÷48	:	Time Zone

Additional info:

Intermediate response parameter:

Name	Type	Default	Description
<time>	string	-	current time as quoted string in the format: "yy/MM/dd,hh:mm:ss±zz"

The Time Zone is applied directly in the Date and Time received by the NTP Server, that is, by definition, GMT+0.

AT#NTP=?

Test command returns the maximum length for <NTPAddr> string, and supported range of values of parameters: <NTPPort>, <updModClock>, <timeout> and <timeZone>.

MAX timeout indicates the time within which the command must be executed. MAX timeout for this command is 10 s.

```
AT#NTP="ntp1.inrim.it",123,1,2,4
#NTP: 12/01/27,14:42:38+04
OK
AT+CCLK?
+CCLK: "12/01/27,14:42:39+04"
OK
```

AT#GDATAVOL - PS Data Volume

The command resets data counters or reports data counts of the PS sessions.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#GDATAVOL=[<mode>]

Execution command reports, for every active PDP context, the amount of data the last PS session received and transmitted, or it will report the total amount of data received and transmitted during all past PS sessions, since last reset.

Parameter:

Name	Type	Default	Description
<mode>	integer	0	resets or reads data counters.

Values:

0 : resets the PS data counter for all available PDP contexts (1-16)

1 : reports the last PS session data counter for the all the set PDP contexts (i.e. all the PDP contexts with APN parameter set using +CGDCONT). Refer to Additional info section to see the format. Not supported.

2 : reports the total PS data counter, since last reset, for the all the set PDP contexts (i.e. all the PDP context with APN parameter set using +CGDCONT).

Refer to Additional info section to see the format.
Not supported.

3 : reports the total PS data counter, since last reset, for the all the PDP contexts in a single line. Refer to Additional info section

Additional info:

▶▶ When <mode>=3, the command returns the following total data counts, since last reset, for the all the PDP contexts in a single line.

The message format is:

#GDATAVOL: ALL,<tot>,<sent>,<received>

Name	Type	Default	Description
<tot>	integer	-	number of bytes either received or transmitted in every PS session since last reset, for all <cid$_n$> PDP context.
<sent>	integer	-	number of bytes transmitted, in every PS session since last reset, for all <cid$_n$> PDP context
<received>	integer	-	number of bytes received, in every PS session since last reset, for all <cid$_n$> PDP context

- Last PS session counters are not saved in NVM so they are lost at power off.
- Total PS session counters are saved on NVM.

AT#GDATAVOL=?

Test command returns the supported range of the <mode> parameter values.

FTPEasy

AT#FTPCLOSE - FTP Close Command

The command purpose is to close the previously open FTP connection.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPCLOSE

Execution command closes an FTP connection.

AT#FTPCLOSE=?

Test command returns the OK result code.

AT#FTPCWD - FTP Change Working Directory

Command to change the working directory on FTP server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPCWD=[<dirname>]

Execution command, issued during an FTP connection, changes the working directory on FTP server.

Parameter:

Name	Type	Default	Description
<dirname>	string	-	Name of the new working directory.

The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

AT#FTPCWD=?

Test command returns the OK result code.

AT#FTPFSIZE - Get File Size from FTP Server

This command returns the size of a file located on a FTP server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPFSIZE=<filename>

Execution command, issued during an FTP connection, permits to get the size of a file located on a FTP server. The response format is:

#FTPFSIZE: <size>

Parameter:

Name	Type	Default	Description
<filename>	string	-	the name of the file that you want to know the size

Additional info:

Parameter:

Name	Type	Default	Description
<size>	integer	-	dimension in bytes of the file located on the FTP server

- AT#FTPTYPE=0 command must be issued before #FTPFSIZE command, to set file transfer type to binary mode.
- When file does not exist on the server, it will return an error.

AT#FTPFSIZE=?

Test command returns OK result code.

AT#FTPGET - FTP Get Command

This command executes the FTP Get function during an FTP connection.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FTPGET=[<filename>]

Execution command opens a data connection and starts getting a file from the FTP server. If the data connection succeeds a CONNECT indication is sent and the file is received on the serial port.

Parameter:

Name	Type	Default	Description
<filename>	string	-	file name to get from server.

The command causes an ERROR result code to be returned in case no FTP connection has been opened yet.

Command closure should always be handled by application. To avoid download stall situations a timeout should be implemented by the application.

AT#FTPGET=?

Test command returns the OK result code.

AT#FTPGETPKT - FTP Get in Command Mode

FTP gets in command mode.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPGETPKT=<fileName>[,<viewMode>]

Execution command, issued during an FTP connection, opens a data connection and starts getting a file from the FTP server while remaining in command mode.

The data port is opened, we remain in command mode and we see the result code OK.
Retrieval from FTP server of <fileName> is started, but data are only buffered in the module.
It is possible to read data afterwards issuing #FTPRECV command.

Parameters:

Name	Type	Default	Description
<fileName>	string	-	file name. Maximum length: 200 characters.
<viewMode>	integer	0	choose the view mode

Values:

0 : text format

1 : hexadecimal format

The command causes an ERROR result code to be returned in case no FTP connection has been opened yet.

Command closure should always be handled by application. To avoid download stall situations a timeout should be implemented by the application.

AT#FTPGETPKT?

Read command reports current download state for <fileName> with <viewMode> chosen, in the format:

#FTPGETPKT: <remotefile>,<viewMode>,<eof>

Additional info:

The following parameter signals the state of the file transmission.

Name	Type	Default	Description
<eof>	integer	N/A	End of file
	Values:		
	0 :	file currently being transferred	
	1 :	complete file has been transferred to FTP client	

AT#FTPGETPKT=?

Test command returns OK result code.

AT#FTPLIST - FTP List

This command is used during a FTP connection.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FTPLIST[=[<name]]

Execution command, issued during an FTP connection, opens a data connection and starts getting from the server the list of contents of the specified directory or the properties of the specified file

Parameter:

Name	Type	Default	Description
<name>	string	-	is the name of the directory or file

The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

Issuing AT#FTPLIST<CR> opens a data connection and starts getting from the server the list of contents of the working directory.

AT#FTPLIST=?

Test command returns the OK result code.

AT#FTPOPEN - FTP Connection Opening

This execution command opens an FTP connection toward the FTP server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPOPEN=[[<server:port>,<username>,<password>,<mode>],<cid>]

Parameters:

Name	Type	Default	Description
<server:port>	string	-	address and port of FTP server (factory default port 21)
<username>	string	-	authentication user identification for FTP
<password>	string	-	authentication password for FTP
<mode>	integer	0	active or passive mode
	Values:		
	0	:	active mode
	1	:	passive mode
<cid>	string	-	PDP context identifier

Before opening an FTP connection the PDP context (or GSM context) must have been activated by AT#SGACT=x,1 command. The context identifier "x" is the one used by FTP, as specified in #PROTOCOLCFG command.

AT#FTPOPEN=?

Test command returns the OK result code

AT#FTPPUT - FTP Send File

This command sends a file to the FTP server.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT#FTPPUT=[<filename>[,<connMode>]]

Execution command, issued during an FTP connection, opens a data connection and starts sending <filename> file to the FTP server.

Parameters:

Name	Type	Default	Description
<filename>	string	-	name of the file (maximum length 200 characters)
<connMode>	integer	0	select online mode: If online mode is selected (default) and the data connection succeeds, a CONNECT indication is sent; afterward a NO CARRIER indication is sent when the socket is closed. If command mode is selected and the data connection succeeds, we remain in command mode and we see the result code OK (instead of CONNECT).

Value:

0	:	online mode
1	:	command mode

- Use the escape sequence +++ to close the data connection.
- The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

AT#FTPPUT=?

Test command reports the maximum length of <filename> and the supported range of values of <connMode>.

Additional info:

- The format is:

  #FTPPUT: <length>, (list of supported <connMode>s)

Name	Type	Default	Description
<length>	integer	-	maximum length of <filename>

AT#FTPRECV - Receive Data in Command Mode

The command permits the user to read a given amount of data already transferred via FTP from a remote file.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#FTPRECV=<blockSize>

Execution command permits the user to transfer at most <blocksize> bytes of remote file, provided that retrieving from the FTP server has been started with a previous #FTPGETPKT command, onto the serial port.

This number is limited to the current number of bytes of the remote file which have been transferred from the FTP server.

Parameter:

Name	Type	Default	Description
<blockSize>	integer	N/A	maximum number of bytes to read
	Value:		
	1÷3000	:	maximum number of bytes to read

- It is necessary to have previously opened FTP data port and started download and buffering of remote file through #FTPGETPKT command.
- Issuing #FTPRECV when there is no FTP data port opened raises an error.
- Data port will stay opened if socket is temporary waiting to receive data (#FTPRECV returns 0 and #FTPGETPTK gives an EOF 0 indication).

AT#FTPRECV?

Read command reports the number of bytes currently transferred from FTP server in the format:

#FTPRECV: <available>

Additional info:

Parameter:

Name	Type	Default	Description
<available>	integer	-	number of transferred bytes and available for reading

AT#FTPRECV=?

Test command returns the supported values for parameter <blocksize>.

```
AT#FTPRECV?
#FTPRECV: 3000
OK
```

Read required part of the buffered data:

```
AT#FTPRECV=400
#FTPRECV:400

Text row number 1 * 111111111111111111111111111 *
Text row number 2 * 222222222222222222222222222 *
Text row number 3 * 333333333333333333333333333 *
Text row number 4 * 444444444444444444444444444 *
Text row number 5 * 555555555555555555555555555 *
Text row number 6 * 666666666666666666666666666 *
Text row number 7 * 777777777777777777777777777 *
Text row number 8 * 88888888888888888888

OK

AT#FTPRECV=200
#FTPRECV:200
```

88888 *
Text row number 9 * 99999999999999999999999999 *
Text row number 10 * AAAAAAAAAAAAAAAAAAAAAAAAAAA*
Text row number 11 * BBBBBBBBBBBBBBBBBBBBBBBBBBB *
Text row number 12 * CCCCCCCCCCCCCCCC

OK

To check when you have received complete file it is possible to use #FTPGETPKT read command:

AT#FTPGETPKT?
#FTPGETPKT:sample.txt,0,1
OK

(you will get <eof> set to 1)

AT#FTPREST - Set Restart Position for FTP GET

Set command sets the restart position for successive #FTPGET (or #FTPGETPKT) command. It permits to restart a previously interrupted FTP download from the selected position in byte.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#FTPREST=[<restartPosition>]

Parameter:

Name	Type	Default	Description
<restartPosition>	integer	-	position in byte of restarting for successive #FTPGET (or #FTPGETPKT)

- It is necessary to issue #FTPTYPE=0 before successive #FTPGET (or #FTPGETPKT command) to set binary file transfer type.
- Setting <restartPosition> takes effect on successive FTP download. After successive successfully initiated #FTPGET (or #FTPGETPKT) command <restartPosition> is automatically reset.
- Value set for <restartPosition> takes effect on next data transfer (data port opened by #FTPGET or #FTPGETPKT). Then <restartPosition> value is automatically assigned to 0 for next download.

AT#FTPREST?

Read command returns the current <restartPosition>:

#FTPREST:<restartPosition>

AT#FTPREST=?

Test command returns the OK result code.

AT#FTPTO - FTP Time Out

Set the FTP time out.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPTO=[<tout>]

Set command sets the time out used when opening either the FTP control channel or the FTP traffic channel.

Parameter:

Name	Type	Default	Description
<tout>	integer	100	time out in 100 milliseconds units

Value:

100÷5000	:	hundreds of milliseconds

AT#FTPTO?

Read command returns the current FTP operations time out in the format:

#FTPTO: <tout>

AT#FTPTO=?

Test command returns the supported values of parameter <tout>.

AT#FTPTYPE - FTP Type

This command sets the FTP file transfer type.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FTPTYPE=[<type>]

Set command, issued during a FTP connection, sets the file transfer type.

Parameter:

Name	Type	Default	Description
<type>	integer	N/A	file transfer type

Values:

0	:	binary
1	:	ASCII

The command causes an ERROR result code to be returned if no FTP connection has been opened yet.

AT#FTPTYPE?

Read command returns the current file transfer type, in the format:

#FTPTYPE: <type>

AT#FTPTYPE=?

Test command returns the range of available values for parameter <type>:

#FTPTYPE: (0,1)

AT#FTPCFG - FTP Configuration

This command sets the time-out used when opening either the FTP control channel or the FTP traffic channel.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPCFG=<tout>,<IPPignoring>[,<FTPSEn>[,<FTPext>]]

Parameters:

Name	Type	Default	Description
<tout>	integer	100	time out in 100 milliseconds units
	Value:		
	100÷5000	:	hundreds of ms
<IPPignoring>	integer	N/A	enable or disable IP private ignoring
	Values:		
	0	:	No IP Private ignoring. During a FTP passive mode connection client uses the IP address received from server, even if it is a private IPV4 address.
	1	:	IP Private ignoring enabled. During a FTP passive mode connection if the server sends a private IPV4 address the client doesn't

consider this and connects with server using the IP address used in #FTPOPEN.

<FTPSEn>	integer	0	disable FTPS security

Value:

0 : disable FTPS security: all FTP commands will perform plain FTP connections

1 : enable FTPS security

<FTPext>	integer	1	PORT/PASV and EPRT/EPSV commands. Option added to pass-through firewall that is unaware of the extended FTP commands for #FTPPUT, #FTPLIST, #FTPAPP, #FTPGET.

Values:

0 : always use EPRT and EPSV commands

1 : if both module and server ipv4 use PORT and PASV commands

If parameter <tout> is omitted the behavior of set command is the same as read command.

AT#FTPCFG?

Read command reports the currently selected parameters in the format:

#FTPCFG: <tout>,<IPPignoring>,<FTPSEn>,<FTPext>

AT#FTPCFG=?

Test command reports the supported range of values for parameter(s): <tout>, <IPPignoring>, <FTPSEn>, and <FTPext>.

AT#FTPAPP - FTP Append

This command is used to append data to an already existing file via FTP during an FTP session.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#FTPAPP=<fileName>[,<connMode>]

Set command, issued during an FTP connection, opens a data connection and appends data to existing <fileName> file.

If the data connection succeeds, a CONNECT indication is sent, afterward a NO CARRIER indication is sent when the socket is closed.

Parameters:

Name	Type	Default	Description
<fileName>	string	-	the file name
<connMode>	integer	0	the connection mode

Values:

0 : online mode

1 : command mode

If <connMode> is set to 1, the data connection is opened, the device remains in command mode and the OK result code is displayed (instead of CONNECT).

- Use the escape sequence +++ to close the data connection.
- The command causes an ERROR result code if no FTP connection has been opened yet.

AT#FTPAPP=?

Test command reports the maximum length of <fileName> and the supported range of values of <connMode>. The format is:

#FTPAPP: <length>, (list of supported <connMode>s)

Additional info:

- Parameter meaning.

Name	Type	Default	Description
<length>	integer	-	is the maximum length of <fileName>

AT#FTPAPPEXT - FTP Append Extended

The command sends data on an FTP data port while the module is in command mode.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPAPPEXT=<bytestosend>[,<eof>]

Execution command sends data on a FTP data port while the module is in command mode.

FTP data port must be previously opened by #FTPPUT (or #FTPAPP) with <connMode> parameter set to command mode connection.

Parameters:

Name	Type	Default	Description
<bytestosend>	integer	N/A	number of bytes to be sent
	Value:		
	1÷1500 : number of bytes		
<eof>	integer	0	data port closure
	Values:		
	0 : normal sending of data chunk		
	1 : close data port after sending data chunk		

Additional info:

The device responds to the command with the prompt <greater_than><space> and waits for the data to send.

When <bytestosend> bytes have been sent, operation is automatically completed. If (all or part of the) data are successfully sent, then the response is:

#FTPAPPEXT:<sentbytes>

OK

Name	Type	Default	Description
<sentbytes>	integer	-	the number of sent bytes

<sentbytes> could be less than <bytestosend>. If data sending fails for some reason, an error code is reported.

AT#FTPAPPEXT=?

Test command reports the supported range of values for parameters <bytestosend> and <eof>.

AT#FTPOPEN="IP",username,password

OK

AT#FTPPUT=<filename>,1

OK

the second param (1) means that we open the connection in command mode

Here data socket will stay opened, but interface will be available (command mode)

AT#FTPAPPEXT=Size

>... write here the binary data. As soon as Size byte are written, data are sent and OK is returned

#FTPAPPEXT:<SentBytes>

OK

.....

Last #FTPAPPEXT will close the data socket, because second (optional) parameter has this meaning:

AT#FTPAPPEXT=Size,1

>... write here the binary data. As soon Size byte are written, data are sent and OK is returned

#FTPAPPEXT:<SentBytes>

OK

If the user has to reopen the data port to send another (or append to the same) file, he can restart with the FTPPUT (or FTPAPP).

Then FTPAPPEXT, ...to send the data chunks on the reopened data port.

NOTE: if while sending the chunks the data port is closed from remote, user will be aware of it because #FTPAPPEXT will indicate ERROR and cause (available if previously issued the command AT+CMEE=2) will indicate that socket has been closed.

Also in this case obviously, data port will have to be reopened with FTPPUT and so on...(same sequence)

AT#FTPMSG - FTP Read Message

This command returns the last response received from the FTP server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#FTPMSG

Execution command returns the last response received from the server during an FTP connection.

AT#FTPMSG=?

Test command returns the OK result code.

HTTP

AT#HTTPCFG - Configure HTTP Parameters

This command sets the parameters needed to the HTTP connection.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#HTTPCFG=<prof_id>[,<server_address>[,<server_port>[,<auth_type>[,<username>[,<password>[,<ssl_enabled>[,<timeout>[,<cid>[,<UNUSED_1>[,<UNUSED_2>]]]]]]]]]]

Parameters:

Name	Type	Default	Description
<prof_id>	integer	N/A	select the profile identifier. Value: 0÷2 : Profile identifier
<server_address>	string	-	IP address of the HTTP server. This parameter can be either: - any valid IP address in the format: "xxx.xxx.xxx.xxx" - any host name to be solved with a DNS query. Default: "" for first and second profile "m2mlocate.telit.com" for third profile
<server_port>	integer	N/A	select TCP remote port. for first and second profile 80, by default for third profile 9978, by default

Value:

1÷65535	:	TCP remote port of the HTTP server to connect to

<auth_type>	integer	0	select HTTP authentication type.

Values:

0	:	no authentication
1	:	basic authentication

<username>	string	-	configure authentication user identification string for HTTP.
<password>	string	-	configure authentication password string for HTTP.
<ssl_enabled>	integer	0	enable/disable SSL encryption.

Values:

0	:	SSL encryption disabled
1	:	SSL encryption enabled

<timeout>	integer	120	timeout, data from HTTPS server.

Value:

	1÷65535	:	time interval in seconds to wait for receiving data from HTTP server.

<cid>	integer	1	PDP context identifier. Value:

	1÷max	:	the value of max is returned by the Test Command

<UNUSED_1>	integer	-	parameter for future use. For Modem, an ERROR is issued if <UNUSED_1> and <UNUSED_2> parameters are set with a value different from 0. For WLAN, <UNUSED_1> is considered as <ssl_verify>, to enable or disable CA certificate validation. <UNUSED_1> parameter can be set to 0 or 1. <UNUSED_2> parameter can be set only to 0.
<UNUSED_2>	integer	-	parameter for future use. For Modem, an ERROR is issued if <UNUSED_1> and <UNUSED_2> parameters are set with a value different from 0. For WLAN, <UNUSED_1> is considered as <ssl_verify>, to enable or disable CA certificate validation. <UNUSED_1> parameter can be set to 0 or 1.

<UNUSED_2> parameter can be set only to 0.

- A special form of the set command, AT#HTTPCFG=<prof_id>, causes the values for profile number <prof_id> to reset to default values.
- Only one profile can use the SSL encryption.
- HTTP AT command supports the connection for IPv6 Server

AT#HTTPCFG?

Read command returns the current settings for each defined profile in the format:

#HTTPCFG:<prof_id>,<server_address>,<server_port>,<auth_type>,<username>,<password>,<ssl_enabled>,<timeout>,<cid>,0,0<CR><LF>[<CR><LF>

#HTTPCFG:<prof_id>,<server_address>,<server_port>,<auth_type>,<username>,<password>,<ssl_enabled>,<timeout>,<cid>,0,0]<CR><LF>[...]]

AT#HTTPCFG=?

Test command returns the supported range of parameters <prof_id>, <server_port>, <auth_type>, <ssl_enabled>,<timeout>, and <cid> and the maximum length of<server_address>, <username> and <password> parameters in the format:

HTTPCFG: (list of supported <prof_id>s),<s_length>,(list of supported <server_port>s), (list of supported <auth_type>s),<u_length>,< p_length>,(list of supported <ssl_enabled>s),(list of supported <timeout>s),(list of supported <cid>s)

Additional info:

- Meaning of the <..._length> parameters:

Name	Type	Default	Description

<s_length>	integer	-	maximum length of parameter <server_address>
<u_length>	integer	-	maximum length of parameter <username>.
<p_length>	integer	-	maximum length of parameter <password>.

```
AT#HTTPCFG=?
#HTTPCFG: (0-2),128,(1,65535),(0,1), 128, 128,(0,1),(1-65535),(1-15),(0),(0)
OK
AT#HTTPCFG=0,47.111.28.211,80,0,123,123,0,125,1
OK
AT#HTTPCFG?
#HTTPCFG: 0,47.111.28.211,80,0,123,123,0,125,1,0,0
#HTTPCFG: 1,,80,0,,,0,120,1,0,0
#HTTPCFG: 2,m2mlocate.telit.com,9978,0,,,0,120,1,0,0
OK
```

AT#HTTPQRY - Send HTTP GET, HEAD or DELETE Request

This command performs a GET, HEAD or DELETE request to HTTP server.

Standard RFC 2616

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#HTTPQRY=<prof_id>,<command>,<resource>[,<extra_header_line>]

Parameters:

Name	Type	Default	Description
<prof_id>	integer	N/A	profile identifier
	Value:		
	0÷2	:	identifier values
<command>	integer	0	identifies command requested to HTTP server
	Values:		
	0	:	GET
	1	:	HEAD
	2	:	DELETE
<resource>	string	-	is the HTTP resource (URI), object of the request.

<extra_header_line>	string	-	is the optional HTTP header line.

Additional info:

- When the HTTP server answer is received, then the following URC is put on the serial port:

  #HTTPRING: <prof_id>,<http_status_code>,<content_type>,<data_size>

  If there are no data from server or the server does not answer within the time interval specified in <timeout> parameter of #HTTPCFG command, then the URC #HTTPRING <http_status_code> parameter has value 0.

Name	Type	Default	Description
<http_status_code>	string	-	is the status code, as received from the server, see RFC 2616
<content_type>	string	-	reports the "Content-Type" header line, as received from the server, see RFC 2616. This field is empty if <http_status_code> has a value out of range (200-206).
<data_size>	string	-	is the byte amount of data received from the server. If the server does not report the "Content-Length:" header line, or <http_status_code> has a value out of range

	(200-206), the parameter value is 0.

If sending ends successfully, the response is OK; otherwise an error code is reported.

The HTTP request header sent with #HTTPQRY always contains the "Connection: close" line, and it cannot be removed.

AT#HTTPQRY=?

Test command reports the supported range of values for the parameters <prof_id> and <command> and the maximum length of <resource> parameter in the format:

#HTTPQRY:(list of supported <prof_id>s),(list of supported <command>s),<r_length>, <m_length>

Additional info:

Meaning of <..._length> parameters:

Name	Type	Default	Description
<r_length>	integer	-	maximum length of parameter <resource>.
<m_length>	integer	-	maximum length of parameter <extra_header_line>.

AT#HTTPCFG=0,www.gsctest.top,80,0,123,123,0,125,0,0

OK

AT#HTTPQRY=0,0,/static/GM510/KFKJ_GM510L1B1_99/update.zip

OK

#HTTPRING: 0,200," application/zip",246496

AT#HTTPSND - Send HTTP POST or PUT request

This command performs a POST or PUT request to HTTP server and starts sending data to the server.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#HTTPSND=<profId>,<command>,<resource>,<dataLen>[,<postParam>[,<extraHeaderLine>]]

After entering AT#HTTPSND=... command, and terminated the command line with <CR>, the module returns the following three characters sequence prompt, and waits for data to send:

<greater_than><greater_than><greater_than> (see IRA character set: 62, 62, 62)

Now, data can be entered from TE, sized <dataLen> bytes. If sending ends successfully the response will be OK, otherwise an error code will be reported.

When the HTTP server answer is received, the URC will be available on the serial port with the following format:

#HTTPRING: <profId>,<httpStatusCode>,<contentType>,<dataSize>

The URC parameters are described in Unsolicited Field section.

Parameters:

Name	Type	Default	Description
<profId>	integer	N/A	profile identifier
	Value:		
	0÷2	:	profile identifier
<command>	integer	N/A	command requested to HTTP server
	Values:		

0	:	POST command
1	:	PUT command

<resource>	string	-	HTTP resource (uri), object of the request
<dataLen>	integer	-	data length to send in bytes
<postParam>	string	N/A	HTTP Content-type identifier, used only for POST command, optionally followed by colon character (:) and a string that extends with sub-types the identifier

Values:

0[:extension]	:	"application/x-www-form-urlencoded" with optional extension
1[:extension]	:	"text/plain" with optional extension
2[:extension]	:	"application/octet-stream" with optional extension
3[:extension]	:	"multipart/form-data" with optional extension
other content	:	free string corresponding to other content type and possible sub-types

<extraHeaderLine>	string	-	optional HTTP header line

Unsolicited fields:

Name	Type	Description

<httpStatusCode>	integer	status code, as received from the server (see RFC 2616)
<contentType>	string	"Content-Type" header line, as received from the server (see RFC 2616). This field is empty if <httpStatusCode> has a value out of range (200-206)
<dataSize>	integer	byte amount of data received from the server. If the server does not report the "Content-Length:" header line or <httpStatusCode> has a value out of range (200-206), the parameter value is 0

- The HTTP request header sent with #HTTPSND always contains the "Connection: close" line, and it cannot be removed.
- If there are no data from server or the server doesn't answer within the time interval specified in <timeout> parameter of #HTTPCFG command, then the URC #HTTPRING <httpStatusCode> parameter will have value 0.
- If sending ends successfully, the response is OK; otherwise an error code is reported.

```
AT#HTTPCFG=0,www.gsctest.top,80,0,123,123,0,125,0,0
OK
AT#HTTPSEND=0,0,"/",15
>>>(123456789012345)
OK
#HTTPRING: 0,403," text/html",1354
```

AT#HTTPRCV - Receive HTTP Server Data

This command permits the user to read data from HTTP server in response to a previous HTTP module request.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#HTTPRCV=<profId>[,<maxByte>]

The module is notified of these data by the #HTTPRING URC.

The device shall prompt a three-character sequence followed by the data:

<less_than><less_than><less_than> (see IRA character set: 60,60,60)

If reading ends successfully, the response is OK; otherwise an error code is reported.

Parameters:

Name	Type	Default	Description
<profId>	integer	N/A	profile identifier
	Value:		
	0÷2	:	profile identifier
<maxByte>	integer	0	max number of bytes to read at a time
	Value:		
	0, 64÷1500	:	0 means infinite size.

- If <maxByte> is unspecified, server data will be transferred all in once.
- If the data are not present or the #HTTPRING <httpStatusCode> parameter has value 0, an error code is reported.
- Use the escape sequence +++ to abort file downloading and close the data connection.

AT#HTTPRCV=?

Test command reports the supported range of values for <profId> parameter in the format:

HTTPRCV: (list of supported <profId>s)

```
AT#HTTPCFG=0,www.gsctest.top,80,0,123,123,0,125,0,0
OK
AT#HTTPQRY=0,0,/static/GM193/2020_1111.txt
OK

#HTTPRING: 0,200," text/plain",17

AT#HTTPRCV=0
<<<123489641687641
```

SSL

AT#SSLCFG - Configure General Parameters of a SSL Socket

This command configures SSL connection parameters.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT#SSLCFG=<SSId>,<cid>,<pktSz>,<maxTo>,<defTo>,<txTo>[,<sslSRingMode>[,<noCarrierMode>[,<skipHostMismatch>[,<UNUSED_4>]]]]

Set command allows configuring SSL connection parameters.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1 : Until now SSL block manages only one socket		
<cid>	integer	-	PDP context identifier, see +CGDCONT command.
<pktSz>	integer	0	packet size to be used by the SSL/TCP/IP stack for data sending
	Values:		

		0 : select automatically default value (300) 1÷1500 : number of bytes	
<maxTo>	integer	90	exchange timeout or socket inactivity timeout. In online mode, if there is no data exchange within this timeout period, the connection is closed.
	Values:		
		0 : no timeout 1÷65535 : timeout in seconds	
<defTo>	integer	100	timeout that will be used by default whenever the corresponding parameter of each command is not set.
	Value:		
		10÷5000 : timeout in tenth of seconds	
<txTo>	integer	50	data sending timeout. In online mode, after this period data are sent also if they are less than max packet size.
	Values:		

0	:	no timeout
1÷255	:	timeout value in hundreds of milliseconds

<sslSRingMode>	integer	0	sslSRING unsolicited mode

Values:

0	:	SSLSRING disabled
1	:	SSLSRING enabled in the format
2	:	SSLSRING enabled in the format

<noCarrierMode>	integer	0	permits to choose NO CARRIER indication format when the secure socket is closed.

Values:

0	:	NO CARRIER (Indication is sent as usual, without additional information)
1	:	NO CARRIER:SSL,<SSId> (Indication of current <SSId> secure socket connection is added. The fixed "SSL" string allows the user to distinguish secure sockets from TCP sockets)
2	:	NO CARRIER:SSL,<SSId>,<cause> (refer to additional info)

<skipHostMismatch>	integer	1	ignores Host Mismatch alert

Values:

0	:	Do not ignore
1	:	Ignore

<UNUSED_4>	integer	-	Reserved 4

Additional info:

- <sslSRingMode> - sslSRing unsolicited mode:

  1 - SSLSRING enabled in the format:

  SSLSRING: <SSId>,<recData>

  where <SSId> is the secure socket identifier and <recData> is the amount of data received and decoded by the SSL socket.

  A new unsolicited is sent whenever the amount of data ready to be read changes. Only a record is decoded at once so, any further record is received and decoded only after the first have been read by the user by means of the #SSLRECV command.

  2 - SSLSRING enabled in the format:

  SSLSRING: <SSId>,<dataLen>,<data>

  where <SSId> is the secure socket identifier, <dataLen> is the length of the current chunk of data (the minimum value between the available bytes and 1300) and <data> is data received (<dataLen> bytes) displayed in ASCII format.

- <noCarrierMode> - this parameter permits to choose NO CARRIER indication format when the secure socket is closed as follows:

  NO CARRIER:SSL,<SSId>,<cause>

  Indication of current <SSId> secure socket connection and closure <cause> are added.

Following the possible <cause> values are listed:

0 - not available (secure socket has not yet been closed)
1 - the remote TCP connection has been closed (RST, or any fatal error in send/receive are all included within this case)
2 - socket inactivity timeout
3 - network deactivation (PDP context deactivation from network)
4 - SSL "Close Notify Alert" message has been received
5 - the remote TCP connection has been closed (FIN) after all data have been retrieved from socket
6 - Closure due to any other SSL alert different from the previous ones.

These parameters cannot be changed if the secure socket is connected.

AT#SSLCFG?

Read command reports the parameters current values in the format:
#SSLCFG: <SSId>,<cid>,<pktSz>,<maxTo>,<defTo><txTo>,<sslSRingMode>,

<noCarrierMode>,<skipHostMismatch>,0

AT#SSLCFG=?

Test command returns the range of supported values for all the parameters.

#SSLCFG:(1), (1-16), (0-1500), (0-65535), (10-5000), (0-255), (0-2), (0-2), (0-1), (0)

AT#SSLSECCFG - Configure Security Parameters of a SSL Socket

Configure security parameters of a SSL socket

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLSECCFG=<SSId>,<CipherSuite>,<auth_mode>[,<cert_format>]

This command allows configuring SSL connection parameters.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1 : Until now only one SSL socket is available		
<CipherSuite>	integer	0	Cipher Suite.
	Values:		
	0 : Chiper Suite is chosen by remote Server.		
	1 : TLS_RSA_WITH_AES_128_CBC_SHA		
	2 : TLS_RSA_WITH_AES_256_CBC_SHA		

3 : TLS_RSA_WITH_AES_128_CBC_SHA256

4 : TLS_RSA_WITH_AES_256_CBC_SHA256

5 : TLS_RSA_WITH_3DES_EDE_CBC_SHA

6 : TLS_RSA_WITH_AES_128_GCM_SHA256

7 : TLS_RSA_WITH_AES_256_GCM_SHA384

8 : TLS_DHE_RSA_WITH_AES_128_CBC_SHA

9 : TLS_DHE_RSA_WITH_AES_256_CBC_SHA

10 : TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA

11 : TLS_DHE_RSA_WITH_AES_128_CBC_SHA256

12 : TLS_DHE_RSA_WITH_AES_256_CBC_SHA256

13 : TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256

14 : TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384

15 : TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256

16 : TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384

17 : TLS_ECDHE_ECDSA_WITH_AES_128_CCM

18 : TLS_ECDHE_ECDSA_WITH_AES_256_CCM

19 : TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256

20 : TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384

21 : TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256

22 : TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384

23 : TLS_DHE_RSA_WITH_AES_128_GCM_SHA256

24 : TLS_DHE_RSA_WITH_AES_256_GCM_SHA384

25 : TLS_DHE_RSA_WITH_AES_128_CCM

26 : TLS_DHE_RSA_WITH_AES_256_CCM

27 : TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256

28 : TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384

29 : TLS_ECDH_ECDSA_WITH_AES_128_GCM_SHA256

30 : TLS_ECDH_ECDSA_WITH_AES_256_GCM_SHA384

31 : TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256

32 : TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384

33 : TLS_ECDH_RSA_WITH_AES_128_GCM_SHA256

34 : TLS_ECDH_RSA_WITH_AES_256_GCM_SHA384

<auth_mode> integer 0 authentication mode

Values:

0 : SSL Verify None

1 : Manage server authentication

	2 : Manage server and client authentication if requested by the remote server		
<cert_format>	integer	1	Selects the format of the certificate to be stored via #SSLSECDATA command
	Values:		
	0 : DER format		
	1 : PEM format		

Assume that the module is just powered on and the #SSLSECCFG command is entered without <cert_format> parameter, the default format is PEM. In this case the #SSLSECCFG? read command does not return the setting of the format in order to meet backward compatibility with other families. Now, let's assume that #SSLSECCFG command is entered again, but using the <cert_format> parameter for the first time: if the read command is entered, it reports the parameter value just used. If subsequently the <cert_format> is omitted, the #SSLSECCFG? read command reports the parameter value entered the last time.

Server CA certificate must be stored through #SSLSECDATA.

If secure socket is not enabled using #SSLEN command, only test command can be used. Read command can be issued if at least a <SSId> is enabled.

If <auth_mode> is different from 0, the module uses its internal time and date to validate the certificate validity period.

If time and date are incorrectly set, the certificate validation may fail. For additional information refer to AT+CCLK, AT#NITZ or AT#NTP.

With =0 cipher suite is chosen by remote server among the following:

TLS_DHE_RSA_WITH_AES_128_CBC_SHA,

TLS_DHE_RSA_WITH_AES_256_CBC_SHA,

TLS_DHE_RSA_WITH_AES_128_CBC_SHA256,

TLS_DHE_RSA_WITH_AES_256_CBC_SHA256,

TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256,

TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,

TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256,

TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384,

TLS_ECDHE_ECDSA_WITH_AES_128_CCM,

TLS_ECDHE_ECDSA_WITH_AES_256_CCM,

TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256,

TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384,

TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,

TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,

TLS_DHE_RSA_WITH_AES_128_GCM_SHA256,

TLS_DHE_RSA_WITH_AES_256_GCM_SHA384,

TLS_DHE_RSA_WITH_AES_128_CCM,

TLS_DHE_RSA_WITH_AES_256_CCM,

TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256,

TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384,

TLS_ECDH_ECDSA_WITH_AES_128_GCM_SHA256,

TLS_ECDH_ECDSA_WITH_AES_256_GCM_SHA384,

TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256,

TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384,

TLS_ECDH_RSA_WITH_AES_128_GCM_SHA256,

TLS_ECDH_RSA_WITH_AES_256_GCM_SHA384

AT#SSLSECCFG?

Read command reports the currently selected parameters in the format:

#SSLSECCFG: <SSId>,<CipherSuite>,<auth_mode>[,<cert_format>]

AT#SSLSECCFG=?

Test command returns the range of supported values for all the parameters.

#SSLSECCFG: (1),(0-34),(0-2),(0,1)

AT#SSLEN - Enable a SSL Socket

This command activates/deactivates a socket secured by SSL.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLEN=<SSId>,<Enable>

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1 : only one socket is available		
<Enable>	integer	0	activate/deactivate secure socket
	Values:		
	0 : deactivate		
	1 : activate		

If the unique available secure socket is not activated, all the commands - belonging to the SSL set (example: #SSLSECDATA, #SSL..., etc.) and different from test commands - return an error message. #SSLS command is an exception, it can be issued also if the socket is deactivated.

If the unique available secure socket is connected, it cannot be deactivated issuing AT#SSLEN=1,0.

AT#SSLEN?

Read command reports the current status of secure socket in the format:

#SSLEN: <SSId>,<Enable>
OK

AT#SSLEN=?

Test command returns the range of supported values for all the parameters:

#SSLEN: (1),(0,1)

AT#SSLD - Open a SSL Socket to a Remote Server

This command opens a remote connection via socket secured through SSL.

SIM Presence	Setting saved	Can be aborted
Required	Other	No

AT#SSLD=<SSId>,<rPort>,<IPAddress>,<ClosureType>[,<connMode>[,<Timeout>]]

Execution command opens a remote connection via socket secured through SSL. Both command and online modes can be used.
In the first case 'OK' is printed on success, and data exchange can be performed by means of #SSLSEND and #SSLRECV commands.
In online mode 'CONNECT' message is printed, and data can be sent/received directly to/by the serial port. Communication can be suspended by issuing the escape sequence (by default +++) and restored with #SSLO command.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1 : Until now only one SSL socket is available		
<rPort>	integer	1	Remote TCP port to contact
	Value:		
	1÷65535 : TCP port number		
<IPAddress>	string	-	address of SSL server.

			This parameter can be either: - any valid IP address in the format: "xxx.xxx.xxx.xxx" - any host name to be solved with a DNS query - any valid IPv6 address in the format: "xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx" or "xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx.xxx"
<ClosureType>	integer	0	Closure type

Value:

0	:	Until now only closure type 0 is supported

<connMode>	integer	1	connection mode

Values:

0	:	online mode connection
1	:	command mode connection

<Timeout>	integer	100	time-out in 100 ms units. It represents the maximum allowed TCP inter-packet delay. It means that, when more data is expected during the handshake, the module awaits <Timeout> * 100 msecs for the next packet. If no more data can be read, the module gives up the handshake and raises an ERROR response

Value:

10÷5000	:	hundreds of ms

- If secure socket is not enabled using #SSLEN only test requests can be made.
- If timeout is not set for SSL connection the default timeout value, set by #SSLCFG, is used.
- In online mode the socket is closed after an inactivity period (configurable with #SSLCFG, with a default value of 90 seconds), and the NO CARRIER message is printed.
- In online mode data are transmitted as soon as the data packet size is reached or as after a transmission timeout. Both these parameters are configurable by using #SSLCFG.
- If there are input data arrived through a connected socket and not yet read because the module entered command mode before reading them (after an escape sequence or after #SSLD has been issued with <connMode> set to command mode connection), these data are buffered and we receive the SSLSRING URC (if any of its presentation formats have been enabled by means the #SSLCFG command); it is possible to read these data afterwards issuing #SSLRECV. Under the same hypotheses it is possible to send data while in command mode issuing #SSLSEND.
- Before opening a SSL connection, the PDP context must have been activated by #SGACT=x,1
- Before opening a SSL connection, make sure to have stored the needed secure data (CA certificate), using AT#SSLSECDATA.
- This module can be supported from at least 2048 bits of Server certificate for Public key length.

AT#SSLD=?

Test command returns the range of supported values for all the parameters:

#SSLD: (1),(1-65535),,(0),(0,1),(10-5000)

AT#SSLO - Restore a SSL Socket after a +++

This command restores a SSL connection (online mode) suspended by an escape sequence (+++).

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSLO=<SSId>

This command restores a SSL connection (online mode) suspended by an escape sequence (+++). After the connection restore, the CONNECT message is printed. Please note that this is possible even if the connection has been started in command mode (#SSLD with <connMode>=1).

Parameter:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier

Value:

1	:	only one socket is available.

- If secure socket has not be enabled through #SSLEN command, only test command can be used.
- Before opening a SSL connection, the PDP context must have been activated by AT#SGACT=X,1.
- If an error occurs during reconnection the socket cannot be reconnected, then a new connection has to be done.

AT#SSLO=?

Test command returns the range of supported values for all the parameters:

#SSLO: (1)

AT#SSLH - Close a SSL Socket

This command allows closing the SSL connection.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SSLH=<SSId>[,<ClosureType>]

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1 : only one socket is available		
<ClosureType>	integer	0	type of socket closure.
	Value:		
	0 : only value 0 is supported		

If secure socket has not be enabled through #SSLEN command, only test command can be used.

AT#SSLH=?

Test command returns the range of supported values for all the parameters:

#SSLH: (1),(0)

AT#SSLSEND - Send Data through a SSL Socket

This command allows sending data through a secure socket.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SSLSEND=<SSId>[,<Timeout>]

The device responds to the command with the prompt '>' and waits for the data to send.

To complete the operation send Ctrl-Z char (0x1A hex); to exit without writing the message send ESC char (0x1B hex).

If data are successfully sent, then the response is OK.

If data sending fails for some reason, an error code is reported.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1 : only one socket is available		
<Timeout>	integer	100	Socket send timeout
	Value:		
	10÷5000 : timeout in 100 ms units		

The maximum number of bytes to send is 1023; trying to send more data will cause the surplus to be discarded and lost

- If secure socket is not enabled using AT#SSLEN only test requests can be made.
- If timeout is not set for SSL connection the default timeout value, set by AT#SSLCFG, is used.
- Before sending data through the SSL connection it has to be established using AT#SSLD

AT#SSLSEND=?

Test command returns the range of supported values for all the parameters.

#SSLSEND: (1),(10-5000)

AT#SSLRECV - Read Data from a SSL Socket

This command reads data from a SSL socket.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#SSLRECV=<SSId>,<MaxNumByte>[,<Timeout>]

Set command allows to receive data, arrived through a connected secure socket. Data has been buffered and not read yet.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	secure socket identifier
	Value:		
	1 : until now only one SSL socket is supported		
<MaxNumByte>	integer	N/A	maximum number of bytes to read
	Value:		
	1÷1000 : maximum number of bytes to read		
<Timeout>	integer	100	time-out in 100 ms units
	Value:		
	1÷5000 : hundreds of ms		

Additional info:

- If data are received, the device responds:

  #SSLRECV: NumByteRead
  ...(Data read)...
  OK

- If no data are received, the device responds:

  #SSLRECV: 0
  TIMEOUT
  OK

- If the remote host closes the connection, the device responds:

  #SSLRECV: 0
  DISCONNECTED
  OK

- If secure socket is not enabled using AT#SSLEN, only test requests can be made.
- If timeout is not set for SSL connection, the default timeout value, set through AT#SSLCFG, is used.
- Before receiving data from the SSL connection, it has to be established using AT#SSLD.

AT#SSLRECV=?

The test command returns the ranges of the parameters values in the form:

#SSLRECV: (1),(1-1000),(1-5000)

AT#SSLS - Report the Status of a SSL Socket

This command reports the status of secure sockets.

SIM Presence	Setting saved	Can be aborted
Required	No	No

- AT#SSLS=<SSId>

  The command reports the status of secure sockets. The response message of the command can have the following formats:

  if secure socket is connected, the format is:
  #SSLS: <SSId>,<ConnectionStatus>,<CipherSuite>

  otherwise:
  #SSLS: <SSId>,<ConnectionStatus>

  The response messages parameters are described in the Additional info section.

  Parameter:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier.

Value:

1 : only one SSL socket is supported

Additional info:

List of the meaning of the response message parameters.

Name	Type	Default	Description
<CipherSuite>	integer	0	Cipher Suite identifier

Values:

0	:	Chiper Suite is chosen by remote Server
1	:	TLS_RSA_WITH_3DES_EDE_CBC_SHA
2	:	TLS_RSA_WITH_AES_128_CBC_SHA
3	:	TLS_RSA_WITH_AES_128_CBC_SHA256
4	:	TLS_RSA_WITH_AES_256_CBC_SHA
5	:	TLS_RSA_WITH_AES_256_CBC_SHA256
6	:	TLS_DHE_RSA_WITH_AES_128_CBC_SHA
7	:	TLS_DHE_RSA_WITH_AES_256_CBC_SHA
8	:	TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA
9	:	TLS_DHE_RSA_WITH_AES_128_CBC_SHA256
10	:	TLS_DHE_RSA_WITH_AES_256_CBC_SHA256

<ConnectionStatus>	integer	N/A	Connection Status identifier

Values:

0	:	socket disabled
1	:	connection closed
2	:	connection open

This command can be issued even if the <SSId> is not enabled.

AT#SSLS=?

Test command returns the ranges of the parameters values in format:

#SSLS: (1)

AT#SSLI - Secure Socket Info

This command is used to get information about secure socket data traffic.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLI[=<SSId>]

Execution command is used to get information about secure socket data traffic. The response is in the format:

#SSLI: <SSId>,<DataSent>,<DataRecv>,<PendingData>,<TCPConnWaitingAck>

Parameter:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier

Value:

1 : only one SSL socket is available

Additional info:

Parameters returned by the response message and not described in the previous sections.

Name	Type	Default	Description

<DataSent>	integer	-	total amount (in bytes) of data sent to the TLS/SSL connection since the beginning of the connection itself (obviously: not yet encoded into TLS/SSL record)
<DataRecv>	integer	-	total number of bytes received from the TLS/SSL connection since the beginning of the connection itself (obviously: already decoded from TLS/SSL record)
<PendingData>	integer	-	number of bytes available to be read from the TLS/SSL record that is currently being processed (obviously: already decoded from TLS/SSL record)
<TCPConnWaitingAck>	integer	N/A	indication of the underlying TCP socket condition, if there are TCP/IP packets sent but not yet

acknowledged or not

Values:

0 : no TCP/IP packets sent waiting for ack

1 : TCP/IP packets sent waiting for ack

AT#SSLI=?

Test command returns the range of supported values for all the parameters.

#SSLI: (1)

AT#SSLSENDEXT - Send Data through a SSL Socket in Command Mode

This command sends data through a secure socket.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#SSLSENDEXT=<SSId>,<bytestosend>[,<Timeout>]

This command allows sending data through a secure socket.
The device responds to the command with the prompt '>' <greater_than><space> and waits for the data to send.
When <bytestosend> bytes have been sent, operation is automatically completed.
If data are successfully sent, then the response is OK.
If data sending fails for some reason, an error code is reported

Parameters:

Name	Type	Default	Description
<SSId>	integer	N/A	Secure Socket Identifier
	Value:		
	1 : only one SSL socket is available.		
<bytestosend>	string	-	number of bytes to be sent. Refer to test command for range
<Timeout>	integer	100	time-out in 100 ms units.
	Value:		
	10÷1500 : hundreds of ms.		

- If secure socket has not be enabled through #SSLEN command, only test command can be used.
- If timeout is not set for SSL connection, is used the timeout value set by #SSLCFG.
- Before sending data through the SSL connection, it must be established using #SSLD.
- All special characters are sent like a generic byte. (For instance: 0x08 is simply sent through the socket and does not behave like a BackSpace, i.e. previous character is not deleted)

AT#SSLSENDEXT=?

Test command returns the range of supported values for parameters

#SSLSENDEXT: (1),(1-1024),(10-5000)

AT#SSLSECDATA - Manage the Security Data

The command stores, reads, and deletes security data (Certificate, CA certificate, private key) in / from NVM.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLSECDATA=<SSId>,<Action>,<DataType>[,<Size>[,<Password>]]

This command allows to store, delete and read security data.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1 : Until now SSL block manages only one socket		
<Action>	integer	N/A	Action to do
	Values:		
	0 : delete data from NVM		
	1 : store data in NVM		
	2 : read data from NVM		
<DataType>	integer	N/A	security data type
	Values:		

0	:	certificate
1	:	CA certificate
2	:	RSA Private key

<Size>	integer	N/A	Size of security data to be stored

Values:

1÷4096	:	for Certificate and RSA Private key
1÷12228	:	for CA certificate

<Password>	string	-	it allows to read RSA key when validated. The max length of this value is 10. Only 1 ..9 / a ..z / A .. Z characters are allowed. Default is the empty string "".

Additional info:

▶▶ If the <Action> parameter is 1 (store data into NVM) the device responds to the command with the prompt '>' and waits for the data to store.

Secured data have to be in PEM or in DER format, depending on <cert_format> chosen with #SSLSECCFG.

If no <cert_format> has been specified with #SSLSECCFG, PEM format is assumed.

PEM format (see #SSLSECCFG command):To complete the operation send Ctrl-Z char (0x1A hex); to exit without writing the message send ESC char (0x1B hex).

DER format (see #SSLSECCFG command): When <size> bytes are entered, the certificate is automatically stored.

ESC or Ctrl-Z don't take effect, because they are considered as possible octets contained in the certificate.

If data are successfully stored, then the response is OK; if it fails for some reason, an error code is reported.

If the <Action> parameter is 2 (read data from NVM), data specified by <DataType> parameter is shown in the following format:

#SSLSECDATA: <connId>,<DataType>

<DATA>

OK

If <DataType> data has not been stored (or it has been deleted) the response has the following format:

#SSLSECDATA: <connId>,<DataType>

No data stored

OK

- <size> parameter is mandatory if the <write> action is issued, but it has to be omitted if <delete> or <read> actions are issued.
- If secure socket is not enabled using AT#SSLEN only test requests can be made.
- If socket is connected an error code is reported.
- Once the <Password> saved, changing password is not allowed. To reset <Password>, it can be used AT#SSLSECDATA=1,0,2.
- Maximum <size> for Certificate/ RSA Private key is 4096 bytes. Only CA Certificate can support up to 12228 bytes in case of PEM format.

  If CA Certificate has a format to DER, it can be limited as 4096 bytes too.

  The CA certificate chain can store 1 Root CA and up to 2 Intermediate CA and it is saved in NVM memory.

This module can support PKCS 1 type of RSA Private key.

AT#SSLSECDATA?

Read command reports what security data are stored in the format:

#SSLSECDATA: <CertId>,<CertIsSet>,<CACertIsSet>,<PrivKeyIsSet>

<CertIsSet>, <CAcertIsSet>, <PrivKeIsset> are 1 if related data are stored into NVM otherwise 0.

AT#SSLSECDATA=?

Test command returns the range of supported values for all the parameters:

#SSLSECDATA: (1),(0-2),(0,2),(1-4096),10

#SSLSECDATA: (1),(0-2),(1),(1-12228),10

AT#SSLSECDATAEXT - Manage the security data extended

Manage the security data extended

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLSECDATAEXT=<SSId>,<Action>,<DataType>,<Index>[,<Size>]

This command allows to store, delete , read and select security data (CA certificate) into NVM.

Parameters:

Name	Type	Default	Description
<SSId>	integer	1	Secure Socket Identifier
	Value:		
	1 : Until now SSL block manages only one socket		
<Action>	integer	N/A	Action to do
	Values:		
	0 : Delete data from NVM		
	1 : Store data into NVM		
	2 : Read data from NVM		
	3 : Select data from NVM		
<DataType>	integer	N/A	Type of Data

Value:

1	:	CA Certificate

<Index>	integer	N/A	Order to be saved

Value:

1÷3	:	Supported index range

<Size>	integer	N/A	Size of security data to be stored

Value:

1÷4096	:	Supported data length

If the <Action> parameter is 1 (store data into NVM) the device responds to the command with the prompt '>' and waits for the data to store.

Secured data have to be in PEM or in DER format, depending on <cert_format> chosen with #SSLSECCFG.

If no <cert_format> has been specified with #SSLSECCFG, PEM format is assumed.

PEM format (see #SSLSECCFG command): to complete the operation send Ctrl-Z char (0x1A hex); to exit without writing the message send ESC char (0x1B hex).

DER format (see #SSLSECCFG command): when <SIze> bytes are entered, the certificate is automatically stored. ESC or Ctrl-Z do not take effect, because they are considered as possible octets contained in the certificate.

If data are successfully stored, then the response is OK; if it fails for some reason, an error code is reported.

If the <Action> parameter is 2 (read data from NVM), data specified by <DataType>,<Index> parameters is shown in the following format:

#SSLSECDATAEXT: <connId>,<DataType>,<Index>
<DATA>
OK

If <DataType>, <Index> data has not been stored (or it has been deleted) the response has the following format:

#SSLSECDATAEXT: <connId>,<DataType>,<Index>
No data stored
OK

- <Size> parameter is mandatory if the <write> action is issued, but it has to be omitted when <delete> or <read> or <select> actions are issued.
- If secure socket is not enabled using AT#SSLEN only test requests can be made.
- If socket is connected an error code is reported.
- Maximum supported number of CA certificate is up to 3.
- To use for H/S procedure, you need to choose <Action> to 3 ("Select data from NVM").
- No chained CA certificates are supported.

AT#SSLSECDATAEXT?

Read command reports what security data are stored in the format:

#SSLSECDATAEXT:
<SSId>,<SelectedIndex>,<CAcert1Set>,<CAcert2Set>,<CAcert3Set>

where

<SelectedIndex>,<CAcert1Set>, <CAcert2Set> and <CAcert3Set> are 1 if related data are stored into NVM, 0 otherwise.

AT#SSLSECDATAEXT=?

Test command returns the range of supported values for all the parameters:

#SSLSECDATAEXT: (1),(0-3),(1),(1-3),(1-4096)

AT#SSLSECCFG2 - Configure Additional Parameters of a SSL Socket

This command allows configuring additional SSL security parameters.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#SSLSECCFG2=<SSId>,<version>[,<unused_A>[,<unused_B>[,<unused_C>[,<unused_D>]]]]

This command allows configuring additional SSL security parameters.

Parameters:

Name	Type	Default	Description
<SSId>	integer	N/A	Secure Socket Identifier
	Value:		
	1 : only one SSL socket is available		
<version>	integer	2	SSL/TLS protocol version
	Values:		
	0÷2 : protocol version TLSv1.2		
<unused_A>	integer	-	reserved for future implementations
<unused_B>	integer	-	reserved for future implementations
<unused_C>	integer	-	reserved for future implementations

<unused_D>	integer	-	reserved for future implementations

If secure socket has not be enabled through #SSLEN command, only test command can be used.

AT#SSLSECCFG2?

Read command reports the currently selected parameters in the format:

#SSLSECCFG2: <SSId>,<version>,0,0,0,0

AT#SSLSECCFG2=?

Test command reports the range of supported values for all the parameters

#SSLSECCFG2: (1),(0-2),(0),(0),(0),(0)

FOTA & OMA

FOTA Legacy

AT+ZFOTADL - Download differential package

This command is used for FOTA differential subcontract download of 8M flash version.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT+ZFOTADL=[<server_url>]

Parameter:

Name	Type	Default	Description
<server_url>	string	-	URL FTP/HTTP server address.

Additional info:

- <server_url> FTP format:

  FTP(S)://<username>:<password>@<serverURL>[:<port>]/<file_path>

Name	Type	Default	Description
<username>	string	-	authentication user identification for FTP.
<password>	string	-	authentication password for FTP.
<serverURL>	string	-	The FTP server URL or IP address.

<port>	integer	21	Connection port. If the parameter is omitted, it will be set to 21 by default. Value: 1÷65535 : Connection port
<file_path>	string	-	Storage path and name of the differential package.

▶▶ <server_url> HTTP format:

HTTP(S)://<serverURL>[:<port>]/<file_path>

Name	Type	Default	Description
<serverURL>	string	-	The HTTP server URL or IP address.
<port>	integer	80	Connection port. If the parameter is omitted, it will be set to 80 by default. Value: 1÷65535 : Connection port

<file_path>	string	-	Storage path and name of the differential package.

- FTP protocol is used to download and transfer data in Passive mode and Binary mode.
- Since parameter <server_url> contains file names (type.bin_1 and.bin_2), multiple subfiles can be placed in the same path.
- If the UART port is used for operation and command interaction, run the AT+IPR command to set the baud rate of the serial port before the upgrade. Otherwise, the upgrade progress cannot be reported in baud rate adaptive mode. During the upgrade, the SYSTEM restarts several times. If the USB port is used, the upgrade progress cannot be reported.
- Upgrade requires two subpackages, but just send the command to download the first subpackage (system_patch.bin_1).
- During the upgrade process, the module will restart several times and enter normal working mode after the upgrade is completed.
- The module will check the downloaded subcontract. If the initial version does not match during the upgrade, the upgrade will fail. If the first subcontract does not match, the system will roll back to the original version after the upgrade fails.
- If the second subcontracting download fails, the upgrade failure information will be reported. The module needs to restart after the upgrade failure message is reported, after which the module continues to download the second subcontract until the upgrade is successful.
- The entire upgrade process takes about 10 minutes. The actual time depends on the network status.

- During the upgrade, the serial port is unavailable, and it is not recommended to use the USB port to send AT commands. Otherwise, module exceptions may occur.

- After executing the FOTA upgrade command, the module will automatically restart and enter the download mode. The module can enter the normal mode only when the differential firmware with extension ". Bin_1" and ". Bin_2 " is downloaded and upgraded successfully. If the module is in download mode and commands are sent through the USB port or the operation is powered off, the module will remain in download mode and only the download and upgrade functions related to the network and supporting subcontracting will be retained.

- If <server_url> is configured with FTPS or HTTPS, TLS1.2 will be used to connect server, and it will leverage TLS configuration for SSId 1.

AT+ZFOTADL=?

Test command returns OK.

- Download subcontracting in FTP mode

  AT+ZFOTADL=FTP://test:test@219.144.130.27/FotaTest/system_patch.bin_1

- Download subcontracting in HTTP mode

  AT+ZFOTADL=HTTP://www.gsctest.top:80/static/FotaTest/system_patch.bin_1

- Download subcontracting in FTP Explicit Secure mode

AT+ZFOTADL=FTPS://test:test@219.144.130.28/FotaTest/
system_patch.bin_1

- Download subcontracting in HTTPS mode

  AT+ZFOTADL=HTTPS://www.gsctest.top:443/static/FotaTest/
  system_patch.bin_1

MQTT

AT#MQEN - Enable MQTT Feature

This command initializes a MQTT client and allocates the necessary resources.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQEN=<instanceNumber>,<enable>

Set command enables/disables the MQTT client for further configuration and usage.

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate
	Value:		
	1÷maxClients	:	client instance range. To know maxClients value use test command
<enable>	integer	N/A	selects if client must be activated or deactivated
	Values:		
	0	:	disable
	1	:	enable

AT#MQEN?

Read command returns the status of the MQTT stack in the format

#MQEN: <instanceNumber>,<enabled>

AT#MQEN=?

Test command reports the available range of values for parameters.

Read command
AT#MQEN?
#MQEN: 1,1
#MQEN: 2,0
OK

Test command
AT#MQEN=?
#MQEN: (1-maxClients),(0-1)
OK

AT#MQCFG - Configure MQTT Parameters

This sets the connection parameters for the selected MQTT client.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQCFG=<instanceNumber>,<hostname>,<port>,<cid>[,<sslEn>,<sslInstance>]

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command Value: 1÷maxClients : client instance range. To know maxClients value use test command
<hostname>	string	-	URL of the MQTT broker. Default value: 0. Maximum length of <hostname> is 512 characters.
<port>	integer	0	TCP port of the MQTT broker Value: 1÷65535 : TCP port range

<cid>	integer	N/A	PDP Context ID to be used for the connection. PDP context should be enabled before giving this command. Default value 0.

Value:

1÷6	:	<cid> range

<sslEn>	integer	0	enable/disable SSL

Values:

0	:	disable
1	:	enable

<sslInstance>	integer	1	selects the ssl instance configuration to be used for this MQTT client. The list of available ssl instances is obtained with the read command.

Value:

1÷maxClients	:	ssl instance range. To know ssl instance value use test command. If the value is not provided it will select ssl instance 1 by default

AT#MQCFG?

Read command returns the configuration of all active MQTT clients in the format:

#MQCFG: <instanceNumber>,<hostname>,<port>,<cid>[,<sslEn>,<sslinstance>]

AT#MQCFG=?

Test command reports the available range of values for parameters.

- - SSL disabled (by default)

  Set command

  AT#MQCFG=1,api.mybroker.com,1883,3
  OK

  Read command

  AT#MQCFG?
  #MQCFG: 1,api.mybroker.com,1883,3,0
  #MQCFG: 2,,1883,1
  OK

  Test command

  AT#MQCFG=?
  #MQCFG: (1-maxClients),512,(1-65535),(1-6),(0-1),(1-maxSSLInstance)
  OK

- - SSL enabled

  Set command

  AT#MQCFG=1,api.mybroker.com,8883,3,1
  OK

  Read command

  AT#MQCFG?
  #MQCFG: 1,api.mybroker.com,8883,3,1
  #MQCFG: 2,,1883,1,0
  OK

  Test command

  AT#MQCFG=?
  #MQCFG: (1-maxClients),512,(1-65535),(1-6),(0-1)),(1-maxSSLInstance)

OK

AT#MQDISC - Log Out and Disconnect from the MQTT Broker

This command performs the logout and disconnection from to the MQTT broker.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQDISC=<instanceNumber>

Disconnects gracefully from the MQTT broker, then closes the network connection

Parameter:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate
	Value:		
	1÷maxClients	:	client instance range. To know maxClients value use test command

AT#MQDISC=?

Test command reports the available range of values for parameters.

AT#MQDISC=1

OK

Test command

AT#MQDISC=?

#MQDISC: (1-maxClients)

OK

AT#MQCONN - Connect and Log in the MQTT Broker

This command performs the connection and login to the MQTT broker.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQCONN=<instanceNumber>,<clientID>,<userName>,<passWord>

Set command performs network connection (using parameters set with #MQCFG) and sends the CONNECT packet to the MQTT broker.

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command. Value: 1÷maxClients : client instance range. To know maxClients value use test command
<clientID>	string	-	identifies each MQTT client that connects to the MQTT broker. Maximum length of <clientID> is 512.
<userName>	string	-	authentication and authorization. Maximum length of <userName> is 512.

<passWord>	string	-	authentication and authorization. Maximum length of <passWord> is 512.

If connection status is other than init state or connected, disconnect using #MQDISC before reconnecting using #MQCONN.

AT#MQCONN?

Read command reports the configuration of active MQTT connections in the format:

#MQCONN=<instanceNumber>,<state>

Additional info:

The following tables shows the <state> values and meanings:

Client status: <state>	Normal/ Failure Events Values
0	client is initialized but not connected
1	client performed MQTT authentication with broker
2	connection closed or reset by the server
3	the answer to the ping request packet was not received
4	the CONNACK packet was not received
5	the CONNECT packet was not delivered

6	failure in the m2mb APIs
7	socket timeout or read error

AT#MQCONN=?

Test command reports the available range of values for parameters.

```
AT#MQCONN=1,myClientID,myUser,myPassword
OK
```

Scenario (1)
Read command

```
AT#MQCONN?
#MQCONN: 1,1
#MQCONN: 2,0
OK
```

Test command

```
AT#MQCONN=?
#MQCONN: (1-maxClient),512,512,512
OK
```

Scenario (2)
Read command

```
AT#MQCONN?
#MQCONN: 1,2
OK
```

Set command

```
AT#MQDISC=1
OK
```

Read command

```
AT#MQCONN?
#MQCONN: 1,0
OK
```

AT#MQCFG2 - Configure Additional MQTT Parameters

This command sets the optional connection parameters for the selected MQTT client.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQCFG2=<instanceNumber>,<keepAlive>,<cleanSession>

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command. Value: 1÷maxClients : client instance range. To know maxClients value use test command
<keepAlive>	integer	60	timeout of periodic packet to keep connection open Value: 60÷3600 : timeout expressed in sec
<cleanSession>	integer	1	Indicates whether a persistent connection is required. Without a persistent connection, when the client is offline all information and messages that

are queued from a previous persistent session are lost.

Values:

1	:	clean session
0	:	persistent session

AT#MQCFG2?

Read command returns the configuration of all active MQTT clients in the format:

#MQCFG2: <instanceNumber>,<keepAlive>,<cleanSession>

AT#MQCFG2=?

Test command reports the available range of values for parameters.

Set command

```
AT# MQCFG2=1,60,1
OK
```

Read command

```
AT#MQCFG2?
#MQCFG2: 1,60,1
#MQCFG2: 2,70,1
OK
```

Test command

```
AT#MQCFG2=?
#MQCFG2: (1-maxClients),(60-3600),(0-1)
```

AT#MQWCFG - Configure MQTT Last Will and Testament

This command sets Last Will and Testament for the selected MQTT client.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQWCFG=<instanceNumber>,<willFlag>[,<willRetain>,<willQos>,<willTopic>,<willMsg>]

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command. Value: 1÷maxClients : client instance range. To know maxClients value use test command
<willFlag>	integer	0	selects whether the client needs to specify a Last Will and Testament. If set to 0, this is the last parameter to be set. Values: 0 : the client does not need to specify a Last Will and Testament 1 : the client needs to specify a Last Will and Testament

<willRetain>	integer	0	selects whether the Last Will message needs to be retained by the server

Values:

0 : the Last Will message does not need to be retained by the server

1 : the Last Will message needs to be retained by the server

<willQos>	integer	0	Quality of Service of the Last Will message

Value:

0÷2 : Quality of Service range

<willTopic>	string	-	Topic to publish the Last Will message to. Unused. Maximum length of <willTopic> is 256.
<willMsg>	string	-	Last Will message. Maximum length of <willMsg> is 1024.

AT#MQWCFG?

Read command returns the Last Will and Testament (if any) of all active MQTT clients in the format:

#MQWCFG:<instanceNumber>,<willFlag>[,<willRetain>,<willQos>,<willTopic>, <willMsg>]

AT#MQWCFG=?

Test command reports the available range of values for parameters.

```
AT#MQWCFG=1,0
OK
```

AT#MQWCFG=1,1,1,2,myLastWillTopic,myLastWillMessage
OK

Read command
AT#MQWCFG?
#MQWCFG: 1,0
#MQWCFG: 2,1,0,2,myTopic,myMessage
OK

Test command
AT#MQWCFG=?
#MQWCFG: (1-maxClients),(0-1),(0-1),(0-2),256,1024
OK

AT#MQTCFG - Configure Timeout Parameters for MQTT Transmission

This command writes the timeout options for the specified client.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQTCFG=<instanceNumber>,<pktTimeout>

Parameters:

Name	Type	Default	Description
<instanceNumber>	string	N/A	selects the client instance. The list of available clients is obtained with AT#MQEN? or the read command. Value: 1÷maxClients : client instance range. To know maxClients value use test command
<pktTimeout>	integer	10	timeout of the packet delivery Value: 1÷60 : timeout range. Value expressed in seconds

AT#MQTCFG?

Read command returns the timeout configuration of all active MQTT clients in the format

#MQTCFG: <instanceNumber>,<pktTimeout>

AT#MQTCFG=?

Test command reports the available range of values for parameters

```
AT#MQTCFG=1,10
OK
```

Read command

```
AT#MQTCFG?
#MQTCFG: 1,5
#MQTCFG: 1,4
OK
```

Test command

```
AT#MQTCFG=?
#MQTCFG: (1-maxClients),(1-60)
OK
```

AT#MQUNS - Unsubscribe from a Topic

This command revokes the subscription to a MQTT topic.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQUNS=<instanceNumber>,<topic>

Parameters:

Name	Type	Default	Description
<instanceNumber>	string	N/A	selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range. To know maxClients value use test command
<topic>	string	-	name of the topic. Maximum length of <topic> is 256.

AT#MQUNS=?

Test command reports the available range of values for parameters.

AT#MQUNS=1,topicToUnsubscribe

OK

Test command

AT#MQUNS=?

#MQUNS: (1-maxClients)

OK

AT#MQSUB - Subscribe to a Topic

This command performs the subscription to a MQTT topic

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQSUB=<instanceNumber>,<topic>[,<qos>]

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range. To know maxClients value use test command
<topic>	string	-	name of the topic. Maximum length of <topic> is 256.
<qos>	integer	1	Configures qos for the subscription. This is an optional parameter. If qos is not specified, qos 1 is taken as default Value:

0÷2	:	qos range

AT#MQSUB=?

Test command reports the available range of values for parameters.

```
AT#MQSUB=1,topicToSubscribe
OK
AT#MQSUB=1,topicToSubscribe_2,0
OK
```

Test command

```
AT#MQSUB=?
#MQSUB: (1-maxClients),256
OK
```

AT#MQPUBS - Publish ASCII String

This command publishes an ASCII string to the specified MQTT topic.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQPUBS=<instanceNumber>,<topic>,<retain>,<qos>,<message>

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range. To know maxClients value use test command
<topic>	string	-	name of the topic. Maximum length of <topic> is 256.
<retain>	integer	N/A	specifies if the broker must retain this message or not Values: 0 : Not retained 1 : Retain

<qos>	integer	N/A	specifies the Quality of Service of this message
	Value:		
	0÷2	:	Quality of Service range
<message>	string	-	message to publish on the topic. Maximum length of <message> can go up to 65536 depending on system memory available.

AT#MQPUBS=?

Test command reports the available range of values for parameters.

```
AT#MQPUBS=1,myTopic,0,0,myMessage
OK
```

Test command

```
AT#MQPUBS=?
#MQPUBS: (1-maxClients),256,(0-1),(0-2),65536
OK
```

AT#MQREAD - Read Messages Received from the MQTT Broker

This command reads the message payload from the queue slot provided.

SIM Presence	Setting saved	Can be aborted
Required	No	No

AT#MQREAD=<instanceNumber>,<mId>

After command line is terminated with <CR>, the module responds sending:

#MQREAD: <instanceNumber>,<topic>,<payload_len>,

then the module prompts the following characters sequence:

<less_than><less_than><less_than><carriage return><line feed> (IRA 60, 60, 60, 13, 10)

followed by the data

Parameters:

Name	Type	Default	Description
<instanceNumber>	string	N/A	selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range. To know maxClients value use test command
<mId>	integer	N/A	message slot Id to be read. The read operation will free the slot resource. Value:

1÷30	:	message slot Id range

Additional info:

UNSOLICITED MESSAGE:

#MQRING - Received data on subscribed topic

When a message is received on the subscribed topic, an URC message is sent to all AT commands interfaces. There are 30 messages slots available for incoming messages, and it is responsibility of the user to keep them empty by reading them with #MQREAD.

If the message queue is full, and a new message arrives, the following URC #MQRING: 0 is received.

Otherwise, for normal messages, the URC format is:

#MQRING: <instanceNumber>,<mId>,<topic>,<len>

In the Unsolicited fields section are described the URC message parameters not described in the previous sections.

Unsolicited fields:

Name	Type	Description
<topic>	string	name of the topic from where the message was received
<len>	integer	length in bytes of the received payload

AT#MQREAD?

Read command returns the unread messages count for each instance number for all active MQTT clients in the format:

#MQREAD: <instanceNumber>,<unread>

AT#MQREAD=?

Test command reports the available range of values for parameters.

```
AT#MQREAD=1,2
#MQREAD: 1,myTopic,10
<<<
0123456789
```

Read command

```
AT#MQREAD?
#MQREAD: 1,0
#MQREAD: 2,3
OK
```

Test command

```
AT#MQREAD=?
#MQREAD: (1-maxClients),(1-30)
OK
```

AT#MQPUBSEXT - Publish Extension command

This command publishes data to the specified MQTT topic.

SIM Presence	Setting saved	Can be aborted
Required	Auto	No

AT#MQPUBSEXT=<instanceNumber>,<topic>,<retain>,<qos>,<dataLen>

After entering AT#MQPUBSEXT=... command, and terminated the command line with <CR>, the module returns the following three characters sequence prompt, and waits for data to send:

<greater_than><greater_than><greater_than> (see IRA character set: 62, 62, 62)

Now, data can be entered from TE, sized <dataLen> bytes. If user does not enters any input for 10 seconds, command is terminated with error.

If sending ends successfully and acknowledgment is received for QoS > 0, the response will be OK, otherwise an error will be reported.

Parameters:

Name	Type	Default	Description
<instanceNumber>	integer	N/A	selects the client instance to activate or deactivate Value: 1÷maxClients : client instance range. To know maxClients value use test command
<topic>	string	-	name of the topic. Maximum length of <topic> is 256

<retain>	integer	N/A	specifies if the broker must retain this message or not

Values:

0	:	Not retained
1	:	Retain

<qos>	integer	N/A	specifies the Quality of Service of this message

Value:

0÷2	:	Quality of Service range

<dataLen>	integer	-	Length of message to publish on the topic. Maximum length of <dataLen> can go up to 65536 depending on system memory available.

AT#MQPUBSEXT=?

Test command reports the available range of values for parameters.

Publish 100 byte message

AT#MQPUBSEXT=1,myTopic,0,0,100

>>>

OK

Test command

AT#MQPUBSEXT=?

#MQPUBSEXT: (1-maxClients),256,(0-1),(0-2),(0-65536)

OK

GNSS

GNSS Configuration

AT$GPSCFG - Set GNSS Configuration Parameters

This command sets the GNSS constellation

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT$GPSCFG=<parameter>,<value>

Parameters:

Name	Type	Default	Description
<parameter>	integer	N/A	selects the configuration parameter to be set Values: 0 : reserved 1 : reserved 2 : set constellation 3 : reserved 4 : reserved
<value>	integer	-	value depends on the first parameter. See Additional info section.

Additional info:

- <parameter>=2, in this case the <value> assumes the meaning of <constellation>.

Name	Type	Default	Description
<constellation>	integer	5	selects the constellation
	Values:		
	0 : reserved		
	1 : reserved		
	2 : reserved		
	3 : reserved		
	4 : GPS+QZSS		
	5 : GPS+GLONASS+GALILEO+QZSS		
	6 : GPS+BDS+GALILEO+QZSS		

AT$GPSCFG?

Read command returns current values of <reserved>, <reserved>, <constellation>, <reserved> and <reserved> in format:

$GPSCFG: <reserved>,<reserved>,<constellation>,<reserved>, <reserved>

AT$GPSCFG=?

Test command returns supported values range of <parameter>,<reserved>, <reserved>, <constellation>, <reserved> and <reserved>

AT$GPSNHZ - GNSS Navigation Update Rate Configuration

This command set the GNSS Navigation Update Rate.

SIM Presence	Setting saved	Can be aborted
Not required	Auto	No

AT$GPSNHZ=<update_rate>

Set command configures the GNSS Navigation Update Rate.

Parameter:

Name	Type	Default	Description
<update_rate>	integer	0	Navigation Update Rate

Values:

0	:	1 Hz (1000 milliseconds)
2	:	5 Hz (200 milliseconds)

AT$GPSNHZ?

Read command returns the currently update rate setting , in the format

$GPSNHZ: <update_rate>

AT$GPSNHZ=?

Test command reports the range of supported values for parameter <update_rate>.

$GPSNHZ: (the supported range of <update_rate>)

AT#GNSSCMD - GNSS Command

This command allows to control the GNSS through direct GNSS proprietary commands.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT#GNSSCMD=<gnssCommandString>

Sends the GNSS proprietary command <gnssCommandString> directly to the GNSS and reports GNSS answer/response in the form:

#GNSSRESP: <gnssResponseString>

followed by

OK

See also Additional info section

Parameter:

Name	Type	Default	Description
<gnssCommandString>	string	-	GNSS proprietary command string

Additional info:

<gnssResponseString> meaning

Name	Type	Default	Description
<gnssResponseString>	string	-	GNSS proprietary command answer/ response string

AT#GNSSCMD=?

Test command returns

OK

GNSS Receiver

AT$GPSP - GNSS Controller Power Management

This command powers on/off GNSS controller .

SIM Presence	Setting saved	Can be aborted
Not required	Other	No

AT$GPSP=<status>

The set command manages the power-up/power-down of the GNSS controller.

Parameter:

Name	Type	Default	Description
<status>	string	0	indicates the power status that has to be set.

Values:

0 : GNSS controller is powered down

1 : GNSS controller is powered up

Power down (AT$GPSP=0) cuts GPS power supply.

GPS operation mode is performed on the base of the current values of $GPSSLSR configuration (see $GPSSLSR).

AT$GPSP?

The read command reports the current value of the <status> parameter, in the format:

$GPSP: <status>

AT$GPSP=?

The test command reports the supported values range for parameter <status>.

GNSS controller is powered down

```
AT$GPSP=0
OK
```

GNSS General Management

AT$GPSSW - GNSS Software Version

This command provides the GNSS module software version.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSSW

Execution command returns the GNSS module software version in the format:

$GPSSW: <swVersion>

The command is available in "controlled mode" only, see $GPSD command.

GNSS module software version is available in few seconds at first GNSS module startup

AT$GPSSW?

Read command has the same behavior as the execution command.

AT$GPSSW=?

Test command returns the OK result code

AT$GPSSW

$GPSSW: R3.4.0.0Build7258

OK

AT$GPSR - Reset the GPS Controller

This command resets the GNSS controller.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSR=<resetType>

Execution command allows to reset the GNSS controller.

Parameter:

Name	Type	Default	Description
<resetType>	integer	3	set the type of GNSS controller reset.

Values:

0 : Factory Reset: this option clears all the GNSS memory including Clock Drift.

1 : Coldstart (No Almanac, No Ephemeris): this option clears all data that is currently stored in the internal memory of the GNSS receiver including Position, Almanac, Ephemeris and Time. The stored Clock Drift is retained.

2 : Warmstart (No ephemeris): this option clears all initialization data in the GNSS receiver and subsequently reloads the data that is currently displayed in the Receiver Initialization Setup screen. The Almanac is retained but the Ephemeris is cleared.

3 : Hotstart (with stored Almanac and Ephemeris): the GNSS receiver restarts by using all data that is currently stored in the internal memory of

the GNSS receiver: validated Ephemeris and Almanac.

- Factory Reset performs the same operation as Coldstart.
- <reset_type> sets the kind of start when GNSS is activated through $GPSP or $GPSSLSR commands.

AT$GPSR?

Read command displays the current <reset_type> value.

AT$GPSR=?

Test command reports the range of supported values for parameter <resetType>.

```
Factory reset
AT$GPSR=0
OK
```

GNSS Positioning Information

AT$GPSNMUN - Unsolicited NMEA Data Configuration

Unsolicited NMEA Data Configuration

[1] NMEA 0183 Standard

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSNMUN=<enable>[,<GGA>,<GLL>,<GSA>,<GSV>,<RMC>,<VTG>]

Set command permits to activate an Unsolicited streaming of GPS data (in NMEA format) through the standard GSM serial port and defines which NMEA sentences will be available.

Parameters:

Name	Type	Default	Description
<enable>	integer	0	Enables unsolicited GNSS data stream and selects one of the available GNSS data stream format display. See Additional info section to have information on GNSS data stream formats.

Values:

0 : NMEA data stream de-activated

1 : enable the first GNSS data stream format

2 : enable the second GNSS data stream format

3 : Dedicated NMEA data stream

<GGA>	integer	0	enables/disables the presence of the Global Positioning System Fix Data NMEA sentence (GGA) in the GNSS data stream.

Values:

0 : disable

1 : enable

<GLL>	integer	0	enable/disable the presence of the Geographic Position - Latitude/Longitude NMEA sentence (GLL) in the GNSS data stream.

Values:

0 : disable

1 : enable

<GSA>	integer	0	enable/disable the presence of the GNSS DOP and Active Satellites NMEA sentence (GSA) in the GNSS data stream.

Values:

0 : disable

	1	:	enable

<GSV>	integer	0	enable/disable the presence of the Satellites in View NMEA sentence (GSV) in the GNSS data stream.

Values:

0	:	disable
1	:	enable

<RMC>	integer	0	enable/disable the presence of the Recommended Minimum Specific GNSS Data NMEA sentence (RMC) in the GNSS data stream.

Values:

0	:	disable
1	:	enable

<VTG>	integer	0	enable/disable the presence of the GNSS Course Over Ground and Ground Speed NMEA sentence (VTG) in the GNSS data stream.

Values:

0	:	disable
1	:	enable

Additional info:

- <enable>=1, GNSS data stream format:

  $GPSNMUN: <NMEA SENTENCE 1><CR><LF>
  ...
  $GPSNMUN: <NMEA SENTENCE N><CR><LF>
  ...

- <enable>=2, GNSS data stream format:

  <NMEA SENTENCE 1><CR><LF>
  ...
  <NMEA SENTENCE N><CR><LF>
  ...

- <enable>=3, in this case, the AT interface port is dedicated to NMEA sentences, it is not possible to send AT commands. Use the escape sequence "+++" to return in command mode. GNSS data stream format:

  <NMEA SENTENCE 1><CR><LF>
  ...
  <NMEA SENTENCE N><CR><LF>
  ...

  The NMEA data stream format is the same as the one selected by <enable>=2.

AT$GPSNMUN?

Read command returns whether the unsolicited GPS NMEA data streaming is currently enabled or not, along with the NMEA sentences availability status, in the format:

$GPSNMUN:<enable>,<GGA>,<GLL>,<GSA>,<GSV>,<RMC>,<VTG >

AT$GPSNMUN=?

Test command returns the supported range of values for parameters:

<enable>, <GGA>, <GLL>, <GSA>, <GSV>, <RMC>, <VTG>.

All NMEA data stream is aligned with NMEA 0183 v4.10.

AT$GPSNMUN=1,0,0,1,0,0,0 OK

These sets the GSA as available sentence in the unsolicited message

AT$GPSNMUN=0 OK

Turn-off the unsolicited mode

AT$GPSNMUN?

$GPSNMUN: 1,0,0,1,0,0,0 OK

Give the current frame selected (GSA) The unsolicited message will be:

$GPSNMUN:

$GPGSA,A,3,23,20,24,07,13,04,02,,,,,,2.4,1.6,1.8°3C

AT$GPSACP - Get Acquired Position

This command returns information about the last GPS position.

[1] NMEA 0183 Standard

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSACP

Execution command returns information about the last GPS position in the format:

$GPSACP: <UTC>,<latitude>,<longitude>,<hdop>,<altitude>,<fix>,<cog>,<spkm>,<spkn>,<date>,<nsat_tot>

Additional info:

Meanings of the parameters returned by the command.

Name	Type	Default	Description
<UTC>	string	-	UTC time (hhmmss.sss) referred to GGA sentence
<latitude>	string	-	latitude in the format ddmm.mmmm N/S (referred to GGA sentence) where: dd: 00..90, degrees mm.mmmm: 00.0000..59.9999, minutes N/S: North/South

<longitude>	string	-	longitude in the format dddmm.mmmm E/W (referred to GGA sentence) where: ddd: 000..180, degrees mm.mmmm: 00.0000..59.9999, minutes E/W: East/West
<hdop>	string	-	Horizontal Dilution of Precision (referred to GGA sentence)
<altitude>	string	-	altitude - mean-sea-level (geoid) in meters (referred to GGA sentence)
<fix>	integer	N/A	fix type Values: 0 : invalid fix 1 : invalid fix 2 : 2D fix 3 : 3D fix
<cog>	string	-	Course over Ground (degrees, True) (referred to VTG sentence) in the format ddd.mm where:

			ddd: 000..360, degrees mm: 00..59, minutes
<spkm>	string	-	speed over ground (Km/hr) (referred to VTG sentence)
<spkn>	string	-	speed over ground (knots) (referred to VTG sentence)
<date>	string	-	date of fix (referred to RMC sentence) in the format ddmmyy where: dd: 01..31, day mm: 01..12, month yy: 00..99, year 2000 to 2099
<nsat_tot>	integer	N/A	total number of satellites used in fix solution (referred to GNGSA sentence)

Value:

0÷24	:	total number of satellites used in fix solution

AT$GPSACP?

Read command has the same behavior as the Execution command.

AT$GPSACP=?

Test command returns the OK result code.

AT$GPSP?

$GPSP: 0

After module is power-up there no aquired position

AT$GPSACP

$GPSACP:

OK

AT$GPSP=1 OK

Until first fix is received, the command will display no fix GPS position

AT$GPSACP

$GPSACP: ,,,,,1,,,,,,

Once fix has been received the command will display actual GPS position

OK

AT$GPSACP

$GPSACP:
135012.00,3913.69494N,00904.15382E,0.65,68.0,3,,0.197,0.106,010822,20

OK

AT$GPSNMUNEX - Unsolicited NMEA Extended Data Configuration

This command permits to activate an unsolicited streaming of GNSS data.

SIM Presence	Setting saved	Can be aborted
Not required	No	No

AT$GPSNMUNEX=<GNGNS>[,<GNGSA>[,<GLGSV>[,<GPGRS>[,<BDGSA>[,<BDGSV>[,<GAGSA>[,<GAGSV>[,<GPDTM>]]]]]]]]

Set command permits to activate an unsolicited streaming of GNSS data (in NMEA extended format) through the NMEA port and defines which NMEA extended sentences will be available.

Parameters:

Name	Type	Default	Description
<GNGNS>	integer	0	Fix data of GNSS receivers
	Value:		
	0	:	disable
<GNGSA>	integer	0	DOP and active satellites of GNSS
	Values:		
	0	:	disable
	1	:	enable
<GLGSV>	integer	0	GLONASS satellites in view

Values:

0	:	disable
1	:	enable

<GPGRS> integer 0 GPS Range Residuals

Value:

0	:	disable

<BDGSA> integer 0 Beidou DOP and Active Satellites

Value:

0	:	disable

<BDGSV> integer 0 Beidou Satellites in View

Values:

0	:	disable
1	:	enable

<GAGSA> integer 0 Galileo DOP and Active Satellites

Value:

0	:	disable

<GAGSV>	integer	0	Galileo Satellites in View

Values:

0	:	disable
1	:	enable

<GPDTM>	integer	0	Datum reference information

Value:

0	:	disable

NMEA extended data is displayed on NMEA port depending on $GPSNMUN <enable> parameter setting.

AT$GPSNMUNEX?

Read command returns the NMEA extended sentences availability status, in the format:

$GPSNMUNEX:
<GNGNS>,<GNGSA>,<GLGSV>,<GPGRS>,<BDGSA>,<BDGSV>,<GAGSA>,<GAGSV>,<GPDTM>

AT$GPSNMUNEX=?

Test command returns the supported range of values for parameters <GNGNS>,<GNGSA>,<GLGSV>,<GPGRS>,<BDGSA>,<BDGSV>,<GAGSA>,<GAGSV> and <GPDTM>.

- When GNSS data is being received, there are four consecutive GNGSA sentences. The sentences are arranged at the following order: gps, galileo, glonass, and beidou
- All NMEA data stream is aligned with NMEA0183 v4.10.

  $BD--- NMEA data stream (for Beidou)

  $GA--- NMEA data stream (for Galileo)
- Galileo satellites are identified by their PRN numbers, ranging 1 to 36
- Beidou satellites are identified by their PRN numbers, ranging 1 to 37
- GNSS System ID (the last field of $--GSA) are identified as GPS: 1, GLONASS: 2, GALILEO: 3, BEIDOU: 4.

AT$GPSNMUN=1

OK

AT$GPSNMUNEX=0,1,0,0,0,0,0,0,0

OK

These sets the GNGSA as available sentence in the unsolicited NMEA sentences.

AT$GPSNMUNEX?

$GPSNMUNEX: 0,1,0,0,0,0,0,0,0

OK

Give the current frame selected (GNGSA)

The unsolicited message will be:

$GPSNMUN: $GNGSA,A,3,05,15,23,24,10,12,,,,,,,2.61,2.22,1.37,1°07

IMS

AT+CIREG - IMS registration state

This command controls the presentation of an unsolicited result code when there is a change in the MT's IMS registration information.

3GPP TS 24.229

3GPP TS 24.173

3GPP TS 24.341

SIM Presence	Setting saved	Can be aborted
Not required	Common profile	No

AT+CIREG=[<mode>]

Unsolicited result code has the following format:

+CIREGU: <reg_info>[,<ext_info>]

Parameter:

Name	Type	Default	Description
<mode>	integer	0	enables or disables reporting of changes in the MT's IMS registration information.

Values:

0 : disable reporting

1 : enable reporting (parameter <reg_info>).

2 : enable extended reporting (parameters <reg_info> and <ext_info>).

Unsolicited fields:

Name	Type	Description
<reg_info>	integer	Indicates the IMS registration status. The UE is seen as registered if one or more of its public user identities are registered with any of its contact addresses, see 3GPP TS 24.229 Values: 0 : not registered. 1 : registered.
<ext_info>	integer	hexadecimal format. The value range is from 1 to FFFFFFFF. It is a sum of hexadecimal values, each representing a particular IMS capability of the MT. The MT can have IMS capabilites not covered by the below list. This parameter is not present if the IMS registration status is "not registered". Values: 1 : RTP-based transfer of voice according to MMTEL. 2 : RTP-based transfer of text according to MMTEL. 4 : SMS using IMS functionality.

8 : RTP-based transfer of video according to MMTEL.

- Parameter <mode> is saved in profile.
- AT+CIREG=6

  returns +CME ERROR: 50

AT+CIREG?

Read command reports the current state of IMS registration in the format:

+CIREG: <mode>,<reg_info>[,<ext_info>]

AT+CIREG=?

Test command returns the supported range of values for parameter <mode>.

Acronyms and Abbreviations

Table 8: Acronyms and Abbreviations

Acronym	Definition
ARFCN	Absolute Radio Frequency Channel Number
AT	Attention command
BA	BCCH Allocation
BCCH	Broadcast Control Channel
CA	Cell Allocation
CBM	Cell Broadcast Message
CBS	Cell Broadcast Service
CCM	Current Call Meter
CLIR	Calling Line Identification Restriction
CTS	Clear To Send
CUG	Closed User Group
DCD	Data Carrier Detect
DCE	Data Communication Equipment
DCS	Digital Cellular System
DGPS	Differential GPS, the use of GPS measurements, which are differentially corrected
DNS	Domain Name System
DSR	Data Set Ready
DTE	Data Terminal Equipment
DTMF	Dual Tone Multi Frequency

DTR	Data Terminal Ready
GGA	GPS Fix data
GLL	Geographic Position - Latitude/Longitude
GLONASS	Global positioning system maintained by the Russian Space Forces
GMT	Greenwich Mean Time
GNSS	Any single or combined satellite navigation system (GPS, GLONASS and combined GPS/GLONASS)
GPRS	Global Packet Radio Service
GPS	Global Positioning System
GSA	GPS DOP and Active satellites
GSM	Global System Mobile
GSV	GPS satellites in view
HDLC	High Level Data Link Control
HDOP	Horizontal Dilution of Precision
IMEI	International Mobile Equipment Identity
IMSI	International Mobile Subscriber Identity
IP	Internet Protocol
IRA	International Reference Alphabet
IWF	Interworking Function
ME	Mobile Equipment
MO	Mobile Originated
MT	either Mobile Terminated or Mobile Terminal
NMEA	National Marine Electronics Association
NVM	Non-Volatile Memory

PCS	Personal Communication Service
PDP	Packet Data Protocol
PDU	Packet Data Unit
PIN	Personal Identification Number
PPP	Point to Point Protocol
PUK	Pin Unblocking Code
RLP	Radio Link Protocol
RMC	Recommended minimum Specific data
RTS	Request To Send
SAP	SIM Access Profile
SCA	Service Center Address
SMS	Short Message Service
SMSC	Short Message Service Center
SMTP	Simple Mail Transport Protocol
TA	Terminal Adapter
TCP	Transmission Control Protocol
TE	Terminal Equipment
UDP	User Datagram Protocol
USSD	Unstructured Supplementary Service Data
UTC	Coordinated Universal Time
VDOP	Vertical dilution of precision
VTG	Course over ground and ground speed
WAAS	Wide Area Augmentation System

Related Documents

Refer to

https://dz.telit.com/

for current documentation and downloads.

Document History

Table 9: Document History

Revision	Date	Changes
7	2025-07-16	Updated Applicability Table Added Commands: AT#CSURV, AT#CSURVF, AT#FTPAPP, AT#FTPAPPEXT, AT#FTPMSG Updated Commands: AT#FTPCFG, AT#FTPPUT, AT#FTPSIZE, AT#HTTPRCV, AT#SSLSECCFG, AT#SSLSECCFG2, AT#TESTMODE, AT#TESTMODE2G, AT+ZFOTADL Removed commands: AT#TXCAL4GD, AT#TXCALD
6	2024-03-07	Updated Applicability Table Added commands: AT#CEER, AT#CFLO, AT#CGMI, AT#CODEC, AT#DNS, AT#GDATAVOL, AT#HFMICG, AT#MQCFG, AT#MQCFG2, AT#MQCONN, AT#MQDISC, AT#MQEN, AT#MQPUBS, AT#MQPUBSEXT, AT#MQREAD, AT#MQSUB, AT#MQTCFG, AT#MQUNS, AT#MQWCFG, AT#NWDNS, AT#WAKE, AT&N, AT&Z, AT+CALA Updated commands: AT#ASIZE, AT#SCFGEXT3, AT+CLCC, AT+CRLP, ATD
5	2024-01-16	Added commands: AT#OAP, AT#TESTMODE, AT#TESTMODE2G Updated commands: AT+VTD, AT+VTS
4	2023-09-25	Updated Applicability Table Added commands: AT#DUALAPN, AT#GNSSCMD, AT$GPSACP, AT$GPSCFG, AT$GPSNHZ, AT$GPSNMUN, AT$GPSNMUNEX, AT$GPSP, AT$GPSR, AT$GPSSW

		Updated commands: AT#DVIEXT, AT#GPIO, AT#USBCFG
3	2023-06-29	New document template Updated Applicability Table Added commands: AT#ADELA, AT#ADELF, AT#AFIND, AT#ALIST, AT#APLAY, AT#ASEND, AT#ASIZE, AT#DIGMICG, AT#DVI, AT#DVICLK, AT#DVIEXT, AT#I2CCF, AT#I2CRD, AT#I2CWR, AT#SI, AT#TSVOL Updated commands: AT#BCCHLOCK, AT#BND, AT#CEERNET, AT#CGMM, AT#CGMR, AT#CIMI, AT#DIALMODE, AT#DTMF, AT#DTR, AT#ECAM, AT#FRWL, AT#GPIO, AT#ICMP, AT#JDR4GCFG, AT#JDRENH2, AT#PING, AT#PORTCFG, AT#PSNT, AT#QDNS, AT#QSS, AT#SGACT, AT#SIMDET, AT#SMSMODE, AT#SSLSECDATA, AT#TEMPCFG, AT#TEMPMON, AT#USBCFG, AT&D, AT&K, AT+CCID, AT+CFUN, AT+CGSMS, AT+CIND, AT+CLCC, AT+CMER, AT+CNMI, AT+CNUM, AT+CPMS, AT+CSIL, AT+CSSN, AT+IFC, AT+ZFOTADL, ATD, ATE
2	2022-12-20	Updated Applicability Table Added commands: AT#BCCHLOCK, AT#BND, AT#CEERNET, AT#CGMM, AT#CGMR, AT#CIMI, AT#DIALMODE, AT#DTMF, AT#DTR, AT#ECAM, AT#FRWL, AT#HWREV, AT#ICMP, AT#PING, AT#PORTCFG, AT#PSNT, AT#QDNS, AT#QSS, AT#RPMCFG, AT#RPMCFGEXT, AT#SI, AT#SMSMODE, AT#TEMPCFG, AT#TEMPMON, AT#USBCFG, AT&K, AT+CCID, AT+CMER, AT+CNUM, AT+CSIL, AT+DUALSIM, AT+ZFOTADL Updated commands: AT#GPIO, AT#HTTPCFG, AT#MSCLASS, AT#SIMDET, AT#TXCAL4GD, AT#TXCALD, AT&D, AT+CFUN, AT+CGDCONT, AT+CIND, AT+CMGD, AT+CMGL, AT+CMGR, AT+CMGS, AT+CMGW, AT+CMSS, AT+CNMI, AT+CSCA, AT+CSCS, AT+IFC
1	2022-09-14	Added commands: AT#E2SMSRI, AT#FTPCWD, AT#FTPGETPKT, AT#FTPLIST, AT#FTPRECV, AT#JDR4GCFG, AT#JDRENH2, AT#MSCLASS, AT#NITZ, AT#PDPAUTH, AT#RFSTS, AT#SGACTCFG, AT#SSLCFG, AT#SSLD, AT#SSLEN, AT#SSLH, AT#SSLI, AT#SSLO, AT#SSLRECV, AT#SSLS, AT#SSLSECCFG, AT#SSLSECCFG2,

		AT#SSLSECDATA, AT#SSLSECDATAEXT, AT#SSLSEND, AT#SSLSENDEXT, AT#STGI, AT#STIA, AT#STSR, AT#TXCAL4GD, AT#TXCALD, AT#V24, AT#V24CFG, AT&C, AT&P, AT&V, AT&W, AT&Y, AT+CIND Updated commands: AT#CMUXMODE, AT#HTTPQRY, AT#HTTPSND, AT#SCFGEXT, AT#SIMDET, AT#SS, AT+CEMODE, AT+CESQ, AT+CFUN, AT+CGDCONT, AT+CGDSCONT, AT+CGEQOSRDP, AT+CPBS, AT+CREG, AT+CRLP, AT+CSCS, AT+IFC, AT+SWITCHSIM, ATD, ATX
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